

# A Physical Introduction to Higher Mathematics

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# 1 Spectral Theory

**Exercise 1.1** (Vector spaces, duality, trace, and dimension). Let  $k$  be a field. A *vector space* over  $k$  is an abelian group  $V$  with scalar multiplication satisfying

$$a(v + w) = av + aw, \quad (a + b)v = av + bv, \quad (ab)v = a(bv), \quad 1v = v.$$

A map  $T : V \rightarrow W$  is *linear* when  $T(av + bw) = aT(v) + bT(w)$ . Its kernel and image are

$$\ker T = \{v : T(v) = 0\}, \quad \operatorname{im} T = \{T(v) : v \in V\}.$$

The *dual space* of  $V$  is  $V^\vee = \operatorname{Hom}_k(V, k)$ . Its canonical evaluation pairing is

$$\operatorname{ev}_V : V^\vee \times V \longrightarrow k, \quad (\varphi, v) \longmapsto \varphi(v).$$

The *tensor product*  $V \otimes W$  is characterized by a bilinear map  $(v, w) \mapsto v \otimes w$  with the following universal property: every bilinear map  $b : V \times W \rightarrow X$  factors uniquely as

$$V \times W \longrightarrow V \otimes W \xrightarrow{\tilde{b}} X.$$

The evaluation pairing induces a linear map  $\operatorname{ev}_V : V^\vee \otimes V \rightarrow k$ .

A vector space  $V$  is *finite-dimensional* when there is a coevaluation map

$$\operatorname{coev}_V : k \longrightarrow V \otimes V^\vee$$

satisfying the snake identities

$$(\operatorname{id}_V \otimes \operatorname{ev}_V) \circ (\operatorname{coev}_V \otimes \operatorname{id}_V) = \operatorname{id}_V,$$

$$(\operatorname{ev}_V \otimes \operatorname{id}_{V^\vee}) \circ (\operatorname{id}_{V^\vee} \otimes \operatorname{coev}_V) = \operatorname{id}_{V^\vee}.$$

Let  $s_{V, V^\vee} : V \otimes V^\vee \rightarrow V^\vee \otimes V$  exchange the factors. For  $T \in \operatorname{End}(V)$ , define

$$\operatorname{tr}(T) = \operatorname{ev}_V \circ s_{V, V^\vee} \circ (T \otimes \operatorname{id}_{V^\vee}) \circ \operatorname{coev}_V.$$

The *dimension* of  $V$  is the scalar

$$\dim_k(V) = \operatorname{tr}(\operatorname{id}_V) \in k.$$

(a) Let  $c, d : k \rightarrow V \otimes V^\vee$  both satisfy the snake identities with  $\operatorname{ev}_V$ . Apply

$$\operatorname{id}_V \otimes \operatorname{ev}_V \otimes \operatorname{id}_{V^\vee}$$

to  $d \otimes c : k \rightarrow V \otimes V^\vee \otimes V \otimes V^\vee$  and use the snake identity for  $d$  to identify the result with  $c$ .

Use the snake identity for  $c$  to identify the same result with  $d$ , and deduce  $c = d$ .

(b) Prove that  $T \mapsto \operatorname{tr}(T)$  is linear and that

$$\operatorname{tr}(ST) = \operatorname{tr}(TS) \quad (S, T \in \operatorname{End}(V)).$$

**Exercise 1.2** (Exterior algebra, rank, and determinant). For  $m \geq 0$ , the  $m$ th *exterior power* of a vector space  $V$  is

$$\Lambda^m V = V^{\otimes m} / \langle v_1 \otimes \cdots \otimes v_m : v_i = v_j \text{ for some } i \neq j \rangle,$$

and the image of  $v_1 \otimes \cdots \otimes v_m$  is written  $v_1 \wedge \cdots \wedge v_m$ . An endomorphism  $T$  induces

$$\Lambda^m T(v_1 \wedge \cdots \wedge v_m) = T v_1 \wedge \cdots \wedge T v_m.$$

The products

$$\Lambda^r V \otimes \Lambda^s V \longrightarrow \Lambda^{r+s} V, \quad (u_1 \wedge \cdots \wedge u_r) \otimes (v_1 \wedge \cdots \wedge v_s) \longmapsto u_1 \wedge \cdots \wedge u_r \wedge v_1 \wedge \cdots \wedge v_s$$

make  $\Lambda^\bullet V = \bigoplus_{m \geq 0} \Lambda^m V$  the *exterior algebra* of  $V$ .

For a finite-dimensional vector space  $V$ , an integer  $n \in \mathbb{Z}_{\geq 0}$  satisfying

$$\Lambda^n V \neq 0, \quad \Lambda^{n+1} V = 0$$

is its *exterior rank*, written  $\text{rank}(V) = n$ , when it exists and is unique. When  $\text{rank}(V) = n$ , the *determinant* of  $T \in \text{End}(V)$  is the scalar  $\det(T)$  satisfying

$$\Lambda^n T = \det(T) \text{id}_{\Lambda^n V}.$$

Let  $V$  be finite-dimensional.

(a) Prove that the exterior algebra is graded-commutative:

$$\xi \wedge \eta = (-1)^{rs} \eta \wedge \xi, \quad \xi \in \Lambda^r V, \quad \eta \in \Lambda^s V.$$

(b) Prove that the exterior rank exists and is unique, that every  $\Lambda^m V$  is finite-dimensional, and that every endomorphism of  $\Lambda^n V$  is a unique scalar multiple of the identity.

(c) Prove

$$\dim_k(V) = n \text{ } 1_k, \quad \dim_k(\Lambda^m V) = \binom{n}{m} 1_k, \quad \Lambda^m V = 0 \quad (m > n).$$

Deduce that  $\dim_k(V)$  determines  $n$  when  $\text{char } k = 0$ , whereas it determines only the residue class of  $n$  modulo  $p$  when  $\text{char } k = p > 0$ .

(d) Prove that the determinant is well-defined. Use functoriality of the highest exterior power to prove

$$\det(ST) = \det(S) \det(T), \quad T \text{ is invertible} \iff \det(T) \neq 0.$$

(e) The coevaluation, trace, and exterior-power constructions extend under scalar extension to the finite free  $k[z]$ -module  $V[z] = V \otimes_k k[z]$ ; in particular,  $\Lambda_{k[z]}^n V[z]$  is free of rank one. Define the determinant of a  $k[z]$ -linear endomorphism through its action on this top exterior power. For  $T \in \text{End}(V)$  prove

$$\det(I + zT) = \sum_{m=0}^n \text{tr}(\Lambda^m T) z^m.$$

Recover  $(1 + z)^n = \sum_m \binom{n}{m} z^m$  by taking  $T = I$ .

**Exercise 1.3** (Rising factorials and the binomial series). For  $a \in \mathbb{C}$  and  $m \in \mathbb{Z}_{\geq 0}$ , define

$$(a)_0 = 1, \quad (a)_m = a(a+1) \cdots (a+m-1) \quad (m \geq 1).$$

For  $|z| < 1$ , define the *binomial series*

$$B_a(z) = \sum_{m=0}^{\infty} \frac{(a)_m}{m!} z^m.$$

Prove that the binomial series converges for  $|z| < 1$  and satisfies

$$(1-z)B'_a(z) = aB_a(z), \quad B_a(0) = 1.$$

For the branch normalized to equal 1 at  $z = 0$ , deduce

$$B_a(z) = (1-z)^{-a}.$$

For  $a = -n$ ,  $n \in \mathbb{Z}_{\geq 0}$ , prove that the series terminates and, after replacing  $z$  by  $-z$ , recover

$$(1+z)^n = \sum_{m=0}^n \binom{n}{m} z^m.$$

**Exercise 1.4** (Hilbert spaces and bounded operators). An *inner product* on a complex vector space is conjugate-linear in its first argument, linear in its second, positive definite, and satisfies  $\langle x, y \rangle = \overline{\langle y, x \rangle}$ . Its norm is  $\|x\| = \sqrt{\langle x, x \rangle}$ . A *Hilbert space* is an inner-product space complete for this norm.

Vectors are *orthogonal* when their inner product is zero. For a subspace  $M \subseteq H$ , define

$$M^\perp = \{x : \langle m, x \rangle = 0 \text{ for every } m \in M\}.$$

A linear map  $T : H \rightarrow K$  between Hilbert spaces is *bounded* when

$$\|T\| = \sup_{\|x\| \leq 1} \|Tx\| < \infty.$$

Write  $B(H, K)$  for the bounded operators and  $B(H) = B(H, H)$ . An *adjoint* of  $T$  is an operator  $T^* : K \rightarrow H$  satisfying

$$\langle x, Ty \rangle = \langle T^*x, y \rangle.$$

An *orthogonal projection* is an operator  $P$  satisfying  $P = P^* = P^2$ .

- (a) Prove the Cauchy–Schwarz inequality, the projection theorem, and the Riesz representation theorem. Deduce the existence of adjoints and prove

$$\|T\|^2 = \|T^*T\|, \quad (\text{im } T)^\perp = \ker T^*.$$

- (b) For every closed subspace  $M \subseteq H$ , prove  $H = M \oplus M^\perp$  and construct the orthogonal projection  $P_M : H \rightarrow M$ .

**Exercise 1.5** ( $L^p$  geometry). For a measure space  $(X, \mathcal{F}, \mu)$  and  $1 \leq p < \infty$ , define

$$L^p(X, \mu) = \left\{ f : X \rightarrow \mathbb{C} \text{ measurable} : \int_X |f|^p d\mu < \infty \right\} / \text{equality a.e.},$$

$$\|f\|_p = \left( \int_X |f|^p d\mu \right)^{1/p}.$$

Let  $L^\infty(X, \mu)$  consist of essentially bounded measurable functions modulo equality almost everywhere, with

$$\|f\|_\infty = \operatorname{ess\,sup}_{x \in X} |f(x)|.$$

Prove Hölder's and Minkowski's inequalities and deduce that  $L^p(X, \mu)$  is complete for  $1 \leq p \leq \infty$ . For  $1 < p < \infty$  with  $p^{-1} + q^{-1} = 1$ , assume that  $(X, \mathcal{F}, \mu)$  is  $\sigma$ -finite and prove that

$$L^q(X, \mu) \longrightarrow L^p(X, \mu)^*, \quad g \longmapsto \left( f \mapsto \int_X \bar{g}f \, d\mu \right)$$

is an isometric isomorphism.

**Exercise 1.6** (Gamma and beta integrals). For  $s, t > 0$ , define

$$\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} \, dx, \quad B(s, t) = \int_0^1 x^{s-1} (1-x)^{t-1} \, dx.$$

Use Tonelli's theorem and the change of variables  $x = ru$ ,  $y = r(1-u)$  to prove

$$B(s, t) = \frac{\Gamma(s)\Gamma(t)}{\Gamma(s+t)}.$$

Use integration by parts to prove  $\Gamma(s+1) = s\Gamma(s)$ , and use the pushforward of Euclidean measure under the norm map to obtain

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \int_{\mathbb{R}^n} e^{-|x|^2/2} \, dx = (2\pi)^{n/2}.$$

State the uses of Tonelli, Fubini, monotone convergence, and dominated convergence in the derivation.

**Exercise 1.7** ( $C^*$ -algebras and spectra). A  $C^*$ -algebra is a complete normed complex algebra  $A$  with a conjugate-linear involution satisfying

$$(ab)^* = b^*a^*, \quad \|ab\| \leq \|a\|\|b\|, \quad \|a^*a\| = \|a\|^2.$$

For a unital  $C^*$ -algebra, define

$$\operatorname{spec}_A(a) = \{\lambda \in \mathbb{C} : a - \lambda 1 \text{ is not invertible}\}.$$

An element is *normal* when  $a^*a = aa^*$ .

Prove that  $\operatorname{spec}_A(a)$  is nonempty and compact and that

$$r(a) = \lim_{n \rightarrow \infty} \|a^n\|^{1/n}.$$

For normal  $a$ , prove  $r(a) = \|a\|$ .

**Exercise 1.8** (Commutative Gelfand theory). For a commutative  $C^*$ -algebra  $A$ , a *character* is a nonzero homomorphism  $\chi : A \rightarrow \mathbb{C}$ . Let  $\hat{A}$  be the space of characters with its weak-\* topology. The *Gelfand transform* is

$$\Gamma : A \longrightarrow C_0(\hat{A}), \quad \Gamma(a)(\chi) = \chi(a).$$

Use as theorem input the commutative Gelfand theorem [11]: every character is continuous,  $\hat{A}$  is locally compact Hausdorff, and the Gelfand transform is an isometric \*-isomorphism  $A \cong C_0(\hat{A})$ . Deduce that  $A$  is unital exactly when  $\hat{A}$  is compact.

**Exercise 1.9** (Continuous functional calculus). An element  $a$  of a  $C^*$ -algebra is *self-adjoint* when  $a = a^*$  and *positive*, written  $a \geq 0$ , when  $a = b^*b$  for some  $b$ .

For every normal element  $a$  of a unital  $C^*$ -algebra, apply commutative Gelfand theory to  $C^*(1, a)$  and prove that there is a unique unital isometric  $*$ -homomorphism

$$C(\text{spec}(a)) \longrightarrow C^*(1, a), \quad f \longmapsto f(a),$$

sending the coordinate function to  $a$ , and prove the continuous spectral mapping theorem. Deduce that positive elements have unique positive square roots and that  $a \geq 0$  if and only if  $a = a^*$  and  $\text{spec}_A(a) \subseteq [0, \infty)$ .

**Exercise 1.10** (Positive functionals and scalar measures). For a locally compact Hausdorff space  $X$ , a linear functional  $\varphi : C_c(X) \rightarrow \mathbb{C}$  is *positive* when  $\varphi(f) \geq 0$  for every real-valued  $f \geq 0$ .

Let  $X$  be locally compact Hausdorff. Use as theorem input the positive Riesz–Markov theorem [18]: every positive linear functional  $\varphi : C_c(X) \rightarrow \mathbb{C}$  that is bounded on functions supported in each compact set has a unique regular Borel measure  $\mu_\varphi$  satisfying

$$\varphi(f) = \int_X f d\mu_\varphi.$$

**Exercise 1.11** (Complex Riesz–Markov representation). Use as theorem input the complex Riesz–Markov theorem [18]: every  $\varphi \in C_0(X)^*$  has a unique finite complex regular measure  $\mu_\varphi$  such that

$$\varphi(f) = \int_X f d\mu_\varphi, \quad \|\varphi\| = |\mu_\varphi|(X).$$

Deduce the positive theorem for functionals bounded in the  $C_0(X)$  norm and characterize positive functionals by positive measures.

**Exercise 1.12** (Spectral distributions of states). For a unital  $C^*$ -algebra, a positive linear functional is one satisfying  $\varphi(a^*a) \geq 0$  for every  $a$ , and a *state* is a positive functional with  $\varphi(1) = 1$ .

Let  $A$  be a unital  $C^*$ -algebra, let  $a = a^* \in A$ , and let  $\varphi$  be a state. Apply the continuous functional calculus and Riesz–Markov theorem to prove that there is a unique Borel probability measure  $\mu_{\varphi, a}$  on  $\text{spec}(a)$  such that

$$\varphi(f(a)) = \int_{\text{spec}(a)} f(\lambda) d\mu_{\varphi, a}(\lambda), \quad f \in C(\text{spec}(a)).$$

Deduce

$$\varphi(a) = \int \lambda d\mu_{\varphi, a}(\lambda), \quad \varphi(a^2) - \varphi(a)^2 = \int (\lambda - \varphi(a))^2 d\mu_{\varphi, a}(\lambda).$$

**Exercise 1.13** ( $C_0(X)$  representations). A *projection-valued measure* on a measurable space  $X$  is a map  $E$  from measurable subsets of  $X$  to orthogonal projections on  $H$  satisfying

$$E(\emptyset) = 0, \quad E(X) = I, \quad E\left(\bigsqcup_{j=1}^{\infty} B_j\right) = \sum_{j=1}^{\infty} E(B_j)$$

for pairwise disjoint measurable sets, where the sum converges in the strong operator topology. The spectral integral  $\int_X f dE$  for bounded measurable  $f$  is defined by uniform approximation from simple functions. It is regular when each scalar measure  $B \mapsto \langle x, E(B)y \rangle$  is a regular Borel measure.

Let  $X$  be locally compact Hausdorff. A representation  $\pi : C_0(X) \rightarrow B(H)$  is *nondegenerate* when  $\overline{\pi(C_0(X))H} = H$ . Prove that every nondegenerate  $*$ -representation determines scalar measures by applying complex Riesz–Markov to  $f \mapsto \langle x, \pi(f)y \rangle$ . Use polarization and multiplicativity to assemble them into a unique regular projection-valued measure  $E_\pi$  satisfying

$$\pi(f) = \int_X f(x) dE_\pi(x), \quad f \in C_0(X).$$

**Exercise 1.14** (Bounded normal spectral theorem). For a normal operator  $T \in B(H)$ , apply the  $C_0(X)$  representation theorem to the continuous functional calculus  $C(\text{spec}(T)) \rightarrow B(H)$  and prove that there is a unique projection-valued measure  $E_T$  on  $\text{spec}(T)$  satisfying

$$T = \int_{\text{spec}(T)} z dE_T(z), \quad f(T) = \int_{\text{spec}(T)} f(z) dE_T(z).$$

Deduce that a normal operator on a finite-dimensional Hilbert space has the intrinsic spectral decomposition

$$H = \bigoplus_{\lambda \in \text{spec}(T)} E_T(\{\lambda\})H, \quad T|_{E_T(\{\lambda\})H} = \lambda I.$$

**Exercise 1.15** (Compact spectral theorem). An operator  $K \in B(H)$  is compact when it maps the closed unit ball to a set with compact closure.

For a compact normal operator  $K$ , prove that every nonzero spectral value is an eigenvalue of finite multiplicity, that the nonzero eigenvalues can accumulate only at zero, and that

$$K = \sum_{\lambda \in \text{spec}(K) \setminus \{0\}} \lambda E_K(\{\lambda\})$$

in operator norm, and prove that the spectral subspaces in the displayed sum are mutually orthogonal.

**Exercise 1.16** (Unbounded self-adjoint operators and the Cayley transform). An unbounded operator is a linear map  $A : \mathcal{D}(A) \rightarrow H$  on a dense domain. Its adjoint has domain

$$\mathcal{D}(A^*) = \{y \in H : x \mapsto \langle y, Ax \rangle \text{ is bounded on } \mathcal{D}(A)\}.$$

For  $y \in \mathcal{D}(A^*)$ , the vector  $A^*y$  is determined by  $\langle A^*y, x \rangle = \langle y, Ax \rangle$  for  $x \in \mathcal{D}(A)$ . The operator is *self-adjoint* when  $A = A^*$ , including equality of domains.

For a self-adjoint operator  $A$ , prove that  $A \pm iI$  are bijective from  $\mathcal{D}(A)$  to  $H$  with bounded inverses and that the Cayley transform

$$C_A = (A - iI)(A + iI)^{-1}$$

is unitary with  $\ker(I - C_A) = 0$ . Prove conversely that every unitary  $C$  with  $\ker(I - C) = 0$  determines a self-adjoint operator through the inverse Cayley transform.

**Exercise 1.17** (Unbounded spectral theorem). Apply the bounded normal spectral theorem to  $C_A$  and transport its projection-valued measure by the inverse Cayley map. Prove that there is a unique projection-valued measure  $E_A$  on  $\mathbb{R}$  satisfying

$$A = \int_{\mathbb{R}} \lambda dE_A(\lambda), \quad \mathcal{D}(A) = \left\{ x : \int_{\mathbb{R}} \lambda^2 d\langle x, E_A(\lambda)x \rangle < \infty \right\}.$$

**Exercise 1.18** (Stone's theorem). Use the measurable functional calculus of  $E_A$  to prove that  $e^{-itA}$  is a strongly continuous unitary group with generator  $-iA$ . Conversely, let  $(U_t)_{t \in \mathbb{R}}$  be a strongly continuous unitary group and define

$$\mathcal{D}(A) = \left\{ x : \lim_{t \rightarrow 0} i \frac{U_t x - x}{t} \text{ exists} \right\}, \quad Ax = \lim_{t \rightarrow 0} i \frac{U_t x - x}{t}.$$

Prove that  $\mathcal{D}(A)$  is dense and that  $A$  is symmetric. Use the strong integrals

$$i \int_0^\infty e^{-t} U_t dt, \quad -i \int_{-\infty}^0 e^t U_t dt$$

to prove that  $A - iI$  and  $A + iI$  are surjective. Deduce that  $A$  is self-adjoint, prove  $U_t = e^{-itA}$ , and prove uniqueness of  $A$ .

**Exercise 1.19** (Spring and string frequencies). Let  $V$  be a finite-dimensional real vector space with positive-definite mass form  $m$  and nonnegative stiffness form  $k$ . Use  $m$  to define the self-adjoint operator  $A$  by  $k(x, y) = m(x, Ay)$ . Prove that solutions of  $\ddot{x} + Ax = 0$  decompose intrinsically through the spectral projections of  $A$  and that the observed angular frequencies are  $\sqrt{\lambda}$  for  $\lambda \in \text{spec}(A) \setminus \{0\}$ . For a stretched string of length  $L$ , wave speed  $c$ , and fixed endpoints, let  $D$  be the self-adjoint operator on  $L^2((0, L); \mathbb{C})$  given by

$$D\phi = -c^2 \phi'',$$

on the domain of functions  $\phi$  for which  $\phi$  and  $\phi'$  are absolutely continuous on  $[0, L]$ ,  $\phi'' \in L^2((0, L))$ , and  $\phi(0) = \phi(L) = 0$ . For the wave equation

$$\partial_t^2 u + Du = 0,$$

prove that a nonzero time-harmonic solution  $u(t, x) = \text{Re}(e^{i\omega t} \phi(x))$ , with  $\omega > 0$ , exists exactly when, for some  $n \geq 1$ ,

$$\omega = \omega_n = \frac{n\pi c}{L}, \quad \phi \in (\mathbb{C}\phi_n) \setminus \{0\}, \quad \phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right),$$

Deduce the harmonic resonance ratios and the nodal locations measured by driving the string at each  $\omega_n$ .

**Exercise 1.20** (Spectral lines and time evolution). Let  $A$  be self-adjoint, let  $B \in B(H)$ , and set  $B(t) = e^{itA/\hbar} B e^{-itA/\hbar}$ . For eigenvalues  $\lambda, \mu$  of  $A$ , write  $P_\lambda = E_A(\{\lambda\})$  and prove

$$P_\mu B(t) P_\lambda = e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

Use the functional calculus to prove the factorized weak identity

$$B(t) = \left( \int_{\mathbb{R}} e^{it\mu/\hbar} dE_A(\mu) \right) B \left( \int_{\mathbb{R}} e^{-it\lambda/\hbar} dE_A(\lambda) \right).$$

When  $A$  has pure-point spectrum, deduce in the weak operator topology that

$$B(t) = \sum_{\lambda, \mu} e^{it(\mu-\lambda)/\hbar} P_\mu B P_\lambda.$$

For an initial state  $\psi$ , define the line strength

$$I_{\mu\lambda}(\psi) = \|P_\mu B P_\lambda \psi\|^2.$$

Prove that a line at  $(\mu - \lambda)/\hbar$  occurs only when this quantity is nonzero, and identify vanishing spectral blocks as selection rules.



**Exercise 1.21** (Hermite polynomials and functions). Define the physicists' Hermite polynomials by

$$e^{2xt-t^2} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}.$$

The normalized *Hermite functions* are

$$h_n(x) = \frac{e^{-x^2/2} H_n(x)}{(2^n n! \sqrt{\pi})^{1/2}}.$$

Use the generating function and the Gaussian integral to prove that  $(h_n)_{n \geq 0}$  is an orthonormal family in  $L^2(\mathbb{R})$ . Derive the Hermite differential equation and recurrence relations.

**Exercise 1.22** (Mehler kernel). For  $|r| < 1$ , use the Hermite generating function and Gaussian integration to derive Mehler's identity

$$\sum_{n=0}^{\infty} r^n h_n(x) h_n(y) = \frac{1}{\sqrt{\pi(1-r^2)}} \exp\left(-\frac{(1+r^2)(x^2+y^2)-4rxy}{2(1-r^2)}\right).$$

**Exercise 1.23** (Hermite completeness). Let  $K_r$  be the integral operator with the Mehler kernel for  $0 < r < 1$ . Use its explicit Gaussian form to prove

$$K_r f \longrightarrow f \quad \text{in } L^2(\mathbb{R}) \quad (r \uparrow 1).$$

Prove that the image of  $K_r$  lies in the closed linear span of the Hermite functions and deduce that  $(h_n)_{n \geq 0}$  is complete in  $L^2(\mathbb{R})$ .

**Exercise 1.24** (Oscillator spectroscopy). The *Schwartz space*  $\mathcal{S}(\mathbb{R})$  consists of the smooth functions  $\psi$  satisfying

$$\sup_{x \in \mathbb{R}} (1 + |x|)^m |\psi^{(r)}(x)| < \infty$$

for all  $m, r \geq 0$ . On  $\mathcal{S}(\mathbb{R})$  let

$$(Q\psi)(x) = x\psi(x), \quad (P\psi)(x) = -i\hbar\psi'(x).$$

For

$$\hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dx^2} + \frac{M\omega^2 x^2}{2},$$

use the Hermite functions to prove that

$$\psi_n(x) = \left(\frac{M\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{M\omega}{\hbar}}x\right), \quad E_n = \hbar\omega\left(n + \frac{1}{2}\right)$$

give the complete spectral decomposition of  $\hat{H}$ . Construct the creation and annihilation operators and compute

$$\langle \psi_k, Q\psi_n \rangle = \sqrt{\frac{\hbar}{2M\omega}} (\sqrt{n+1} \delta_{k,n+1} + \sqrt{n} \delta_{k,n-1}).$$

Deduce the spectral-line prediction  $|E_{n \pm 1} - E_n|/\hbar = \omega$  and the relative line strengths for a linearly coupled oscillator. For  $\alpha \in \mathbb{C}$ , prove convergence and normalization of

$$\psi_\alpha = e^{-|\alpha|^2/2} \sum_{n \geq 0} \frac{\alpha^n}{\sqrt{n!}} \psi_n,$$

show that its number statistics are Poisson with mean  $|\alpha|^2$ , and derive its sinusoidal position expectation. Analytically continue Mehler's identity with the usual  $t - i0$  prescription to obtain the oscillator propagator for  $t \notin (\pi/\omega)\mathbb{Z}$ ,

$$K_t(x, y) = \sqrt{\frac{M\omega}{2\pi i \hbar \sin(\omega t)}} \exp\left[\frac{iM\omega}{2\hbar \sin(\omega t)}((x^2 + y^2) \cos(\omega t) - 2xy)\right].$$

Determine the Maslov phase obtained by continuation across each caustic. Prove

$$e^{-2\pi i \hat{H}/(\hbar\omega)} = -I, \quad e^{-4\pi i \hat{H}/(\hbar\omega)} = I,$$

and distinguish the projective and observable revival period  $2\pi/\omega$  from the exact operator period  $4\pi/\omega$ . Use these formulas to predict the spectral spacing, coherent oscillation frequency, occupation-number distribution, and revival period measured in an oscillator experiment.

**Exercise 1.25** (Airy functions and quantum-bouncer spectroscopy). The Airy function  $\text{Ai}$  is the unique real solution of

$$\text{Ai}''(x) = x \text{Ai}(x)$$

normalized by

$$\text{Ai}(x) \sim \frac{x^{-1/4}}{2\sqrt{\pi}} e^{-2x^{3/2}/3} \quad (x \rightarrow +\infty).$$

Let  $a_n > 0$  be determined by  $\text{Ai}(-a_n) = 0$ , ordered increasingly.

On  $L^2((0, \infty), dz)$  consider the self-adjoint Dirichlet realization

$$\hat{H} = -\frac{\hbar^2}{2M} \frac{d^2}{dz^2} + Mgz, \quad \psi(0) = 0.$$

With

$$\ell_g = \left( \frac{\hbar^2}{2M^2 g} \right)^{1/3},$$

derive

$$E_n = Mg\ell_g a_n, \quad \psi_n(z) = C_n \text{Ai}\left(\frac{z}{\ell_g} - a_n\right).$$

Let a weak drive  $\epsilon B \cos(\Omega t)$  act for an observation time  $T_{\text{obs}}$ . In first-order time-dependent perturbation theory, use the spectral projections of  $\hat{H}$  to derive, for  $k \neq n$ ,

$$c_{k \leftarrow n}^{(1)}(T_{\text{obs}}) = -\frac{i\epsilon B_{kn}}{2\hbar} \left[ \frac{e^{i(\omega_{kn} + \Omega)T_{\text{obs}}} - 1}{i(\omega_{kn} + \Omega)} + \frac{e^{i(\omega_{kn} - \Omega)T_{\text{obs}}} - 1}{i(\omega_{kn} - \Omega)} \right],$$

where  $\omega_{kn} = (E_k - E_n)/\hbar$  and  $B_{kn} = \langle \psi_k, B\psi_n \rangle$ . Prove that this amplitude depends only on  $E_{\hat{H}}(\{E_k\})BE_{\hat{H}}(\{E_n\})$  and that its squared modulus has peaks of width  $O(T_{\text{obs}}^{-1})$  centered at

$$\Omega_{kn} = \frac{Mg\ell_g}{\hbar} |a_k - a_n|.$$

It vanishes to this order when the displayed spectral block is zero. Separate this first-order, single-photon prediction from strong-drive and multiphoton effects. Deduce the classical turning heights

$z_n = \ell_g a_n$ . Model an absorber with lower face at height  $h$ , width  $w$ , and loss strength  $\Gamma > 0$  by the potential  $-i\Gamma \mathbf{1}_{[h, h+w]}$ . In first-order decay theory, derive

$$\gamma_n(h) = \frac{2\Gamma}{\hbar} \int_h^{h+w} |\psi_n(z)|^2 dz$$

and the transmission factor  $e^{-T\gamma_n(h)}$  after flight time  $T$ . Use the Airy asymptotic to prove that the onset occurs across a height interval of order  $\ell_g$  near  $z_n$ , rather than as a sharp step. State how measured onset heights and resonance frequencies determine  $g$ , and separate the predicted Airy scales from absorber roughness, finite width, and loss.

## 2 Fourier Analysis

**Exercise 2.1** (Invariant Riemannian geometry). A *smooth manifold* of dimension  $n$  is a Hausdorff, second-countable space with a maximal atlas of homeomorphisms to open subsets of  $\mathbb{R}^n$  whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth; it is a *diffeomorphism* when it has a smooth inverse.

The *tangent space*  $T_p M$  is the vector space of derivations  $v : C^\infty(M) \rightarrow \mathbb{R}$  at  $p$  satisfying

$$v(fg) = f(p)v(g) + g(p)v(f).$$

For a smooth map  $F : M \rightarrow N$ , define

$$dF_p : T_p M \longrightarrow T_{F(p)} N, \quad dF_p(v)(f) = v(f \circ F).$$

Put  $TM = \coprod_{p \in M} T_p M$ . A *smooth vector bundle* is a smooth map  $\pi : E \rightarrow M$  locally isomorphic by fiberwise linear maps to  $U \times W \rightarrow U$  for a finite-dimensional real vector space  $W$ . A section is a smooth map  $s : M \rightarrow E$  satisfying  $\pi s = \text{id}_M$ . The differentials of chart maps give  $TM$  its smooth vector-bundle structure; write  $\Gamma(E)$  for the space of sections, and call a section of  $TM$  a *vector field*.

A *Lie group* is a topological group with a smooth manifold structure for which multiplication and inversion are smooth. Its Lie algebra  $\mathfrak{g} = T_e G$  has the bracket induced by left-invariant vector fields.

A *Riemannian metric* on an  $n$ -manifold  $M$  is a smooth family of positive-definite inner products  $g_p$  on  $T_p M$ . Its Riemannian density is characterized by

$$|\text{vol}_g|_p(v_1 \wedge \cdots \wedge v_n) = \det(g_p(v_i, v_j))^{1/2},$$

and determines the Riemannian measure  $d\text{vol}_g$ . The gradient is defined by

$$g(\text{grad}_g f, X) = Xf.$$

The divergence is determined by

$$\int_M Xf d\text{vol}_g = - \int_M f \text{div}_g X d\text{vol}_g$$

for compactly supported  $f$ , and the Laplace–Beltrami operator is  $\Delta_g = \text{div}_g \text{grad}_g$ .

An isometry  $F : (M, g) \rightarrow (N, h)$  is a diffeomorphism satisfying  $h_{F(p)}(dF_p v, dF_p w) = g_p(v, w)$ . For a Lie group  $G$  and  $g \in G$ , write  $L_g(h) = gh$ . A Riemannian metric on  $G$  is *left-invariant* when  $L_g$  is an isometry for every  $g$ .

- (a) Prove that an isometry preserves the Riemannian density, gradient, divergence, and Laplace–Beltrami operator.
- (b) Prove that a left-invariant Riemannian metric on a Lie group has left-invariant Riemannian measure.
- (c) For the Euclidean metric, recover Lebesgue measure and  $\Delta = \sum_j \partial_j^2$ .

**Exercise 2.2** (Connections, curvature, and holonomy). A *connection* on a vector bundle  $E \rightarrow M$  is an  $\mathbb{R}$ -bilinear map  $\nabla : \Gamma(TM) \times \Gamma(E) \rightarrow \Gamma(E)$ , linear over smooth functions in the first variable and satisfying  $\nabla_X(fs) = X(f)s + f\nabla_X s$ . Along a curve  $\gamma$  it induces a covariant derivative  $D/dt$ ; its solutions  $Ds/dt = 0$  define *parallel transport* between endpoint fibers.

The *curvature* of  $\nabla$  is the  $\text{End}(E)$ -valued two-form

$$R^\nabla(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

The *holonomy group* at  $p$  consists of the automorphisms of  $E_p$  obtained by parallel transport around loops based at  $p$ .

Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

**Exercise 2.3** (Levi–Civita connections, geodesics, and the exponential map). Let  $(M, g)$  be a Riemannian manifold. For a connection on  $TM$ , define its torsion by

$$T_\nabla(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y].$$

Call the connection metric-compatible when

$$Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z).$$

Show that torsion-freeness and metric compatibility force the Koszul formula

$$\begin{aligned} 2g(\nabla_X Y, Z) &= Xg(Y, Z) + Yg(Z, X) - Zg(X, Y) \\ &\quad - g(X, [Y, Z]) + g(Y, [Z, X]) + g(Z, [X, Y]), \end{aligned}$$

and use it to prove existence and uniqueness of the Levi–Civita connection.

Define a geodesic to be a curve  $\gamma$  satisfying  $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$ . For a fixed-endpoint variation, derive the first-variation formula for

$$E(\gamma) = \frac{1}{2} \int g(\dot{\gamma}, \dot{\gamma}) dt$$

and prove that its critical curves are precisely the constant-speed geodesics. Prove local existence and uniqueness from initial data  $(p, v) \in TM$ , define

$$\exp_p(v) = \gamma_v(1),$$

and show that  $d(\exp_p)_0$  is the identity on  $T_p M$ . Deduce that short geodesics uniquely minimize length by proving the Gauss lemma for radial geodesics in a normal neighborhood.

**Exercise 2.4** (Curvature, Jacobi fields, and geodesic deviation). For the Levi–Civita connection, let  $R$  be the curvature endomorphism defined above. Derive its skew symmetries and the first Bianchi identity. Define

$$\text{Ric}(X, Y) = \text{Tr}(Z \mapsto R(Z, X)Y), \quad \mathcal{R} = \text{Tr}_g \text{Ric},$$

and, for a two-plane spanned by  $v, w$ , define its sectional curvature by

$$K(v, w) = \frac{g(R(v, w)w, v)}{g(v, v)g(w, w) - g(v, w)^2}.$$

For a variation through geodesics, let  $J$  be its variation field and prove the Jacobi equation

$$\frac{D^2 J}{dt^2} + R(J, \dot{\gamma})\dot{\gamma} = 0.$$

On a manifold of constant sectional curvature  $K$ , solve this equation for a perpendicular field with  $J(0) = 0$  and  $DJ/dt(0) = w$  to obtain the factors

$$\frac{\sin(\sqrt{K}t)}{\sqrt{K}}, \quad t, \quad \frac{\sinh(\sqrt{-K}t)}{\sqrt{-K}}$$

for  $K > 0$ ,  $K = 0$ , and  $K < 0$ . Relate their zeros or growth to conjugate points and the focusing or spreading of neighboring rays.

**Exercise 2.5** (Haar measure, convolution, and regular representations). Every locally compact group below is equipped with a chosen nonzero left Haar measure  $\mu$ . Haar measure on a compact group is normalized by  $\mu(G) = 1$ .

Let  $G$  be a locally compact group with left Haar measure  $\mu$ . For  $f, g \in C_c(G)$ , define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) d\mu(y).$$

The *left regular representation* on  $L^2(G)$  is

$$(\lambda_s \xi)(t) = \xi(s^{-1}t).$$

Identify Haar measure on  $\mathbb{R}^n$ ,  $\mathbb{Z}^n$ , and  $\mathbb{T}^n$ . Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the  $L^1$  convolution inequality, and  $\|f * g\|_2 \leq \|f\|_1 \|g\|_2$ . Prove that  $g \mapsto \lambda_g$  is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over  $G$  to obtain an invariant one.

**Exercise 2.6** (Pontryagin duality and Abelian Fourier analysis). For a locally compact abelian group  $G$ , its *Pontryagin dual*  $\widehat{G}$  is the group of continuous characters  $\chi : G \rightarrow U(1)$ , with pointwise multiplication and the compact-open topology. The *double-dual map* sends  $x \in G$  to the character  $\chi \mapsto \chi(x)$  of  $\widehat{G}$ .

Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \quad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \quad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. For a linear subspace  $W \subseteq \mathbb{R}^n$  and a full-rank lattice  $\Gamma \subseteq \mathbb{R}^n$ , compute their annihilators and identify the duals of  $\mathbb{R}^n/W$  and  $\mathbb{R}^n/\Gamma$ .

**Exercise 2.7** (Fourier series). On  $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ , write

$$dm_{\mathbb{T}}(x) = \frac{dx}{2\pi}$$

for normalized Haar measure and define

$$\widehat{f}(k) = \int_{\mathbb{T}} f(x) e^{-ikx} dm_{\mathbb{T}}(x), \quad k \in \mathbb{Z}.$$

The *Fourier series* of  $f$  is  $\sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ikx}$ .

Prove that  $(e^{ikx})_{k \in \mathbb{Z}}$  is a complete orthonormal family in  $L^2(\mathbb{T})$  and derive

$$f = \sum_{k \in \mathbb{Z}} \widehat{f}(k) e^{ikx}, \quad \|f\|_2^2 = \sum_{k \in \mathbb{Z}} |\widehat{f}(k)|^2.$$

**Exercise 2.8** (Euclidean Fourier inversion). The *Schwartz space*  $\mathcal{S}(V)$  consists of the smooth functions  $f$  satisfying

$$\sup_{q \in V} (1 + \|q\|)^m \|d^r f(q)\| < \infty$$

for all  $m, r \geq 0$ , where the norm on the symmetric covariant tensors is induced by the Euclidean structure. Let  $V$  be an  $n$ -dimensional real Euclidean space with Lebesgue measure  $dq$ . Equip  $V^*$  with its dual Euclidean structure and the resulting Lebesgue measure  $dp$ . For  $f \in \mathcal{S}(V)$ , define

$$(\mathcal{F}_h f)(p) = (2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q) dq, \quad \mathcal{F} = \mathcal{F}_1.$$

Prove Fourier inversion on  $\mathcal{S}(V)$  and prove that  $\mathcal{F}_h$  extends uniquely to a unitary operator on  $L^2(V)$ . For  $v \in V$ , prove

$$\mathcal{F}_h(-i\hbar d_v) \mathcal{F}_h^{-1} \widehat{f}(p) = p(v) \widehat{f}(p).$$

**Exercise 2.9** (Free-particle dispersion). Substitute the plane wave  $\psi(q, t) = e^{i(k(q) - \omega t)}$ , with  $k \in V^*$ , into the free Schrödinger equation and prove

$$\hbar\omega = \frac{\hbar^2 \|k\|^2}{2m}.$$

Set  $p = \hbar k$  and  $E = \hbar\omega$ . Derive

$$E(p) = \frac{\|p\|^2}{2m}, \quad v_{\text{group}} = \frac{p^\sharp}{m},$$

where  $p^\sharp \in V$  is the Euclidean-dual vector to  $p$ . Deduce the time-of-flight displacement of a wave packet concentrated near  $p_0$  and the de Broglie relation between its measured wavelength and momentum.

**Exercise 2.10** (Bessel functions and radial Fourier analysis). For  $\alpha > -1$ , the Bessel function of the first kind is

$$J_\alpha(z) = \sum_{m=0}^{\infty} \frac{(-1)^m}{m! \Gamma(m + \alpha + 1)} \left(\frac{z}{2}\right)^{2m + \alpha},$$

with a branch of  $z^\alpha$  fixed when  $\alpha \notin \mathbb{Z}$ .

Prove

$$z^2 J_\alpha''(z) + z J_\alpha'(z) + (z^2 - \alpha^2) J_\alpha(z) = 0$$

and the recurrence identities

$$J_{\alpha-1}(z) + J_{\alpha+1}(z) = \frac{2\alpha}{z} J_\alpha(z), \quad J_{\alpha-1}(z) - J_{\alpha+1}(z) = 2J_\alpha'(z).$$

Average the Euclidean Fourier kernel over spheres and express the Fourier transform of a radial Schwartz function on  $\mathbb{R}^n$  as a Hankel transform with kernel  $J_{n/2-1}$ . Derive the corresponding inversion identity.

**Exercise 2.11** (Bessel diffraction). Let  $j_{\nu,k}$  be the  $k$ th positive zero of  $J_\nu$ . Separate the wave equation on a circular membrane of radius  $R$  with Dirichlet boundary and prove that its normal-mode frequencies are

$$\omega_{m,k} = \frac{c}{R} j_{|m|,k},$$

with radial factor  $J_{|m|}(j_{|m|,k}r/R)$ . Deduce the nodal circles and angular nodal lines and state how measured resonance ratios determine the ratios of Bessel zeros without knowledge of  $c/R$ .

For a uniformly illuminated circular aperture of radius  $a$ , average the Fraunhofer kernel over the aperture, put  $k = 2\pi/\lambda$ , and derive

$$\frac{I(\theta)}{I(0)} = \left[ \frac{2J_1(ka \sin \theta)}{ka \sin \theta} \right]^2.$$

Prove that the dark-ring angles satisfy  $ka \sin \theta_k = j_{1,k}$  and obtain the small-angle prediction  $\theta_1 \sim j_{1,1}\lambda/(2\pi a)$ . Identify which membrane resonances and diffraction minima measure zeros of the same Bessel functions.

**Exercise 2.12** (Lattices and Poisson summation). A *lattice* in  $\mathbb{R}^n$  is a subgroup  $\Gamma = A\mathbb{Z}^n$  for some invertible linear map  $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ . Its covolume is  $\text{vol}(\Gamma) = |\det A|$ , and a fundamental cell is  $F = A[0, 1)^n$ . Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi \cdot \gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

For  $f \in \mathcal{S}(\mathbb{R}^n)$ , its  $\Gamma$ -periodization is

$$P_\Gamma f(x) = \sum_{\gamma \in \Gamma} f(x + \gamma).$$

Prove that  $P_\Gamma f$  converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\text{vol}(\Gamma)} \sum_{\xi \in \Gamma^*} \mathcal{F}f(\xi) e^{i\xi \cdot x}.$$

Set  $x = 0$  to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

**Exercise 2.13** (Crystal diffraction and Bragg peaks). For a finite motif  $B \subset \mathbb{R}^n$  with scattering weights  $c_b$ , its *structure factor* is

$$S_B(q) = \sum_{b \in B} c_b e^{-iq \cdot b}.$$

Let  $\Gamma_N$  be a finite rectangular portion of  $\Gamma$ . In the single-scattering approximation, prove that the amplitude at momentum transfer  $q$  factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma \in \Gamma_N} e^{-iq \cdot \gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at  $q \in \Gamma^*$  as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude  $2\pi/\lambda$ , derive the Laue condition and, for planes spaced by  $d$ , the Bragg relation

$$2d \sin \theta = m\lambda.$$

Show how zeros of  $S_B$  produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

### 3 Compact Harmonic Analysis

**Exercise 3.1** (Finite representations and Maschke's theorem). For a group  $G$ , a representation on a vector space  $V$  is a homomorphism  $\rho : G \rightarrow \text{GL}(V)$ . A subspace is invariant when it is preserved by every  $\rho(g)$ , and a nonzero representation is irreducible when it has no proper nonzero invariant subspace. A *unitary representation* on a Hilbert space  $H$  is a homomorphism to  $U(H)$ ; for a topological group it is required to be strongly continuous.

Let  $G$  be finite, let the characteristic of  $k$  not divide  $|G|$ , and let  $V$  be a finite-dimensional representation. Average an arbitrary projection over  $G$  to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Let  $H$  be a finite-dimensional complex Hilbert space carrying a unitary representation of  $G$ , and let  $A \in \text{End}(H)$  be self-adjoint and commute with the action. Prove that every spectral subspace of  $A$  is invariant and decomposes into irreducible representations. If  $A_s$  is a norm-continuous family of such operators and a spectral cluster remains isolated, use its spectral projection to prove that the multiplicity of every irreducible type in the cluster is locally constant. Deduce that a change in these multiplicities requires a collision with another spectral cluster.

**Exercise 3.2** (Schur's lemma). An intertwiner  $T : (V, \rho) \rightarrow (W, \sigma)$  between representations satisfies  $T\rho(g) = \sigma(g)T$ .

Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

**Exercise 3.3** (Characters and finite-group harmonic analysis). The *character* of a finite-dimensional representation  $V$  is the class function  $\chi_V(g) = \text{Tr}(\rho_V(g))$ . For class functions on a finite group, define

$$\langle f, h \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} h(g).$$

Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that  $\langle \chi_U, \chi_V \rangle_G = \dim \text{Hom}_G(U, V)$ . Deduce the multiplicity formula and the character table constraints.

**Exercise 3.4** (One-parameter subgroups and the exponential map). Prove existence and uniqueness of the one-parameter subgroup generated by  $X \in \mathfrak{g}$ . For a closed matrix group, prove that it is  $t \mapsto e^{tX}$ , identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

**Exercise 3.5** (Compact convolution operators). Let  $G$  be compact with normalized Haar measure and  $k \in C(G)$ . Prove that

$$(T_k f)(x) = \int_G k(y^{-1}x) f(y) dy$$

defines a compact operator on  $L^2(G)$  by uniformly approximating its continuous kernel by finite sums of product functions. Prove that  $T_k$  commutes with left translations and that  $k(g^{-1}) = \overline{k(g)}$  makes  $T_k$  self-adjoint. Deduce that every nonzero eigenspace of such a self-adjoint convolution operator is a finite-dimensional invariant subspace of the left regular representation.

**Exercise 3.6** (Density of matrix coefficients). For a unitary representation  $\pi : G \rightarrow U(H_\pi)$ , its matrix coefficients are the functions  $g \mapsto \langle y, \pi(g)x \rangle$ .



Construct continuous convolution kernels that approximate the identity on  $C(G)$ . Apply the compact spectral theorem to their self-adjoint symmetrizations and prove that the linear span of matrix coefficients of finite-dimensional unitary representations is uniformly dense in  $C(G)$  and dense in  $L^2(G)$ .

**Exercise 3.7** (Schur orthogonality). For irreducible finite-dimensional unitary representations  $\pi$  and  $\sigma$ , prove that

$$A \mapsto \int_G \pi(g) A \sigma(g)^* dg$$

is zero when  $\pi \not\cong \sigma$  and is  $A \mapsto \text{tr}(A) I_{H_\pi} / \dim H_\pi$  when  $\pi = \sigma$ . Deduce that the coefficient map identifies  $H_\pi \otimes H_\pi^\vee$ , after multiplication by  $\sqrt{\dim H_\pi}$ , with an isometric subspace of  $L^2(G)$ , and that distinct irreducible coefficient spaces are orthogonal, where

$$C_\pi : H_\pi \otimes H_\pi^\vee \longrightarrow L^2(G), \quad C_\pi(x \otimes \varphi)(g) = \varphi(\pi(g)x).$$

**Exercise 3.8** (Peter–Weyl decomposition). The unitary dual  $\widehat{G}$  is the set of unitary-equivalence classes of irreducible finite-dimensional unitary representations of  $G$ .

Use density and Schur orthogonality to prove

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}} H_\pi \otimes H_\pi^\vee},$$

with the left regular representation acting as  $\pi \otimes I_{H_\pi^\vee}$  on the  $\pi$ -summand. For  $f \in L^2(G) \subseteq L^1(G)$ , define  $\widehat{f}(\pi) = \int_G f(g) \pi(g)^* dg$  and prove

$$f = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \text{tr}(\widehat{f}(\pi) \pi(\cdot)), \quad \|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim H_\pi) \text{tr}(\widehat{f}(\pi)^* \widehat{f}(\pi)),$$

where the first sum converges in  $L^2(G)$ . Prove uniform convergence on  $C(G)$  for the summability means supplied by the approximate identities in the density exercise.

**Exercise 3.9** (Character inversion). Let  $G$  be compact with normalized Haar measure, and let  $f$  be a finite linear combination of irreducible characters. For an irreducible unitary representation  $\pi$ , set

$$\widehat{f}(\pi) = \int_G f(s) \pi(s^{-1}) ds.$$

If  $f$  is central, use Schur's lemma to write  $\widehat{f}(\pi) = c_\pi I_{d_\pi}$ . Derive

$$f(e) = \sum_{\pi \in \widehat{G}} d_\pi \text{Tr}(\widehat{f}(\pi)) = \sum_{\pi \in \widehat{G}} d_\pi^2 c_\pi$$

and prove that convolution by  $f$  acts by the scalar  $c_\pi$  on every copy of  $\pi$  in  $L^2(G)$ .

**Exercise 3.10** (Irreducible representations of  $SU(2)$ ). Define

$$SU(2) = \{U \in \text{End}(\mathbb{C}^2) : U^*U = \text{id}, \det U = 1\}.$$

For a complex vector space  $V$ , set  $\text{Sym}^0(V) = \mathbb{C}$  and, for  $N \geq 1$ , define

$$\text{Sym}^N(V) = V^{\otimes N} / \langle w - \sigma w : w \in V^{\otimes N}, \sigma \in S_N \rangle,$$

where  $S_N$  permutes the tensor factors. For  $j \in \{0, \frac{1}{2}, 1, \dots\}$ , put

$$V_j = \text{Sym}^{2j}(\mathbb{C}^2)$$

with the induced  $SU(2)$  action. Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is isomorphic to exactly one  $V_j$ , that it has dimension  $2j + 1$ , and that it has  $J_3$ -eigenvalues  $m\hbar$  for  $m = -j, -j + 1, \dots, j$ . Compute the spectrum of  $J^2 = J_1^2 + J_2^2 + J_3^2$ . For  $j = \frac{1}{2}$ , relate the two  $J_3$  outcomes to the two Stern–Gerlach beams.

**Exercise 3.11** (Clebsch–Gordan decomposition). Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of  $V_{1/2} \otimes V_{1/2}$ .

**Exercise 3.12** (Gauss hypergeometric functions). For  $c \notin \mathbb{Z}_{\leq 0}$  the *Gauss hypergeometric function* is

$${}_2F_1\left(\begin{matrix} a, b \\ c \end{matrix}; z\right) = \sum_{m=0}^{\infty} \frac{(a)_m (b)_m}{(c)_m} \frac{z^m}{m!}.$$

It is interpreted as a finite sum when a numerator parameter is a nonpositive integer and otherwise converges for  $|z| < 1$ . The Gauss hypergeometric operator is

$$\mathcal{L}_{a,b,c} = z(1-z) \frac{d^2}{dz^2} + (c - (a+b+1)z) \frac{d}{dz} - ab.$$

Prove that  ${}_2F_1(a, b; c; z)$  is the solution of  $\mathcal{L}_{a,b,c} f = 0$  analytic at  $z = 0$  with  $f(0) = 1$ . Derive Euler's integral

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du,$$

and Gauss's summation

$${}_2F_1(a, b; c; 1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}.$$

State the parameter conditions for the integral and the summation.

**Exercise 3.13** (Kummer, Whittaker, and Laguerre functions). For  $b \notin \mathbb{Z}_{\leq 0}$ , define the Kummer function

$$M(a, b, z) = {}_1F_1(a; b; z) = \sum_{m=0}^{\infty} \frac{(a)_m}{(b)_m} \frac{z^m}{m!}.$$

For  $2\mu + 1 \notin \mathbb{Z}_{\leq 0}$ , define

$$M_{\lambda, \mu}(z) = e^{-z/2} z^{\mu+1/2} M\left(\mu - \lambda + \frac{1}{2}, 2\mu + 1, z\right),$$

with a branch of  $z^{\mu+1/2}$  fixed. The Whittaker function  $W_{\lambda, \mu}$  is the solution of

$$w'' + \left(-\frac{1}{4} + \frac{\lambda}{z} + \frac{1/4 - \mu^2}{z^2}\right) w = 0$$

normalized by  $W_{\lambda,\mu}(z) \sim e^{-z/2}z^\lambda$  as  $z \rightarrow +\infty$ .

For  $\alpha > -1$  and  $n \in \mathbb{Z}_{\geq 0}$ , define

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha}e^x}{n!} \frac{d^n}{dx^n} (e^{-x}x^{n+\alpha}) = \frac{(\alpha+1)_n}{n!} M(-n, \alpha+1, x).$$

Write  $L_n = L_n^{(0)}$ .

Prove locally uniformly in  $z$  that

$$\lim_{B \rightarrow +\infty} {}_2F_1\left(\begin{matrix} a, B \\ c \end{matrix}; \frac{z}{B}\right) = M(a, c, z),$$

and derive Kummer's equation

$$zy'' + (b-z)y' - ay = 0.$$

Apply the substitution  $w(z) = e^{-z/2}z^{\mu+1/2}y(z)$  to obtain the Whittaker equation with  $a = \mu - \lambda + \frac{1}{2}$  and  $b = 2\mu + 1$ .

Derive the equality of the two displayed definitions of  $L_n^{(\alpha)}$ . Use the Rodrigues formula and integration by parts to prove

$$\int_0^\infty L_m^{(\alpha)}(x) L_n^{(\alpha)}(x) x^\alpha e^{-x} dx = \frac{\Gamma(n+\alpha+1)}{n!} \delta_{mn},$$

and derive

$$(n+1)L_{n+1}^{(\alpha)}(x) = (2n+\alpha+1-x)L_n^{(\alpha)}(x) - (n+\alpha)L_{n-1}^{(\alpha)}(x).$$

Prove

$$\sum_{n \geq 0} L_n^{(\alpha)}(x) t^n = (1-t)^{-\alpha-1} \exp\left(-\frac{xt}{1-t}\right), \quad |t| < 1,$$

and obtain the corresponding coefficient contour integral around  $t = 0$ .

**Exercise 3.14** (Jacobi polynomials and their spectral equation). For  $\alpha, \beta > -1$ , the *Jacobi polynomial* is

$$P_n^{(\alpha,\beta)}(x) = \frac{(\alpha+1)_n}{n!} {}_2F_1\left(\begin{matrix} -n, n+\alpha+\beta+1 \\ \alpha+1 \end{matrix}; \frac{1-x}{2}\right).$$

Prove

$$\left( (1-x^2) \frac{d^2}{dx^2} + (\beta - \alpha - (\alpha + \beta + 2)x) \frac{d}{dx} \right) P_n^{(\alpha,\beta)} = -n(n + \alpha + \beta + 1) P_n^{(\alpha,\beta)}.$$

Derive Jacobi orthogonality on  $[-1, 1]$  with weight  $(1-x)^\alpha(1+x)^\beta$  and derive the three-term recurrence.

**Exercise 3.15** (Spherical harmonics and atomic orbitals). Identify  $S^2$  with  $SO(3)/SO(2)$  and  $L^2(S^2)$  with the right  $SO(2)$ -invariant subspace of  $L^2(SO(3))$ . Use Peter–Weyl to obtain

$$L^2(S^2) \cong \bigoplus_{\ell=0}^{\infty} V_\ell$$

For normalized  $J_3$ -eigenstates  $|\ell m\rangle$ , define the Wigner matrix coefficients

$$D_{mn}^\ell(g) = \langle \ell m | \pi_\ell(g) | \ell n \rangle.$$

Prove that the functions  $Y_{\ell m} \propto D_{m0}^\ell$ ,  $-\ell \leq m \leq \ell$ , form a complete orthogonal family in  $V_\ell$ . For the unit round sphere, set

$$\Delta_{S^2} = \frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{\sin^2 \theta} \partial_\phi^2.$$

Derive  $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$  and completeness of the spherical harmonics. Derive the Jacobi realization

$$Y_{\ell m}(\theta, \phi) = C_{\ell m} e^{im\phi} (\sin \theta)^{|m|} P_{\ell-|m|}^{(|m|, |m|)}(\cos \theta)$$

with  $C_{\ell m}$  fixed by unit norm. On

$$L^2(\mathbb{R}^3, d^3x) \cong L^2((0, \infty), r^2 dr) \otimes L^2(S^2, d\Omega),$$

consider the Coulomb Hamiltonian

$$\hat{H}_C = -\frac{\hbar^2}{2M} \Delta_{\mathbb{R}^3} - \frac{\kappa}{r}, \quad M, \kappa > 0.$$

Separate  $\hat{H}_C$  in spherical coordinates and prove that its radial operator on the  $\ell$ th angular sector is

$$-\frac{\hbar^2}{2M} \left( \frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{\ell(\ell+1)}{r^2} \right) - \frac{\kappa}{r}$$

on  $L^2((0, \infty), r^2 dr)$ . Recover  $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$ . Put

$$a_0 = \frac{\hbar^2}{M\kappa}, \quad \beta = \frac{\sqrt{-2ME}}{\hbar}, \quad \rho = 2\beta r$$

for  $E < 0$ , and set  $u(r) = rR(r)$ . Reduce the radial eigenvalue equation to the Whittaker equation with

$$\lambda = \frac{1}{a_0\beta}, \quad \mu = \ell + \frac{1}{2}.$$

Use regularity at the origin and square-integrability at infinity to prove that

$$\ell + 1 - \lambda = -n_r, \quad n_r \in \mathbb{Z}_{\geq 0}.$$

Writing  $n = n_r + \ell + 1$ , deduce

$$E_n = -\frac{M\kappa^2}{2\hbar^2 n^2}, \quad R_{n\ell}(r) \propto e^{-r/(na_0)} \left( \frac{2r}{na_0} \right)^\ell L_{n-\ell-1}^{(2\ell+1)} \left( \frac{2r}{na_0} \right).$$

Prove that the spatial multiplicity of  $E_n$  is  $n^2$  and derive the spectral-line frequencies

$$\omega_{n_i \rightarrow n_f} = \frac{M\kappa^2}{2\hbar^3} \left| \frac{1}{n_f^2} - \frac{1}{n_i^2} \right|.$$

Determine the angular nodes of the  $s$ ,  $p$ , and  $d$  orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare the predicted momentum-space nodal suppression with Weiß et al., *Nature Communications* **6** (2015), and distinguish the orbital-symmetry prediction, plane-wave approximation, measured intensity, and final-state scattering corrections.

**Exercise 3.16** (Clebsch–Gordan and Wigner coefficients). Choose the normalized  $J_3$ -eigenstates in each irreducible angular-momentum representation by the Condon–Shortley ladder convention

$$J_-|jm\rangle = \hbar\sqrt{(j+m)(j-m+1)}|j, m-1\rangle.$$

Define

$$|jm\rangle = \sum_{m_1+m_2=m} C_{j_1m_1, j_2m_2}^{jm} |j_1m_1\rangle \otimes |j_2m_2\rangle,$$

with all coefficients real and with the coefficient of  $|j_1, j_1\rangle \otimes |j_2, j-j_1\rangle$  in  $|jj\rangle$  positive, and set

$$\begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & m_3 \end{pmatrix} = \frac{(-1)^{j_1-j_2-m_3}}{\sqrt{2j+1}} C_{j_1m_1, j_2m_2}^{j, -m_3}.$$

Define the Wigner  $6j$  symbol by

$$|(j_1j_2)j_{12}, j_3; jm\rangle = \sum_{j_{23}} (-1)^{j_1+j_2+j_3+j} \sqrt{(2j_{12}+1)(2j_{23}+1)} \begin{Bmatrix} j_1 & j_2 & j_{12} \\ j_3 & j & j_{23} \end{Bmatrix} |j_1, (j_2j_3)j_{23}; jm\rangle.$$

For a rotation through angle  $\beta$  about the second axis, set

$$d_{m'm}^j(\beta) = \langle jm' | e^{-i\beta J_2/\hbar} | jm \rangle.$$

Use the  $SU(2)$  differential equation and the value at  $\beta = 0$  to express  $d_{m'm}^j(\beta)$  as a normalized product of powers of  $\sin(\beta/2)$  and  $\cos(\beta/2)$  with a Jacobi polynomial in  $\cos\beta$ . Deduce its orthogonality from Peter–Weyl. Use the Clebsch–Gordan decomposition to derive the linearization formula for a product of two Wigner coefficients and identify its coefficients as products of Clebsch–Gordan coefficients.

**Exercise 3.17** (Angular amplitudes). Derive the orthogonality relations for the Clebsch–Gordan coefficients and the triangle and magnetic selection rules for the  $3j$  symbols. Prove the Gaunt identity

$$\int_{S^2} \overline{Y_{\ell_f m_f}} Y_{1q} Y_{\ell_i m_i} d\Omega = (-1)^{m_f} \sqrt{\frac{3(2\ell_i+1)(2\ell_f+1)}{4\pi}} \begin{pmatrix} \ell_i & 1 & \ell_f \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \ell_i & 1 & \ell_f \\ m_i & q & -m_f \end{pmatrix}.$$

Deduce the electric-dipole rules  $\ell_f = \ell_i \pm 1$  and  $m_f = m_i + q$ , and prove that the relative Zeeman-resolved line intensities are the squared absolute values of the second  $3j$  symbol after a common reduced matrix element is removed. Use the  $6j$  identity to derive the corresponding relative intensities when orbital and spin angular momenta are coupled in different orders. For an oriented initial state, express the dipole radiation pattern as the squared absolute value of the resulting spherical-harmonic amplitude and identify its symmetry-enforced dark directions and absent spectral lines.

## 4 Coadjoint Orbits

**Exercise 4.1** (Differential forms and Cartan calculus). The cotangent space at  $p$  is  $T_p^*M = (T_pM)^*$ , and  $T^*M = \coprod_{p \in M} T_p^*M$  has the dual vector-bundle structure.

A *differential  $k$ -form* on  $M$  is a smooth section of  $\Lambda^k T^*M$ ; write  $\Omega^k(M)$  for the space of differential  $k$ -forms. The exterior products on the cotangent spaces define

$$\wedge : \Omega^r(M) \times \Omega^s(M) \longrightarrow \Omega^{r+s}(M).$$

For a smooth map  $F : M \rightarrow N$ , define

$$(F^*\alpha)_p(v_1, \dots, v_k) = \alpha_{F(p)}(dF_p v_1, \dots, dF_p v_k).$$

For a vector field  $X$ , define contraction by

$$(\iota_X \alpha)_p(v_2, \dots, v_k) = \alpha_p(X_p, v_2, \dots, v_k).$$

The *exterior derivative* is the family  $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$  satisfying  $df(X) = Xf$ , the graded Leibniz rule,  $d^2 = 0$ , and  $F^*d = dF^*$ . A *local flow* of  $X$  is a smooth local family  $\Phi_t$  satisfying  $\Phi_0 = \text{id}$ ,  $\Phi_{t+s} = \Phi_t \Phi_s$  whenever both sides are defined, and  $\frac{d}{dt}\big|_{t=0} \Phi_t(p) = X_p$ . Define

$$\mathcal{L}_X \alpha = \frac{d}{dt}\bigg|_{t=0} \Phi_t^* \alpha.$$

Verify that the differentials of chart maps give  $TM$  and  $T^*M$  the stated smooth vector-bundle structures and that  $d(G \circ F) = dG \circ dF$ . Construct  $d$  locally and prove its uniqueness. For contraction  $\iota_X$ , prove Cartan's identity

$$\mathcal{L}_X = d\iota_X + \iota_X d.$$

For  $A \in \Omega^1(M)$ , put  $F = dA$ . Prove that replacing  $A$  by  $A + d\chi$ , with  $\chi \in C^\infty(M)$ , leaves  $F$  unchanged. Use Cartan's identity to derive

$$\mathcal{L}_X A = \iota_X F + d(\iota_X A).$$

**Exercise 4.2** (Local Poincaré lemma). Let  $U$  be a star-shaped open neighborhood of 0 in a finite-dimensional real vector space. For  $\alpha \in \Omega^k(U)$  with  $k > 0$ , define

$$(K\alpha)_x(v_1, \dots, v_{k-1}) = \int_0^1 t^{k-1} \alpha_{tx}(x, v_1, \dots, v_{k-1}) dt.$$

Use Cartan's identity for the radial flow to prove

$$dK + Kd = \text{id}$$

on positive-degree forms. Deduce that every closed form of positive degree on  $U$  is exact, with a primitive vanishing at the origin.

**Exercise 4.3** (Integration of forms and Stokes' theorem). An *orientation* of an  $n$ -manifold is a continuous choice of component of the nonzero elements in  $\Lambda^n T_p^*M$ . The integral  $\int_M \omega$  of a compactly supported top form is defined using orientation-preserving charts and a partition of unity. Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form  $B$  and electric one-form  $E$ , prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt} \int_S B \quad \text{and} \quad dE = -\partial_t B,$$

recovering the observed flux-emf law.

**Exercise 4.4** (de Rham cohomology). A differential form  $\omega$  is *closed* if  $d\omega = 0$  and *exact* if  $\omega = d\eta$  for some form  $\eta$ . Since  $d^2 = 0$ , the *de Rham cohomology* of  $M$  is

$$H_{\text{dR}}^k(M) = \frac{\ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M))}{\text{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes  $H_{\text{dR}}^*(M)$  a graded-commutative algebra.

Use the local Poincaré lemma and Stokes' theorem to prove that an exact  $k$ -form has zero integral over every closed oriented  $k$ -manifold mapped smoothly into  $M$ . Prove  $H_{\text{dR}}^1(S^1) \cong \mathbb{R}$ , with the isomorphism given by integration.

On  $\mathbb{R}^2 \setminus \{0\}$ , show that

$$\alpha = \frac{-y dx + x dy}{x^2 + y^2}$$

is closed and has integral  $2\pi$  around the unit circle, hence is not exact. Compute  $H_{\text{dR}}^1(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$ . Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential  $F = dA$  exactly when  $[F] = 0$  in de Rham cohomology.

**Exercise 4.5** (Cotangent phase space). For the projection  $\pi : T^*Q \rightarrow Q$ , define the tautological one-form by

$$\theta_\alpha(v) = \alpha(d\pi(v)), \quad \alpha \in T^*Q, \quad v \in T_\alpha(T^*Q).$$

The canonical two-form on  $T^*Q$  is  $\omega = d\theta$ .

For a finite-dimensional real vector space  $V$ , construct the canonical identifications

$$TV \cong V \times V, \quad T^*V \cong V \times V^*.$$

Prove that a phase  $S \in C^\infty(Q)$  determines the momentum  $p = dS_q \in T_q^*Q$ . For a diffeomorphism  $F : Q \rightarrow Q$ , construct its cotangent lift and prove that it preserves  $\theta$  and  $d\theta$ . Under  $T^*V \cong V \times V^*$ , prove

$$\omega((q, p), (q', p')) = p(q') - p'(q).$$

**Exercise 4.6** (Symplectic manifolds and Hamiltonian fields). A *symplectic form* on a smooth manifold  $M$  is a closed nondegenerate two-form  $\omega$ . A *symplectic vector space* is a finite-dimensional real vector space equipped with a nondegenerate alternating bilinear form; it carries the corresponding constant symplectic form. For  $f \in C^\infty(M)$ , define

$$\iota_{X_f}\omega = -df, \quad \{f, g\} = -\omega(X_f, X_g).$$

The local flow of  $X_f$  is the *Hamiltonian flow* of  $f$ . A diffeomorphism  $F : (M, \omega_M) \rightarrow (N, \omega_N)$  is a *canonical transformation* when  $F^*\omega_N = \omega_M$ .

Prove that the canonical two-form  $d\theta$  on  $T^*Q$  is nondegenerate and hence symplectic. Prove that the Poisson bracket is antisymmetric, satisfies the Leibniz rule and Jacobi identity, and is preserved by canonical transformations. For the Hamiltonian flow  $\Phi_t$  of  $H$ , prove

$$\frac{d}{dt}f(\Phi_t(x)) = \{f, H\}(\Phi_t(x)).$$

**Exercise 4.7** (Liouville's theorem). On a  $2n$ -dimensional symplectic manifold, prove that Hamiltonian flow preserves  $\omega$ , the volume form  $\omega^n/n!$ , and every density depending only on the Hamiltonian. On a symplectic vector space, prove that the infinitesimal generator of a linear Hamiltonian flow has trace zero and that the flow has determinant one.

**Exercise 4.8** (Moment maps and Noether charges). For a smooth left action of a Lie group  $G$  on  $M$  and  $\xi \in \mathfrak{g}$ , define the fundamental vector field

$$\xi_M(x) = d(g \mapsto gx)_e(\xi).$$

Define

$$\text{Ad}_g = d(h \mapsto ghg^{-1})_e, \quad (\text{Ad}_g^* \ell)(\xi) = \ell(\text{Ad}_{g^{-1}} \xi).$$

An action of  $G$  on  $(M, \omega)$  is *Hamiltonian* when it preserves  $\omega$  and has an equivariant map  $\mu : M \rightarrow \mathfrak{g}^*$  satisfying

$$\iota_{\xi_M} \omega = -d\mu^\xi, \quad \mu^\xi(x) = \langle \mu(x), \xi \rangle, \quad \mu(gx) = \text{Ad}_g^* \mu(x).$$

The map  $\mu$  is the *moment map*.

Prove

$$X_{\mu^\xi} = \xi_M, \quad \{\mu^\xi, \mu^\eta\} = \mu^{[\xi, \eta]}.$$

If a Hamiltonian  $H$  is  $G$ -invariant, prove Noether's conservation law

$$\frac{d}{dt} \mu^\xi(\Phi_t(x)) = 0$$

for its Hamiltonian flow  $\Phi_t$ .

Let  $G$  act on a manifold  $Q$  and lift the action to  $T^*Q$ . Prove that the lifted action preserves the tautological form and is Hamiltonian with

$$\langle \mu(\alpha_q), \xi \rangle = \alpha_q(\xi_Q(q)).$$

For the distinguished Euclidean configuration space  $Q = \mathbb{R}^3$ , identify the moment maps for translations and rotations with linear momentum  $p$  and angular momentum  $q \times p$ . Deduce conservation of  $p$  for a translation-invariant Hamiltonian and conservation of  $q \times p$  for

$$H(q, p) = \frac{\|p\|^2}{2m} + V(\|q\|).$$

**Exercise 4.9** (Symplectic reduction). Let a compact Lie group  $G$  act on  $(M, \omega)$  with equivariant moment map  $\mu : M \rightarrow \mathfrak{g}^*$ . Suppose 0 is a regular value, meaning  $d\mu_x$  is surjective for every  $x \in \mu^{-1}(0)$ , and suppose  $G$  acts freely on  $\mu^{-1}(0)$ . Use as theorem input [34] that

$$M//G = \mu^{-1}(0)/G$$

is a smooth manifold and the quotient map  $\pi : \mu^{-1}(0) \rightarrow M//G$  is a submersion.

For the inclusion  $i : \mu^{-1}(0) \hookrightarrow M$ , prove that  $i^* \omega$  is invariant and vanishes on the tangent directions to the  $G$ -orbits. Prove that there is a unique symplectic form  $\omega_{\text{red}}$  such that

$$\pi^* \omega_{\text{red}} = i^* \omega, \quad \dim(M//G) = \dim M - 2 \dim G.$$

If  $H$  is  $G$ -invariant, prove that its restriction descends to  $H_{\text{red}}$  and that the Hamiltonian flow of  $H$  on  $\mu^{-1}(0)$  projects to the Hamiltonian flow of  $H_{\text{red}}$ . Write the reduced equations governing motion at the imposed value of the conserved Noether charges.



**Exercise 4.10** (Coadjoint orbits and the Kirillov form). For  $\ell \in \mathfrak{g}^*$ , define

$$\mathcal{O}_\ell = \text{Ad}^*(G)\ell, \quad G_\ell = \{g \in G : \text{Ad}_g^* \ell = \ell\},$$

and

$$\mathfrak{g}_\ell = \{X \in \mathfrak{g} : \ell([X, Y]) = 0 \text{ for every } Y \in \mathfrak{g}\}.$$

For  $X \in \mathfrak{g}$ , set

$$(\text{ad}_X^* \ell)(Y) = -\ell([X, Y]).$$

On  $T_\ell \mathcal{O}_\ell$ , define

$$\omega_\ell(\text{ad}_X^* \ell, \text{ad}_Y^* \ell) = \ell([X, Y]).$$

For  $\eta = \text{Ad}_g^* \ell$ , transport  $\omega_\ell$  by the coadjoint action to define  $\omega_\eta$ .

Prove

$$T_\ell \mathcal{O}_\ell = \{\text{ad}_X^* \ell : X \in \mathfrak{g}\} \cong \mathfrak{g}/\mathfrak{g}_\ell.$$

Prove that the Kirillov form is well-defined, alternating, nondegenerate, and  $G$ -invariant. Prove the cyclic identity

$$\ell([X, [Y, Z]]) + \ell([Y, [Z, X]]) + \ell([Z, [X, Y]]) = 0$$

and prove directly that it is the cyclic identity satisfied by the transported field  $\omega$  on  $\mathcal{O}_\ell$ . Prove that

$$\mu_{\mathcal{O}} : \mathcal{O}_\ell \longrightarrow \mathfrak{g}^*, \quad \mu_{\mathcal{O}}(\eta) = -\eta,$$

is an equivariant moment map with the sign conventions above.

For the distinguished group

$$SO(3) = \{R \in \text{End}(\mathbb{R}^3) : R^* R = \text{id}, \det R = 1\},$$

use the Euclidean inner product and orientation to identify  $\mathfrak{so}(3)$  and its dual with  $\mathbb{R}^3$ , with bracket  $[X, Y] = X \times Y$ . Prove that  $\mathcal{O}_\ell$  is the sphere of radius  $L = \|\ell\|$ , that

$$\omega_\ell(X \times \ell, Y \times \ell) = \ell \cdot (X \times Y),$$

and that the Kirillov form is  $L^{-1}$  times the oriented Euclidean area form on this sphere. Deduce

$$\int_{\mathcal{O}_\ell} \omega = 4\pi L.$$

**Exercise 4.11** (Principal connections and gauge covariance). A *principal  $G$ -bundle* is a smooth map  $P \rightarrow M$  with a free right  $G$ -action, transitive on each fiber, and local  $G$ -equivariant trivializations. For a homomorphism  $\rho : G \rightarrow \text{GL}(W)$ , its associated vector bundle is  $P \times_G W = (P \times W)/((pg, w) \sim (p, \rho(g)w))$ . A *principal connection* is an equivariant  $\mathfrak{g}$ -valued one-form  $A$  on  $P$  reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is  $F_A = dA + \frac{1}{2}[A \wedge A]$ . A *gauge transformation* is a  $G$ -equivariant bundle automorphism covering  $\text{id}_M$ .

Prove  $d_A F_A = 0$ . For a gauge transformation  $\Phi$ , prove  $F_{\Phi^* A} = \Phi^* F_A$  and that holonomy changes by conjugacy. Prove that a flat connection has trivial holonomy on contractible loops and is gauge-equivalent to a product connection on a connected base exactly when its holonomy homomorphism is trivial. For  $G = U(1)$ , compute the holonomy around a generator of the fundamental group and exhibit the obstruction to removing a flat connection globally.

**Exercise 4.12** (Flat-connection reduction and Goldman symplectic geometry). Let  $\Sigma$  be a closed oriented surface and let  $\mathcal{A}$  be the affine space of connections on the product principal  $SU(2)$ -bundle over  $\Sigma$ . Its tangent space at every connection is  $\Omega^1(\Sigma; \mathfrak{su}(2))$ . Let  $\mathcal{G} = C^\infty(\Sigma, SU(2))$  be the gauge group, use the invariant pairing

$$\langle X, Y \rangle = -\text{Tr}(XY),$$

and define

$$\Omega_A(a, b) = \int_{\Sigma} \langle a \wedge b \rangle.$$

- (a) Prove that  $\Omega$  is closed, gauge invariant, and nondegenerate. For  $\xi \in \Omega^0(\Sigma; \mathfrak{su}(2))$ , identify the infinitesimal gauge action at  $A$  with  $d_A \xi$ . Using

$$\left. \frac{d}{dt} \right|_{t=0} F_{A+ta} = d_A a$$

and integration by parts, prove that curvature is the moment map:

$$\iota_{d_A \xi} \Omega = -d \left( \int_{\Sigma} \langle F_A, \xi \rangle \right).$$

Use as theorem input the smooth-locus gauge-reduction theorem [22]: on the irreducible locus, the quotient

$$\mathcal{M}_{\text{flat}}(\Sigma, SU(2)) = \{A \in \mathcal{A} : F_A = 0\} / \mathcal{G}$$

is a finite-dimensional symplectic space with the form induced by  $\Omega$ . Use flat parallel transport to identify it with

$$\text{Hom}(\pi_1(\Sigma), SU(2)) / SU(2),$$

where  $SU(2)$  acts by conjugation.

- (b) For the complexified holonomy model, put

$$\mathcal{R}(\Sigma, SL_2) = \text{Hom}(\pi_1(\Sigma), SL_2(\mathbb{C}))$$

and let

$$\mathcal{M}_{\text{flat}}(\Sigma, SL_2)^{\text{irr}} = \mathcal{R}(\Sigma, SL_2)^{\text{irr}} / SL_2(\mathbb{C}).$$

Let its character algebra be  $\mathbb{C}[\mathcal{R}(\Sigma, SL_2)]^{SL_2}$ . For an oriented loop  $\gamma$ , define the Wilson trace

$$W_{\gamma}(\rho) = \text{tr} \rho(\gamma).$$

Prove that it is conjugation invariant, depends only on the free homotopy class of  $\gamma$ , and satisfies  $W_{\gamma^{-1}} = W_{\gamma}$ .

For  $\Sigma = T^2$ , write

$$x = \text{tr} A, \quad y = \text{tr} B, \quad z = \text{tr}(AB)$$

for a commuting pair  $(A, B)$ . Use Cayley–Hamilton to prove

$$\text{tr} A \text{ tr} B = \text{tr}(AB) + \text{tr}(AB^{-1})$$

and

$$\text{tr}[A, B] = x^2 + y^2 + z^2 - xyz - 2.$$

Use as theorem input the rank-two  $SL_2$  trace-generation theorem: the simultaneous-conjugation invariant algebra is generated by these three traces. Deduce that the torus trace quotient is the Fricke cubic

$$x^2 + y^2 + z^2 - xyz - 4 = 0.$$

Determine directly the points at which this cubic is not smooth.

- (c) Let  $\rho : \pi_1(\Sigma) \rightarrow SL_2(\mathbb{C})$  be irreducible and let  $\text{Ad } \rho$  be the local system with fiber  $\mathfrak{sl}_2(\mathbb{C})$ . Use as theorem input [22]

$$T_{[\rho]} \mathcal{M}_{\text{flat}}(\Sigma, SL_2)^{\text{irr}} \cong H^1(\Sigma; \text{Ad } \rho).$$

For local-system-valued one-forms representing cohomology classes, define

$$\omega_G([\alpha], [\beta]) = \int_{\Sigma} \text{Tr}(\alpha \wedge \beta).$$

Prove that this is independent of representatives and use Poincaré duality with local coefficients as theorem input to prove nondegeneracy. For immersed oriented loops in transverse position, use Goldman's trace-bracket formula [22]

$$\{W_{\alpha}, W_{\beta}\} = \sum_{p \in \alpha \cap \beta} \epsilon_p \left( W_{\alpha_p \beta_p} - \frac{1}{2} W_{\alpha} W_{\beta} \right)$$

and the trace identity above to rewrite each summand as

$$\frac{\epsilon_p}{2} \left( W_{\alpha_p \beta_p} - W_{\alpha_p \beta_p^{-1}} \right).$$

Use as theorem input the algebraic extension of this bracket to the torus trace algebra. For the torus trace coordinates, derive

$$\{x, y\} = z - \frac{1}{2}xy, \quad \{y, z\} = x - \frac{1}{2}yz, \quad \{z, x\} = y - \frac{1}{2}zx,$$

verify that the Fricke cubic is a Casimir, and identify the Hamiltonian flow of a Wilson trace with the twist flow along its loop.

**Exercise 4.13** (Integral curvature and circle cocycles). A *good cover* of a manifold is an open cover  $\{U_i\}_{i \in I}$  for which every nonempty finite intersection is contractible. A *smooth complex vector bundle* is a smooth vector bundle locally modeled on a finite-dimensional complex vector space by fiberwise complex-linear product maps; it is a *complex line bundle* when its complex rank is one. A *Hermitian vector bundle* carries a smoothly varying positive-definite Hermitian form, and a connection on it is *unitary* when it preserves that form. Let  $F \in \Omega^2(M; i\mathbb{R})$  be closed and choose, on a good cover,

$$dA_i = F|_{U_i}.$$

On  $U_i \cap U_j$ , choose  $f_{ij} : U_i \cap U_j \rightarrow i\mathbb{R}$  with  $A_j - A_i = df_{ij}$  and  $f_{ji} = -f_{ij}$ . Prove that

$$c_{ijk} = f_{ij} + f_{jk} + f_{ki}$$

is constant on every nonempty triple intersection.

Use the following good-cover form of the integral de Rham theorem as theorem input [7]:

$$\frac{1}{2\pi i} \int_{\Sigma} h^* F \in \mathbb{Z} \quad \text{for every smooth } h : \Sigma \rightarrow M$$

from a closed oriented surface if and only if the  $f_{ij}$  may be chosen so that  $c_{ijk} \in 2\pi i\mathbb{Z}$ . Under this condition put

$$g_{ij} = e^{f_{ij}}.$$

Prove the cocycle identities for the  $g_{ij}$ , construct the associated Hermitian line bundle, and show that the local operators  $d + A_i$  define a unitary connection of curvature  $F$ . Conversely, compare local unitary frames of a Hermitian line bundle with connection and recover the integral period condition from its transition functions.

**Exercise 4.14** (Prequantum line bundles on coadjoint orbits). A *prequantum line bundle* over  $(M, \omega)$  is a Hermitian line bundle  $L \rightarrow M$  with a unitary connection  $\nabla$  satisfying

$$F_{\nabla} = -\frac{i}{\hbar}\omega.$$

Prove that a prequantum line bundle exists if and only if

$$\frac{1}{2\pi\hbar} \int_{\Sigma} f^*\omega \in \mathbb{Z}$$

for every smooth map  $f : \Sigma \rightarrow M$  from a closed oriented surface. Apply the circle-cocycle construction above to  $F = -i\omega/\hbar$ , identify its local connection forms with symplectic potentials, and prove that

$$A_i \mapsto A_i + d\phi_i, \quad f_{ij} \mapsto f_{ij} + \phi_j - \phi_i$$

produces an isomorphic prequantum line bundle with connection. Show that two prequantum line bundles with the same curvature differ by a flat Hermitian line bundle, so curvature alone need not determine the prequantization up to isomorphism. For the  $SO(3)$  coadjoint orbit of radius  $L$ , prove from the Kirillov period that prequantization exists exactly when  $2L/\hbar$  is an integer.

**Exercise 4.15** (Kostant–Souriau operators). For  $f \in C^\infty(M)$ , define

$$\mathcal{Q}_{\text{pre}}(f) = -i\hbar\nabla_{X_f} + f.$$

Use the curvature identity for  $\nabla$  and the convention for Hamiltonian vector fields to prove

$$[\mathcal{Q}_{\text{pre}}(f), \mathcal{Q}_{\text{pre}}(g)] = i\hbar\mathcal{Q}_{\text{pre}}(\{f, g\}).$$

For a Hamiltonian  $G$ -action with moment map  $\mu$ , define

$$d\pi_{\text{pre}}(\xi) = \frac{1}{i\hbar}\mathcal{Q}_{\text{pre}}(\mu^\xi).$$

Prove

$$[d\pi_{\text{pre}}(\xi), d\pi_{\text{pre}}(\eta)] = d\pi_{\text{pre}}([\xi, \eta]),$$

and call  $d\pi_{\text{pre}}$  the prequantum Lie-algebra representation on sections of  $L$ .

**Exercise 4.16** (Prequantum circle bundles). Let  $Y$  be a smooth manifold of dimension  $2n + 1$ . A one-form  $\alpha$  is a *contact form* when  $\alpha \wedge (d\alpha)^n$  is nowhere zero. Its *Reeb vector field*  $R_\alpha$  is defined by

$$\alpha(R_\alpha) = 1, \quad \iota_{R_\alpha}d\alpha = 0.$$

Let  $\pi : S(L) \rightarrow M$  be the unit-circle bundle of a prequantum line bundle, let  $A \in \Omega^1(S(L); i\mathbb{R})$  be its connection form normalized by  $A(\xi) = i$  on the circle generator  $\xi$ , and put  $\alpha = -iA$ . Prove

$$d\alpha = -\frac{1}{\hbar}\pi^*\omega,$$

that  $\alpha$  is a contact form with Reeb vector field  $\xi$ , and that sections of  $L$  correspond to functions  $F : S(L) \rightarrow \mathbb{C}$  satisfying

$$F(ue^{i\theta}) = e^{-i\theta}F(u).$$

**Exercise 4.17** (Coadjoint-orbit quantization, coherent-state harmonic analysis, and spin spectroscopy). On the distinguished Hermitian space  $\mathbb{C}^2$ , let the Pauli endomorphisms  $\sigma_1, \sigma_2, \sigma_3$  be normalized by

$$\sigma_a \sigma_b = \delta_{ab} I + i \sum_{c=1}^3 \varepsilon_{abc} \sigma_c, \quad \text{Tr}(\sigma_a \sigma_b) = 2\delta_{ab}.$$

- (a) Let  $SU(2)$  act by conjugation on the real vector space of traceless Hermitian endomorphisms of  $\mathbb{C}^2$ . Use  $x \mapsto x_1 \sigma_1 + x_2 \sigma_2 + x_3 \sigma_3$  to identify this Euclidean space with  $\mathbb{R}^3$  and prove that the resulting homomorphism

$$SU(2) \longrightarrow SO(3)$$

is surjective with kernel  $\{\pm \text{id}\}$ .

- (b) Let  $\mathcal{O}_L$  be the coadjoint sphere of radius  $L > 0$  with its Kirillov form, and regard it as an  $SU(2)$  orbit through the spin covering. Put

$$N = \frac{2L}{\hbar}.$$

Assume  $N \in \mathbb{Z}_{>0}$  and use as theorem input [47] that  $\mathcal{O}_L$  has an  $SU(2)$ -invariant complex structure  $J$ , an endomorphism of its tangent bundle satisfying  $J^2 = -\text{id}$ . Let  $T^{0,1}\mathcal{O}_L$  be the  $-i$ -eigenspace of the complex-linear extension of  $J$ . Call a section  $s$  of the prequantum line bundle *polarized* when  $\nabla_X s = 0$  for every  $X \in T^{0,1}\mathcal{O}_L$ . Use as theorem input [47] that the polarized sections form the  $SU(2)$  representation

$$V_j = \text{Sym}^N(\mathbb{C}^2), \quad j = \frac{N}{2}.$$

Call this polarized-section space the *geometric quantization* of  $\mathcal{O}_L$ . The action on  $V_j$  is induced from the tensor-power action on  $(\mathbb{C}^2)^{\otimes N}$ .

- (c) For

$$u_\theta = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix},$$

derive the character identity

$$\chi_j(u_\theta) = \sum_{m=-j}^j e^{im\theta} = \frac{\sin((2j+1)\theta/2)}{\sin(\theta/2)} = U_{2j}(\cos(\theta/2)).$$

Define the Chebyshev polynomials of the second kind by this identity and derive

$$U_0(t) = 1, \quad U_1(t) = 2t, \quad U_{n+1}(t) = 2tU_n(t) - U_{n-1}(t).$$

- (d) Write  $\rho_j$  for the action on  $V_j$ . Let  $P_j$  be the orthogonal projection onto its highest-weight line, let  $d\mu$  be the invariant probability measure on  $\mathcal{O}_L \cong SU(2)/U(1)$ , and define

$$Q_{gU(1)} = \rho_j(g)P_j\rho_j(g)^*.$$

Prove that  $Q$  is well defined on the orbit and use Schur's lemma to derive the coherent-state resolution and transform identities

$$(2j+1) \int_{\mathcal{O}_L} Q_x d\mu(x) = I, \quad \psi = (2j+1) \int_{\mathcal{O}_L} Q_x \psi d\mu(x),$$

$$(2j+1) \int_{\mathcal{O}_L} \|Q_x \psi\|^2 d\mu(x) = \|\psi\|^2.$$

If  $n_x, n_y$  are the corresponding unit orbit directions, prove that the reproducing kernel is

$$\text{Tr}(Q_x Q_y) = \left( \frac{1 + n_x \cdot n_y}{2} \right)^{2j}.$$

- (e) Normalize the infinitesimal generators by  $[J_a, J_b] = i\hbar \varepsilon_{abc} J_c$ . Prove that  $J_3$  has values  $m\hbar$  and that a  $2\pi$  rotation acts by  $(-1)^{2j}$ , whereas a  $4\pi$  rotation acts trivially. For the Zeeman Hamiltonian  $\hat{H}_B = -\gamma B J_3$ , derive

$$E_m = -\gamma B m \hbar, \quad m = -j, -j+1, \dots, j.$$

Deduce the  $2j+1$  resolved Stern–Gerlach outcomes, the Zeeman spacing  $|\gamma|B\hbar$ , and the  $4\pi$  restoration measured by spin interferometry, stating the field-gradient, weak-coupling, beam-resolution, and coherence assumptions separately.

## 5 Heisenberg Group

**Exercise 5.1** (Weyl multipliers). A *normalized multiplier* on a locally compact group  $G$  is a continuous map  $\sigma : G \times G \rightarrow \mathbb{T}$  satisfying

$$\sigma(x, e) = \sigma(e, x) = 1, \quad \sigma(x, y)\sigma(xy, z) = \sigma(y, z)\sigma(x, yz).$$

Let  $(E, \omega)$  be a symplectic vector space. For  $\lambda \in \mathbb{R}$ , define

$$\sigma_{\lambda\omega}(z, z') = e^{i\lambda\omega(z, z')/2}.$$

Prove that  $\sigma_{\lambda\omega}$  is a normalized multiplier for every  $\lambda \in \mathbb{R}$ .

**Exercise 5.2** (Commutator bicharacter). For a multiplier  $\sigma$  on an abelian group  $A$ , define

$$c_\sigma(x, y) = \sigma(x, y) \overline{\sigma(y, x)}.$$

Multipliers  $\sigma$  and  $\sigma'$  are cohomologous when there is a continuous map  $b : A \rightarrow \mathbb{T}$ ,  $b(e) = 1$ , with

$$\sigma'(x, y) = b(x)b(y)\overline{b(x+y)}\sigma(x, y).$$

For  $E = V \oplus V^*$ , call  $\sigma_{\omega/\hbar}$  the Weyl multiplier.

Prove that  $c_\sigma$  is an alternating bicharacter and depends only on the cohomology class of  $\sigma$ . For the Weyl multiplier on  $V \oplus V^*$ , prove

$$c_{\sigma_{\omega/\hbar}}((q, p), (q', p')) = e^{i(p(q') - p'(q))/\hbar}.$$

Prove

$$c_{\sigma_{\lambda\omega}}(z, z') = e^{i\lambda\omega(z, z')}.$$

**Exercise 5.3** (Heisenberg groups and central characters). A  $\sigma$ -projective unitary representation of a group  $E$  is a strongly continuous map  $U : E \rightarrow U(H)$  satisfying  $U_z U_{z'} = \sigma(z, z') U_{z+z'}$  and  $U_0 = I$ .

For a symplectic vector space  $(E, \omega)$ , its *Heisenberg group* is

$$\mathbb{H}(E, \omega) = E \times \mathbb{R}$$

with multiplication

$$(z, s)(z', s') = (z + z', s + s' + \frac{1}{2}\omega(z, z')).$$

Write  $\mathbb{H}_n = \mathbb{H}(\mathbb{R}^n \oplus (\mathbb{R}^n)^*, \omega)$  for the canonical symplectic form on  $\mathbb{R}^n \oplus (\mathbb{R}^n)^*$ . Its Lie algebra is  $\mathfrak{h}(E, \omega) = E \oplus \mathbb{R}Z$  with

$$[(z, sZ), (z', s'Z)] = \omega(z, z')Z.$$

Prove that  $\mathbb{H}(E, \omega)$  is a locally compact group with center  $\{0\} \times \mathbb{R}$ . Prove that projective representations of  $E$  with multiplier  $e^{i\lambda\omega/2}$  correspond to representations of  $\mathbb{H}(E, \omega)$  with central character  $e^{i\lambda s}$ .

**Exercise 5.4** (Heisenberg coadjoint orbits and Schrödinger polarization). For a connected simply connected nilpotent group  $G$  and  $\ell \in \mathfrak{g}^*$ , a *polarization* is a Lie subalgebra  $\mathfrak{p} \subseteq \mathfrak{g}$  satisfying

$$\ell([\mathfrak{p}, \mathfrak{p}]) = 0, \quad \dim \mathfrak{p} = \frac{\dim \mathfrak{g} + \dim \mathfrak{g}_\ell}{2}.$$

For  $P = \exp(\mathfrak{p})$ , set  $\chi_\ell(\exp X) = e^{i\ell(X)}$ . The induced representation  $\text{Ind}_P^G \chi_\ell$  acts by left translation on the  $L^2$ -sections over  $G/P$  satisfying  $F(gp) = \chi_\ell(p)^{-1}F(g)$ .

Identify

$$\mathfrak{h}(E, \omega)^* = E^* \oplus \mathbb{R}Z^*.$$

For  $(\xi, \lambda) \in E^* \oplus \mathbb{R}Z^*$ , prove

$$\text{Ad}_{(z, s)}^*(\xi, \lambda) = (\xi - \lambda\omega(z, \cdot), \lambda).$$

Deduce that the orbits with  $\lambda = 0$  are points and that, for  $\lambda \neq 0$ ,

$$\mathcal{O}_\lambda = \{(\xi, \lambda) : \xi \in E^*\}.$$

Prove that

$$E \longrightarrow \mathcal{O}_\lambda, \quad z \longmapsto (-\lambda\omega(z, \cdot), \lambda),$$

pulls the Kirillov form back to  $\lambda\omega$ .

For  $E = V \oplus V^*$  and  $\mathfrak{p} = V^* \oplus \mathbb{R}Z$ , prove that  $\mathfrak{p}$  is a polarization at  $(0, \lambda)$ . Identify  $\mathbb{H}(E, \omega)/\exp(\mathfrak{p})$  with  $V$  and compute the induced action to obtain

$$(\pi_\lambda(q, p, s)\psi)(x) = e^{i\lambda(s+p(x-q/2))}\psi(x-q).$$

Prove that this representation has central character  $e^{i\lambda s}$ . For  $\lambda = 0$ , recover every character of  $E$  from a point orbit.

**Exercise 5.5** (Moyal algebra of a Heisenberg orbit). Let  $(E, \omega)$  be a symplectic vector space, let  $\Pi : E^* \rightarrow E$  be the inverse of  $v \mapsto \omega(v, -)$ , and define

$$\{f, g\} = dg(\Pi(df)), \quad \Pi(k, \ell) = \ell(\Pi(k)).$$

For  $\lambda \neq 0$ , use the orbit map  $E \rightarrow \mathcal{O}_\lambda$  to identify  $\omega$  with the pullback of  $\lambda^{-1}$  times the Kirillov form.

A *differential formal deformation* of  $C^\infty(E)$  is an associative  $\mathbb{C}[[\hbar]]$ -bilinear product on  $C^\infty(E)[[\hbar]]$  of the form

$$f \star g = fg + \sum_{r \geq 1} \hbar^r C_r(f, g),$$

where the  $C_r$  are bidifferential operators, 1 is a unit, and

$$C_1(f, g) - C_1(g, f) = i\{f, g\}.$$

Let  $D_\Pi$  be the constant bidifferential operator characterized on decomposable functions by

$$D_\Pi(f \otimes g)(x, y) = d_y g(\Pi(dx f)).$$

Define

$$f \star g = m \circ \exp\left(\frac{i\hbar}{2} D_\Pi\right)(f \otimes g),$$

where  $m$  restricts a function on  $E \times E$  to the diagonal. Prove associativity from the commutation of the induced constant bidifferential operators on  $E^3$ , and prove

$$f \star g = fg + \frac{i\hbar}{2} \{f, g\} + O(\hbar^2).$$

For  $k, \ell \in E^*$  put  $e_k(x) = e^{ik(x)}$  and derive

$$e_k \star e_\ell = e^{-i\hbar \Pi(k, \ell)/2} e_{k+\ell}.$$

Check the multiplier identity directly and prove that every affine symplectic transformation of  $E$  preserves  $\star$ . Call this intrinsic formal algebra the *Moyal algebra* of the Heisenberg orbit.

**Exercise 5.6** (Canonical commutation and Weyl relations). For  $\hbar > 0$ ,  $\ell \in V^*$ , and  $v \in V$ , define on  $\mathcal{S}(V)$

$$Q_\ell \psi(x) = \ell(x) \psi(x), \quad P_v \psi(x) = -i\hbar d\psi_x(v).$$

For

$$(T_q \psi)(x) = \psi(x - q), \quad (M_p^\hbar \psi)(x) = e^{ip(x)/\hbar} \psi(x),$$

prove that the corresponding one-parameter groups have infinitesimal generators  $-iP_v/\hbar$  and  $iQ_\ell/\hbar$ , respectively, and prove

$$M_p^\hbar T_q = e^{ip(q)/\hbar} T_q M_p^\hbar.$$

For

$$W_\hbar(q, p) = e^{-ip(q)/(2\hbar)} M_p^\hbar T_q,$$

prove

$$W_\hbar(z) W_\hbar(z') = e^{i\omega(z, z')/(2\hbar)} W_\hbar(z + z')$$

and identify  $W_\hbar(z)$  with  $\pi_{1/\hbar}(z, 0)$  under the canonical set-theoretic section  $z \mapsto (z, 0)$ . Prove that this section is not a homomorphism and that its failure to preserve multiplication is exactly the displayed multiplier. Differentiate the Weyl relations on  $\mathcal{S}(V)$  to prove

$$[Q_\ell, P_v] = i\hbar \ell(v) I.$$

**Exercise 5.7** (Irreducibility of the Schrödinger representation). Prove that an operator on  $L^2(V)$  commuting with every translation and every modulation is scalar: first use the Fourier transform to identify the commutant of translations, and then impose commutation with modulations. Deduce that  $\pi_\lambda$  is irreducible for  $\lambda \neq 0$ .



**Exercise 5.8** (Stone–von Neumann theorem). Let  $U$  be a strongly continuous irreducible unitary representation of  $\mathbb{H}(V \oplus V^*, \omega)$  with central character  $U(0, s) = e^{i\lambda s}I$ , where  $\lambda \neq 0$ . Integrate Schwartz functions on  $V \oplus V^*$  against the canonical section of  $U$ . Verify in the Schrödinger model that Fourier inversion sends such an integrated operator with Schwartz symbol to an integral operator with Schwartz kernel. Use as theorem input the Stone–von Neumann theorem [17]: every such  $U$  is unitarily equivalent to  $\pi_\lambda$ . Deduce that the nonzero central character  $\lambda$  determines the irreducible representation up to unitary equivalence.

**Exercise 5.9** (Trace and Hilbert–Schmidt classes). For  $x, y \in H$ , let

$$\theta_{x,y}(\xi) = \langle y, \xi \rangle x.$$

The *trace class*  $S_1(H)$  consists of operators admitting a representation

$$A = \sum_{j=1}^{\infty} \theta_{x_j, y_j}, \quad \sum_{j=1}^{\infty} \|x_j\| \|y_j\| < \infty,$$

where the series converges in operator norm, with norm the infimum of the displayed sums. Define

$$\text{Tr}(A) = \sum_{j=1}^{\infty} \langle y_j, x_j \rangle.$$

The *Hilbert–Schmidt class* is

$$S_2(H) = \{A \in B(H) : A^*A \in S_1(H)\}, \quad \|A\|_2^2 = \text{Tr}(A^*A).$$

Prove that  $S_1(H)$ ,  $\text{Tr}$ , and  $S_2(H)$  are well-defined. Prove that  $\langle A, B \rangle_{S_2} = \text{Tr}(A^*B)$  makes  $S_2(H)$  a Hilbert space and that the pairing  $(A, T) \mapsto \text{Tr}(AT)$  identifies  $S_1(H)^*$  isometrically with  $B(H)$ .

**Exercise 5.10** (Coefficient and Fourier–Wigner transforms). Let  $V$  be an  $n$ -dimensional real vector space with Lebesgue measure  $dq$ , let  $\hbar > 0$ , and equip  $V^*$  with the dual measure  $dp$  for which  $(2\pi\hbar)^{-n/2} \int_V e^{-ip(q)/\hbar} f(q) dq$  is unitary on  $L^2$ . For  $q \in V$  and  $p \in V^*$ , define

$$(T_q \xi)(x) = \xi(x - q), \quad (M_p^h \xi)(x) = e^{ip(x)/\hbar} \xi(x).$$

For  $z = (q, p) \in V \oplus V^*$ , define

$$W_h(z) = e^{-ip(q)/(2\hbar)} M_p^h T_q.$$

The *Fourier–Wigner transform* is

$$\mathcal{W}_h : S_1(L^2(V)) \longrightarrow C_b(V \oplus V^*), \quad \mathcal{W}_h(A)(z) = (2\pi\hbar)^{-n/2} \text{Tr}(AW_h(z)).$$

Prove

$$W_h(z)^* = W_h(-z).$$

For  $A, B \in S_1(L^2(V)) \cap S_2(L^2(V))$ , prove the Fourier–Wigner orthogonality identity

$$\int_{V \oplus V^*} \overline{\mathcal{W}_h(A)(z)} \mathcal{W}_h(B)(z) dz = \text{Tr}(A^*B).$$

Deduce that  $\mathcal{W}_h$  extends uniquely to a unitary map

$$S_2(L^2(V)) \longrightarrow L^2(V \oplus V^*)$$

and, whenever  $\mathcal{W}_h(A) \in L^1(V \oplus V^*)$ , prove

$$A = (2\pi\hbar)^{-n/2} \int_{V \oplus V^*} \mathcal{W}_h(A)(z) W_h(z)^* dz$$

in the weak operator topology.

**Exercise 5.11** (Schrödinger character). For  $V = \mathbb{R}^n$ , let  $f \in C^\infty(\mathbb{H}_n)$  satisfy

$$(q, p, s) \mapsto f(\exp(q, p, s)) \in \mathcal{S}(\mathbb{R}^{2n+1})$$

under the canonical identification  $\mathfrak{h}_n = \mathbb{R}^n \oplus (\mathbb{R}^n)^* \oplus \mathbb{R}$ . With Haar measure  $dq dp ds$ , put

$$\pi_\lambda(f) = \int_{\mathbb{H}_n} f(g) \pi_\lambda(g) dg.$$

Prove

$$\text{Tr } \pi_\lambda(f) = (2\pi)^n |\lambda|^{-n} \int_{\mathbb{R}} f(0, 0, s) e^{i\lambda s} ds.$$

Derive this identity directly from the Schrödinger kernel and identify the factor  $|\lambda|^{-n}$  with the symplectic-volume scaling of  $\mathcal{O}_\lambda$ .

**Exercise 5.12** (Heisenberg Plancherel and inversion). For  $f \in L^1(\mathbb{H}_n) \cap L^2(\mathbb{H}_n)$ , use Haar measure  $dq dp ds$  and the scalar Fourier transform  $\hat{h}(\lambda) = \int_{\mathbb{R}} h(s) e^{-i\lambda s} ds$ , and define

$$\hat{f}(\lambda) = \int_{\mathbb{H}_n} f(g) \pi_\lambda(g)^* dg.$$

Compute the integral kernel of  $\hat{f}(\lambda)$  and apply Euclidean Plancherel in the central and phase-space variables. Deduce the Plancherel measure  $(2\pi)^{-(n+1)} |\lambda|^n d\lambda$  and prove

$$\|f\|_2^2 = (2\pi)^{-(n+1)} \int_{\mathbb{R} \setminus \{0\}} \|\hat{f}(\lambda)\|_{\text{HS}}^2 |\lambda|^n d\lambda.$$

Prove the inversion formula and the distributional orthogonality of Weyl operators. Prove that representations with trivial central character have zero Plancherel measure.

**Exercise 5.13** (Weyl quantization and Wigner distributions). For  $a \in \mathcal{S}(V \oplus V^*)$ , define  $\text{Op}_h^W(a) : \mathcal{S}(V) \rightarrow \mathcal{S}(V)$  by

$$(\text{Op}_h^W(a)\psi)(x) = \frac{1}{(2\pi\hbar)^n} \int_{V \times V^*} e^{ip(x-y)/\hbar} a\left(\frac{x+y}{2}, p\right) \psi(y) dy dp.$$

For polynomial symbols, use the same formula as an oscillatory integral on  $\mathcal{S}(V)$ .

For  $\psi, \varphi \in \mathcal{S}(V)$ , define

$$W_h(\psi, \varphi)(q, p) = \frac{1}{(2\pi\hbar)^n} \int_V e^{-ip(y)/\hbar} \overline{\psi(q - y/2)} \varphi(q + y/2) dy.$$

Write  $W_h[\psi] = W_h(\psi, \psi)$ .

For  $\ell \in V^*$  and  $v \in V$ , prove

$$\text{Op}_h^W((q, p) \mapsto \ell(q)) = Q_\ell, \quad \text{Op}_h^W((q, p) \mapsto p(v)) = P_v.$$

Prove

$$\text{Op}_h^W(a)^* = \text{Op}_h^W(\bar{a}), \quad \|\text{Op}_h^W(a)\|_{\text{HS}} = (2\pi\hbar)^{-n/2} \|a\|_2,$$

and

$$\langle \psi, \text{Op}_h^W(a)\varphi \rangle = \int_{V \oplus V^*} a(q, p) W_h(\psi, \varphi)(q, p) dq dp.$$

Prove that for  $a, b \in \mathcal{S}(V \oplus V^*)$  there is a unique  $a \star_{\hbar} b \in \mathcal{S}(V \oplus V^*)$  satisfying

$$\mathrm{Op}_{\hbar}^W(a \star_{\hbar} b) = \mathrm{Op}_{\hbar}^W(a) \mathrm{Op}_{\hbar}^W(b).$$

Derive the oscillatory-integral formula for  $\star_{\hbar}$  and prove

$$a \star_{\hbar} b = ab + \frac{i\hbar}{2}\{a, b\} + O(\hbar^2), \quad \frac{a \star_{\hbar} b - b \star_{\hbar} a}{i\hbar} = \{a, b\} + O(\hbar^2)$$

in the Schwartz-symbol topology. Prove that the complete asymptotic expansion agrees coefficient by coefficient with the Moyal product. First prove that every coefficient of that product preserves Schwartz functions, so for  $E = V \oplus V^*$  it restricts to a formal product on  $\mathcal{S}(V \oplus V^*)[[\hbar]]$ . For  $k \in (V \oplus V^*)^*$  put  $e_k(z) = e^{ik(z)}$ . Extend Weyl quantization to these oscillatory symbols and prove that  $\mathrm{Op}_{\hbar}^W(e_k)$  is unitary and that

$$\mathrm{Op}_{\hbar}^W(e_k) \mathrm{Op}_{\hbar}^W(e_{\ell}) = e^{-i\hbar\Pi(k, \ell)/2} \mathrm{Op}_{\hbar}^W(e_{k+\ell}).$$

Deduce the operator realization of the Moyal plane-wave identity.

**Exercise 5.14** (Groenewold–van Hove obstruction). Specialize to  $V = \mathbb{R}$  and write  $Q = Q_{\mathrm{id}_{\mathbb{R}}}$  and  $P = P_1$ . Prove that the polynomials of degree at most two form a Lie subalgebra for the Poisson bracket and that Weyl quantization is an exact Lie-algebra quantization on this subalgebra. Quantize  $q^2 p^2$  using

$$q^2 p^2 = \frac{1}{9}\{q^3, p^3\} = \frac{1}{3}\{q^2 p, qp^2\}$$

and prove that there is no linear quantization of the polynomial Poisson algebra satisfying

$$\mathcal{Q}(1) = I, \quad \mathcal{Q}(q) = Q, \quad \mathcal{Q}(p) = P, \quad \mathcal{Q}(q^m) = Q^m, \quad \mathcal{Q}(p^m) = P^m,$$

together with

$$\mathcal{Q}(\{f, g\}) = (i\hbar)^{-1}[\mathcal{Q}(f), \mathcal{Q}(g)]$$

on a common invariant domain.

**Exercise 5.15** (Classical and quantum evolution). Let  $H \in C^\infty(V \oplus V^*, \mathbb{R})$  have bounded derivatives of order at least two. Define its Hamiltonian field by

$$da(X_H) = \{a, H\}$$

for every smooth  $a$ , let  $\Phi_t$  be the flow of  $X_H$ , and let  $\mathrm{Op}_{\hbar}^W(H)$  denote its self-adjoint realization. Define

$$U_{\hbar}(t) = e^{-it \mathrm{Op}_{\hbar}^W(H)/\hbar}.$$

For  $a \in \mathcal{S}(V \oplus V^*)$ , define

$$a_t = a \circ \Phi_t, \quad A_{\hbar}(t) = U_{\hbar}(t)^* \mathrm{Op}_{\hbar}^W(a) U_{\hbar}(t).$$

Prove

$$\frac{d}{dt} a_t = \{a_t, H\}, \quad \frac{d}{dt} A_{\hbar}(t) = \frac{i}{\hbar} [\mathrm{Op}_{\hbar}^W(H), A_{\hbar}(t)].$$

Use as theorem input [17] the semiclassical operator estimate that the Moyal commutator remainder obtained from symbols with the displayed derivative bounds has operator norm  $O(\hbar)$ , uniformly for  $|t| \leq T$ , after division by  $i\hbar$ . Apply Duhamel's formula to prove, for every  $T > 0$ , Egorov's theorem

$$\sup_{|t| \leq T} \|A_{\hbar}(t) - \mathrm{Op}_{\hbar}^W(a \circ \Phi_t)\| = O_T(\hbar).$$

Prove exact equality for quadratic  $H$ . Derive Ehrenfest's equations for the expectations of  $Q_{\ell}$  and  $P_v$  for every  $\ell \in V^*$  and  $v \in V$ .

**Exercise 5.16** (Linear symplectic covariance). Let  $S$  be a linear symplectic automorphism of  $V \oplus V^*$ . Construct unitary implementers for cotangent lifts of linear automorphisms of  $V$ , for symplectic shears defined by quadratic forms, and for the exchange of position and momentum implemented by the Fourier transform. Prove for these generators that

$$\mu(S) \text{Op}_h^W(a) \mu(S)^* = \text{Op}_h^W(a \circ S^{-1}), \quad W_h[\mu(S)\psi] = W_h[\psi] \circ S^{-1}.$$

Deduce the covariance for their products and prove that each implementer is determined up to phase. Show that coherent choices form the metaplectic double cover, whose two lifts of a fixed symplectic map differ by sign. Use the Groenewold–van Hove obstruction to show why exact covariance cannot extend to every nonlinear canonical transformation.

**Exercise 5.17** (Wigner–Laguerre identity). Use the generalized Laguerre polynomials and their orthogonality. For

$$H_{\text{cl}}(q, p) = \frac{p^2}{2m} + \frac{m\omega^2 q^2}{2}, \quad \psi_n(q) = \left(\frac{m\omega}{\hbar}\right)^{1/4} h_n\left(\sqrt{\frac{m\omega}{\hbar}} q\right),$$

prove

$$W_h[\psi_n](q, p) = \frac{(-1)^n}{\pi \hbar} e^{-2H_{\text{cl}}(q, p)/(\hbar\omega)} L_n\left(\frac{4H_{\text{cl}}(q, p)}{\hbar\omega}\right).$$

Verify the position and momentum marginals and identify the zeros and sign changes of the Wigner distribution from those of  $L_n$ .

**Exercise 5.18** (Laguerre tomography). For the dimensionless quadrature

$$X_\theta = \sqrt{\frac{m\omega}{\hbar}} Q \cos \theta + \frac{P}{\sqrt{m\hbar\omega}} \sin \theta,$$

prove that its probability density in a state  $\psi$  is the Radon transform of  $W_h[\psi]$  along the corresponding phase-space lines. Derive the Fourier-slice inversion formula reconstructing the Wigner distribution from the family of quadrature densities. For the oscillator state  $\psi_n$ , use the Wigner–Laguerre identity to prove that the reconstructed radial profile is

$$\frac{(-1)^n}{\pi \hbar} e^{-2H_{\text{cl}}/(\hbar\omega)} L_n\left(\frac{4H_{\text{cl}}}{\hbar\omega}\right),$$

while every quadrature density is a scaled  $e^{-x^2} H_n(x)^2$ . Deduce the predicted number of quadrature nodes, the radial zero circles determined by  $L_n$ , and the sign at the phase-space origin. State how homodyne quadrature data distinguish adjacent number states and detect Wigner negativity after inverse Radon reconstruction.

## 6 Poincaré Group

**Exercise 6.1** (Minkowski momentum orbits and relativistic dispersion). A *time-oriented Minkowski space* is a real four-dimensional vector space  $M$  with a symmetric nondegenerate form  $g$  of signature  $(1, 3)$  and a chosen component of the cone  $\{v : g(v, v) > 0\}$ . Let  $SO^+(M, g)$  be the group preserving  $g$ , the orientation, and the time orientation. Define the proper orthochronous Poincaré group by

$$\mathcal{P}^+(M, g) = M \rtimes SO^+(M, g), \quad (a, A)(b, B) = (a + Ab, AB).$$

Prove that  $(a, A)$  acts on the affine space underlying  $M$  by  $x \mapsto a + Ax$ , that this identifies  $\mathcal{P}^+(M, g)$  with its orientation- and time-orientation-preserving affine isometries, and that the induced action on momentum covectors factors through  $SO^+(M, g)$ . Use  $g$  to identify  $M^\vee$  with  $M$ . For  $m \geq 0$ , define the positive mass shell

$$\mathcal{O}_m^+ = \{p \in M^\vee : g^{-1}(p, p) = m^2 c^2, p^\sharp \text{ is future-pointing}\},$$

with the zero covector removed when  $m = 0$ .

Let  $u$  be a future-pointing observer with  $g(u, u) = c^2$ . Define

$$E_u(p) = g(p^\sharp, u), \quad \mathbf{p}_u = p^\sharp - \frac{E_u(p)}{c^2} u, \quad |\mathbf{p}_u|^2 = -g(\mathbf{p}_u, \mathbf{p}_u).$$

Prove

$$E_u(p)^2 = c^2 |\mathbf{p}_u|^2 + m^2 c^4$$

and show that  $SO^+(M, g)$  acts transitively on each  $\mathcal{O}_m^+$ . For  $m > 0$ , prove that the stabilizer of a point is the rotation group of its negative-definite orthogonal complement and hence is isomorphic to  $SO(3)$ . For  $m = 0$ , show that the stabilizer preserves the Euclidean plane  $p^\perp/\mathbb{R}p$  and is isomorphic to  $SO(2) \ltimes \mathbb{R}^2$ .

Put  $Q(p) = g^{-1}(p, p)$ . For a Lorentz-invariant volume form  $dV$  on  $M^\vee$ , choose along the shell a vector field  $N$  satisfying  $dQ(N) = 1$ . Define

$$(\iota_N dV)(v_1, v_2, v_3) = dV(N, v_1, v_2, v_3)$$

and restrict this three-form to the shell. Prove that the restriction is independent of  $N$  and constructs an invariant measure  $d\mu_m$ , unique up to scale. Show that in the observer decomposition it is proportional to

$$\frac{d^3 \mathbf{p}_u}{E_u(\mathbf{p})}.$$

Deduce that the mass-shell relation and this measure are independent of the observer used to express them.

**Exercise 6.2** (The Lorentz spin cover and finite-dimensional fields). Let  $S$  be a two-dimensional complex vector space equipped with a nonzero alternating form  $\epsilon : \Lambda^2 S \rightarrow \mathbb{C}$ . Let  $\text{Herm}(S)$  be the fixed real subspace of  $S \otimes \bar{S}$  under conjugation followed by exchange of the factors, and use  $\epsilon \otimes \bar{\epsilon}$  to define its determinant. Prove that

$$q(X) = \det X$$

has signature  $(1, 3)$ , with the positive-definite forms selecting a time orientation. For  $A \in SL(S)$ , define

$$X \mapsto (A \otimes \bar{A})X.$$

Prove that this action preserves  $q$  and the chosen orientations and induces an exact sequence

$$1 \longrightarrow \{\pm I\} \longrightarrow SL(S) \longrightarrow SO^+(\text{Herm}(S), q) \longrightarrow 1.$$

Use as theorem input [21] the finite-dimensional Lorentz-representation classification: every irreducible finite-dimensional complex representation of  $SL(S)$ , regarded as the spin cover of the real Lorentz group, is uniquely of the form

$$\text{Sym}^{2j_L} S \otimes \text{Sym}^{2j_R} \bar{S}, \quad j_L, j_R \in \frac{1}{2}\mathbb{Z}_{\geq 0}.$$

Identify the scalar, left- and right-handed Weyl-spinor, Lorentz-vector, and self-dual and anti-self-dual two-form representations. Restrict to the massive little group  $SU(2) \subset SL(S)$  and derive

$$(j_L, j_R)|_{SU(2)} \cong \bigoplus_{j=|j_L-j_R|}^{j_L+j_R} V_j.$$

Prove that no nontrivial representation in the displayed family is unitary for the full Lorentz group, and distinguish these covariant field representations from the unitary one-particle representations constructed below.

**Exercise 6.3** (Wigner induction and the Poincaré particle classification). Let

$$\tilde{\mathcal{P}} = M \rtimes SL(S)$$

be the spin Poincaré group. Fix a Lorentz orbit  $\mathcal{O} \subset M^\vee$ , a point  $p_0 \in \mathcal{O}$ , its little group  $K \subset SL(S)$ , and a unitary representation  $\sigma : K \rightarrow U(V_\sigma)$ . Define the associated bundle

$$\mathcal{E}_\sigma = SL(S) \times_K V_\sigma \longrightarrow \mathcal{O}$$

as the quotient by  $(A, v) \sim (Ak, \sigma(k)^{-1}v)$ , with  $[A, v]$  lying over  $Ap_0$ . Define the Hilbert space  $L^2(\mathcal{O}, \mathcal{E}_\sigma; d\mu)$  of its square-integrable sections. For  $(a, A) \in \tilde{\mathcal{P}}$ , define intrinsically

$$[U(a, A)\psi](p) = e^{-ip(a)/\hbar} A \cdot \psi(A^{-1}p),$$

where  $A \cdot$  is the induced map between the associated fibers. Prove that this is a strongly continuous unitary representation and that the translation spectral measure is supported on  $\mathcal{O}$ .

Call a unitary Poincaré representation *positive-energy* when its translation spectral measure is supported in the closed future cone. Use as theorem input the Mackey–Wigner imprimitivity theorem [33, 45]: every irreducible positive-energy unitary representation whose translation spectrum is supported on one orbit is obtained by this construction, and two such representations are equivalent exactly when their orbit and little-group representation data are equivalent. Deduce that a massive particle is labelled by  $m > 0$  and an irreducible  $SU(2)$  representation  $V_j$ , so it has  $2j + 1$  rest-frame polarizations. For a massless orbit, show that little-group representations trivial on the translation subgroup are labelled by helicity  $h \in \frac{1}{2}\mathbb{Z}$ , while nontrivial translation action produces the continuous-spin family and an additional positive scale.

On the common invariant space of smooth vectors, let  $P$  be the translation generator and let  $J$  be the Lorentz bivector generator. Using the Minkowski volume form, define the Pauli–Lubanski covector by

$$W = *(P \wedge J).$$

With this sign convention for the Minkowski form, prove

$$P^2 = m^2 c^2 I, \quad W^2 = -m^2 c^2 \hbar^2 j(j+1)I$$

in the massive representation and

$$P^2 = 0, \quad W = \hbar h P$$

in a finite-helicity massless representation. Explain why this classification constrains masses, spins, helicities, and polarization counts but does not determine the numerical masses or interactions.

**Exercise 6.4** (Covariant wave equations and helicity polarizations). Use units  $c = 1$  in this exercise. Let  $\square$  be the constant-coefficient d'Alembertian whose symbol is  $-g^{-1}(p, p)$ . For  $\hat{\phi} \in C_c^\infty(\mathcal{O}_m^+)$ , put

$$\phi(x) = \int_{\mathcal{O}_m^+} e^{-ip(x)/\hbar} \hat{\phi}(p) d\mu_m(p).$$

Prove

$$(\square + m^2/\hbar^2)\phi = 0$$

and show that completion in the mass-shell norm realizes the positive energy spin-zero Wigner representation.

Let  $\text{Cl}(M, g)$  be the Clifford algebra defined by

$$vw + wv = 2g(v, w)I.$$

A *complex spin module* is a four-dimensional  $\mathbb{Z}/2\mathbb{Z}$ -graded irreducible module  $\Sigma = \Sigma^+ \oplus \Sigma^-$  for the complexified Clifford algebra in which multiplication by  $M$  exchanges the two graded summands. Let  $\mathcal{D}$  be Clifford multiplication followed by differentiation. Prove  $\mathcal{D}^2 = \square$  and show that

$$(i\hbar\mathcal{D} - m)\psi = 0$$

implies the Klein–Gordon equation and that its positive-frequency solution space realizes the massive spin-1/2 mass shell. At  $m = 0$ , prove that the two chiral positive-frequency solution spaces carry helicities  $\pm\frac{1}{2}$ .

Put  $M_{\mathbb{C}} = M \otimes_{\mathbb{R}} \mathbb{C}$ . For a null covector  $p$ , define the complex transverse-polarization space

$$\mathcal{T}_p = \{\varepsilon \in M_{\mathbb{C}}^\vee : g^{-1}(p, \varepsilon) = 0\}/\mathbb{C}p.$$

Interpret  $\varepsilon$  as the amplitude of a plane-wave vector potential and the quotient by  $\mathbb{C}p$  as its gauge equivalence. Show that rotations in the massless little group split  $\mathcal{T}_p$  into helicities  $+1$  and  $-1$ , giving the two plane-wave polarizations of the source-free Maxwell equations. Show that the transverse traceless symmetric square has precisely the two helicity lines  $+2$  and  $-2$ . State the gauge, positive-energy, and finite-helicity assumptions required for these polarization counts, without invoking the spin–statistics relation.

**Exercise 6.5** (Spin-weighted spherical harmonics and radiation patterns). Write  $S^2 = SU(2)/U(1)$ . The *spin-weight- $s$  line bundle* is the bundle associated to the character of  $U(1)$  under which rotation of an oriented tangent frame through  $\psi$  acts by  $e^{-is\psi}$ . Write it as

$$\mathcal{L}_s = SU(2) \times_{U(1)} \mathbb{C}_s \longrightarrow S^2.$$

For integers  $\ell \geq |s|$  and  $-\ell \leq m \leq \ell$ , define

$${}_sY_{\ell m}(\theta, \phi) = (-1)^s \sqrt{\frac{2\ell+1}{4\pi}} \overline{D_{m, -s}^\ell(R_z(\phi)R_y(\theta))},$$

using the Wigner coefficients in the Condon–Shortley convention. Verify the frame-equivariance law, prove

$$\int_{S^2} {}_sY_{\ell m} \overline{{}_sY_{\ell' m'}} d\Omega = \delta_{\ell\ell'} \delta_{mm'},$$

and prove completeness in  $L^2(S^2, \mathcal{L}_s)$ . Use the Jacobi-polynomial formula for the Wigner  $d$ -functions to express every  ${}_sY_{\ell m}$  as a phase times powers of  $\sin(\theta/2)$  and  $\cos(\theta/2)$  times a Jacobi polynomial.

Deduce that helicity- $h$  radiation has no angular multipoles with  $\ell < |h|$ . In the spin-weight  $-2$  convention, derive

$$_{-2}Y_{2,2} = \sqrt{\frac{5}{64\pi}}(1 + \cos\theta)^2 e^{2i\phi}, \quad _{-2}Y_{2,-2} = \sqrt{\frac{5}{64\pi}}(1 - \cos\theta)^2 e^{-2i\phi}.$$

For a prepared pure quadrupole helicity mode, deduce the normalized angular intensity

$$\frac{I_{\pm}(\theta)}{I_{\pm}^{\max}} = \left( \frac{1 \pm \cos\theta}{2} \right)^4$$

and its symmetry-enforced dark direction. State the linearized massless-spin-2, far-field, single-mode, polarization-frame, and angular-resolution assumptions under which this pattern and the absence of monopole and dipole spin-2 radiation are tested.

**Exercise 6.6** (Bessel mass-shell kernels and relativistic signal propagation). Use units  $c = \hbar = 1$ , put  $E_p = \sqrt{|p|^2 + m^2}$ , and let  $\Theta$  be the Heaviside function. For a test function  $f$  on spacetime, define the Dirac distribution at the origin by  $\langle \delta_0, f \rangle = f(0)$ . Define the retarded Klein–Gordon kernel by

$$G_{\text{ret}}(t, x) = \Theta(t) \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{\sin(E_p t)}{E_p} e^{ip \cdot x}.$$

Using the radial Fourier–Bessel transform, prove

$$(\square + m^2)G_{\text{ret}} = \delta_0$$

and, for  $r = |x|$  and  $\tau = \sqrt{t^2 - r^2}$ ,

$$G_{\text{ret}}(t, r) = \Theta(t) \left[ \frac{\delta(t^2 - r^2)}{2\pi} - \Theta(t^2 - r^2) \frac{m J_1(m\tau)}{4\pi\tau} \right].$$

Deduce exact vanishing outside the light cone, a light-cone impulse, and the massive interior tail. Show that the tail kernel has proper-time zeros  $m\tau = j_{1,n}$ , where  $j_{1,n}$  are the positive zeros of  $J_1$ .

For  $\text{Re } z > 0$ , define the modified Bessel function by

$$K_\nu(z) = \int_0^\infty e^{-z \cosh u} \cosh(\nu u) du.$$

For the positive-frequency mass-shell kernel

$$\Delta_+(t, x) = \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3} \frac{1}{2E_p} e^{-iE_p t + ip \cdot x},$$

prove that at spacelike separation  $\rho = \sqrt{r^2 - t^2} > 0$ ,

$$\Delta_+(t, r) = \frac{m}{4\pi^2 \rho} K_1(m\rho).$$

Derive

$$K_1(z) \sim \sqrt{\frac{\pi}{2z}} e^{-z} \quad (z \rightarrow +\infty)$$

by Laplace’s method. Restore  $c$  and  $\hbar$  to identify the Compton length  $\hbar/(mc)$  and the invariant tail-node lengths  $j_{1,n}\hbar/(mc)$ . Distinguish this nonzero spacelike positive-frequency correlation from the causal response represented by  $G_{\text{ret}}$ .



For a wave packet concentrated near spatial momentum  $p_0$ , derive

$$v_{\text{group}} = \frac{c^2 p_0}{\sqrt{c^2 |p_0|^2 + m^2 c^4}}, \quad \Delta t \sim \frac{L m^2 c}{2 |p_0|^2} \quad (|p_0| \gg mc)$$

for its delay relative to a light signal over distance  $L$ . State the flat-spacetime, free-field, point-impulse, wave-packet, and detector-smearing assumptions under which the causal front, Bessel tail nodes, exponential spacelike decay, and time-of-flight delay are physical predictions.

## 7 Potential Scattering

**Exercise 7.1** (Outgoing Green kernels and the Lippmann–Schwinger equation). Let  $\mu > 0$  be a reduced mass, let  $k > 0$ , put

$$E = \frac{\hbar^2 k^2}{2\mu}, \quad H = -\frac{\hbar^2}{2\mu} \Delta + V,$$

and initially take  $V$  to be a real Schwartz function on  $\mathbb{R}^3$ . For an incident direction  $\omega \in S^2$ , call a solution of  $(H - E)\psi = 0$  *outgoing* when

$$\psi(x) = e^{ik\omega \cdot x} + f(\hat{x}, \omega; k) \frac{e^{ik|x|}}{|x|} + o(|x|^{-1}), \quad \hat{x} = \frac{x}{|x|},$$

and call  $f$  its *scattering amplitude*. For outgoing direction  $\omega'$ , define the scattering angle by  $\cos \theta = \omega' \cdot \omega$ .

For  $\phi \in C_c^\infty(\mathbb{R}^3)$ , let  $\delta_0$  be the Dirac distribution at the origin, and interpret an identity of distributions after pairing both sides with every such  $\phi$ . Prove in the sense of distributions that

$$G_k^+(x) = \frac{e^{ik|x|}}{4\pi|x|}, \quad (-\Delta - k^2)G_k^+ = \delta_0,$$

and verify the Sommerfeld condition  $(\partial_r - ik)G_k^+ = o(r^{-1})$ . Derive the Lippmann–Schwinger equation

$$\psi(x) = e^{ik\omega \cdot x} - \frac{2\mu}{\hbar^2} \int_{\mathbb{R}^3} G_k^+(x - y) V(y) \psi(y) dy.$$

First take  $V$  compactly supported and use

$$G_k^+(x - y) = \frac{e^{ik|x|}}{4\pi|x|} e^{-ik\hat{x} \cdot y} + O(|x|^{-2})$$

for bounded  $y$  to prove

$$f(\omega', \omega; k) = -\frac{\mu}{2\pi\hbar^2} \int_{\mathbb{R}^3} e^{-ik\omega' \cdot y} V(y) \psi(y) dy.$$

Extend the formula to Schwartz  $V$  by controlling the tails with its decay.

**Exercise 7.2** (Probability current, Born series, and screened-Coulomb counts). For the probability current

$$j_\psi = \frac{\hbar}{\mu} \text{Im}(\bar{\psi} \nabla \psi),$$

divide the outgoing radial flux through  $r^2 d\Omega$  by the incident flux and derive

$$\frac{d\sigma}{d\Omega} = |f(\omega', \omega; k)|^2.$$

Replace  $\psi$  in the amplitude by the incident plane wave to obtain the first Born term

$$f_B(\omega', \omega; k) = -\frac{\mu}{2\pi\hbar^2} \int_{\mathbb{R}^3} e^{-iQ \cdot y} V(y) dy, \quad Q = k(\omega' - \omega).$$

On any Banach space  $X$  containing the incident plane wave and on which

$$K_k \phi = \frac{2\mu}{\hbar^2} G_k^+ * (V\phi)$$

is bounded, prove that  $\|K_k\|_{X \rightarrow X} < 1$  implies the convergent Born series

$$\psi = \sum_{n \geq 0} (-K_k)^n e^{ik\omega \cdot x}.$$

Writing  $q = \|K_k\|_{X \rightarrow X}$ , prove the geometric remainder bounds

$$\|\psi - e^{ik\omega \cdot x}\|_X \leq \frac{q}{1-q} \|e^{ik\omega \cdot x}\|_X,$$

$$\|\psi - (I - K_k)e^{ik\omega \cdot x}\|_X \leq \frac{q^2}{1-q} \|e^{ik\omega \cdot x}\|_X,$$

and explain which bound controls the first and second Born approximations to the amplitude when the amplitude integral is a bounded functional on  $X$ . For the screened Coulomb potential

$$V_\alpha(x) = g \frac{e^{-\alpha|x|}}{|x|}, \quad \alpha > 0,$$

justify the same calculation by approximation, prove

$$\int_{\mathbb{R}^3} e^{-iQ \cdot x} \frac{e^{-\alpha|x|}}{|x|} dx = \frac{4\pi}{\|Q\|^2 + \alpha^2}, \quad \|Q\|^2 = 4k^2 \sin^2 \frac{\theta}{2},$$

and derive

$$\frac{d\sigma_B}{d\Omega} = \frac{4\mu^2 g^2}{\hbar^4 (4k^2 \sin^2(\theta/2) + \alpha^2)^2}.$$

State the weak-coupling, single-scattering, wave-packet, far-field, and angular resolution assumptions under which this differential cross section predicts detector counts.

**Exercise 7.3** (Scattering operators and elastic unitarity). For a real radial potential supported in  $\{x : \|x\| \leq R\}$ , define the on-shell scattering operator  $S(k)$  on angular wave packets by sending the incoming far-field coefficient to the outgoing one, with the free propagation phases normalized so that  $V = 0$  gives  $S(k) = I$ . Use conservation of probability current to prove that  $S(k)$  is unitary, and use rotation covariance together with the multiplicity-free decomposition

$$L^2(S^2) = \widehat{\bigoplus_{\ell \geq 0} \mathcal{H}_\ell}$$

where  $\mathcal{H}_\ell$  is the degree- $\ell$  spherical-harmonic summand, to prove

$$S(k)|_{\mathcal{H}_\ell} = S_\ell(k)I, \quad S_\ell(k) = e^{2i\delta_\ell(k)}$$

for real phase shifts  $\delta_\ell$ , determined modulo  $\pi$ .

**Exercise 7.4** (Partial waves, Legendre amplitudes, and the optical theorem). Put

$$j_\ell(z) = \sqrt{\frac{\pi}{2z}} J_{\ell+1/2}(z).$$

Project the Euclidean Fourier kernel onto the spherical-harmonic summands and derive the distinguished expansion

$$e^{ikr \cos \theta} = \sum_{\ell \geq 0} i^\ell (2\ell + 1) j_\ell(kr) P_\ell(\cos \theta),$$

where  $P_\ell = P_\ell^{(0,0)}$  is the Legendre polynomial. Separate the radial equation and show that the regular radial function  $u_\ell$  satisfies

$$u_\ell''(r) + \left( k^2 - \frac{\ell(\ell+1)}{r^2} - \frac{2\mu}{\hbar^2} V(r) \right) u_\ell(r) = 0,$$

with

$$u_\ell(r) \sim \sin \left( kr - \frac{\ell\pi}{2} + \delta_\ell(k) \right) \quad (r \rightarrow \infty).$$

Match incoming and outgoing spherical waves to derive

$$f(\theta; k) = \frac{1}{2ik} \sum_{\ell \geq 0} (2\ell + 1) (S_\ell(k) - 1) P_\ell(\cos \theta) = \frac{1}{k} \sum_{\ell \geq 0} (2\ell + 1) e^{i\delta_\ell} \sin \delta_\ell P_\ell(\cos \theta).$$

Use Legendre orthogonality and  $P_\ell(1) = 1$  to prove

$$\sigma_{\text{el}}(k) = \frac{4\pi}{k^2} \sum_{\ell \geq 0} (2\ell + 1) \sin^2 \delta_\ell(k), \quad \sigma_{\text{el}}(k) = \frac{4\pi}{k} \text{Im } f(0; k),$$

and obtain the partial-wave unitarity bound

$$\sigma_\ell(k) \leq \frac{4\pi}{k^2} (2\ell + 1).$$

Identify exactly where reality of the potential and absence of inelastic channels enter the proof.

**Exercise 7.5** (Pöschl–Teller Jost functions and reflectionless transmission). For  $k > 0$  and a nonnegative integer  $\nu$ , consider

$$-\psi''(x) - \nu(\nu+1) \text{sech}^2 x \psi(x) = k^2 \psi(x).$$

Put  $z = (1 - \tanh x)/2$  and prove that the right Jost solution normalized by  $\phi_{\nu,k}(x) \sim e^{ikx}$  as  $x \rightarrow +\infty$  is

$$\phi_{\nu,k}(x) = e^{ikx} {}_2F_1(-\nu, \nu+1; 1-ik; z(x)).$$

Use the hypergeometric connection formula, interpreted by continuation from generic parameters, to determine the coefficients

$$\phi_{\nu,k}(x) \sim A_\nu(k) e^{ikx} + B_\nu(k) e^{-ikx} \quad (x \rightarrow -\infty).$$

Show that polynomial termination gives

$$B_\nu(k) = 0, \quad A_\nu(k) = \prod_{j=1}^{\nu} \frac{k - ij}{k + ij}.$$

For unit incoming amplitude, define  $T_\nu(k) = A_\nu(k)^{-1}$  and  $R_\nu(k) = B_\nu(k)A_\nu(k)^{-1}$ . Prove

$$R_\nu(k) = 0, \quad T_\nu(k) = \prod_{j=1}^{\nu} \frac{k + ij}{k - ij}, \quad |T_\nu(k)| = 1.$$

Continue  $T_\nu$  meromorphically in  $k$  and identify its poles at  $k = ij$ ,  $1 \leq j \leq \nu$ , with the bound-state energies  $E = -j^2$ . Thus transmission measurements test both reflectionlessness and the hypergeometric pole set.

**Exercise 7.6** (Hard-sphere phase shifts and effective-range scattering). Define the spherical Neumann function by

$$n_\ell(z) = (-1)^{\ell+1} \sqrt{\frac{\pi}{2z}} J_{-\ell-1/2}(z).$$

For scattering from an impenetrable sphere of radius  $a$ , impose the Dirichlet condition at  $r = a$  and write the exterior radial factor as

$$\cos \delta_\ell j_\ell(kr) - \sin \delta_\ell n_\ell(kr).$$

Prove

$$\tan \delta_\ell(k) = \frac{j_\ell(ka)}{n_\ell(ka)}.$$

Use the Bessel series to show that  $\delta_0 = -ka$  modulo  $\pi$  and  $\delta_\ell = O((ka)^{2\ell+1})$  for  $\ell > 0$ . Deduce the low-energy predictions

$$f(\theta; k) \longrightarrow -a, \quad \sigma_{\text{el}}(k) \longrightarrow 4\pi a^2 \quad (ka \downarrow 0).$$

For a short-range  $s$ -wave phase shift, define the effective-range parameters by

$$k \cot \delta_0(k) = -\frac{1}{a_s} + \frac{1}{2} r_e k^2 + O(k^4).$$

For the hard sphere, use  $\delta_0 = -ka$  to prove

$$a_s = a, \quad r_e = \frac{2a}{3}.$$

State how low-energy measurements at two or more momenta distinguish the scattering length from the effective range.

**Exercise 7.7** (Square-well thresholds and Breit–Wigner resonances). For the attractive square well

$$V(r) = \begin{cases} -V_0, & 0 < r < a, \\ 0, & r > a, \end{cases} \quad V_0 > 0,$$

put  $K = (k^2 + 2\mu V_0/\hbar^2)^{1/2}$  and match the logarithmic derivatives of the regular  $s$ -wave solutions to prove

$$K \cot(Ka) = k \cot(ka + \delta_0(k)).$$

Define the scattering length by  $\delta_0(k) = -ka_s + o(k)$  and derive

$$a_s = a - \frac{\tan(\gamma a)}{\gamma}, \quad \gamma = \frac{\sqrt{2\mu V_0}}{\hbar}.$$

Show that  $a_s$  diverges when  $\gamma a = (n + \frac{1}{2})\pi$ ,  $n \in \mathbb{Z}_{\geq 0}$ , and identify these values with the appearance of an  $s$ -wave state at zero binding energy.

If an elastic phase shift has a simple resonance at  $E_R$  with

$$\cot \delta_\ell(E) = \frac{2(E_R - E)}{\Gamma} + O\left(\frac{(E - E_R)^2}{\Gamma^2}\right), \quad \Gamma > 0,$$

derive the Breit–Wigner line shape

$$\sigma_\ell(E) = \frac{4\pi}{k^2} (2\ell + 1) \frac{\Gamma^2/4}{(E - E_R)^2 + \Gamma^2/4}$$

to leading order, prove that its full width at half maximum is  $\Gamma$ , and show that the Wigner delay  $2\hbar d\delta_\ell/dE$  equals  $4\hbar/\Gamma$  at the resonance. State the hard-core or square-well model, elastic-channel, energy-resolution, and finite-range assumptions needed to infer a radius, scattering length, resonance energy, or width from measured angular distributions.

**Exercise 7.8** (Coulomb radial functions and phase shifts). For the repulsive Coulomb potential  $V(r) = \kappa/r$ , with  $\kappa > 0$ , put

$$\rho = kr, \quad \eta = \frac{\mu\kappa}{\hbar^2 k}, \quad C_\ell(\eta) = \frac{2^\ell e^{-\pi\eta/2} |\Gamma(\ell + 1 + i\eta)|}{(2\ell + 1)!}.$$

Verify that the regular radial solution is

$$F_\ell(\eta, \rho) = C_\ell(\eta) \rho^{\ell+1} e^{-i\rho} M(\ell + 1 - i\eta, 2\ell + 2, 2i\rho).$$

Derive from the Kummer series and the beta integral that

$$M(a, b, z) = \frac{\Gamma(b)}{\Gamma(a)\Gamma(b-a)} \int_0^1 e^{zt} t^{a-1} (1-t)^{b-a-1} dt$$

when  $\operatorname{Re} b > \operatorname{Re} a > 0$ . Analyze its two endpoints for the displayed Coulomb parameters and prove

$$F_\ell(\eta, \rho) = \sin\left(\rho - \eta \log(2\rho) - \frac{\ell\pi}{2} + \sigma_\ell(\eta)\right) + o(1), \quad \sigma_\ell(\eta) = \arg \Gamma(\ell + 1 + i\eta),$$

where the argument is chosen continuously from  $\sigma_\ell(0) = 0$ . Deduce the Coulomb partial-wave factor

$$S_\ell^C = e^{2i\sigma_\ell} = \frac{\Gamma(\ell + 1 + i\eta)}{\Gamma(\ell + 1 - i\eta)}.$$

**Exercise 7.9** (Coulomb amplitude and Rutherford detector counts). Because the Coulomb potential is not short-range, use the logarithmically modified radial phase above and define its nonforward amplitude by Abel regularization of the partial-wave expression.

For  $\operatorname{Re} s > 0$ , use the hypergeometric form of  $P_\ell$  to prove

$$\int_{-1}^1 (1-x)^{s-1} P_\ell(x) dx = 2^s (-1)^\ell \frac{\Gamma(s)^2}{\Gamma(s-\ell)\Gamma(s+\ell+1)}.$$

Continue in  $s$  and use the Legendre generating function to obtain, for  $-1 < x < 1$ , the Abel-summed identity

$$\lim_{r \uparrow 1} \sum_{\ell \geq 0} (2\ell + 1) (S_\ell^C - 1) r^\ell P_\ell(x) = -2i\eta 2^{i\eta} \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)} (1-x)^{-1-i\eta}.$$

With  $x = \cos \theta$ , derive the exact Coulomb amplitude away from the forward direction,

$$f_C(\theta) = -\frac{\eta}{2k \sin^2(\theta/2)} \frac{\Gamma(1+i\eta)}{\Gamma(1-i\eta)} \exp\left(-i\eta \log \sin^2 \frac{\theta}{2}\right),$$

and hence the Rutherford law

$$\frac{d\sigma_C}{d\Omega} = \left( \frac{\kappa}{4E \sin^2(\theta/2)} \right)^2.$$

For incident particle rate  $\dot{N}_{\text{in}}$ , target areal density  $n_T$ , acquisition time  $T$ , detector efficiency  $\epsilon(\theta)$ , and a narrow solid-angle bin  $\Delta\Omega$  away from  $\theta = 0$ , derive

$$N(\theta; \Delta\Omega) \simeq \dot{N}_{\text{in}} n_T T \epsilon(\theta) \frac{d\sigma_C}{d\Omega} \Delta\Omega.$$

State separately the assumptions of nonrelativistic spinless two-body motion, a thin target, pointlike unscreened charges, single scattering, negligible inelasticity, far-field detection, and sufficiently fine angular bins. State how screening, finite charge radius, recoil calibration, multiple scattering, and detector acceptance alter the comparison with measured counts.

**Exercise 7.10** (Schwarzschild geometry and gravitational redshift). A *Lorentzian metric* on a four-manifold is a smooth nondegenerate symmetric form of signature  $(-, +, +, +)$ . Call a tangent vector timelike, null, or spacelike according as its squared length is negative, zero, or positive. Use the Koszul formula to prove that such a metric has a unique torsion-free metric-compatible connection. Let

$$r_s = \frac{2G_N M}{c^2}, \quad f(r) = 1 - \frac{r_s}{r},$$

and on  $(r_s, \infty) \times \mathbb{R} \times S^2$  consider

$$g = -f(r)c^2 dt^2 + f(r)^{-1} dr^2 + r^2 g_{S^2}.$$

Use as theorem input Birkhoff's theorem [44]: every spherically symmetric Lorentzian vacuum solution with  $\text{Ric} = 0$  is locally isometric to a member of this family. Compute its Ricci tensor and verify  $\text{Ric} = 0$ . Compute the invariant

$$R_{abcd} R^{abcd} = \frac{48G_N^2 M^2}{c^4 r^6}$$

and use it to distinguish the curvature singularity at  $r = 0$  from the coordinate singularity at  $r = r_s$ .

Prove that time translation preserves  $g$ . For its timelike Killing field  $\partial_t$ , prove that  $-g(k, \partial_t)$  is conserved along a null geodesic with tangent  $k$ . Compare it with the frequency measured by static observers to derive

$$\frac{\nu_{\text{rec}}}{\nu_{\text{emit}}} = \sqrt{\frac{f(r_{\text{emit}})}{f(r_{\text{rec}})}}.$$

In particular, for reception at infinity predict  $\nu_{\infty}/\nu_r = \sqrt{1 - r_s/r}$ . State the static-emitter, static-receiver, geometric-optics, negligible-backreaction, and spectral-resolution assumptions under which a clock or spectral-line measurement tests the formula.

**Exercise 7.11** (Weierstrass elliptic functions and Schwarzschild light rays). Restrict a Schwarzschild null geodesic to an equatorial great circle. Let  $\mathcal{E} = f(r)c^2 \dot{t}$  and  $L = r^2 \dot{\phi}$  be its conserved energy and angular momentum per affine parameter, put  $b = cL/\mathcal{E}$ , and set  $u = 1/r$ . Derive

$$\left( \frac{du}{d\phi} \right)^2 = \frac{1}{b^2} - u^2 + r_s u^3, \quad \frac{d^2 u}{d\phi^2} + u = \frac{3}{2} r_s u^2.$$

Define the Weierstrass elliptic function by

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3, \quad \wp(z) = z^{-2} + O(z^2),$$

and prove that every nonradial orbit is obtained by analytic continuation of

$$u(\phi) = \frac{1/3 + 4\wp(\phi - \phi_0; g_2, g_3)}{r_s}, \quad g_2 = \frac{1}{12}, \quad g_3 = \frac{1}{216} - \frac{r_s^2}{16b^2}.$$

Show that its nontrivial finite-impact-parameter discriminant degeneration occurs at

$$r_{\text{ph}} = \frac{3}{2}r_s = \frac{3G_{\text{N}}M}{c^2}, \quad b_c = \frac{3\sqrt{3}G_{\text{N}}M}{c^2},$$

the unstable circular photon orbit and critical capture impact parameter. Expand the orbit equation for  $r_s/b \ll 1$  to derive the deflection angle

$$\delta\phi = \frac{4G_{\text{N}}M}{bc^2} + O\left(\frac{G_{\text{N}}^2M^2}{b^2c^4}\right).$$

Predict the far-field shadow angular radius  $b_c/D$  for an observer at distance  $D$ . State the vacuum, nonrotating, geometric-optics, thin-lens, far-observer, and source-model assumptions separately from the bending and shadow predictions.

**Exercise 7.12** (Regge–Wheeler equations and confluent-Heun scattering). Define the tortoise coordinate by

$$r_* = r + r_s \log\left(\frac{r}{r_s} - 1\right).$$

For integer spin  $s \in \{0, 1, 2\}$  and  $\ell \geq s$ , use as theorem input [37, 8] the scalar, electromagnetic, and axial gravitational separation theorem: after expansion in modes with time dependence  $e^{-i\omega t}$  and the corresponding spherical harmonics, the radial master field satisfies

$$\frac{d^2\psi}{dr_*^2} + \left(\frac{\omega^2}{c^2} - V_{\ell,s}(r)\right)\psi = 0,$$

$$V_{\ell,s}(r) = f(r) \left(\frac{\ell(\ell+1)}{r^2} + (1-s^2)\frac{r_s}{r^3}\right).$$

Show that  $r = r_s$  and  $r = \infty$  are respectively regular and irregular singular endpoints. After putting  $z = r/r_s$  and factoring the ingoing or outgoing exponential and power behavior, transform the radial equation to

$$y'' + \left(\alpha + \frac{\beta+1}{z} + \frac{\gamma+1}{z-1}\right)y' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right)y = 0.$$

Define its locally normalized solutions to be the confluent Heun functions and determine  $\alpha, \beta, \gamma, \mu, \nu$  from  $\omega r_s/c$ ,  $\ell$ , and  $s$  for the chosen asymptotic factors.

Define the horizon-ingoing and infinity-outgoing solutions by

$$\psi_{\text{in}} \sim e^{-i\omega r_*/c} \quad (r_* \rightarrow -\infty), \quad \psi_{\text{out}} \sim e^{+i\omega r_*/c} \quad (r_* \rightarrow +\infty),$$

and define the connection determinant

$$D_{\ell,s}(\omega) = W(\psi_{\text{in}}, \psi_{\text{out}}).$$

Use constancy of the Wronskian to express reflection and transmission amplitudes as confluent-Heun connection coefficients and prove flux conservation for real  $\omega$ .

**Exercise 7.13** (Black-hole resonances, photon spheres, and ringdown). Define a Schwarzschild quasinormal frequency to be a zero of the confluent-Heun connection determinant  $D_{\ell,s}(\omega)$  in the lower half-plane. Show that it is equivalently a mode ingoing at the horizon and outgoing at infinity, and that its time dependence is a damped oscillation.

For  $\ell \rightarrow \infty$ , locate the maximum of the Regge–Wheeler potential at the photon sphere. Apply the leading parabolic-barrier WKB connection formula, with the Langer replacement  $\ell(\ell+1) \sim (\ell + \frac{1}{2})^2$ , to derive

$$\omega_{\ell n} = \Omega_{\text{ph}} \left( \ell + \frac{1}{2} \right) - i\lambda_{\text{ph}} \left( n + \frac{1}{2} \right) + O(\Omega_{\text{ph}} \ell^{-1}),$$

$$\Omega_{\text{ph}} = \lambda_{\text{ph}} = \frac{c^3}{3\sqrt{3}G_N M}, \quad n = 0, 1, 2, \dots$$

Relate  $\Omega_{\text{ph}}$  to the critical impact parameter by  $\Omega_{\text{ph}} = c/b_c$ . If the source has cosmological redshift  $z_{\text{cos}}$  and angular-diameter distance  $D_A$ , predict the observed ringdown frequency, e-folding time, and shadow angular radius

$$f_{\ell n}^{\text{obs}} = \frac{\text{Re } \omega_{\ell n}}{2\pi(1 + z_{\text{cos}})}, \quad \tau_{\ell n}^{\text{obs}} = \frac{1 + z_{\text{cos}}}{|\text{Im } \omega_{\ell n}|}, \quad \theta_{\text{sh}} = \frac{b_c}{D_A},$$

all from the same photon-sphere geometry. State the isolated, nonrotating, asymptotically flat, linear-perturbation, eikonal, detector-frame, cosmological-redshift, and mode-identification assumptions under which shadow and ringdown measurements test these linked predictions.

## 8 Determinantal Kernels

**Exercise 8.1** (Scattering projections and fermionic correlations). Let

$$H_R = -\frac{\hbar^2}{2\mu} \Delta + V$$

be the Dirichlet realization of a real Schrödinger operator on the ball  $B_R \subset \mathbb{R}^3$ . Fix a Fermi energy  $E_F$  outside its spectrum and let

$$P_R = \mathbf{1}_{(-\infty, E_F]}(H_R)$$

have rank  $N$ . Suppose its integral kernel  $K_R$  is continuous. Prove intrinsically from  $P_R^2 = P_R = P_R^*$  that

$$\int_{B_R} K_R(x, z) K_R(z, y) dz = K_R(x, y), \quad \int_{B_R} K_R(x, x) dx = N,$$

and that every matrix  $[K_R(x_i, x_j)]$  is positive semidefinite.

The position density of the state filling this spectral subspace with  $N$  identical spinless fermions is

$$p_R(x_1, \dots, x_N) = \frac{1}{N!} \det[K_R(x_i, x_j)]_{i,j=1}^N.$$

Use the projection identity and determinant expansion to prove that  $p_R$  is a probability density and that, for  $1 \leq k \leq N$ ,

$$\frac{N!}{(N-k)!} \int_{B_R^{N-k}} p_R(x_1, \dots, x_N) dx_{k+1} \cdots dx_N = \det[K_R(x_i, x_j)]_{i,j=1}^k.$$



Assume now that  $V$  is radial and supported in a fixed ball. In angular momentum channel  $\ell$ , use the exterior asymptotic

$$u_\ell(r; k) \sim \sin\left(kr - \frac{\ell\pi}{2} + \delta_\ell(k)\right)$$

and the Dirichlet condition at  $r = R$  to derive

$$kR - \frac{\ell\pi}{2} + \delta_\ell(k) = n\pi + o(1) \quad (R \rightarrow \infty).$$

Where  $\delta_\ell$  is differentiable and no level crosses the chosen energy window, deduce that the shift of the radial density of states from its free value is  $\delta'_\ell(k)/\pi$ . Include the multiplicity  $2\ell + 1$  and state the finite-volume, elastic, threshold, angular-momentum cutoff, and detector-resolution assumptions under which fermion counts below  $E_F$  test the phase-shift prediction.

**Exercise 8.2** (Hermite Gaussian moments). Let  $\mathbb{P}$  be the probability measure on  $\mathbb{R}$  with density  $(2\pi)^{-1/2}e^{-x^2/2}$ , let  $X(x) = x$ , and write  $\mathbb{E}f(X) = \int_{\mathbb{R}} f(x) d\mathbb{P}(x)$  whenever the integral exists.

For this  $X$ , prove

$$\mathbb{E}[e^{tX}] = e^{t^2/2}.$$

Define the probabilists' Hermite polynomials by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}$$

and prove

$$\mathbb{E}[\text{He}_m(X) \text{He}_n(X)] = n! \delta_{mn}, \quad \mathbb{E}[X^{2n}] = \frac{(2n)!}{2^n n!}.$$

Relate  $\text{He}_n$  to the oscillator Hermite functions.

**Exercise 8.3** (Dyson ensembles and unitary orbit Jacobians). For  $\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$ , put  $\beta_D = \dim_{\mathbb{R}} \mathbb{F} \in \{1, 2, 4\}$ . The Gaussian Dyson ensemble on the self-adjoint operators on  $\mathbb{F}^N$  has density

$$d\mathbb{P}_{N, \beta_D}(H) = C_{N, \beta_D} \exp\left[-\frac{\beta_D N}{4\sigma^2} \text{Tr}(H^2)\right] dH$$

for the invariant Hilbert–Schmidt volume. For  $\lambda = (\lambda_1, \dots, \lambda_N)$  define

$$\Delta(\lambda) = \det[\lambda_i^{j-1}]_{i,j=1}^N.$$

Use the exterior-power determinant to prove

$$\Delta(\lambda) = \prod_{i < j} (\lambda_j - \lambda_i).$$

Compute the Jacobian of the unitary conjugation-orbit map on the regular spectral locus, including the stabilizer and permutation factors, and derive

$$p_{N,2}(\lambda) = \frac{1}{Z_{N,2}} e^{-N \sum_i \lambda_i^2 / (2\sigma^2)} |\Delta(\lambda)|^2.$$

**Exercise 8.4** (Dyson level repulsion). For the real and quaternionic classes, show that each off-diagonal spectral direction has real multiplicity  $\beta_D$  and deduce

$$p_{N,\beta_D}(\lambda) = \frac{1}{Z_{N,\beta_D}} e^{-\beta_D N \sum_i \lambda_i^2 / (4\sigma^2)} |\Delta(\lambda)|^{\beta_D}.$$

For  $2 \times 2$  GOE and GUE, separate the trace from the traceless operator, rescale the gap to unit mean, and obtain

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \quad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Identify the exponent  $\beta_D$  as the small-gap repulsion prediction and state the symmetry-sector separation required in spectral data.

**Exercise 8.5** (Fredholm determinants and trace identities). Let  $A$  be a positive compact operator on a Hilbert space. It is *trace class* when its nonzero spectral values, repeated with multiplicity, satisfy  $\sum_j \lambda_j(A) < \infty$ . Define

$$\det(I + zA) = \prod_j (1 + z\lambda_j(A)), \quad \text{Tr}(A^m) = \sum_j \lambda_j(A)^m.$$

Prove absolute local uniform convergence of the defining product and

$$\det(I - zA) = \exp\left(-\sum_{m \geq 1} \frac{z^m}{m} \text{Tr}(A^m)\right) \quad (|z| < \|A\|^{-1}).$$

Prove that the determinant is entire, identify its zeros, and show that its finite-rank specialization agrees with the exterior-power identity.

**Exercise 8.6** (Point processes and determinantal statistics). Let  $X$  be a locally compact second-countable Hausdorff space and let  $\mu$  be a Radon measure on  $X$ . A simple point process is a random locally finite counting measure  $\Xi = \sum_i \delta_{x_i}$  with distinct points almost surely. When its  $k$ th factorial moment measure is absolutely continuous with respect to  $\mu^{\otimes k}$ , its density  $\rho_k$  is defined, for every nonnegative compactly supported measurable function  $f$ , by

$$\mathbb{E} \sum_{i_1, \dots, i_k}^{\neq} f(x_{i_1}, \dots, x_{i_k}) = \int_{X^k} f \rho_k d\mu^{\otimes k}.$$

A *determinantal point process* with measurable Hermitian kernel  $K$  is one for which

$$\rho_k(x_1, \dots, x_k) = \det[K(x_i, x_j)]_{i,j=1}^k.$$

For the existence and counting formulas below, the integral operator defined by  $K$  is a locally trace-class positive contraction on  $L^2(X, \mu)$ .

For a relatively compact measurable set  $B$ , put  $N_B = \Xi(B)$  and  $K_B = \mathbf{1}_B K \mathbf{1}_B$ . Use factorial moments and the Fredholm series to prove

$$\mathbb{E}[z^{N_B}] = \det(I + (z - 1)K_B), \quad \mathbb{P}(N_B = 0) = \det(I - K_B).$$

Random variables are independent when their joint law is the product of their marginal laws. Prove that  $N_B$  is distributed as a sum of independent Bernoulli variables whose parameters are the eigenvalues of  $K_B$ .

**Exercise 8.7** (Unitary polynomial ensembles and Christoffel–Darboux kernels). Let  $\mu$  be a Radon measure on  $\mathbb{R}$  such that  $w(x)d\mu(x)$  has infinite support and finite moments, and let  $P_j$  be its monic orthogonal polynomials, normalized by

$$\int P_j(x)P_k(x)w(x)d\mu(x) = H_j\delta_{jk}, \quad H_j > 0.$$

The associated unitary polynomial ensemble has density, with respect to  $\mu^{\otimes N}$ ,

$$\mathfrak{p}_N(x_1, \dots, x_N) = \frac{1}{N! \prod_{j=0}^{N-1} H_j} \Delta(x)^2 \prod_{i=1}^N w(x_i)$$

and kernel

$$K_N(x, y) = \sqrt{w(x)w(y)} \sum_{j=0}^{N-1} \frac{P_j(x)P_j(y)}{H_j}.$$

Prove Andréief’s identity

$$\int_{X^N} \det[f_i(x_j)] \det[g_i(x_j)] \prod_j d\mu(x_j) = N! \det \left[ \int_X f_i g_j d\mu \right].$$

Deduce that the unitary polynomial ensemble is determinantal with kernel  $K_N$ . Derive the Christoffel–Darboux formula from the three-term recurrence and prove

$$\int K_N(x, z)K_N(z, y)d\mu(z) = K_N(x, y), \quad \int K_N(x, x)d\mu(x) = N.$$

**Exercise 8.8** (Simple and coalescing saddle estimates). Let

$$I_N(u) = \frac{1}{2\pi i} \int_{\mathcal{C}_N(u)} e^{N\Phi_u(z)} a_u(z) dz,$$

where  $u$  ranges over a compact set, the phases and amplitudes are holomorphic on fixed neighborhoods of the contours, and the contour lengths grow at most polynomially in  $N$ . Assume that the phases, amplitudes, contour parametrizations, and all derivatives used below are uniformly bounded on these neighborhoods.

Suppose first that  $\mathcal{C}_N(u)$  passes through finitely many nondegenerate critical points  $z_j(u)$  along steepest directions. Assume that the critical points remain uniformly separated, that  $|\Phi_u''(z_j(u))|$  is bounded below by a positive constant, and that, outside fixed neighborhoods of these points,

$$\operatorname{Re} \Phi_u(z) \leq \max_j \operatorname{Re} \Phi_u(z_j(u)) - \delta$$

for one  $\delta > 0$  independent of  $u$ . Rescale each fixed local neighborhood by  $z - z_j = N^{-1/2}w$ , use Taylor’s theorem and Gaussian domination there, and use the displayed gap on the complement to derive the sum of the Gaussian saddle contributions with absolute remainder  $O(N^{-3/2}e^{NM(u)})$ , where  $M(u) = \max_j \operatorname{Re} \Phi_u(z_j(u))$ . Deduce an  $O(N^{-1})$  relative error whenever the normalized leading saddle sum is bounded away from zero, uniformly in  $u$ .

Suppose instead that two critical points coalesce at  $u = u_*$  and that a local analytic variable  $w$  has been exhibited for which

$$\Phi_u(z) = c(u) + \frac{w^3}{3} - s(u)w, \quad s(u_*) = 0,$$

with the variable, its inverse, and their first three derivatives uniformly bounded. Assume that the transformed amplitude and its first two derivatives are uniformly bounded, that the local contour deforms to the Airy rays without crossing a singularity, and that the complement satisfies a uniform real-part gap and contains no other contributing saddle. Prove, uniformly for bounded  $N^{2/3}s(u)$ , that  $I_N(u)$  is the corresponding linear combination of

$$N^{-1/3} \text{Ai}\left(N^{2/3}s(u)\right), \quad N^{-2/3} \text{Ai}'\left(N^{2/3}s(u)\right),$$

with a uniform  $O(N^{-1})$  remainder after the common exponential factor is removed. Prove exponentially weighted bounds when  $N^{2/3}s(u)$  lies on the positive real half-line. State precisely how the contour gap and amplitude bounds make both conclusions uniform.

**Exercise 8.9** (Hermite steepest descent). Write  $\psi_j^H$  for the normalized oscillator Hermite functions. From the generating function derive

$$H_j(x) = \frac{j!}{2\pi i} \oint e^{2xz-z^2} z^{-j-1} dz$$

and the resulting contour integral for  $\psi_j^H$ . After the GUE scaling used below, show that the critical-point equation is quadratic, has two simple conjugate roots for  $E \in (-2\sigma, 2\sigma)$ , and has a double root at  $E = 2\sigma$ . Exhibit the local Gaussian coordinates in the bulk and the cubic Airy coordinate at the edge. Apply the simple- and coalescing-saddle estimates to establish uniform Plancherel–Rotach formulas on compact bulk windows and uniform exponentially weighted Airy formulas on edge half-lines, including the same estimates for the neighboring indices  $j = N - 1, N$  needed in the Christoffel–Darboux kernel.

**Exercise 8.10** (GUE bulk statistics). For  $E \in (-2\sigma, 2\sigma)$ , define

$$\rho_{\text{sc}}(E) = \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - E^2}, \quad K_{\text{sin}}(u, v) = \frac{\sin \pi(u - v)}{\pi(u - v)}.$$

Define

$$\phi_{j,N}(x) = \left(\frac{N}{2\sigma^2}\right)^{1/4} \psi_j^H\left(\sqrt{\frac{N}{2\sigma^2}}x\right), \quad K_N^{\text{GUE}}(x, y) = \sum_{j=0}^{N-1} \phi_{j,N}(x) \phi_{j,N}(y).$$

Insert the bulk Hermite asymptotics into the Christoffel–Darboux identity and prove

$$\frac{1}{N\rho_{\text{sc}}(E)} K_N^{\text{GUE}}\left(E + \frac{u}{N\rho_{\text{sc}}(E)}, E + \frac{v}{N\rho_{\text{sc}}(E)}\right) \longrightarrow K_{\text{sin}}(u, v)$$

locally uniformly. For the limiting sine process prove

$$\text{Var } N_{[0,L]} = \int_0^L K_{\text{sin}}(u, u) du - \int_0^L \int_0^L |K_{\text{sin}}(u, v)|^2 du dv = \frac{1}{\pi^2} \log L + O(1).$$

State the unfolding, symmetry-sector separation, and finite-window assumptions required when testing the spacing and number-variance predictions with resonance or Hamiltonian spectra.

**Exercise 8.11** (GUE edge statistics). Define

$$K_{\text{Ai}}(u, v) = \frac{\text{Ai}(u) \text{Ai}'(v) - \text{Ai}'(u) \text{Ai}(v)}{u - v}, \quad F_2(s) = \det(I - K_{\text{Ai}})_{L^2(s, \infty)},$$

with diagonal values defined by continuity.

Insert the turning-point asymptotics into the Christoffel–Darboux identity and prove

$$\frac{\sigma}{N^{2/3}} K_N^{\text{GUE}} \left( 2\sigma + \frac{\sigma u}{N^{2/3}}, 2\sigma + \frac{\sigma v}{N^{2/3}} \right) \longrightarrow K_{\text{Ai}}(u, v).$$

Use the exponentially weighted bounds to pass to the Fredholm series and deduce the GUE largest-eigenvalue law

$$\mathbb{P} \left( \frac{N^{2/3}}{\sigma} (\lambda_{\max} - 2\sigma) \leq s \right) \longrightarrow F_2(s).$$

State the centering, scale calibration, symmetry class, and finite-size assumptions required when testing this prediction with extreme resonance or Hamiltonian levels.

**Exercise 8.12** (Painlevé form of the Tracy–Widom law). The Hastings–McLeod function is the real pole-free solution  $q$  of

$$q''(s) = sq(s) + 2q(s)^3, \quad q(s) \sim \text{Ai}(s) \quad (s \rightarrow +\infty).$$

Let  $K_s$  be the Airy-kernel operator on  $L^2((s, \infty))$  and define

$$Q_s = (I - K_s)^{-1} \text{Ai}, \quad P_s = (I - K_s)^{-1} \text{Ai}'.$$

With the inner product on  $L^2((s, \infty))$ , put

$$q(s) = Q_s(s), \quad p(s) = P_s(s), \quad u(s) = \langle Q_s, \text{Ai} \rangle, \quad v(s) = \langle P_s, \text{Ai} \rangle.$$

Use the integrable form of  $K_{\text{Ai}}$ , the Airy equation, and differentiation of the moving endpoint to prove

$$q' = p - qu, \quad p' = sq + pu - 2qv, \quad u' = -q^2, \quad v' = -pq.$$

Prove that  $u^2 - 2v - q^2$  is constant and vanishes as  $s \rightarrow +\infty$ . Deduce

$$q'' = sq + 2q^3, \quad q(s) \sim \text{Ai}(s) \quad (s \rightarrow +\infty).$$

Use the positive integral representation

$$K_{\text{Ai}}(x, y) = \int_0^\infty \text{Ai}(x+t) \text{Ai}(y+t) dt$$

to prove that  $I - K_s$  is invertible for every real  $s$ . Conclude that the resolvent construction gives the real pole-free Hastings–McLeod solution. For the resolvent  $R_s = K_s(I - K_s)^{-1}$ , prove

$$\frac{d}{ds} \log \det(I - K_s) = R_s(s, s) = u(s) = \int_s^\infty q(x)^2 dx.$$

Conclude that the Fredholm and Painlevé definitions agree:

$$F_2(s) = \exp \left[ - \int_s^\infty (x-s) q(x)^2 dx \right].$$

Use the Airy asymptotic of the Hastings–McLeod solution to derive

$$1 - F_2(s) \sim \frac{e^{-4s^{3/2}/3}}{16\pi s^{3/2}} \quad (s \rightarrow +\infty).$$

State how exceedance counts for centered and edge-scaled extreme eigenvalues or resonance levels test this right-tail prediction, including the finite-size range over which the asymptotic is used.

**Exercise 8.13** (Semistandard tableaux and Schur polynomials). For a partition  $\lambda$  of length at most  $N$ , a *semistandard Young tableau* of shape  $\lambda$  is a filling  $T$  of its Young diagram by entries in  $\{1, \dots, N\}$  that is weakly increasing along rows and strictly increasing down columns. If  $m_i(T)$  is the number of entries equal to  $i$ , define

$$s_\lambda(x_1, \dots, x_N) = \sum_T \prod_i x_i^{m_i(T)}.$$

Define the complete symmetric polynomials by

$$h_r(x_1, \dots, x_N) = \sum_{1 \leq i_1 \leq \dots \leq i_r \leq N} x_{i_1} \cdots x_{i_r}, \quad h_0 = 1,$$

and put  $h_r = 0$  for  $r < 0$ . Encode  $h_r$  by lattice paths directed east and south whose east steps at height  $i$  have weight  $x_i$ . For sources  $A_i = (i - \lambda_i, N)$  and sinks  $B_j = (j, 1)$ , expand  $\det[h_{\lambda_i - i + j}]_{i,j=1}^N$  as a signed sum of path families and pair every intersecting family with the family obtained by exchanging the two tails after its first intersection. Show that the remaining nonintersecting families correspond weight-preservingly to semistandard tableaux by reading each path's east-step heights from right to left, and deduce

$$s_\lambda(x_1, \dots, x_N) = \det[h_{\lambda_i - i + j}]_{i,j=1}^N.$$

Call this the *Schur polynomial*. Derive the alternant formula

$$s_\lambda(x) = \frac{\det[x_i^{\lambda_j + N - j}]_{i,j=1}^N}{\det[x_i^{N - j}]_{i,j=1}^N}$$

and prove  $s_{(r)} = h_r$ .

**Exercise 8.14** (RSK, last-passage values, and the finite Schur measure). To insert an integer  $x$  into a semistandard Young tableau, replace the first entry strictly larger than  $x$  in the first row and insert the replaced entry in the next row; append when no such entry exists. For  $W = (W_{ij}) \in M_{m,n}(\mathbb{Z}_{\geq 0})$ , form the lexicographically ordered two-line word containing  $W_{ij}$  copies of  $(i, j)$ , insert the lower letters, and record the upper letters in the created boxes. The resulting pair  $(P, Q)$  of tableaux of common shape  $\lambda$  is the RSK image of  $W$ .

For nonnegative weights  $W = (W_{ij})$ , define

$$G(m, n) = \max_{\pi: (1,1) \nearrow (m,n)} \sum_{(i,j) \in \pi} W_{ij}.$$

For parameters  $\alpha_i, \gamma_j \geq 0$  with  $\alpha_i \gamma_j < 1$ , the geometric model has independent weights

$$\mathbb{P}(W_{ij} = k) = (1 - \alpha_i \gamma_j)(\alpha_i \gamma_j)^k, \quad k \in \mathbb{Z}_{\geq 0}.$$

Prove that row insertion is reversible and hence that RSK is a bijection. Show that the length of the first row of its shape is the maximal weight of an up-right path:

$$\lambda_1 = G(m, n).$$

Use the contents of the two RSK tableaux and the geometric weight law to derive

$$\mathbb{P}(\lambda) = \prod_{i,j} (1 - \alpha_i \gamma_j) s_\lambda(\alpha) s_\lambda(\gamma)$$

and the Cauchy identity

$$\sum_\lambda s_\lambda(\alpha) s_\lambda(\gamma) = \prod_{i,j} (1 - \alpha_i \gamma_j)^{-1}.$$

**Exercise 8.15** (Interlacing partitions and Schur paths). For partitions  $\mu$  and  $\lambda$ , write  $\mu \prec \lambda$  when

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots.$$

Prove the branching formula

$$s_\lambda(\alpha_1, \dots, \alpha_m) = \sum_{\emptyset = \lambda^{(0)} \prec \dots \prec \lambda^{(m)} = \lambda} \prod_{k=1}^m \alpha_k^{|\lambda^{(k)}| - |\lambda^{(k-1)}|}.$$

Encode each interlacing chain by nonintersecting paths on the integer lattice. For a finite directed acyclic graph with compatible sources  $u_i$  and sinks  $v_i$ , let  $Z(u_i, v_j)$  be the weighted sum of paths from  $u_i$  to  $v_j$ . Use the first intersection and tail exchange as a sign-reversing involution to prove

$$\sum_{\substack{(\pi_1, \dots, \pi_r) \\ \text{vertex-disjoint}}} \prod_i \text{wt}(\pi_i) = \det[Z(u_i, v_j)]_{i,j=1}^r.$$

Apply this identity to the interlacing-chain encoding of  $s_\lambda$ .

**Exercise 8.16** (Finite Schur kernels). For finite alphabets, set

$$\Phi(z) = \frac{\prod_j (1 - \gamma_j/z)}{\prod_i (1 - \alpha_i z)}.$$

Choose concentric contours with

$$\max_j \gamma_j < |w| < |z| < \min_i \alpha_i^{-1}.$$

For  $r, s \in \mathbb{Z} + \frac{1}{2}$ , define

$$K_{\alpha, \gamma}(r, s) = \frac{1}{(2\pi i)^2} \oint \oint \frac{\Phi(z)}{\Phi(w)} \frac{1}{z - w} z^{-r-1/2} w^{s-1/2} dz dw.$$

Encode  $\lambda$  by

$$\mathcal{X}(\lambda) = \{\lambda_i - i + \frac{1}{2} : i \geq 1\}.$$

Glue the two nonintersecting path ensembles associated with  $s_\lambda(\alpha)$  and  $s_\lambda(\gamma)$  along their common boundary  $\lambda$ . Apply the Andréief identity successively to prove that the occupation correlations of the glued ensemble are determinants. Evaluate the resulting one-particle propagators by coefficient extraction and derive the kernel  $K_{\alpha, \gamma}$ . Deduce the exact finite Fredholm formula

$$\mathbb{P}(G(m, n) \leq L) = \det(I - K_{\alpha, \gamma})_{\ell^2(\{L+\frac{1}{2}, L+\frac{3}{2}, \dots\})}.$$

State how repeated last-passage or growth experiments with the specified geometric weights test this distribution without invoking an asymptotic limit.

**Exercise 8.17** (Exponential last-passage percolation). Let  $E_{ij}$  be independent rate-one exponential random variables and define

$$G_{\text{exp}}(m, n) = \max_{\pi: (1,1) \nearrow (m,n)} \sum_{(i,j) \in \pi} E_{ij}.$$

The variables  $E_{ij}$  are independent when their joint law is the product of their marginal laws. For random variables  $Y_\varepsilon, Y$ , write  $Y_\varepsilon \Rightarrow Y$  when their laws converge against bounded continuous test functions. Set  $\alpha_i = \gamma_j = e^{-\varepsilon/2}$  in the  $n \times n$  geometric model and prove that  $\varepsilon W_{ij}$  converges in distribution to a rate-one exponential variable as  $\varepsilon \downarrow 0$ . Take this limit in the finite Schur Fredholm formula and identify the resulting finite- $n$  kernel with the Laguerre Christoffel–Darboux kernel.

**Exercise 8.18** (Laguerre edge limit). Use the generating function to derive

$$L_n^{(\alpha)}(x) = \frac{1}{2\pi i} \oint \frac{\exp(-xz/(1-z))}{(1-z)^{\alpha+1} z^{n+1}} dz.$$

For the Laguerre kernel, write its scaled phase explicitly and locate the double saddle at the upper spectral edge. Exhibit the cubic coordinate and contours satisfying the gap hypotheses of the coalescing-saddle exercise. Apply that exercise to derive convergence of the rescaled Laguerre kernel to  $K_{\text{Ai}}$  with uniform exponentially weighted bounds, and use those bounds to pass termwise to the Fredholm series.

**Exercise 8.19** (Last-passage fluctuations). Combine the exponential last-passage determinant with the Laguerre edge limit to prove

$$\frac{G_{\text{exp}}(n, n) - 4n}{2^{4/3} n^{1/3}} \implies \chi_2, \quad \mathbb{P}(\chi_2 \leq s) = F_2(s).$$

Deduce the  $n^{1/3}$  height-fluctuation scale. Balance this scale against the quadratic curvature of the deterministic limit shape to predict the transverse  $n^{2/3}$  scale. State the independent-weight, boundary, centering, scale-calibration, and finite-size assumptions required when the Tracy–Widom prediction is compared with interface-height or directed transport data.

## 9 Transfer Matrices

**Exercise 9.1** (Hard-hexagon transfer matrices). Let  $\mathcal{R}_N$  be a finite set of allowed width- $N$  configurations and let  $T_N(z)$  be an operator on the configuration space whose matrix element between  $a, b \in \mathcal{R}_N$  is the weight of adjoining  $b$  to  $a$ . For a height- $M$  periodic lattice define

$$Z_{N,M}(z) = \text{Tr}(T_N(z)^M).$$

When  $T_N(z)$  has nonnegative entries, write  $\lambda_N(z)$  for its spectral radius.

Let

$$\mathcal{R}_N = \{a \in \{0, 1\}^{\mathbb{Z}/N\mathbb{Z}} : a_i a_{i+1} = 0 \text{ for every } i\}.$$

Two rows  $a, b \in \mathcal{R}_N$  are compatible when

$$a_i b_i = 0, \quad a_i b_{i+1} = 0 \quad (i \in \mathbb{Z}/N\mathbb{Z}).$$

For  $z \geq 0$ , with  $|a| = \sum_i a_i$ , define

$$T_N(z)_{ab} = \mathbf{1}_{\{a, b \text{ compatible}\}} z^{(|a|+|b|)/2}.$$

Identify the horizontal, vertical, and diagonal nearest-neighbor edges encoded by the two compatibility conditions. Prove

$$\text{Tr}(T_N(z)^M) = \sum_I z^{|I|},$$

where the sum is over independent vertex sets of the specified triangular  $N \times M$  torus. Prove that every coefficient is an exact state count. Let  $0$  denote the empty row. For  $z > 0$ , prove

$$(T_N(z)^2)_{ab} \geq T_N(z)_{a0} T_N(z)_{0b} > 0.$$



Use as theorem input [38] the finite Perron–Frobenius statement that a strictly positive matrix has a positive simple spectral radius and every other spectral value has strictly smaller modulus. Apply it to show that  $\lambda_N(z)$  is simple, that every other spectral value has smaller modulus, and that

$$\lim_{M \rightarrow \infty} \frac{1}{NM} \log Z_{N,M}(z) = \frac{1}{N} \log \lambda_N(z).$$

Put  $\eta = \log z = \beta_{\text{th}} \mu$  and

$$\Psi_{N,M}(\eta) = \log Z_{N,M}(e^\eta).$$

Give the independent sets the probability  $\mathbb{P}_\eta(I) = e^{\eta|I|} / Z_{N,M}(e^\eta)$  and write  $\mathbb{E}_\eta$  for expectation with respect to this measure. For a square-integrable random variable  $Y$ , write  $\text{Var}_\eta(Y) = \mathbb{E}_\eta|Y - \mathbb{E}_\eta Y|^2$ . Set  $A = \partial_\eta \Psi_{N,M}$  and prove

$$A = \mathbb{E}_\eta |I|, \quad \partial_\eta A = \text{Var}_\eta(|I|),$$

and derive the mean occupied fraction and isothermal finite-size susceptibility

$$\rho_{N,M} = \frac{A}{NM}, \quad \chi_{N,M} = \frac{\partial \rho_{N,M}}{\partial \mu} = \frac{\beta_{\text{th}}}{NM} \text{Var}_\eta(|I|).$$

State the lattice geometry, boundary conditions, and finite  $N, M$  limitations required when these quantities are compared with adsorption or exclusion-process data.

**Exercise 9.2** (The planar Ising model and Kramers–Wannier duality). Let  $G = (V, E)$  be a finite connected graph with a fixed planar embedding. For couplings  $K_e > 0$ , define

$$Z_G^f(K) = \sum_{s: V \rightarrow \{-1, 1\}} \exp \left( \sum_{e=uv} K_e s_u s_v \right).$$

Let  $G^*$  be the planar dual, fix the spin of its outer-face vertex to  $+1$ , and denote the corresponding partition function by  $Z_{G^*}^+(K^*)$ . Let  $\mathcal{E}(G)$  be the family of edge subsets in which every vertex has even degree. The dual couplings are determined by

$$e^{-2K_{e^*}^*} = \tanh K_e.$$

Expand each edge factor as

$$e^{K_e s_u s_v} = \cosh K_e (1 + s_u s_v \tanh K_e)$$

and prove

$$Z_G^f(K) = 2^{|V|} \prod_{e \in E} \cosh K_e \sum_{\gamma \in \mathcal{E}(G)} \prod_{e \in \gamma} \tanh K_e.$$

Identify  $\mathcal{E}(G)$  with the domain walls of dual spin configurations whose outer-face spin is fixed, and deduce the exact planar duality

$$Z_G^f(K) = 2^{|V|} \prod_{e \in E} (\cosh K_e e^{-K_{e^*}^*}) Z_{G^*}^+(K^*).$$

For horizontal and vertical square-lattice couplings  $K_h, K_v$ , show that the duality interchanges the directions and that its fixed locus is

$$\sinh(2K_h) \sinh(2K_v) = 1.$$

Assuming the infinite-volume ferromagnetic model has a single transition and that duality interchanges its high- and low-temperature phases, deduce in the isotropic case

$$K_c = \frac{1}{2} \log(1 + \sqrt{2}).$$

State separately the exact finite-graph identity, the infinite-volume assumptions, and the conclusion about the critical temperature.

**Exercise 9.3** (Ising star–triangle transformation). For  $K_1, K_2, K_3 > 0$  and boundary spins  $s_1, s_2, s_3$ , prove that there are unique  $A > 0$  and real  $L_{12}, L_{23}, L_{31}$  satisfying

$$\sum_{s_0=\pm 1} e^{K_1 s_0 s_1 + K_2 s_0 s_2 + K_3 s_0 s_3} = A e^{L_{12} s_1 s_2 + L_{23} s_2 s_3 + L_{31} s_3 s_1}.$$

Put

$$\begin{aligned} w_0 &= 2 \cosh(K_1 + K_2 + K_3), & w_1 &= 2 \cosh(-K_1 + K_2 + K_3), \\ w_2 &= 2 \cosh(K_1 - K_2 + K_3), & w_3 &= 2 \cosh(K_1 + K_2 - K_3). \end{aligned}$$

Derive

$$\begin{aligned} A &= (w_0 w_1 w_2 w_3)^{1/4}, \\ e^{4L_{12}} &= \frac{w_0 w_3}{w_1 w_2}, & e^{4L_{23}} &= \frac{w_0 w_1}{w_2 w_3}, & e^{4L_{31}} &= \frac{w_0 w_2}{w_1 w_3}. \end{aligned}$$

Show that replacing every star of one sublattice of the honeycomb model by a triangle changes the partition function only by the product of the local factors  $A$ . In the isotropic case derive

$$e^{4L} = \frac{\cosh(3K)}{\cosh K}.$$

Combine this transformation with planar duality and, under the single-transition hypothesis, obtain

$$K_c^{\text{tri}} = \frac{1}{4} \log 3, \quad K_c^{\text{hex}} = \frac{1}{2} \log(2 + \sqrt{3}).$$

Identify the coupling normalization and lattice geometry needed when these critical temperatures are compared with an Ising magnet or a programmable spin array.

**Exercise 9.4** (Ising transfer matrices and free fermions). Let  $\mathcal{S}_N = \{-1, 1\}^{\mathbb{Z}/N\mathbb{Z}}$  and  $K_h, K_v > 0$ . For row configurations  $s, t \in \mathcal{S}_N$  define

$$T_N^{\text{Is}}(K_h, K_v)_{s,t} = \exp \left[ \frac{K_h}{2} \sum_i (s_i s_{i+1} + t_i t_{i+1}) + K_v \sum_i s_i t_i \right].$$

Let  $R_N$  be cyclic row translation. On  $\mathcal{H}_N = \ell^2(\mathcal{S}_N)$  let

$$(Z_j f)(s) = s_j f(s), \quad (X_j f)(s) = f(s^{(j)}), \quad Y_j = i X_j Z_j,$$

where  $s^{(j)}$  is obtained by reversing  $s_j$ . Put

$$P = \prod_{j=1}^N X_j, \quad c_{2j-1} = \left( \prod_{\ell < j} X_\ell \right) Z_j, \quad c_{2j} = \left( \prod_{\ell < j} X_\ell \right) Y_j.$$

Let  $K_v^* > 0$  satisfy  $e^{-2K_v^*} = \tanh K_v$ , and set

$$\mathcal{K}_+ = \left\{ \frac{(2m+1)\pi}{N} : 0 \leq m < N \right\}, \quad \mathcal{K}_- = \left\{ \frac{2m\pi}{N} : 0 \leq m < N \right\}.$$

Write  $a_x$  and  $a_t$  for the spatial and transfer-step lattice spacings and  $L = N a_x$  for the circumference.

(a) Verify

$$R_N T_N^{\text{Is}}(K_h, K_v) = T_N^{\text{Is}}(K_h, K_v) R_N.$$

(b) Prove

$$T_N^{\text{Is}}(K_h, K_v) = (2 \sinh(2K_v))^{N/2} e^{\frac{K_h}{2} \sum_j Z_j Z_{j+1}} e^{K_v^* \sum_j X_j} e^{\frac{K_h}{2} \sum_j Z_j Z_{j+1}}.$$

(c) Verify the Clifford relations

$$c_r c_s + c_s c_r = 2\delta_{rs} I, \quad X_j = i c_{2j-1} c_{2j}, \quad Z_j Z_{j+1} = i c_{2j} c_{2j+1} \quad (j < N),$$

and

$$Z_N Z_1 = -i P c_{2N} c_1.$$

(d) On the  $P = +1$  and  $P = -1$  sectors use, respectively, the distinguished Fourier modes with momenta in  $\mathcal{K}_+$  and  $\mathcal{K}_-$  to diagonalize the induced action on the Clifford generators. Derive

$$\cosh \gamma(k) = \cosh(2K_h) \cosh(2K_v^*) - \sinh(2K_h) \sinh(2K_v^*) \cos k, \quad \gamma(k) \geq 0,$$

and prove that the transfer eigenvalues are

$$\Lambda = (2 \sinh(2K_v))^{N/2} \exp \left( \frac{1}{2} \sum_{k \in \mathcal{K}_{\pm}} \epsilon_k \gamma(k) \right), \quad \epsilon_k \in \{-1, 1\},$$

with the fermion-parity condition determined by the chosen  $P$  sector.

(e) At  $K_h = K_v = K_c$  prove

$$\gamma(k) = 2 \operatorname{arsinh} |\sin(k/2)| = |k| + O(|k|^3).$$

After subtracting the sector ground energy and measuring gaps in units  $2\pi(a_x/a_t)/L$ , use the parity projections to obtain the chiral scaling products

$$\begin{aligned} \chi_0(q) &= \frac{q^{-1/48}}{2} \left[ \prod_{m \geq 0} (1 + q^{m+1/2}) + \prod_{m \geq 0} (1 - q^{m+1/2}) \right], \\ \chi_{1/2}(q) &= \frac{q^{-1/48}}{2} \left[ \prod_{m \geq 0} (1 + q^{m+1/2}) - \prod_{m \geq 0} (1 - q^{m+1/2}) \right], \\ \chi_{1/16}(q) &= q^{1/24} \prod_{m \geq 1} (1 + q^m). \end{aligned}$$

Use the Jacobi triple product to recover the theta-eta formulas in the Ising character exercise.

**Exercise 9.5** (Theta-character transfer spectra). Let  $\Lambda_0(L) > 0$  be the Perron eigenvalue of  $T_N^{\text{Is}}(K_c, K_c)$  and let  $\Lambda_{a,r,\bar{r}}(L)$  denote its low-lying symmetry-resolved eigenvalues. Define

$$E_{a,r,\bar{r}}(L) = -\frac{1}{a_t} \log |\Lambda_{a,r,\bar{r}}(L)|.$$

Use the critical Clifford spectrum and Euler–Maclaurin summation to prove the finite-size asymptotic

$$E_{a,r,\bar{r}}(L) = e_{\infty} L + \frac{2\pi v}{L} \left( h_a + \bar{h}_a + r + \bar{r} - \frac{c + \bar{c}}{24} \right) + o(L^{-1}).$$

If  $R_N$  acts on the same state by  $e^{ip a_x}$ , prove

$$\mathbf{p} = \frac{2\pi}{L}(h_a - \bar{h}_a + r - \bar{r}) \pmod{2\pi/a_x}.$$

Prove that  $v = a_x/a_t$  for the displayed isotropic transfer normalization. Use the three Ising character products to prove that the coefficient of  $q^{h_a+r-c/24}\bar{q}^{\bar{h}_a+\bar{r}-\bar{c}/24}$  gives the predicted multiplicity at these energy and momentum values. Express every excitation gap directly as

$$E_j(L) - E_0(L) = -\frac{1}{a_t} \log \left| \frac{\Lambda_j(L)}{\Lambda_0(L)} \right|.$$

Compute the first spectral gaps and degeneracies in the three Ising sectors. Deduce the universal gap ratios and the  $L^{-1}$  Casimir-energy correction. State how finite-size transfer eigenvalues determine  $c$ , the primary weights, and  $v$ , including the symmetry-sector resolution, anisotropy corrections, and irrelevant-operator corrections required for the comparison.

**Exercise 9.6** (Harper monodromy and  $SL_2(\mathbb{R})$  band geometry). Let  $\mathcal{E} = V \oplus V^*$  be a symplectic phase space with inverse Poisson tensor  $\Pi$ . Choose  $k, \ell \in \mathcal{E}^*$  with

$$-\hbar \Pi(k, \ell) = 2\pi\alpha$$

and let  $U$  and  $V$  be the Weyl quantizations of  $e_k$  and  $e_\ell$ . Use the Moyal plane-wave identity to derive

$$UV = e^{2\pi i \alpha} VU.$$

Interpret  $\alpha$  as the magnetic flux through one lattice cell in units of the flux quantum, and define the magnetic nearest-neighbor Hamiltonian

$$H_\alpha = U + U^* + V + V^*.$$

In the position realization on  $\ell^2(\mathbb{Z})$  with transverse phase  $\theta$ , take

$$(U\psi)_m = \psi_{m+1}, \quad (V\psi)_m = e^{i(2\pi\alpha m + \theta)} \psi_m$$

and derive the Harper operator

$$(H_{\alpha,\theta}\psi)_m = \psi_{m+1} + \psi_{m-1} + 2\cos(2\pi\alpha m + \theta)\psi_m.$$

Prove that the commutator phase is the holonomy around one lattice cell. On an  $N_x \times N_y$  periodic sample, show that  $N_x N_y \alpha \notin \mathbb{Z}$  obstructs consistent torus boundary phases, whereas integral total flux makes their corner cocycle trivial. Prove that nonzero integral total flux still obstructs a single globally periodic vector potential.

Rewrite  $H_{\alpha,\theta}\psi = E\psi$  as

$$\begin{pmatrix} \psi_{m+1} \\ \psi_m \end{pmatrix} = A_m(E, \theta) \begin{pmatrix} \psi_m \\ \psi_{m-1} \end{pmatrix}, \quad A_m(E, \theta) = \begin{pmatrix} E - 2\cos(2\pi\alpha m + \theta) & -1 \\ 1 & 0 \end{pmatrix}.$$

For coprime integers  $p, q$  with  $q > 0$ , put  $\alpha = p/q$  and define the  $q$ -step monodromy and discriminant by

$$M_{p/q}(E, \theta) = A_{q-1}(E, \theta) \cdots A_0(E, \theta), \quad \Delta_{p/q}(E, \theta) = \text{Tr } M_{p/q}(E, \theta).$$

Define

$$SL_2(\mathbb{R}) = \{A \in \text{End}(\mathbb{R}^2) : \det A = 1\}.$$

Prove that every  $A_m(E, \theta)$  and  $M_{p/q}(E, \theta)$  belongs to this group. For  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ , prove that

$$A \cdot z = \frac{az + b}{cz + d}$$

preserves the upper half-plane. Show that the eigenvalues of a monodromy matrix satisfy

$$\lambda^2 - \Delta_{p/q}(E, \theta)\lambda + 1 = 0.$$

Call the monodromy elliptic when  $|\Delta_{p/q}| < 2$ , parabolic when  $|\Delta_{p/q}| = 2$  and  $M_{p/q} \neq \pm I$ , and hyperbolic when  $|\Delta_{p/q}| > 2$ ; treat  $M_{p/q} = \pm I$  as the central limiting case. Prove that the Bloch condition  $\psi_{m+q} = e^{iq\kappa}\psi_m$  is equivalent to

$$\Delta_{p/q}(E, \theta) = 2 \cos(q\kappa),$$

that band interiors have elliptic monodromy, open band edges have parabolic monodromy, closed gaps have central monodromy, and open gaps have hyperbolic monodromy with growth exponent

$$\gamma(E, \theta) = \text{arcosh}\left(\frac{|\Delta_{p/q}(E, \theta)|}{2}\right).$$

For any chosen transfer period  $q$  at zero flux, use

$$T_q(\cos t) = \cos(qt)$$

to derive

$$\Delta_{0,q}(E, \theta) = 2T_q\left(\frac{E - 2 \cos \theta}{2}\right).$$

At half a flux quantum, compute the two bands

$$E_{\pm}(\kappa, \theta) = \pm 2\sqrt{\cos^2 \kappa + \cos^2 \theta}.$$

Deduce the flux-dependent band edges, touching energy, and exponential attenuation per period in every gap. State the nearest-neighbor, uniform-flux, periodicity, contact-matching, coherence, and finite-size assumptions under which microwave, photonic, or electronic lattice spectroscopy tests the Chebyshev discriminant, half-flux splitting, and hyperbolic attenuation exponent.

## 10 $SL_2(\mathbb{R})$

**Exercise 10.1** (The hyperbolic plane and rank-one decompositions). Let

$$G = SL_2(\mathbb{R}), \quad K = SO(2),$$

and let  $G$  act on the upper half-plane by fractional linear transformations. Define

$$a_r = \begin{pmatrix} e^{r/2} & 0 \\ 0 & e^{-r/2} \end{pmatrix}, \quad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix},$$

with  $r, x \in \mathbb{R}$ . Give the upper half-plane the metric  $ds^2 = (dx^2 + dy^2)/y^2$  and fix its curvature radius to be one.

Use the polar and QR decompositions of an element of  $SL_2(\mathbb{R})$  to prove

$$K \times \mathbb{R} \times \mathbb{R} \longrightarrow G, \quad (k, r, x) \longmapsto ka_r n_x$$

is a diffeomorphism and

$$G = K\{a_r : r \geq 0\}K.$$

Prove that  $K$  is the stabilizer of  $i$  and that the orbit map identifies  $G/K$  isometrically with the hyperbolic plane. Identify the two-point Weyl group and prove that a  $K$ -bi-invariant function is determined by its value on  $a_r$ ,  $r \geq 0$ .

Derive geodesic polar coordinates about  $i$ ,

$$ds^2 = dr^2 + \sinh^2 r \, d\theta^2, \quad d\text{vol}_{\mathbb{H}^2} = \sinh r \, dr \, d\theta.$$

Normalize  $dk$  to have total mass one and derive the Iwasawa and radial integration formulas

$$\begin{aligned} \int_G F(g) \, dg &= \int_K \int_{\mathbb{R}} \int_{\mathbb{R}} F(ka_r n_x) e^r \, dx \, dr \, dk, \\ \int_{\mathbb{H}^2} f(z) \, d\text{vol}(z) &= 2\pi \int_0^\infty f(a_r K) \sinh r \, dr \end{aligned}$$

for radial  $f$ .

**Exercise 10.2** (Hyperbolic Gauss–Bonnet and Weyl asymptotics). For  $R > 0$ , let  $\mathbb{H}_R^2$  be the upper half-plane with the metric

$$ds_R^2 = R^2 \frac{dx^2 + dy^2}{y^2},$$

of constant curvature  $-R^{-2}$ . Let  $\Gamma$  be a torsion-free cocompact subgroup of  $PSL_2(\mathbb{R}) = SL_2(\mathbb{R})/\{\pm I\}$  and let  $S = \Gamma \backslash \mathbb{H}_R^2$  be a closed oriented surface of genus  $g$ .

For a geodesic triangle  $D$  with interior angles  $\alpha, \beta, \gamma$ , use the Levi–Civita connection one-form, Stokes' theorem, and the exterior turns at the vertices to prove

$$\text{Area}(D) = R^2(\pi - \alpha - \beta - \gamma).$$

Choose a geodesic triangulation of  $S$ , sum this identity over its faces, and use

$$3F = 2E, \quad V - E + F = \chi(S) = 2 - 2g$$

to derive

$$\int_S K \, dA = 2\pi\chi(S), \quad \text{Area}(S) = 4\pi(g - 1)R^2.$$

Let

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots$$

be the spectral values of the positive Laplacian  $-\Delta_S$ , repeated with multiplicity, and put

$$N_S(\Lambda) = \#\{j : \lambda_j \leq \Lambda\}.$$

Use as theorem input the two-dimensional Weyl law [26]

$$N_S(\Lambda) = \frac{\text{Area}(S)}{4\pi} \Lambda + o(\Lambda)$$

to prove

$$N_S(\Lambda) = (g - 1)R^2\Lambda + o(\Lambda).$$

For a membrane or wave circuit with propagation speed  $c_s$  and modal frequencies  $\omega_j = c_s\sqrt{\lambda_j}$ , derive

$$\#\{j : \omega_j \leq \Omega\} = (g - 1)\frac{R^2\Omega^2}{c_s^2} + o(\Omega^2).$$

State the compact-quotient, constant-curvature, boundary-free, high-frequency, propagation-speed calibration, loss, and mode-resolution assumptions under which the measured mode count tests both the topological area normalization and Weyl asymptotics.

**Exercise 10.3** (Spherical convolution and the rank-one Gelfand pair). For  $K$ -bi-invariant functions  $f_1, f_2 \in C_c(G)$ , define

$$(f_1 * f_2)(g) = \int_G f_1(h)f_2(h^{-1}g) dh.$$

Use the Cartan decomposition to prove that  $g$  and  $g^{-1}$  belong to the same double coset  $KgK$ . Apply the anti-involution  $g \mapsto g^{-1}$  to prove

$$f_1 * f_2 = f_2 * f_1.$$

Thus  $(G, K)$  is a *Gelfand pair*.

A normalized continuous  $K$ -bi-invariant function  $\varphi$  is *spherical* when

$$\int_K \varphi(gkh) dk = \varphi(g)\varphi(h).$$

Show that integration against  $\varphi$  is a character of the commutative convolution algebra, and prove the converse for characters represented by continuous functions. Deduce that spherical functions are the distinguished special functions diagonalizing radial convolution.

**Exercise 10.4** (Poisson kernels and rank-one spherical functions). For the sign convention

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\text{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

For  $\lambda \in \mathbb{R}$ , define

$$\varphi_\lambda(r) = \frac{1}{2\pi} \int_0^{2\pi} (\cosh r - \sinh r \cos \theta)^{-1/2-i\lambda} d\theta.$$

Interpret the integrand as a complex power of the hyperbolic Poisson kernel. Prove that  $\varphi_\lambda$  is real-valued, even in  $\lambda$ , normalized at  $r = 0$ , and satisfies the spherical product formula. Differentiate under the integral to prove

$$\Delta_{\mathbb{H}^2}\varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \quad \varphi_\lambda(eK) = 1.$$

Regarded as a  $K$ -bi-invariant function on  $G$  and written in the radial coordinate  $a_r K$ , it is determined by

$$\varphi_\lambda''(r) + \coth r \varphi_\lambda'(r) = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda(r), \quad \varphi_\lambda(0) = 1, \quad \varphi_\lambda'(0) = 0.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta \varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \quad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

**Exercise 10.5** (Finite-radius conical orthogonality). For real  $\lambda, \mu$ , derive the Wronskian identity

$$(\mu^2 - \lambda^2) \int_0^R \varphi_\lambda(r) \varphi_\mu(r) \sinh r \, dr = \left[ \sinh r (\varphi_\mu \varphi_\lambda' - \varphi_\lambda \varphi_\mu') \right]_0^R.$$

Impose the Dirichlet condition  $\varphi_\lambda(R) = 0$ . Use the radial Sturm–Liouville operator to prove that its positive roots  $\lambda_j$  are real and simple and that the corresponding conical functions are orthogonal in  $L^2((0, R), \sinh r \, dr)$ . Compare the large- $r$  equation with a constant-coefficient equation and show how the discrete parameters become continuous as  $R \rightarrow \infty$ .

**Exercise 10.6** (Abel transforms and Euclidean Fourier reduction). For a smooth compactly supported radial function  $f$  on  $\mathbb{H}^2$ , define its Abel transform by

$$(\mathcal{A}f)(t) = e^{t/2} \int_{\mathbb{R}} f(a_t n_x K) \, dx.$$

Use the Iwasawa integration formula and the Poisson integral for  $\varphi_\lambda$  to prove

$$\tilde{f}(\lambda) := \int_{\mathbb{H}^2} f(z) \varphi_\lambda(z) \, d\text{vol}(z) = \int_{\mathbb{R}} (\mathcal{A}f)(t) e^{-i\lambda t} \, dt.$$

Prove that  $\mathcal{A}$  intertwines radial convolution on  $G/K$  with ordinary convolution on  $\mathbb{R}$ . Deduce injectivity of the spherical transform from Euclidean Fourier inversion and recover the commutativity of radial convolution.

For  $f$  supported in a geodesic disk of radius  $R$ , determine the support and parity properties of  $\mathcal{A}f$ . Derive the rank-one formulas

$$(\mathcal{A}f)(t) = \sqrt{2} \int_{|t|}^{\infty} \frac{f(r) \sinh r}{\sqrt{\cosh r - \cosh t}} \, dr,$$

$$f(r) = -\frac{1}{\pi\sqrt{2}} \int_r^{\infty} \frac{(\mathcal{A}f)'(t)}{\sqrt{\cosh t - \cosh r}} \, dt,$$

and use them to express  $f(r)$  in terms of the inverse Euclidean Fourier transform of  $\tilde{f}$ .

**Exercise 10.7** (Harish–Chandra normalization). For real  $\lambda \neq 0$ , define  $c(\lambda)$  by the asymptotic normalization

$$\varphi_\lambda(r) = c(\lambda) e^{(i\lambda-1/2)r} + c(-\lambda) e^{(-i\lambda-1/2)r} + o(e^{-r/2}).$$

Apply endpoint asymptotics to the Poisson integral and evaluate the resulting beta integrals to derive

$$c(\lambda) = \frac{\Gamma(i\lambda)}{\sqrt{\pi} \Gamma(\frac{1}{2} + i\lambda)}, \quad |c(\lambda)|^{-2} = \pi \lambda \tanh(\pi \lambda).$$



**Exercise 10.8** (Mehler–Fock inversion). For the spherical transform defined by the Abel transform, use radial integration to obtain

$$\tilde{f}(\lambda) = 2\pi \int_0^\infty f(r) \varphi_\lambda(r) \sinh r \, dr.$$

For  $\Phi \in C_c^\infty((0, \infty)^2)$ , define the diagonal distribution by

$$\iint \Phi(\lambda, \mu) \delta(\lambda - \mu) \, d\lambda \, d\mu = \int_0^\infty \Phi(\lambda, \lambda) \, d\lambda.$$

Interpret the following identity after pairing both sides with such test functions: truncate the radial integral at  $R$ , use the Harish–Chandra asymptotics in the Wronskian identity, and apply the distributional limit of  $\sin(R(\lambda - \mu))/(\lambda - \mu)$  to prove

$$2\pi \int_0^\infty \varphi_\lambda(r) \varphi_\mu(r) \sinh r \, dr = \frac{2\pi}{\lambda \tanh(\pi\lambda)} \delta(\lambda - \mu).$$

Deduce the Mehler–Fock inversion and Plancherel formulas

$$f(r) = \frac{1}{2\pi^2} \int_0^\infty \tilde{f}(\lambda) P_{-1/2+i\lambda}(\cosh r) |c(\lambda)|^{-2} \, d\lambda,$$

$$\|f\|_2^2 = \frac{1}{2\pi^2} \int_0^\infty |\tilde{f}(\lambda)|^2 |c(\lambda)|^{-2} \, d\lambda.$$

**Exercise 10.9** (Associated Legendre functions and hyperbolic-disk modes). In geodesic polar coordinates, derive the full Laplace–Beltrami operator

$$\Delta_{\mathbb{H}^2} = \frac{\partial^2}{\partial r^2} + \coth r \frac{\partial}{\partial r} + \frac{1}{\sinh^2 r} \frac{\partial^2}{\partial \theta^2}.$$

For the intrinsic  $SO(2)$  character  $e^{i\ell\theta}$ , derive the radial equation

$$R'' + \coth r \, R' - \frac{\ell^2}{\sinh^2 r} R = -(\lambda^2 + \tfrac{1}{4}) R.$$

Show that the solution regular at the origin is proportional to

$$P_{-1/2+i\lambda}^{-|\ell|}(\cosh r)$$

and determine its normalization from  $R(r) \sim r^{|\ell|}$  as  $r \downarrow 0$ .

For the geodesic disk  $D_R$ , prove that its Dirichlet eigenmodes are

$$e^{i\ell\theta} P_{-1/2+i\lambda_{n\ell}}^{-|\ell|}(\cosh r), \quad P_{-1/2+i\lambda_{n\ell}}^{-|\ell|}(\cosh R) = 0.$$

Establish orthogonality for distinct pairs  $(n, \ell)$  and identify the  $\ell = 0$  family with the conical modes in Mehler–Fock theory.

**Exercise 10.10** (Harish–Chandra wave kernels and geodesic phase fronts). Let  $u_{tt} = v^2 \Delta_{\mathbb{H}^2} u$  have radial initial data  $u(0, r) = f(r)$  and  $u_t(0, r) = g(r)$  in  $C_c^\infty$ . Put  $\omega_\lambda = v(\lambda^2 + 1/4)^{1/2}$ . Use Mehler–Fock inversion to prove

$$u(t, r) = \frac{1}{2\pi^2} \int_0^\infty \left( \cos(t\omega_\lambda) \tilde{f}(\lambda) + \frac{\sin(t\omega_\lambda)}{\omega_\lambda} \tilde{g}(\lambda) \right) P_{-1/2+i\lambda}(\cosh r) |c(\lambda)|^{-2} \, d\lambda.$$

For a point source at  $z_0$ , define the point distribution by  $\langle \delta_{z_0}, \phi \rangle = \phi(z_0)$ . Take  $f = 0$ ,  $g = \delta_{z_0}$ , use  $G$ -invariance to replace  $r$  by the hyperbolic distance  $d(z, z_0)$ , and derive the impulse kernel

$$K_t(d) = \frac{1}{2\pi^2} \int_0^\infty \frac{\sin(t\omega_\lambda)}{\omega_\lambda} P_{-1/2+i\lambda}(\cosh d) |c(\lambda)|^{-2} d\lambda.$$

Use as theorem input [40] finite propagation speed for the wave equation and the one-dimensional stationary-phase estimate for a nondegenerate critical point, including its uniform form for compact parameter sets. Apply them to show that the leading phase fronts are the geodesic circles  $d(z, z_0) = vt$ . Derive the high-frequency spectral density from  $|c(\lambda)|^{-2} = \pi\lambda \tanh(\pi\lambda)$  and state the interior tail present for the two-dimensional wave equation.

**Exercise 10.11** (Hyperbolic-drum spectra and circuit phase fronts). Let  $D_R \subset \mathbb{H}^2$  be the geodesic disk of radius  $R$  in curvature  $-1$ , impose Dirichlet boundary conditions, and let the wave speed be  $v$ . For  $n \geq 1$  and  $\ell \in \mathbb{Z}$ , prove that the normal modes are indexed by the positive solutions of

$$P_{-1/2+i\lambda_{n\ell}}^{-|\ell|}(\cosh R) = 0$$

and have angular frequencies

$$\omega_{n\ell} = v\sqrt{\lambda_{n\ell}^2 + \frac{1}{4}}.$$

For  $\ell = 0$ , use the  $c$ -function asymptotics to derive the large- $R$  quantization law

$$\lambda_{n0}R + \arg c(\lambda_{n0}) = \left(n - \frac{1}{2}\right)\pi + o(1),$$

with the indexing convention stated explicitly. Use the associated Legendre equations to compare the  $(n, \ell)$  branches and deduce the curvature-dependent reordering of radial and angular modes. Order the pairs  $(n, \ell)$  by increasing continuum eigenvalue, with multiplicity, and write the resulting parameter as  $\lambda_j$ .

For the  $\{3, 7\}$  tessellation used in Lenggenhager et al., *Nature Communications* **13** (2022), let  $Q_R$  be the graph Laplacian in their sign convention, put  $L_R = -Q_R \geq 0$ , and write  $\Lambda_j(L_R)$  for its eigenvalues. The experiment uses curvature  $-4$  and lattice parameter  $h = \tanh(0.545275)$ . At an interior vertex  $p$ , define

$$(Q_R f)(p) = \sum_{q \sim p} (f(q) - f(p)).$$

For the seven tangent displacements  $v_j$  to neighboring vertices, treat  $h$  as the long-wavelength expansion parameter, use the continuum normalization  $\|v_j\| = h$ , and verify the isotropic moment identities

$$\sum_{j=1}^7 v_j = 0, \quad \sum_{j=1}^7 v_j \otimes v_j = \frac{7}{2} h^2 I.$$

Apply the covariant Taylor expansion of  $f(\exp_p v_j)$  to prove the long-wavelength calibration

$$\Lambda_j(L_R) = \frac{7}{4} h^2 \nu_j^{(-4)} + O(h^3), \quad \nu_j^{(-4)} = 4\nu_j^{(-1)},$$

where  $\nu_j^{(-1)} = \lambda_j^2 + \frac{1}{4}$  is the curvature- $-1$  continuum eigenvalue. Prove that the metric and geodesic-radius normalizations satisfy  $g_{-1} = 4g_{-4}$  and  $R_{-1} = 2R_{-4}$ .

For circuit capacitance  $C$  and grounding inductance  $L$ , derive from the grounded circuit Laplacian

$$J(\omega) = -i\omega C Q_R + \frac{1}{i\omega L} I$$

that the measured circuit resonance satisfies

$$\omega_j^{\text{circ}} = \frac{1}{\sqrt{LC \Lambda_j(L_R)}}, \quad \nu_j^{(-1)} \sim \frac{\Lambda_j(L_R)}{7h^2} = \frac{1}{28\pi^2 h^2 LC (f_j^{\text{circ}})^2}.$$

Use this calibration to compare the inferred parameters  $\lambda_{n\ell}$  and measured voltage eigenmodes with the associated Legendre zeros and their predicted angular-momentum reordering. For the time-resolved experiment, plot the measured phase against  $d_{\mathbb{H}^2}(z, z_0)$  and against Euclidean distance, and compare the result with the Harish–Chandra impulse kernel and its geodesic phase fronts. State separately the curvature rescaling, continuum prediction, circuit resonance law, measured quantities, and errors caused by boundary shape, loss, finite frequency resolution, and discretization.

## 11 Automorphic Forms

**Exercise 11.1** (Lattice theta series). A *Euclidean lattice*  $L \subset \mathbb{R}^d$  is a discrete subgroup for which  $\mathbb{R}^d/L$  is compact. Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y \rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

For  $\tau$  in the upper half-plane and  $q = e^{2\pi i\tau}$ , its theta series is

$$\Theta_L(\tau) = \sum_{x \in L} q^{\|x\|^2/2}.$$

Its coefficient of  $q^n$  counts lattice vectors of squared length  $2n$ .

Apply Exercise 2.12 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\text{vol}(L)} \Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives  $\Theta_L(\tau+1) = \Theta_L(\tau)$ . Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

**Exercise 11.2** (Modular forms, zeta functions, and Eisenstein series). The modular group  $SL_2(\mathbb{Z})$  acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \quad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by  $S : \tau \mapsto -1/\tau$  and  $T : \tau \mapsto \tau + 1$ . A *modular form of integral weight  $k$*  for  $SL_2(\mathbb{Z})$  is a holomorphic function  $f$  on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion  $f(\tau) = \sum_{n \geq 0} a_n q^n$  has no negative powers. It is a *cusp form* if  $a_0 = 0$ . Write  $M_k$  for the vector space of modular forms of weight  $k$ .

For  $\text{Re } s > 1$ , define the Riemann zeta function by

$$\zeta(s) = \sum_{n \geq 1} n^{-s}.$$

For a  $d$ -dimensional Euclidean lattice  $L$ , with  $d \geq 1$ , and  $\operatorname{Re} s > d/2$ , define its Epstein zeta function by

$$Z_L(s) = \sum_{x \in L \setminus \{0\}} \|x\|^{-2s}.$$

For  $m \geq 2$  and  $\tau$  in the upper half-plane, define

$$G_{2m}(\tau) = \sum_{(c,d) \in \mathbb{Z}^2 \setminus \{(0,0)\}} \frac{1}{(c\tau + d)^{2m}}, \quad E_{2m}(\tau) = \frac{G_{2m}(\tau)}{2\zeta(2m)}.$$

(a) Put  $\theta_L(t) = \Theta_L(it)$ . For  $\operatorname{Re} s > d/2$ , prove

$$\pi^{-s}\Gamma(s)Z_L(s) = \int_0^\infty (\theta_L(t) - 1)t^{s-1} dt.$$

Split the integral at 1 and use Poisson duality to prove the meromorphic continuation

$$\begin{aligned} \pi^{-s}\Gamma(s)Z_L(s) &= \int_1^\infty (\theta_L(t) - 1)t^{s-1} dt \\ &\quad + \frac{1}{\operatorname{vol}(L)} \int_1^\infty (\theta_{L^*}(t) - 1)t^{d/2-s-1} dt + \frac{1}{\operatorname{vol}(L)(s - d/2)} - \frac{1}{s}. \end{aligned}$$

Deduce

$$\pi^{-s}\Gamma(s)Z_L(s) = \frac{1}{\operatorname{vol}(L)} \pi^{-(d/2-s)} \Gamma(d/2 - s) Z_{L^*}(d/2 - s),$$

$Z_L(0) = -1$ , and

$$\operatorname{res}_{s=d/2} Z_L(s) = \frac{\pi^{d/2}}{\operatorname{vol}(L)\Gamma(d/2)}.$$

(b) For  $L = \mathbb{Z}$ , use  $Z_{\mathbb{Z}}(s) = 2\zeta(2s)$  to derive the functional equation of  $\zeta$ . Use Fourier series and Parseval to prove

$$\zeta(2) = \frac{\pi^2}{6}, \quad \zeta(4) = \frac{\pi^4}{90},$$

and then deduce

$$\zeta(-1) = -\frac{1}{12}, \quad \zeta(-3) = \frac{1}{120}.$$

**Exercise 11.3** (Holomorphic Eisenstein series). Prove absolute local uniform convergence of  $G_{2m}$  and prove

$$G_{2m}\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^{2m} G_{2m}(\tau)$$

for every  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ . Using Poisson summation in one lattice direction, derive

$$G_{2m}(\tau) = 2\zeta(2m) + \frac{(2\pi i)^{2m}}{(2m-1)!} \sum_{n \geq 1} \sigma_{2m-1}(n) q^n, \quad \sigma_r(n) = \sum_{d|n} d^r.$$

Deduce that  $E_{2m}$  is a modular form of weight  $2m$  and that

$$E_4(\tau) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n, \quad E_6(\tau) = 1 - 504 \sum_{n \geq 1} \sigma_5(n) q^n.$$

**Exercise 11.4** (Theta and eta functions). Set

$$\begin{aligned}\vartheta_2(\tau) &= \sum_{n \in \mathbb{Z}} q^{(n+1/2)^2/2}, \\ \vartheta_3(\tau) &= \sum_{n \in \mathbb{Z}} q^{n^2/2}, & \eta(\tau) &= q^{1/24} \prod_{m \geq 1} (1 - q^m). \\ \vartheta_4(\tau) &= \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2/2},\end{aligned}$$

Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2} \vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2} \vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2} \vartheta_2(\tau).$$

Use the branch positive for  $\tau$  on the positive imaginary axis. Prove directly from the series that

$$\vartheta_2(\tau + 1) = e^{\pi i/4} \vartheta_2(\tau), \quad \vartheta_3(\tau + 1) = \vartheta_4(\tau), \quad \vartheta_4(\tau + 1) = \vartheta_3(\tau).$$

Write these  $S$ - and  $T$ -laws as a vector-valued transformation law of weight  $\frac{1}{2}$  for  $(\vartheta_2, \vartheta_3, \vartheta_4)^T$  and prove that it is not an instance of a scalar integral-weight transformation law.

**Exercise 11.5** (Jacobi triple product and eta transformations). Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n \in \mathbb{Z}} z^n q^{n^2/2} = \prod_{m \geq 1} (1 - q^m)(1 + zq^{m-1/2})(1 + z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Deduce

$$2\eta(\tau)^3 = \vartheta_2(\tau)\vartheta_3(\tau)\vartheta_4(\tau).$$

Use the theta transformations and the branch positive on the positive imaginary axis to derive

$$\eta(\tau + 1) = e^{\pi i/12} \eta(\tau), \quad \eta(-1/\tau) = (-i\tau)^{1/2} \eta(\tau).$$

Prove that these laws give  $\eta$  weight  $\frac{1}{2}$  with a nontrivial scalar multiplier, rather than an integral-weight law with trivial multiplier. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

**Exercise 11.6** (Modular functions, the discriminant, and the  $J$ -function). Put  $PSL_2(\mathbb{R}) = SL_2(\mathbb{R})/\{\pm I\}$  and define  $PSL_2(\mathbb{Z})$  similarly. Let  $\Gamma \subset PSL_2(\mathbb{R})$  be commensurable with  $PSL_2(\mathbb{Z})$  and suppose that the cusp at infinity has width one. A *modular function* for  $\Gamma$  is a  $\Gamma$ -invariant function meromorphic on the upper half-plane and at the cusps. Write  $X_\Gamma$  for the compactified quotient. A *normalized Hauptmodul* is a modular function inducing an isomorphism  $X_\Gamma \cong \mathbb{P}^1(\mathbb{C})$  and having Fourier expansion

$$q^{-1} + O(q)$$

at infinity.

Define

$$\Delta(\tau) = \eta(\tau)^{24}, \quad j(\tau) = \frac{E_4(\tau)^3}{\Delta(\tau)}, \quad J(\tau) = j(\tau) - 744 = \sum_{n \geq -1} c(n)q^n.$$

Let  $\mathbb{H}$  be the upper half-plane, let  $\rho = e^{2\pi i/3}$ , and let  $\mathcal{F}$  be the standard fundamental domain for  $PSL_2(\mathbb{Z})$ . Apply the argument principle to a truncated copy of  $\mathcal{F}$ , pair its boundary arcs by  $S$  and  $T$ , and prove that every nonzero modular form  $f$  of weight  $k$  satisfies

$$v_\infty(f) + \frac{1}{2}v_i(f) + \frac{1}{3}v_\rho(f) + \sum_{\substack{[z] \in PSL_2(\mathbb{Z}) \setminus \mathbb{H} \\ [z] \neq [i], [\rho]}} v_z(f) = \frac{k}{12}.$$

Use the eta transformation laws and product to show that  $\Delta$  is a weight-12 cusp form with a simple zero at infinity and no zero in the upper half-plane. Deduce

$$M_{12} = \mathbb{C}E_4^3 \oplus \mathbb{C}\Delta, \quad \Delta = \frac{E_4^3 - E_6^2}{1728}.$$

Show that  $j$  has one simple pole on  $X_{PSL_2(\mathbb{Z})}$  and hence induces an isomorphism with  $\mathbb{P}^1(\mathbb{C})$ . Expand the products and prove that

$$J(\tau) = q^{-1} + 196884q + 21493760q^2 + 864299970q^3 + O(q^4),$$

so  $J$  is the normalized Hauptmodul for  $PSL_2(\mathbb{Z})$ .

**Exercise 11.7** (Leech theta series and its first diffraction shell). For a Euclidean lattice  $L$ , call  $L$  integral when  $L \subseteq L^*$ , unimodular when  $L = L^*$ , and even when  $\|x\|^2 \in 2\mathbb{Z}$  for every  $x \in L$ . If the same lattice is used as a crystal lattice, its physical reciprocal lattice is  $L_{\text{physical}}^* = 2\pi L^*$ .

Call a vector of squared length 2 in an integral Euclidean lattice a *root*. Take the rootless case of the Niemeier classification as theorem input [12]: there is a unique isometry class of even unimodular Euclidean lattices of rank 24 having no roots. Call its members *Leech lattices*, and fix one representative  $\Lambda$ . Use Poisson duality to prove that  $\Theta_\Lambda$  is a modular form of weight 12. Use the absence of roots and the description of  $M_{12}$  to derive

$$\Theta_\Lambda = E_4^3 - 720\Delta = 1 + 196560q^2 + O(q^3).$$

For the idealized 24-dimensional crystal with lattice  $\Lambda$  and a one-point motif, deduce that the first nonzero reciprocal-lattice shell has radius  $4\pi$  in the physical reciprocal convention and consists of 196560 Bragg peaks. State how a nontrivial motif changes their intensities without changing the shell radius.

**Exercise 11.8** (Theta diffraction). Let  $T_L = \mathbb{R}^d/L$  carry its flat metric, and normalize  $L^*$  by  $\langle x, \xi \rangle \in \mathbb{Z}$ . Let  $K_t(x, y)$  be the heat kernel of the flat Laplacian and prove

$$\int_{T_L} K_t(x, x) dx = \sum_{\xi \in L^*} e^{-4\pi^2 t \|\xi\|^2} = \Theta_{L^*}(4\pi i t).$$

Apply Poisson duality to derive the short-time image sum and its leading term  $\text{vol}(T_L)(4\pi t)^{-d/2}$ . For a periodic array with motif  $B$ , prove that the elastic intensity at reciprocal vector  $2\pi\xi$  is proportional to

$$|S_B(2\pi\xi)|^2.$$

Prove that the heat-trace exponents and diffraction-peak radii determine the same reciprocal squared lengths  $\|\xi\|^2$ . Deduce the predicted peak locations, systematic absences, and thermal decay rates from the theta coefficients of  $L^*$ .

**Exercise 11.9** (Theta heat kernels, spectral zeta functions, and Casimir pressure). For a massless scalar field between parallel Dirichlet plates of area  $A$  and separation  $a$ , neglect edge effects and take the normal-mode frequencies to be

$$\omega_{n,k} = c\sqrt{|k|^2 + \left(\frac{\pi n}{a}\right)^2}, \quad n \geq 1, \quad k \in \mathbb{R}^2.$$

Define the normal-direction theta sum and the heat trace per unit area by

$$\Theta_a(t) = \sum_{n \in \mathbb{Z}} e^{-\pi^2 t n^2 / a^2},$$

$$\frac{K_D(t)}{A} = \sum_{n \geq 1} \int_{\mathbb{R}^2} \frac{d^2 k}{(2\pi)^2} e^{-t(|k|^2 + \pi^2 n^2 / a^2)} = \frac{\Theta_a(t) - 1}{8\pi t}.$$

(a) Apply the Jacobi theta transformation to prove

$$\Theta_a(t) = \frac{a}{\sqrt{\pi t}} \sum_{m \in \mathbb{Z}} e^{-a^2 m^2 / t}$$

and hence

$$\frac{K_D(t)}{A} = \frac{a}{8\pi^{3/2} t^{3/2}} - \frac{1}{8\pi t} + \frac{a}{4\pi^{3/2} t^{3/2}} \sum_{m \geq 1} e^{-a^2 m^2 / t}.$$

Identify the first term as the infinite-volume contribution, the second as plate-local, and the remaining image sum as the interaction term.

(b) For  $\mu > 0$  and  $\text{Re } s$  sufficiently large, define the regulated vacuum energy density by

$$\frac{\mathcal{E}_D(s)}{A} = \frac{\hbar c}{2} \mu^{2s} \sum_{n \geq 1} \int_{\mathbb{R}^2} \frac{d^2 k}{(2\pi)^2} \left( |k|^2 + \frac{\pi^2 n^2}{a^2} \right)^{1/2-s}.$$

Use the Mellin identity

$$\lambda^{1/2-s} = \frac{1}{\Gamma(s - \frac{1}{2})} \int_0^\infty t^{s-3/2} e^{-t\lambda} dt$$

to express  $\mathcal{E}_D(s)$  through the heat trace. Remove the bulk and plate-local terms identified above and evaluate the image integrals to continue the interaction energy as

$$\frac{\mathcal{E}_D^{\text{int}}(s)}{A} = \frac{\hbar c \mu^{2s} a^{2s-3}}{8\pi^{3/2}} \frac{\Gamma(2-s)}{\Gamma(s - \frac{1}{2})} \zeta(4-2s).$$

Evaluate at  $s = 0$  and prove

$$\frac{\mathcal{E}_D}{A} = -\frac{\pi^2 \hbar c}{1440 a^3}.$$

(c) For the ideal electromagnetic field, account for the two polarizations and deduce

$$\frac{\mathcal{E}_{\text{EM}}}{A} = -\frac{\pi^2 \hbar c}{720 a^3}, \quad P = -\frac{\partial(\mathcal{E}_{\text{EM}}/A)}{\partial a} = -\frac{\pi^2 \hbar c}{240 a^4}.$$

State how a force measurement tests the  $a^{-4}$  law and its coefficient, separating the ideal prediction from finite conductivity, temperature, roughness, alignment, and edge corrections.

## 12 Modular Characters

**Exercise 12.1** (Bosonic and fermionic trace generating functions). For a finite-dimensional complex vector space  $V$ , an endomorphism  $T$  induces  $\text{Sym}^m T$  and  $\Lambda^m T$ . Define

$$Z_B(T; z) = \sum_{m \geq 0} \text{tr}(\text{Sym}^m T) z^m, \quad Z_F(T; z) = \sum_{m \geq 0} \text{tr}(\Lambda^m T) z^m.$$

An *integer partition* of  $n \geq 0$  is a finite nonincreasing sequence of positive integers with sum  $n$ , with the empty sequence allowed for  $n = 0$ . Let  $p(n)$  be the number of integer partitions of  $n$ , so  $p(0) = 1$ .

- (a) Let  $V$  be a finite-dimensional complex vector space and  $T \in \text{End}(V)$ . Using the determinant defined by the action on the highest exterior power, prove

$$Z_B(T; z) = \frac{1}{\det(I - zT)}, \quad Z_F(T; z) = \det(I + zT).$$

The first identity holds for  $|z|$  below the reciprocal spectral radius; the second is a polynomial identity. Prove them first for a direct sum of generalized spectral subspaces and then by polynomial continuation, without choosing vectors in those subspaces. If  $\lambda_1, \dots, \lambda_N$  are the eigenvalues counted with algebraic multiplicity, prove, with the empty products interpreted as 1,

$$\text{tr}(\text{Sym}^m T) = \sum_{1 \leq i_1 \leq \dots \leq i_m \leq N} \lambda_{i_1} \cdots \lambda_{i_m}, \quad \text{tr}(\Lambda^m T) = \sum_{1 \leq i_1 < \dots < i_m \leq N} \lambda_{i_1} \cdots \lambda_{i_m}.$$

- (b) Let the one-particle Hamiltonian have one-dimensional spectral subspaces at energies  $j\hbar\omega$ ,  $j \geq 1$ . Apply the identities to the finite spectral truncation and then pass coefficientwise to the limit to derive

$$\prod_{j \geq 1} \frac{1}{1 - q^j} = \sum_{n \geq 0} p(n) q^n, \quad \prod_{j \geq 1} (1 + q^j) = \sum_{n \geq 0} p_{\text{dist}}(n) q^n,$$

where  $p_{\text{dist}}(n)$  counts partitions into distinct parts. Construct the multiplicity map separating powers of two and prove

$$\prod_{j \geq 1} (1 + q^j) = \prod_{j \geq 1} \frac{1}{1 - q^{2j-1}}.$$

Interpret the coefficient of  $q^n$  as the exact degeneracy at excitation energy  $n\hbar\omega$  for the corresponding bosonic or fermionic system.

- (c) For the bosonic system put  $q = e^{-\beta_{\text{th}}\hbar\omega}$ , where  $\beta_{\text{th}} = (k_B T)^{-1}$ , and define

$$Z(\beta_{\text{th}}) = \prod_{j \geq 1} \frac{1}{1 - e^{-\beta_{\text{th}} j \hbar \omega}}, \quad \Psi(\beta_{\text{th}}) = \log Z(\beta_{\text{th}}).$$

Define

$$U = -\frac{d\Psi}{d\beta_{\text{th}}}.$$

For

$$\mathcal{S} = k_B(\Psi + \beta_{\text{th}} U), \quad C = k_B \beta_{\text{th}}^2 \frac{d^2 \Psi}{d\beta_{\text{th}}^2},$$

prove  $d\mathcal{S} = k_B \beta_{\text{th}} dU$  as  $\beta_{\text{th}}$  varies and express  $d^2 \Psi / d\beta_{\text{th}}^2$  as the energy variance. Derive

$$U = \hbar\omega \sum_{j \geq 1} \frac{j}{e^{\beta_{\text{th}} j \hbar \omega} - 1}, \quad C = k_B (\beta_{\text{th}} \hbar\omega)^2 \sum_{j \geq 1} \frac{j^2 e^{\beta_{\text{th}} j \hbar \omega}}{(e^{\beta_{\text{th}} j \hbar \omega} - 1)^2}.$$



- (d) State how resolved spectroscopy tests the integer-spaced levels and the partition degeneracies, while equilibrium calorimetry tests the displayed energy and heat-capacity curves. Separate the independent-mode, harmonic, thermal-equilibrium, finite-resolution, and high-energy-cutoff assumptions from these predictions.

**Exercise 12.2** (Virasoro symmetry and chiral finite-size levels). The Virasoro algebra has generators  $L_n$ ,  $n \in \mathbb{Z}$ , and a central generator  $C$ , with

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{1}{12}(m^3 - m)\delta_{m+n,0}C, \quad [C, L_n] = 0.$$

A module has central charge  $c$  when  $C$  acts by  $cI$ . A highest-weight vector  $|h\rangle$  satisfies  $L_0|h\rangle = h|h\rangle$  and  $L_n|h\rangle = 0$  for  $n > 0$ . For a chiral system on a circle of circumference  $\ell$  and propagation speed  $u$ , define the finite-size Hamiltonian on such a module by

$$H_{\text{cyl}} = \frac{2\pi\hbar u}{\ell}(L_0 - c/24).$$

Order the negative Virasoro modes by their indices. Use the commutation relations to rewrite every word as an ordered word. Check every three-letter overlap and use the Jacobi identity to prove that the rewriting is confluent. Filter by word length and deduce that the associated graded algebra is the symmetric algebra on the negative modes. Conclude that the ordered monomials in the negative modes are linearly independent in the universal highest-weight module. Prove that  $L_{-n_1} \cdots L_{-n_r}|h\rangle$  has excitation energy  $2\pi\hbar u(n_1 + \cdots + n_r)/\ell$  above the highest-weight state and identify its exact degeneracy with a partition count before null-vector relations are imposed.

**Exercise 12.3** (Verma and vacuum characters). For  $\tau$  in the upper half-plane put  $q = e^{2\pi i\tau}$ . If  $H_a$  is a positive-energy module with finite-dimensional  $L_0$  spectral subspaces, define

$$\chi_a(\tau) = \text{Tr}_{H_a} q^{L_0 - c/24}$$

when the displayed operator is trace class; the same expression denotes the formal graded trace otherwise.

Use the ordered descendants to prove that the Verma-module character is

$$\chi_{c,h}^{\text{Verma}}(\tau) = q^{h-c/24} \prod_{m \geq 1} (1 - q^m)^{-1}.$$

For the vacuum impose  $L_{-1}|0\rangle = 0$  and derive the corresponding generic vacuum product.

**Exercise 12.4** (Null-vector quotients). Let a singular vector of degree  $r$  in a Verma module generate a submodule isomorphic to the Verma module of highest weight  $h + r$ . Prove that the quotient character is

$$\chi_{c,h}^{\text{Verma}} - \chi_{c,h+r}^{\text{Verma}}.$$

For several singular vectors, prove that their submodules must be combined by inclusion-exclusion, and show that independent subtraction is valid only when the intersections of their descendant submodules have been accounted for.

**Exercise 12.5** (Unitary modular character systems and Ising characters). A finite character vector  $\chi$  is a *unitary modular character system* when

$$\chi(-1/\tau) = S\chi(\tau), \quad \chi(\tau + 1) = T\chi(\tau)$$

for unitary matrices  $S, T$ . For a matrix  $M$  with nonnegative integral entries define

$$Z_M(\tau, \bar{\tau}) = \sum_{a,b} \overline{\chi_a(\tau)} M_{ab} \chi_b(\tau).$$

It is *modular invariant* when

$$S^* M S = M, \quad T^* M T = M.$$

For the Ising character system with  $c = 1/2$  and weights  $0, 1/2, 1/16$ , choose the square roots whose  $q$ -expansions have the displayed positive leading coefficients and put

$$\begin{aligned} \chi_0 &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/2} &= \frac{1}{2} \left( \sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}} \right), \\ \chi_{1/16} &= \frac{1}{\sqrt{2}} \sqrt{\frac{\vartheta_2}{\eta}}. \end{aligned}$$

Use the theta and eta transformation laws to derive

$$S = \frac{1}{2} \begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \quad T_{aa} = e^{2\pi i(h_a - c/24)}.$$

Expand each character as  $q^{h-c/24}$  times a power series, identify its coefficients with exact descendant counts, and verify the modular invariance of  $\sum_a |\chi_a|^2$ .

**Exercise 12.6** (Eta asymptotics and bosonic thermodynamics). Use Exercise 11.5 to prove

$$\prod_{j \geq 1} (1 - e^{-tj})^{-1} \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \quad (t \downarrow 0).$$

For  $t = \beta_{\text{th}} \hbar \omega$ , differentiate the transformed eta-product identity to strengthen this to

$$\Psi(\beta_{\text{th}}) = \frac{\pi^2}{6t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{t}{24} + O(e^{-4\pi^2/t})$$

and derive

$$\begin{aligned} U &= \hbar \omega \left( \frac{\pi^2}{6t^2} - \frac{1}{2t} + \frac{1}{24} + o(1) \right), \\ C &= k_B \left( \frac{\pi^2}{3t} - \frac{1}{2} + o(1) \right), \quad S = k_B \left( \frac{\pi^2}{3t} + \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{2} + o(1) \right). \end{aligned}$$

**Exercise 12.7** (Partition saddle). Insert this estimate into Cauchy's coefficient integral, locate its positive real saddle, prove that its equation is

$$n = \frac{U}{\hbar \omega},$$

and identify the leading saddle exponent with the Legendre transform  $S/k_B$ . Expand the phase through quadratic order and derive the major-arc contribution

$$\frac{1}{4n\sqrt{3}} \exp\left(\pi \sqrt{\frac{2n}{3}}\right).$$

**Exercise 12.8** (Partition minor arcs). Let  $r_n$  be the positive saddle radius and use a Farey dissection of the circle  $|q| = r_n$  whose principal major arc contains  $q = r_n$ . Use as theorem input the Hardy–Ramanujan minor-arc estimate [23]: for some  $\delta > 0$ , the coefficient integral over the complement of the principal major arc is

$$O\left(\exp\left[\left(\pi\sqrt{\frac{2}{3}} - \delta\right)\sqrt{n}\right]\right).$$

Combine this estimate with the major-arc calculation to deduce

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

**Exercise 12.9** (Ising state counts). For the diagonal Ising system, define nonnegative integers  $d_a(N)$  by

$$\chi_a(\tau) = q^{h_a - c/24} \sum_{N \geq 0} d_a(N) q^N$$

and define the resolved two-sided degeneracy

$$\rho_{\text{Ising}}(N, \bar{N}) = \sum_{a \in \{0, 1/2, 1/16\}} d_a(N) d_a(\bar{N}).$$

Use the explicit theta-product formulas and the  $S$ -transformation to control each character in a complex neighborhood of the positive real saddle. Bound the complementary arcs as in the partition calculation and prove, as  $N, \bar{N} \rightarrow \infty$  with  $N/\bar{N}$  in a fixed compact subset of  $(0, \infty)$ ,

$$\log \rho_{\text{Ising}}(N, \bar{N}) \sim 2\pi\sqrt{\frac{cN}{6}} + 2\pi\sqrt{\frac{c\bar{N}}{6}}, \quad c = \frac{1}{2}.$$

Derive the resulting asymptotic number of states below a specified high energy on a circle and the corresponding entropy. State the circumference, propagation speed, energy window, vacuum assumption, sector resolution, saddle-point regime, and finite-size limitations required when comparing these predictions with measured or transfer-matrix spectra.

**Exercise 12.10** (Lattice Fock characters). For an even positive-definite lattice  $L$  of rank  $d$ , let  $\varepsilon : L \times L \rightarrow \{\pm 1\}$  satisfy

$$\varepsilon(\lambda, 0) = \varepsilon(0, \lambda) = 1, \quad \varepsilon(\lambda, \mu)\varepsilon(\lambda + \mu, \nu) = \varepsilon(\mu, \nu)\varepsilon(\lambda, \mu + \nu),$$

$$\frac{\varepsilon(\lambda, \mu)}{\varepsilon(\mu, \lambda)} = (-1)^{\langle \lambda, \mu \rangle}.$$

Let  $\mathbb{C}_\varepsilon[L]$  have symbols  $e^\lambda$ ,  $\lambda \in L$ , with  $e^\lambda e^\mu = \varepsilon(\lambda, \mu) e^{\lambda + \mu}$ , and define

$$\mathcal{F}_L = \text{Sym}\left(\bigoplus_{n \geq 1} (L \otimes_{\mathbb{Z}} \mathbb{C}) t^{-n}\right) \otimes \mathbb{C}_\varepsilon[L],$$

where  $\text{Sym}(W) = \bigoplus_{r \geq 0} \text{Sym}^r(W)$ . Give  $(L \otimes_{\mathbb{Z}} \mathbb{C}) t^{-n}$  degree  $n$ , give the element  $e^\lambda$  of the twisted group algebra degree  $\|\lambda\|^2/2$ , and let  $L_0$  act by total degree.

The standard lattice vertex-operator construction equips  $\mathcal{F}_L$  with a central-charge- $d$  vertex operator algebra structure.

(a) Use the symmetric-power trace formula and the definition of the lattice theta series to prove

$$\mathrm{Tr}_{\mathcal{F}_L} q^{L_0-d/24} = \frac{\Theta_L(\tau)}{\eta(\tau)^d}.$$

(b) For the Leech lattice, use Exercise 11.7 to derive

$$\mathrm{Tr}_{\mathcal{F}_\Lambda} q^{L_0-1} = \frac{\Theta_\Lambda}{\eta^{24}} = J + 24.$$

(c) Show directly from the graded construction that the coefficient 24 counts the degree-one oscillator states and that

$$196884 = 196560 + 324, \quad 324 = 24 + \dim \mathrm{Sym}^2(\Lambda \otimes_{\mathbb{Z}} \mathbb{C}),$$

splits the degree-two states into nonzero lattice momenta and oscillator states.

**Exercise 12.11** (McKay–Thompson series and monstrous moonshine). Let  $V = \bigoplus_{m \geq 0} V_m$  be a positive-energy vertex operator algebra with finite-dimensional graded pieces and central charge  $c$ , and let  $L_0|_{V_m} = m \mathrm{id}$ . Let  $G$  act by grading-preserving automorphisms of  $V$ . For  $g \in G$ , define its *McKay–Thompson series* by

$$T_g(\tau) = \mathrm{Tr}_V (g q^{L_0-c/24}).$$

Let  $\mathbb{M}$  denote the Monster sporadic simple group.

Take the Frenkel–Lepowsky–Meurman construction and Borchers’s monstrous moonshine theorem as theorem input [6]. If  $\theta$  lifts  $\lambda \mapsto -\lambda$  to the Leech-lattice vertex operator algebra and  $V_\Lambda^{T,+}$  is the  $+1$  eigenspace in its  $\theta$ -twisted module, the orbifold

$$V^\natural = V_\Lambda^+ \oplus V_\Lambda^{T,+}$$

is a central-charge-24 vertex operator algebra with

$$\mathrm{Aut}(V^\natural) = \mathbb{M}, \quad T_1(\tau) = J(\tau),$$

and, for every  $g \in \mathbb{M}$ , the series  $T_g$  is the normalized Hauptmodul for the Conway–Norton genus-zero group  $\Gamma_g$ . Deduce

$$\dim V_0^\natural = 1, \quad \dim V_1^\natural = 0, \quad \dim V_2^\natural = 196884, \quad \dim V_3^\natural = 21493760.$$

If  $\mathbf{d}$  denotes the indicated irreducible Monster representation of dimension  $d$ , verify the first decompositions supplied by the Monster character table,

$$V_2^\natural \cong \mathbf{1} \oplus \mathbf{196883}, \quad V_3^\natural \cong \mathbf{1} \oplus \mathbf{196883} \oplus \mathbf{21296876}.$$

With  $p = e^{2\pi i \sigma}$ , use Borchers’s denominator identity

$$p^{-1} \prod_{m>0, n \in \mathbb{Z}} (1 - p^m q^n)^{c(mn)} = J(\sigma) - J(\tau)$$

as a formal product in the region  $|p| < |q| < 1$ . Compare coefficients through order  $p$  and derive the first replication identity

$$\frac{1}{2} \left( J(\tau)^2 - \sum_{n \in \mathbb{Z}} c(n) q^{2n} \right) - \sum_{n \in \mathbb{Z}} c(2n) q^n = 196884.$$

**Exercise 12.12** (Monster-resolved finite-size spectrum). Let  $\chi_\rho$  be the character of an irreducible Monster representation  $\rho$ . Use finite-group character orthogonality to prove that its multiplicity in  $V_m^h$  is

$$\mu_\rho(m) = \frac{1}{|\mathbb{M}|} \sum_{g \in \mathbb{M}} \overline{\chi_\rho(g)} [q^{m-1}] T_g(\tau).$$

Deduce that the symmetry-resolved series

$$Z_\rho(\tau) = \frac{\dim(\rho)}{|\mathbb{M}|} \sum_{g \in \mathbb{M}} \overline{\chi_\rho(g)} T_g(\tau)$$

has coefficient  $\dim(\rho)\mu_\rho(m)$  at  $q^{m-1}$ . Model an exactly Monster-symmetric chiral edge of circumference  $\ell$  and propagation speed  $u$  by

$$H = \frac{2\pi\hbar u}{\ell} (L_0 - 1).$$

Derive the excitation energies

$$E_m - E_0 = \frac{2\pi\hbar u}{\ell} m$$

and predict a missing level at  $2\pi\hbar u/\ell$ , followed by levels of degeneracy 196884 and 21493760 at  $4\pi\hbar u/\ell$  and  $6\pi\hbar u/\ell$ . Express the symmetry-resolved degeneracy at level  $m$  as  $\dim(\rho)\mu_\rho(m)$ . State the assumptions of exact Monster symmetry, linear chiral dispersion, periodic boundary conditions, access to the twined sectors, and the low-energy finite-size regime. Explain why a purely chiral central-charge-24 theory must be realized as an anomalous edge or accompanied by a compensating opposite-chirality sector, and distinguish the derived spectral prediction from evidence for a particular physical realization.

## 13 Quantum Groups

**Exercise 13.1** (Poisson–Lie groups and classical Yang–Baxter equations). A *Poisson–Lie group* is a Lie group  $G$  with a Poisson structure for which multiplication  $G \times G \rightarrow G$  is Poisson. A *Lie bialgebra* is a Lie algebra  $\mathfrak{g}$  with a linear map

$$\delta : \mathfrak{g} \rightarrow \mathfrak{g} \wedge \mathfrak{g}$$

that is a 1-cocycle for the adjoint action and whose transpose defines a Lie bracket on  $\mathfrak{g}^*$ . It is *coboundary* when

$$\delta(x) = [x \otimes 1 + 1 \otimes x, r]$$

for some  $r \in \mathfrak{g} \otimes \mathfrak{g}$ .

Define

$$\text{CYB}(r) = [r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}].$$

Differentiate a Poisson–Lie structure at the identity and prove that it defines a Lie bialgebra. Prove that the co-Jacobi identity is equivalent to  $\text{CYB}(r)$  being invariant under the diagonal adjoint action. Under the additional triangular hypothesis  $\text{CYB}(r) = 0$ , recover the classical Yang–Baxter equation; state the modified classical Yang–Baxter equation as  $\text{CYB}(r) = \Omega$  for an invariant tensor  $\Omega$  and distinguish the case  $\Omega \neq 0$ .

**Exercise 13.2** (Hopf algebras and  $U_v(\mathfrak{sl}_2)$ ). A *bialgebra* is a unital associative algebra  $H$  with unital algebra homomorphisms

$$\Delta : H \rightarrow H \otimes H, \quad \varepsilon : H \rightarrow \mathbb{C}$$

satisfying coassociativity and the counit identities. A *Hopf algebra* is a bialgebra with a linear antipode  $S : H \rightarrow H$  satisfying

$$m(S \otimes \text{id})\Delta = \eta\varepsilon = m(\text{id} \otimes S)\Delta.$$

Fix a quantum-group parameter  $0 < v < 1$ .

For  $0 < v < 1$ ,  $U_v(\mathfrak{sl}_2)$  is generated by  $E, F, K, K^{-1}$  with

$$KEK^{-1} = v^2E, \quad KFK^{-1} = v^{-2}F, \quad [E, F] = \frac{K - K^{-1}}{v - v^{-1}},$$

and

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

Its counit, antipode, and compact real-form involution are

$$\begin{aligned} \varepsilon(E) = \varepsilon(F) = 0, \quad \varepsilon(K) = 1, \quad S(E) = -K^{-1}E, \quad S(F) = -FK, \quad S(K) = K^{-1}, \\ K^* = K, \quad E^* = KF, \quad F^* = EK^{-1}. \end{aligned}$$

Verify that the coproduct preserves the defining relations and that the displayed counit, antipode, and involution make  $U_v(\mathfrak{sl}_2)$  a Hopf  $*$ -algebra. In the  $\hbar$ -adic completion put

$$v = e^{-\hbar}, \quad K = v^H = e^{-\hbar H}.$$

Prove that

$$[H, E] = 2E, \quad [H, F] = -2F, \quad \frac{K - K^{-1}}{v - v^{-1}} = H + O(\hbar^2),$$

and that the relations and coproduct tend to those of  $U(\mathfrak{sl}_2)$  as  $\hbar \rightarrow 0$ . Expand the coproduct through first order and compute the resulting coboundary Lie bialgebra.

**Exercise 13.3** (Gaussian binomial coefficients). For an indeterminate  $u$ ,  $n \geq 0$ , and  $0 \leq r \leq n$ , define

$$\begin{bmatrix} n \\ r \end{bmatrix}_u = \prod_{j=1}^r \frac{1 - u^{n-r+j}}{1 - u^j},$$

with value 1 for  $r = 0$  and value 0 outside this range.

Derive

$$\begin{bmatrix} n \\ r \end{bmatrix}_u = \begin{bmatrix} n-1 \\ r \end{bmatrix}_u + u^{n-r} \begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_u$$

and prove

$$\prod_{j=0}^{n-1} (1 + tu^j) = \sum_{r=0}^n u^{r(r-1)/2} \begin{bmatrix} n \\ r \end{bmatrix}_u t^r.$$

For  $YX = uXY$ , prove the noncommutative identity

$$(X + Y)^n = \sum_{r=0}^n \begin{bmatrix} n \\ r \end{bmatrix}_u X^r Y^{n-r}.$$

Set  $u = v^2$  and apply it to  $X = E \otimes 1$  and  $Y = K \otimes E$  in  $U_v(\mathfrak{sl}_2)$  to obtain an explicit formula for  $\Delta(E^n)$ . Show that the limit  $v \rightarrow 1$  recovers the ordinary binomial theorem and the undeformed coproduct.

**Exercise 13.4** (Gaussian-binomial state counts). For an integer partition  $\lambda$ , write  $|\lambda|$  for the sum of its parts and  $\ell(\lambda)$  for the number of nonzero parts. Its Young diagram has one left-justified row for each part. Apply the Gaussian binomial theorem. Prove that the coefficient of  $u^m$  in  $\begin{bmatrix} n \\ r \end{bmatrix}_u$  counts partitions of  $m$  whose diagrams fit inside an  $r \times (n - r)$  rectangle. For  $n$  fermionic levels with energies  $j\hbar\omega$ ,  $0 \leq j < n$ , prove that the coefficient of  $t^r u^{m+r(r-1)/2}$  in

$$\prod_{j=0}^{n-1} (1 + tu^j)$$

is the exact number of  $r$ -particle states at excitation energy  $m\hbar\omega$  above the filled lowest configuration, and identify it with the coefficient of  $u^m$  in  $\begin{bmatrix} n \\ r \end{bmatrix}_u$ .

**Exercise 13.5** (Compact quantum groups and  $SU_v(2)$ ). A *compact quantum group*  $\mathbb{G}$  is a unital  $C^*$ -algebra  $C(\mathbb{G})$  with a unital  $*$ -homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where  $\otimes_{\min}$  is the closure of the algebraic tensor product in  $B(H \otimes K)$  for faithful representations on  $H$  and  $K$ . The coproduct satisfies

$$(\Delta \otimes \text{id})\Delta = (\text{id} \otimes \Delta)\Delta$$

and

$$\overline{\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))} = C(\mathbb{G}) \otimes_{\min} C(\mathbb{G}) = \overline{\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)}.$$

Here a state on a unital  $C^*$ -algebra is a linear functional  $h$  with  $h(a^*a) \geq 0$  and  $h(1) = 1$ . A *Haar state* is a state  $h$  satisfying

$$(h \otimes \text{id})\Delta(a) = h(a)1 = (\text{id} \otimes h)\Delta(a).$$

Let  $\text{Pol}(SU_v(2))$  be the universal unital  $*$ -algebra generated by  $\alpha, \gamma$  so that

$$u = \begin{pmatrix} \alpha & -v\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary. Equivalently, its relations are

$$\begin{aligned} \alpha^* \alpha + \gamma^* \gamma &= 1, & \alpha \alpha^* + v^2 \gamma \gamma^* &= 1, & \gamma^* \gamma &= \gamma \gamma^*, \\ \alpha \gamma &= v \gamma \alpha, & \alpha \gamma^* &= v \gamma^* \alpha, \end{aligned}$$

Define  $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$ , and let  $C(SU_v(2))$  be the universal  $C^*$ -completion.

Prove that  $\Delta$  is a coassociative  $*$ -homomorphism satisfying the cancellation conditions. On  $\ell^2(\mathbb{Z}_{\geq 0})$ , define for  $\xi \in \mathbb{T}$

$$\pi_\xi(\gamma)e_n = \xi v^n e_n, \quad \pi_\xi(\alpha)e_n = (1 - v^{2n})^{1/2} e_{n-1}, \quad e_{-1} = 0.$$

Verify the relations and prove that

$$h(a) = (1 - v^2) \sum_{n \geq 0} v^{2n} \int_{\mathbb{T}} \langle e_n, \pi_\xi(a)e_n \rangle dm_{\mathbb{T}}(\xi)$$

is the Haar state.

**Exercise 13.6** (Quantum-group coefficient pairings). Call a finite-dimensional  $U_v(\mathfrak{sl}_2)$ -module  $V$  *type-1* when  $K$  is diagonalizable and

$$V = \bigoplus_{m \in \mathbb{Z}} V[m], \quad K|_{V[m]} = v^m \text{id}_{V[m]}.$$

On the distinguished Hermitian space  $V_1 = \mathbb{C}e_+ \oplus \mathbb{C}e_-$  define the fundamental type-1 module by

$$Ke_{\pm} = v^{\pm 1}e_{\pm}, \quad Ee_- = v^{-1/2}e_+, \quad Ee_+ = 0, \quad Fe_+ = v^{1/2}e_-, \quad Fe_- = 0.$$

For a finite-dimensional type-1 module  $V$ ,  $\xi \in V$ , and  $\varphi \in V^\vee$ , define

$$c_{\varphi, \xi}(x) = \varphi(x\xi) \quad (x \in U_v(\mathfrak{sl}_2)).$$

Let  $\mathcal{O}_v$  be the Hopf algebra generated by these coefficient functionals. Its pairing with  $U_v(\mathfrak{sl}_2)$  is

$$\langle x, c_{\varphi, \xi} \rangle = c_{\varphi, \xi}(x),$$

with multiplication and coproduct characterized by

$$\langle x, ab \rangle = \langle \Delta x, a \otimes b \rangle, \quad \langle xy, a \rangle = \langle x \otimes y, \Delta a \rangle.$$

Prove that the coefficient map of a tensor product is the product of the coefficient maps, that the coefficient map of a dual module is obtained from the antipode, and that the displayed pairing is a nondegenerate Hopf pairing. Apply it to the fundamental module and prove that sending its coefficient corepresentation to the displayed matrix  $u$  extends to a Hopf  $*$ -isomorphism

$$\mathcal{O}_v \longrightarrow \text{Pol}(SU_v(2)).$$

**Exercise 13.7** ( $SU_v(2)$  Peter–Weyl decomposition). For a finite-dimensional unitary corepresentation  $U$ , a coefficient is  $u_\omega = (\omega \otimes \text{id})(U)$  for  $\omega \in B(H_U)^*$ . The algebra  $\text{Pol}(SU_v(2))$  is the unital  $*$ -subalgebra generated by all such coefficients.

Construct the irreducible type-1 unitary  $U_v(\mathfrak{sl}_2)$ -modules  $V_n$  from their highest-weight lines and form their coefficient corepresentations of  $\text{Pol}(SU_v(2))$  using the Hopf pairing. Use as theorem input the compact-quantum-group Peter–Weyl theorem [48]: irreducible coefficient spaces are orthogonal for the Haar state and their algebraic direct sum is  $\text{Pol}(SU_v(2))$ . Use the explicit Haar state to compute the orthogonality constants and deduce

$$\text{Pol}(SU_v(2)) = \bigoplus_{n \geq 0} C(V_n)$$

algebraically and as a Hilbert direct sum in the Haar GNS space.

**Exercise 13.8** ( $C^*$ -tensor categories and corepresentations). A finite-dimensional unitary corepresentation of a compact quantum group  $\mathbb{G}$  on  $H_U$  is a unitary  $U \in B(H_U) \otimes C(\mathbb{G})$  with  $(\text{id} \otimes \Delta)(U) = U_{12}U_{13}$ .

A  $\mathbb{C}$ -linear category  $\mathcal{C}$  consists of objects, complex vector spaces  $\text{Hom}_{\mathcal{C}}(X, Y)$ , bilinear associative compositions

$$\text{Hom}_{\mathcal{C}}(Y, Z) \otimes \text{Hom}_{\mathcal{C}}(X, Y) \longrightarrow \text{Hom}_{\mathcal{C}}(X, Z), \quad (g, f) \longmapsto gf,$$



and identity morphisms  $\text{id}_X$ . A  $\mathbb{C}$ -linear tensor category also has a bilinear tensor product on objects and morphisms, a tensor unit  $\mathbf{1}$ , and isomorphisms

$$a_{X,Y,Z} : (X \otimes Y) \otimes Z \longrightarrow X \otimes (Y \otimes Z), \quad l_X : \mathbf{1} \otimes X \longrightarrow X, \quad r_X : X \otimes \mathbf{1} \longrightarrow X.$$

The tensor product satisfies

$$(f'f) \otimes (g'g) = (f' \otimes g')(f \otimes g), \quad \text{id}_X \otimes \text{id}_Y = \text{id}_{X \otimes Y},$$

and the structure maps commute with morphisms; in particular,

$$a_{X',Y',Z'}((f \otimes g) \otimes h) = (f \otimes (g \otimes h))a_{X,Y,Z}.$$

They satisfy the pentagon and triangle identities

$$\begin{aligned} a_{W,X,Y \otimes Z} a_{W \otimes X,Y,Z} &= (\text{id}_W \otimes a_{X,Y,Z}) a_{W,X \otimes Y,Z} (a_{W,X,Y} \otimes \text{id}_Z), \\ (\text{id}_X \otimes l_Y) a_{X,\mathbf{1},Y} &= r_X \otimes \text{id}_Y. \end{aligned}$$

A  $C^*$ -category has Banach morphism spaces and conjugate-linear maps  $f \mapsto f^*$  with

$$\begin{aligned} f^* : Y \longrightarrow X, \quad (f^*)^* &= f, \quad (gf)^* = f^* g^*, \\ \|gf\| &\leq \|g\| \|f\|, \quad \|f^* f\| = \|f\|^2. \end{aligned}$$

A  $C^*$ -tensor category is a  $\mathbb{C}$ -linear tensor category with this structure, with

$$(f \otimes g)^* = f^* \otimes g^*, \quad a^* = a^{-1}, \quad l^* = l^{-1}, \quad r^* = r^{-1}.$$

Associators and unit maps are suppressed in subsequent tensor composites.

For a compact quantum group  $\mathbb{G}$ , let  $\text{Rep}(\mathbb{G})$  have finite-dimensional unitary corepresentations as objects and

$$\text{Hom}_{\mathbb{G}}(U, V) = \{A : H_U \rightarrow H_V : (A \otimes 1)U = V(A \otimes 1)\}$$

as morphism spaces. Composition and adjoints are those of Hilbert-space operators. Define

$$U \boxtimes V = U_{13} V_{23} \quad \text{on } H_U \otimes H_V,$$

and take the trivial corepresentation on  $\mathbb{C}$  as tensor unit, with the Hilbert-space associator and unit maps.

- Prove that the adjoint of an intertwiner is an intertwiner, that  $U \boxtimes V$  is a unitary corepresentation, and that tensor products of intertwiners are intertwiners.
- Verify the pentagon, triangle, and  $C^*$ -identities and deduce that  $\text{Rep}(\mathbb{G})$  is a  $C^*$ -tensor category.
- For  $C(\mathbb{G}) = \mathbb{C}$ , identify it with  $\text{Hilb}_{\text{fd}}$ , the category of finite-dimensional Hilbert spaces and all linear maps.

**Exercise 13.9** (Rigidity, conjugates, and quantum dimension). An orthogonal direct sum  $X = \bigoplus_{i=1}^r X_i$  is specified by morphisms  $u_i : X_i \rightarrow X$  satisfying

$$u_i^* u_j = \delta_{ij} \text{id}_{X_i}, \quad \sum_{i=1}^r u_i u_i^* = \text{id}_X.$$

In a  $\mathbb{C}$ -linear tensor category, a *left dual* of  $V$  is an object  $V^\vee$  with morphisms

$$\text{ev}_V : V^\vee \otimes V \longrightarrow \mathbf{1}, \quad \text{coev}_V : \mathbf{1} \longrightarrow V \otimes V^\vee$$

satisfying

$$\begin{aligned} (\text{id}_V \otimes \text{ev}_V)(\text{coev}_V \otimes \text{id}_V) &= \text{id}_V, \\ (\text{ev}_V \otimes \text{id}_{V^\vee})(\text{id}_{V^\vee} \otimes \text{coev}_V) &= \text{id}_{V^\vee}. \end{aligned}$$

A *right dual*  ${}^\vee V$  has maps

$$\widetilde{\text{ev}}_V : V \otimes {}^\vee V \longrightarrow \mathbf{1}, \quad \widetilde{\text{coev}}_V : \mathbf{1} \longrightarrow {}^\vee V \otimes V$$

satisfying

$$\begin{aligned} (\widetilde{\text{ev}}_V \otimes \text{id}_V)(\text{id}_V \otimes \widetilde{\text{coev}}_V) &= \text{id}_V, \\ (\text{id}_{{}^\vee V} \otimes \widetilde{\text{ev}}_V)(\widetilde{\text{coev}}_V \otimes \text{id}_{{}^\vee V}) &= \text{id}_{{}^\vee V}. \end{aligned}$$

In a  $C^*$ -tensor category, a *conjugate* of an object  $V$  is an object  $\overline{V}$  with morphisms

$$R_V : \mathbf{1} \longrightarrow \overline{V} \otimes V, \quad \overline{R}_V : \mathbf{1} \longrightarrow V \otimes \overline{V}$$

satisfying

$$(\overline{R}_V^* \otimes \text{id}_V)(\text{id}_V \otimes R_V) = \text{id}_V, \quad (R_V^* \otimes \text{id}_{\overline{V}})(\text{id}_{\overline{V}} \otimes \overline{R}_V) = \text{id}_{\overline{V}}.$$

Its dual maps are

$$\text{ev}_V = R_V^*, \quad \text{coev}_V = \overline{R}_V, \quad \widetilde{\text{ev}}_V = \overline{R}_V^*, \quad \widetilde{\text{coev}}_V = R_V.$$

A solution of the conjugate equations is *standard* when

$$R_V^*(\text{id}_{\overline{V}} \otimes T)R_V = \overline{R}_V^*(T \otimes \text{id}_{\overline{V}})\overline{R}_V \quad (T \in \text{End}(V)).$$

For a standard solution, define

$$\text{tr}_{\mathcal{C}}(T) = R_V^*(\text{id}_{\overline{V}} \otimes T)R_V, \quad \dim_{\mathcal{C}}(V) = \text{tr}_{\mathcal{C}}(\text{id}_V).$$

For  $\mathcal{C} = \text{Rep}(SU_v(2))$ , write  $\text{tr}_v$  and  $\dim_v$ .

- (a) Let two standard solutions of the conjugate equations be given for  $V$ . Use the snake identities and standardness to construct the unitary isomorphism between their conjugate objects and prove that they define the same trace.
- (b) For  $S : V \rightarrow W$  and  $T : W \rightarrow V$ , prove

$$\text{tr}_{\mathcal{C}}(TS) = \text{tr}_{\mathcal{C}}(ST), \quad \text{tr}_{\mathcal{C}}(T^*T) \geq 0.$$

- (c) Construct standard solutions for tensor products, conjugates, and existing orthogonal direct sums and prove

$$\dim_{\mathcal{C}}(V \otimes W) = \dim_{\mathcal{C}}(V) \dim_{\mathcal{C}}(W), \quad \dim_{\mathcal{C}}(\overline{V}) = \dim_{\mathcal{C}}(V),$$

$$\dim_{\mathcal{C}}(V \oplus W) = \dim_{\mathcal{C}}(V) + \dim_{\mathcal{C}}(W).$$

**Exercise 13.10** (Finite Hilbert-space trace). For a finite-dimensional Hilbert space  $V$ , let  $j_V : \bar{V} \rightarrow V^\vee$  be the Riesz isomorphism  $j_V(\bar{v}) = \langle v, - \rangle$ . From the coevaluation and symmetry put

$$\bar{R}_V = (\text{id}_V \otimes j_V^{-1}) \text{coev}_V : \mathbb{C} \longrightarrow V \otimes \bar{V}, \quad R_V = s_{V, \bar{V}} \bar{R}_V : \mathbb{C} \longrightarrow \bar{V} \otimes V.$$

Prove the conjugate equations and standardness. Prove, without choosing a basis, that

$$\text{tr}_{\text{Hilb}_{\text{fd}}}(T) = \text{tr}(T), \quad \dim_{\text{Hilb}_{\text{fd}}}(V) = \dim_{\mathbb{C}}(V),$$

where the right-hand sides are the ordinary trace and dimension.

**Exercise 13.11** ( $SU_v(2)$  fusion and quantum dimensions). Let  $V_n$  be the irreducible type-1  $U_v(\mathfrak{sl}_2)$ -module of highest weight  $n$ . Construct its conjugate equations and prove

$$V_m \otimes V_n \cong \bigoplus_{k=0}^{\min(m,n)} V_{m+n-2k}.$$

Normalize the conjugate solutions by the compact involution and compute

$$\dim_v(V_n) = [n+1]_v = \frac{v^{n+1} - v^{-(n+1)}}{v - v^{-1}}.$$

Prove that finite orthogonal direct sums of the  $V_n$  are closed under tensor products and conjugates and form a rigid  $C^*$ -tensor category. Prove

$$\begin{aligned} \dim_{\mathbb{C}}(V_1) &= 2, & \dim_v(V_1) &= v + v^{-1}, \\ \dim_{\mathbb{C}}(V_n) &= n+1, & \lim_{v \rightarrow 1} \dim_v(V_n) &= n+1, \\ & & \dim_v(V_n) &> n+1 \quad (0 < v < 1, n \geq 1). \end{aligned}$$

**Exercise 13.12** (Basic hypergeometric series and summations). For  $m \in \mathbb{N}$ , define

$$\begin{aligned} (a; Q)_m &= \prod_{j=0}^{m-1} (1 - aQ^j), & (a; Q)_\infty &= \prod_{j=0}^{\infty} (1 - aQ^j), \\ (a_1, \dots, a_r; Q)_m &= \prod_{i=1}^r (a_i; Q)_m, & (a_1, \dots, a_r; Q)_\infty &= \prod_{i=1}^r (a_i; Q)_\infty. \end{aligned}$$

The *basic hypergeometric series* is

$${}_r\phi_s \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; Q, z \right) = \sum_{m=0}^{\infty} \frac{(a_1, \dots, a_r; Q)_m}{(Q, b_1, \dots, b_s; Q)_m} \left( (-1)^m Q^{m(m-1)/2} \right)^{1+s-r} z^m.$$

It is terminating when one numerator parameter is  $Q^{-n}$ .

Prove the  $Q$ -Gauss identity

$${}_2\phi_1 \left( \begin{matrix} a, b \\ c \end{matrix}; Q, \frac{c}{ab} \right) = \frac{(c/a, c/b; Q)_\infty}{(c, c/(ab); Q)_\infty}, \quad \left| \frac{c}{ab} \right| < 1,$$

and the terminating  $Q$ -Pfaff-Saalschütz identity

$${}_3\phi_2 \left( \begin{matrix} a, b, Q^{-n} \\ c, abQ^{1-n}/c \end{matrix}; Q, Q \right) = \frac{(c/a, c/b; Q)_n}{(c, c/(ab); Q)_n}$$

by comparing coefficients after multiplication by the denominator  $Q$ -products in the first identity, and by showing that the two sides of the second identity satisfy the same recurrence in  $n$  and have the same initial value.

**Exercise 13.13** (Quantum coupling coefficients and XXZ symmetry). Use the distinguished weight-line realization  $V_1 = \mathbb{C}e_+ \oplus \mathbb{C}e_-$  of the fundamental type-1 module. Define the Pauli endomorphisms, relative to the ordered weight lines, by

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and let  $\sigma_j^a$  act as  $\sigma^a$  on the  $j$ th tensor factor and as the identity on the others. On  $V_1^{\otimes L}$  let

$$H_{\text{XXZ}} = J \sum_{j=1}^{L-1} \left( \sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \frac{v + v^{-1}}{2} (\sigma_j^z \sigma_{j+1}^z - I) + \frac{v - v^{-1}}{2} (\sigma_j^z - \sigma_{j+1}^z) \right).$$

For  $J \in \mathbb{R}$  and  $0 < v < 1$ , prove that this open-chain Hamiltonian is self-adjoint and commutes with the iterated coproduct action of  $U_v(\mathfrak{sl}_2)$ .

**Exercise 13.14** (Dual  $Q$ -Hahn coupling coefficients). For irreducible type-1 modules  $V_a, V_b, V_c$ , let

$$C_{ab}^c : V_c \longrightarrow V_a \otimes V_b$$

be the intertwining isometry normalized by its positive highest-weight coefficient. Its matrix coefficients between the distinguished weight lines are the  $v$ -Clebsch–Gordan coefficients.

Decompose two coupled irreducible spin sectors by the projections  $C_{ab}^c (C_{ab}^c)^*$  and derive the orthogonality relations for their  $v$ -Clebsch–Gordan coefficients. Prove that their weight dependence satisfies the dual  $Q$ -Hahn recurrence with  $Q = v^2$  and normalized amplitudes with a terminating  ${}_3\phi_2$  of base  $Q$ .

**Exercise 13.15** (XXZ transition strengths). Let the weak drive be an irreducible tensor operator  $A_\mu^{(s)}$  for the  $U_v(\mathfrak{sl}_2)$  action. Prove the  $v$ -Wigner–Eckart factorization of its matrix elements into a reduced matrix element and a  $v$ -Clebsch–Gordan coefficient. Derive the relative transition strengths at fixed reduced matrix element as squared  $v$ -Clebsch–Gordan coefficients. State the chain length, open boundary terms, resolved weights, energy scale  $J$ , and nondegeneracy assumptions under which the measured line positions and strengths determine  $v$ .

## 14 Braided Symmetry

**Exercise 14.1** (Universal  $R$ -matrices and braided tensor categories). A *universal  $R$ -matrix* for a Hopf algebra  $H$  is an invertible element  $\mathcal{R}$  in a completion of  $H \otimes H$  on which the finite-dimensional tensor-product representations under consideration extend, satisfying

$$\mathcal{R} \Delta(x) \mathcal{R}^{-1} = \Delta^{\text{op}}(x), \quad (\Delta \otimes \text{id})(\mathcal{R}) = \mathcal{R}_{13} \mathcal{R}_{23}, \quad (\text{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13} \mathcal{R}_{12}.$$

For finite-dimensional modules  $V, W$ , define

$$c_{V,W} = \tau \circ (\pi_V \otimes \pi_W)(\mathcal{R}) : V \otimes W \longrightarrow W \otimes V.$$

A *braiding* on a  $\mathbb{C}$ -linear tensor category  $\mathcal{C}$  is a family of isomorphisms

$$c_{X,Y} : X \otimes Y \longrightarrow Y \otimes X$$

satisfying, for  $f : X \rightarrow X'$  and  $g : Y \rightarrow Y'$ ,

$$(g \otimes f)c_{X,Y} = c_{X',Y'}(f \otimes g),$$

and the two hexagon identities

$$c_{X,Y \otimes Z} = (\text{id}_Y \otimes c_{X,Z})(c_{X,Y} \otimes \text{id}_Z),$$

$$c_{X \otimes Y,Z} = (c_{X,Z} \otimes \text{id}_Y)(\text{id}_X \otimes c_{Y,Z}).$$

The category is *braided* when it has a braiding.

Prove that  $c_{V,W}$  is a module intertwiner satisfying naturality and that the two coproduct identities for  $\mathcal{R}$  give the two hexagon identities, so that the finite-dimensional module category is braided. Derive

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

and, on  $V^{\otimes 3}$ ,

$$c_{12}c_{23}c_{12} = c_{23}c_{12}c_{23}.$$

Prove that the operators  $c_{j,j+1}$  define braid-group representations on tensor powers of every finite-dimensional type-1 module.

**Exercise 14.2** (Ribbon categories and ribbon elements). Let  $\mathcal{C}$  be a braided rigid  $\mathbb{C}$ -linear tensor category. A *twist* is a family of isomorphisms  $\theta_X : X \rightarrow X$  satisfying

$$\theta_Y f = f \theta_X \quad (f : X \rightarrow Y)$$

and

$$\theta_1 = \text{id}_1, \quad \theta_{X \otimes Y} = c_{Y,X} c_{X,Y} (\theta_X \otimes \theta_Y).$$

The category is *ribbon* when

$$\theta_{X^\vee} = (\theta_X)^\vee$$

for every object  $X$ .

For the universal  $R$ -matrix, set

$$u = m(S \otimes \text{id})(\mathcal{R}_{21}).$$

Let  $\nu_R \in H$  be central and invertible and satisfy

$$\nu_R^2 = uS(u), \quad S(\nu_R) = \nu_R, \quad \varepsilon(\nu_R) = 1, \quad \Delta(\nu_R) = (\mathcal{R}_{21}\mathcal{R})^{-1}(\nu_R \otimes \nu_R).$$

Prove that

$$\theta_V = \pi_V(\nu_R^{-1})$$

defines a ribbon twist on the category of finite-dimensional  $H$ -modules. Derive invariance of the quantum trace under framed ribbon isotopy.

**Exercise 14.3** (Kauffman bracket skein algebra). Let  $\Sigma$  be an oriented surface and let  $\mathcal{L}(\Sigma)$  be the set of framed-isotopy classes of unoriented links in  $\Sigma \times [0, 1]$ , including the empty link. For  $v \in \mathbb{C}^\times$ , define  $K_v(\Sigma)$  as the complex span of  $\mathcal{L}(\Sigma)$  modulo the following local relations. If  $L_\times, L_0, L_\infty$  agree outside a disk and differ inside it by a crossing and its two smoothings, impose

$$[L_\times] = v[L_0] + v^{-1}[L_\infty].$$

If  $\bigcirc$  is a contractible component disjoint from  $L$ , impose

$$[L \sqcup \bigcirc] = -(v^2 + v^{-2})[L].$$

Define multiplication by stacking links in the interval direction.

Prove that the relations are preserved by framed Reidemeister moves and that stacking makes  $K_v(\Sigma)$  an associative algebra with the empty link as identity. For two simple closed curves meeting once, resolve their crossing to express both multiplication orders in terms of the two smoothings and compute their commutator.

At  $v = -1$ , let  $\rho : \pi_1(\Sigma) \rightarrow SL_2(\mathbb{C})$  and assign to an unoriented loop  $\gamma$  the function

$$[\gamma] \mapsto -W_\gamma(\rho) = -\text{tr } \rho(\gamma),$$

extending multiplicatively over disjoint components. Use Cayley–Hamilton to prove

$$W_\alpha W_\beta = W_{\alpha\beta} + W_{\alpha\beta^{-1}}$$

and show that the assignment respects both skein relations. Deduce that the classical specialization is commutative and maps to the algebra of Wilson-trace functions on flat  $SL_2$  connections.

For  $\Sigma = T^2$ , put

$$x = W_{1,0}, \quad y = W_{0,1}, \quad z = W_{1,1}.$$

Use successive skein resolutions and the Euclidean algorithm to express every  $W_{p,q}$  as a polynomial in  $x, y, z$ , reduced modulo

$$x^2 + y^2 + z^2 - xyz - 4 = 0.$$

Use the commutator computed above to prove that  $K_v(T^2)$  is noncommutative for generic  $v$  and that its specialization at  $v = -1$  maps to the torus Wilson-trace algebra.

**Exercise 14.4** (Chebyshev quantum dimensions). For the irreducible  $SU_v(2)$  objects  $V_n$ , use  $V_1 \otimes V_n \cong V_{n+1} \oplus V_{n-1}$  to prove

$$d_{n+1} = (v + v^{-1})d_n - d_{n-1}, \quad d_0 = 1, \quad d_1 = v + v^{-1},$$

where  $d_n = \dim_v(V_n)$ . Using the Chebyshev polynomial  $U_n$ , deduce

$$\dim_v(V_n) = U_n\left(\frac{v + v^{-1}}{2}\right) = \frac{v^{n+1} - v^{-(n+1)}}{v - v^{-1}}.$$

**Exercise 14.5** (Quantum recoupling coefficients). The matrix coefficients of the recoupling map

$$(V_a \otimes V_b) \otimes V_c \longrightarrow V_a \otimes (V_b \otimes V_c)$$

between fixed intermediate irreducible summands are the  $v$ -6j coefficients.

Compute the change between the two coupling orders of  $V_a \otimes V_b \otimes V_c$  and express its entries as terminating  ${}_4\phi_3$  series with base  $Q = v^2$ . Identify them with  $Q$ -Racah polynomials and derive their orthogonality from unitarity of the recoupling map for the compact real form. Prove the pentagon identity by comparing the five coupling orders of four factors. For three resolved XXZ spin sectors, derive the recoupling probabilities as squared  $v$ -6j coefficients and state the unitarity and spectral-resolution assumptions required for this probability interpretation.

**Exercise 14.6** (Type-A Hecke actions). For  $v \neq 0$ , the Hecke algebra  $\mathcal{H}_v(S_r)$  is generated by  $T_1, \dots, T_{r-1}$  with

$$T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}, \quad T_i T_j = T_j T_i \quad (|i - j| > 1), \\ (T_i - v)(T_i + v^{-1}) = 0.$$

Let  $V_1$  be the fundamental  $U_v(\mathfrak{sl}_2)$ -module. On  $V_1 \otimes V_1 \cong V_2 \oplus V_0$ , let  $P_2, P_0$  be the invariant orthogonal projections, and define

$$T = vP_2 - v^{-1}P_0, \quad \rho_r(T_i) = \text{id}^{\otimes(i-1)} \otimes T \otimes \text{id}^{\otimes(r-i-1)}.$$

Prove that the coproduct action of  $U_v(\mathfrak{sl}_2)$  commutes with  $\rho_r(T_i)$ . Derive the braid relation from the  $Q$ -Racah recoupling identity and the Hecke relation from the two displayed spectral projections. Put  $e_i = (v - T_i)/(v + v^{-1})$  and prove that the action factors through the Temperley–Lieb relations

$$e_i^2 = e_i, \quad e_i e_{i \pm 1} e_i = (v + v^{-1})^{-2} e_i.$$

**Exercise 14.7** (Hecke Poincaré identity). Put  $\tau = v^2$ . For  $w \in S_r$ , let  $\ell(w)$  be the least number of adjacent transpositions in an expression for  $w$ , and let  $T_w = T_{i_1} \cdots T_{i_{\ell(w)}}$  for any reduced expression  $w = s_{i_1} \cdots s_{i_{\ell(w)}}$ . Prove from the braid relations that  $T_w$  is well-defined and that

$$\sum_{w \in S_r} \tau^{\ell(w)} = \prod_{j=1}^r \frac{1 - \tau^j}{1 - \tau}.$$

Prove that

$$e_+ = \frac{\sum_{w \in S_r} v^{\ell(w)} T_w}{\sum_{w \in S_r} v^{2\ell(w)}}, \quad e_- = \frac{\sum_{w \in S_r} (-v^{-1})^{\ell(w)} T_w}{\sum_{w \in S_r} v^{-2\ell(w)}}$$

are idempotents satisfying  $T_i e_+ = v e_+$  and  $T_i e_- = -v^{-1} e_-$ . Under the fundamental Hecke action, identify their images with the quantum symmetric and exterior state spaces and interpret the Poincaré polynomial as their graded state count.

**Exercise 14.8** (Affine Hecke algebras and Baxterization). The extended affine Hecke algebra  $\mathcal{H}_v^{\text{aff}}(GL_r)$  is generated by  $T_1, \dots, T_{r-1}$  and commuting invertible elements  $X_1, \dots, X_r$ , subject to the finite Hecke relations and

$$T_i X_i T_i = X_{i+1}, \quad T_i X_j = X_j T_i \quad (j \neq i, i+1).$$

Equivalently, it is the Hecke quotient of the extended affine braid group of  $\mathbb{Z}^r \rtimes S_r$ .

For an invertible spectral parameter  $u$ , define

$$\check{R}_i(u) = uT_i - u^{-1}T_i^{-1}.$$

Its eigenvalues on the  $v$  and  $-v^{-1}$  eigenspaces of  $T_i$  are

$$uv - u^{-1}v^{-1}, \quad -uv^{-1} + u^{-1}v.$$

Use the braid and Hecke relations for  $T_i$  to prove

$$\check{R}_i(u) \check{R}_{i+1}(uv) \check{R}_i(v) = \check{R}_{i+1}(v) \check{R}_i(uv) \check{R}_{i+1}(u).$$

Prove that  $\check{R}_i(1)$  is a scalar multiple of the identity and compute  $\check{R}_i(u) \check{R}_i(u^{-1})$ . For  $|u| = 1$ , prove that  $\check{R}_i(u)^* = \check{R}_i(u^{-1})$  and normalize the operators to be unitary wherever the scalar product is nonzero. Prove that their two eigenvalues depend on  $u$  and hence that the spectral family itself does not satisfy one parameter-independent Hecke polynomial.

**Exercise 14.9** (Six-vertex transfer matrices and the XXZ spectrum). On the fundamental module  $V_1$ , put

$$\check{R}(u) = uT - u^{-1}T^{-1}, \quad R(u) = \tau \circ \check{R}(u),$$

where  $T = vP_2 - v^{-1}P_0$  and  $\tau$  is the flip. For an auxiliary copy  $V_a \cong V_1$  and a periodic row of length  $N$ , define

$$\mathsf{T}_N(u) = \text{Tr}_{V_a}(R_{aN}(u) \cdots R_{a1}(u)) \quad \text{on } V_1^{\otimes N},$$

and for an  $N \times M$  periodic lattice define

$$Z_{N,M}^{6V}(u) = \text{Tr}_{V_1^{\otimes N}} \mathsf{T}_N(u)^M.$$

Use invariance of  $P_2$  and  $P_0$  under the Cartan action to prove

$$[R(u), K \otimes K] = 0.$$

Relative to the distinguished weight lines of  $V_1$ , deduce that exactly the six arrow configurations preserving total weight can have nonzero local weights, and compute those weights from  $P_2$  and  $P_0$ . Prove the spectral Yang–Baxter equation for  $R$  from the Baxterization identity. Apply it to an auxiliary pair and take their intrinsic traces to prove

$$[\mathsf{T}_N(u), \mathsf{T}_N(w)] = 0.$$

Show that  $Z_{N,M}^{6V}(u)$  is the sum of products of the six local weights over arrow configurations on the periodic square lattice.

Use  $\check{R}(1) = (v - v^{-1})I$  and compute the logarithmic derivative at the regular point. Prove that

$$\hat{H}_N = J \left[ (v - v^{-1}) \mathsf{T}'_N(1) \mathsf{T}_N(1)^{-1} - N(v + v^{-1})I \right]$$

is the periodic XXZ Hamiltonian, with  $\Delta = (v + v^{-1})/2$  and with the telescoping boundary term equal to zero. On the sector with one reversed spin, apply the distinguished Fourier modes and derive

$$E(k) = 4J(\cos k - \Delta), \quad k = \frac{2\pi m}{N}, \quad 0 \leq m < N.$$

Deduce the finite-size line spacings and the band edges. State the energy calibration, periodic-boundary, polarization-sector, and spectral-resolution assumptions required when these values are compared with an XXZ spin-chain simulator, and determine the scalar gauge and real spectral-parameter regime under which the same six weights are nonnegative Boltzmann weights.

**Exercise 14.10** ( $q$ KZ transport). Let  $v$  be the quantum-group parameter and let  $\kappa \neq 0$  be an independent shift parameter. Let  $D \in \text{End}(V)$  be invertible and satisfy  $[D \otimes D, \check{R}(u)] = 0$ . Let  $D_1$  act on the transported tensor factor and define

$$\mathcal{K}_1(z_1, \dots, z_r) = D_1 \check{R}_{1r}(z_1/z_r) \cdots \check{R}_{12}(z_1/z_2).$$

The first  $q$ KZ equation is

$$\Psi(z_2, \dots, z_r, \kappa z_1) = \mathcal{K}_1(z_1, \dots, z_r) \Psi(z_1, \dots, z_r),$$

and the remaining equations are obtained by cyclic conjugation.

Use Baxterization to prove consistency of the adjacent exchange rules and the  $q$ KZ difference equations. Identify the constant exchanges and winding operators with the generators of  $\mathcal{H}_v^{\text{aff}}(GL_r)$ ,



keeping  $v$  and  $\kappa$  independent. For the unitary normalization, prepare and detect three resolved XXZ spin sectors and prove that the two factorizations of a three-spin exchange give the same output probabilities. Express this equality as the  $Q$ -Racah recoupling identity. Derive the phase accumulated in one  $\kappa$ -shift cycle and state the twist, unitarity, preparation, and detection assumptions under which it is an interference observable.

**Exercise 14.11** (Izergin determinant and domain-wall ice entropy). Use the additive trigonometric normalization of the six-vertex Yang–Baxter family. For horizontal rapidities  $x = (x_1, \dots, x_N)$ , vertical rapidities  $y = (y_1, \dots, y_N)$ , and crossing parameter  $\eta$ , put

$$a(x_i, y_j) = \sin(x_i - y_j + \eta), \quad b(x_i, y_j) = \sin(x_i - y_j), \quad c = \sin \eta.$$

Impose domain-wall boundary arrows, with the top and bottom arrows pointing inward and the left and right arrows pointing outward, and let  $Z_N^{\text{DW}}(x; y)$  be the sum of products of these local weights over all allowed arrow configurations.

Use the Yang–Baxter equation to prove that  $Z_N^{\text{DW}}$  is separately symmetric in the  $x_i$  and in the  $y_j$ . At  $x_N = y_N$ , show that the vertex at their intersection has weight  $c$ , that its row and column freeze, and that

$$Z_N^{\text{DW}}(x; y) \Big|_{x_N=y_N} = c \prod_{i=1}^{N-1} a(x_i, y_N) \prod_{j=1}^{N-1} a(x_N, y_j) Z_{N-1}^{\text{DW}}(\hat{x}_N; \hat{y}_N).$$

Here a hat means that the indicated rapidity is omitted. With  $X_i = e^{2ix_i}$  and  $Y_j = e^{2iy_j}$ , prove that  $X_i^{(N-1)/2} Z_N^{\text{DW}}$  is a polynomial of degree  $N-1$  in  $X_i$  and that the analogous statement holds for every  $Y_j$ . Use as theorem input the Korepin interpolation lemma [29]: separate symmetry, these degree bounds, the displayed freezing recursion, and  $Z_1^{\text{DW}} = c$  uniquely determine the domain-wall partition function.

Verify that the determinant below has these properties and deduce the Izergin identity

$$Z_N^{\text{DW}}(x; y) = \frac{\prod_{i,j=1}^N a(x_i, y_j) b(x_i, y_j)}{\prod_{1 \leq i < j \leq N} \sin(x_i - x_j) \sin(y_j - y_i)} \det \left[ \frac{c}{a(x_i, y_j) b(x_i, y_j)} \right]_{i,j=1}^N.$$

Take the homogeneous ice limit

$$x_i - y_j = \eta = \frac{\pi}{3}, \quad a = b = c = \frac{\sqrt{3}}{2},$$

and divide every local weight by  $\sqrt{3}/2$ . Let  $A_N$  be the resulting unit-weight partition function. When every  $x_i = x$  and every  $y_j = y$ , put  $z = x - y$ ,  $a(z) = \sin(z + \eta)$ ,  $b(z) = \sin z$ , and write the unnormalized partition function as  $Z_N^{\text{hom}}(z; \eta)$ . Take the confluent limit of the Izergin determinant and prove

$$Z_N^{\text{hom}}(z; \eta) = \frac{[a(z)b(z)]^{N^2}}{\prod_{j=0}^{N-1} (j!)^2} \det \left[ \frac{d^{i+j}}{dz^{i+j}} \frac{c}{a(z)b(z)} \right]_{i,j=0}^{N-1}.$$

Use as theorem input [29] the homogeneous determinant evaluation obtained from its Desnanot–Jacobi recurrence at  $z = \eta = \pi/3$ :

$$\frac{A_N}{A_{N-1}} = \frac{(3N-2)!(N-1)!}{(2N-1)!(2N-2)!}, \quad N \geq 2, \text{quad} A_1 = 1,$$

and deduce

$$A_N = \prod_{j=0}^{N-1} \frac{(3j+1)!}{(N+j)!}.$$

Compute  $A_N$  for  $1 \leq N \leq 5$  and use Stirling's formula to derive

$$\lim_{N \rightarrow \infty} \frac{1}{N^2} \log A_N = \log \frac{3\sqrt{3}}{4}.$$

Deduce the exact finite-size state count and the residual configurational entropy per vertex

$$s_{\text{DW}} = k_B \log \frac{3\sqrt{3}}{4}$$

for a unit-weight domain-wall ice sample. State the domain-wall preparation, equal-weight calibration, equilibration, and finite-size assumptions under which configuration counts or free-energy measurements test the determinant and product identities.

**Exercise 14.12** (Elliptic gamma functions and the deformation hierarchy). For  $|p|, |q| < 1$  and  $z \in \mathbb{C}^\times$ , define

$$(z; p)_\infty = \prod_{j=0}^{\infty} (1 - zp^j), \quad \theta(z; p) = (z; p)_\infty (p/z; p)_\infty,$$

and use the abbreviations

$$\theta(z_1, \dots, z_s; p) = \prod_{j=1}^s \theta(z_j; p),$$

$$(a; q, p)_m = \prod_{j=0}^{m-1} \theta(aq^j; p),$$

$$(a_1, \dots, a_s; q, p)_m = \prod_{i=1}^s (a_i; q, p)_m,$$

and

$$\Gamma(z; p, q) = \prod_{j,k \geq 0} \frac{1 - p^{j+1} q^{k+1} / z}{1 - p^j q^k z}.$$

Prove

$$\theta(pz; p) = -z^{-1} \theta(z; p), \quad \theta(z; 0) = 1 - z,$$

and that the zeros of  $\theta(\cdot; p)$  are the points  $p^n$ ,  $n \in \mathbb{Z}$ , with multiplicity one when  $p \neq 0$ . Prove

$$\Gamma(qz; p, q) = \theta(z; p) \Gamma(z; p, q),$$

$$\Gamma(pz; p, q) = \theta(z; q) \Gamma(z; p, q), \quad \Gamma(z; p, q) \Gamma(pq/z; p, q) = 1.$$

Prove

$$(a; q, 0)_m = (a; q)_m, \quad \Gamma(z; 0, q) = \frac{1}{(z; q)_\infty}.$$

**Exercise 14.13** (The Frenkel–Turaev sum). For  $a^2 q^{n+1} = bcde$ , define

$$\begin{aligned} {}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) &= \sum_{m=0}^n \frac{\theta(aq^{2m}; p)}{\theta(a; p)} \\ &\quad \times \frac{(a, b, c, d, e, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq^{n+1}; q, p)_m} q^m. \end{aligned}$$

(a) Prove the multiplicative theta addition formula

$$\begin{aligned} \theta(xz, x/z, yw, y/w; p) - \theta(xw, x/w, yz, y/z; p) \\ = \frac{y}{z} \theta(xy, x/y, zw, z/w; p) \end{aligned}$$

by comparing quasi-periodicity and zeros.

(b) Use it to telescope the difference between the  $n$ th and  $(n-1)$ st terminating sums and prove

$${}_{10}V_9(a; b, c, d, e, q^{-n}; q, p) = \frac{(aq, aq/(bc), aq/(bd), aq/(cd); q, p)_n}{(aq/b, aq/c, aq/d, aq/(bcd); q, p)_n}.$$

(c) Take  $p \rightarrow 0$  and identify the terminating very-well-poised basic hypergeometric summation.

**Exercise 14.14** (Elliptic face weights and vertex–face intertwiners). Fix logarithms of  $p, q$  with  $|p|, |q| < 1$  and put

$$[x]_{p,q} = q^{(1-x)/2} \frac{\theta(q^x; p)}{\theta(q; p)}.$$

For integer heights  $i, j, k, l$  satisfying

$$|i - j| = |j - k| = |k - l| = |l - i| = 1,$$

define  $W(i, j, k, l \mid x)$  by

$$\begin{aligned} W(l \pm 1, l \pm 2, l \pm 1, l \mid x) &= [x + 1]_{p,q}, \\ W(l \mp 1, l, l \pm 1, l \mid x) &= \frac{[l + 1 \pm 1]_{p,q} [x]_{p,q}}{[l + 1]_{p,q}}, \\ W(l \pm 1, l, l \pm 1, l \mid x) &= \frac{[l + 1 \mp x]_{p,q}}{[l + 1]_{p,q}}, \end{aligned}$$

whenever the denominators are nonzero, and set all other weights equal to zero. On  $V = \mathbb{C}e_+ \oplus \mathbb{C}e_-$  let  $he_{\pm} = \pm e_{\pm}$  and define  $R(x, \lambda) \in \text{End}(V \otimes V)$  at  $\lambda = l + 1$  by

$$\langle e_{j-k} \otimes e_{k-l}, R(x, l+1)(e_{i-l} \otimes e_{j-i}) \rangle = W(i, j, k, l \mid x),$$

using the orthonormal distinguished weight lines and extending meromorphically in  $\lambda$ .

Let  $R^{\text{vert}}(u) \in \text{End}(V \otimes V)$  be a spectral exchange operator. For each admissible oriented height edge  $|a - b| = 1$ , let  $\psi(a, b \mid u) \in V$  be meromorphic in  $u$ . The pair  $(R^{\text{vert}}, \psi)$  satisfies the vertex–face relation for  $W$  when

$$\begin{aligned} R^{\text{vert}}(u - v)(\psi(a, b \mid u) \otimes \psi(b, c \mid v)) \\ = \sum_{\substack{b' \\ |a-b'|=|b'-c|=1}} W(a, b, c, b' \mid u - v) \psi(a, b' \mid v) \otimes \psi(b', c \mid u). \end{aligned}$$

The intertwiners are nondegenerate when the maps from admissible height-path spaces induced by their tensor products are injective for generic spectral parameters.

(a) Prove  $[-x]_{p,q} = -[x]_{p,q}$  and rewrite the theta addition identity as the corresponding four-term identity for the brackets  $[x]_{p,q}$ .

(b) Verify that  $R(x, \lambda)$  has weight zero and prove the dynamical Yang–Baxter equation

$$\begin{aligned} R_{12}(x, \lambda + h^{(3)})R_{13}(x + y, \lambda)R_{23}(y, \lambda + h^{(1)}) \\ = R_{23}(y, \lambda)R_{13}(x + y, \lambda + h^{(2)})R_{12}(x, \lambda) \end{aligned}$$

by reducing every nontrivial weight component to that bracket identity. Translate the two sides into the two sums over the admissible internal height of a three-face patch. Show directly that the shifts by  $h^{(1)}, h^{(2)}, h^{(3)}$  are the changes of the adjacent face height and that  $\lambda$ , unlike  $x$  and  $y$ , is therefore the dynamical parameter.

(c) For periodic admissible height paths, form the row-to-row transfer maps by multiplying the displayed face weights and summing over the intermediate path. Use the face Yang–Baxter relation to prove commutation of transfer maps with distinct spectral parameters. For fixed boundary heights of a three-face patch with two admissible internal heights, compute their normalized probabilities from the two products of  $W$  and prove that a star–triangle move preserves the boundary partition function. State the real parameter and gauge conditions under which the normalized weights are positive statistical-mechanical probabilities.

**Exercise 14.15** (Vertex–face transfer correspondence). Assume nondegenerate vertex–face intertwiners. Apply the ordinary spectral Yang–Baxter equation for  $R^{\text{vert}}$  to

$$\psi(a, b \mid u) \otimes \psi(b, c \mid v) \otimes \psi(c, d \mid w)$$

and use the vertex–face relation in the two orders. Prove that equality of the resulting coefficients is the face Yang–Baxter relation and hence the dynamical Yang–Baxter equation. Conversely, use nondegeneracy to prove the ordinary Yang–Baxter equation on the image of the height–path map from the face relation.

For inhomogeneities  $z_1, \dots, z_N$ , tensor the edge intertwiners around a periodic row to obtain the map  $\Phi_u$  from periodic height paths to the corresponding vertex sector. Prove the row-transfer intertwining identity

$$\mathsf{T}_N^{\text{vert}}(u)\Phi_u = \Phi_u\mathsf{T}_N^{\text{face}}(u)$$

with the boundary twist dictated by the net height change. Deduce equality of the matched transfer spectra and periodic-strip partition functions. In the trigonometric degeneration  $p \rightarrow 0$ , identify the vertex transfer operator with the six-vertex family and the face transfer operator with the trigonometric solid-on-solid model. State the boundary-sector, normalization, injectivity, and face-gauge conditions required when their finite-size spectra are compared.

**Exercise 14.16** (Terminating elliptic  $6j$ -symbols). For  $bcdefg = a^3q^{n+2}$ , define

$$\begin{aligned} {}_{12}V_{11}(a; b, c, d, e, f, g, q^{-n}; q, p) \\ = \sum_{m=0}^n \frac{\theta(aq^{2m}; p)}{\theta(a; p)} \frac{(a, b, c, d, e, f, g, q^{-n}; q, p)_m}{(q, aq/b, aq/c, aq/d, aq/e, aq/f, aq/g, aq^{n+1}; q, p)_m} q^m. \end{aligned}$$

Let

$$\lambda = \frac{a^2q}{bcd}, \quad g = \frac{\lambda aq^{n+1}}{ef},$$

and assume no displayed denominator vanishes. Expand the series on the right, interchange the

resulting finite sums, and apply the Frenkel–Turaev  ${}_{10}V_9$  summation to prove

$$\begin{aligned} & {}_{12}V_{11}(a; b, c, d, e, f, g, q^{-n}; q, p) \\ &= \frac{(aq, aq/(ef), \lambda q/e, \lambda q/f; q, p)_n}{(aq/e, aq/f, \lambda q/(ef), \lambda q; q, p)_n} \\ & \quad \times {}_{12}V_{11}\left(\lambda; \frac{\lambda b}{a}, \frac{\lambda c}{a}, \frac{\lambda d}{a}, e, f, g, q^{-n}; q, p\right). \end{aligned}$$

Fuse  $n$  consecutive elementary face weights by summing over admissible internal height paths. Factor the endpoint theta products and show that the remaining path sum has the displayed  ${}_{12}V_{11}$  term ratio. Identify the two sides of the transformation with the two fusion orders and hence with the two recouplings of the four external height paths. Deduce the tetrahedral change of channel for the resulting elliptic  $6j$  coefficients and recover the face Yang–Baxter relation for fused weights.

Choose a numerator parameter so that the transformed series reduces to its zeroth term and recover the  ${}_{10}V_9$  summation. Take  $p \rightarrow 0$  to obtain Bailey’s terminating very-well-poised  ${}_{10}\phi_9$  transformation, and then take the spectral degeneration to the  $Q$ -Racah quantum  $6j$  coefficients. For a finite fused face strip, express the relative Boltzmann probabilities of two resolved intermediate heights as ratios of normalized fused face weights. Separately, in a unitary recoupling specialization, express channel probabilities as squared normalized elliptic  $6j$  coefficients. State the positivity, face-gauge, boundary-height, unitarity, and spectral-resolution assumptions required for the two interpretations.

**Exercise 14.17** (Elliptic beta kernels). Write

$$\Gamma(tz^{\pm 1}; p, q) = \Gamma(tz; p, q)\Gamma(t/z; p, q), \quad \kappa_{p,q} = \frac{(p; p)_{\infty}(q; q)_{\infty}}{2}.$$

For  $|t_j| < 1$  with  $\prod_{j=1}^6 t_j = pq$ , define

$$\mathcal{B}(t_1, \dots, t_6) = \kappa_{p,q} \int_{\mathbb{T}} \frac{\prod_{j=1}^6 \Gamma(t_j z^{\pm 1}; p, q)}{\Gamma(z^{\pm 2}; p, q)} \frac{dz}{2\pi i z}.$$

Use the elliptic-gamma difference equations to derive the parameter-shift recurrences of  $\mathcal{B}$ .

**Exercise 14.18** (Elliptic beta integral). Evaluate the residue degeneration in which two parameters multiply to  $pq$  and normalize the product side by the same degeneration. Use as theorem input the elliptic-shift uniqueness lemma [39]: a meromorphic function of the balanced parameters that satisfies the two independent  $p$ - and  $q$ -shift recurrences and has the specified residue normalization is uniquely determined. Apply the lemma and analytic continuation to prove

$$\mathcal{B}(t_1, \dots, t_6) = \prod_{1 \leq i < j \leq 6} \Gamma(t_i t_j; p, q).$$

Take  $p \rightarrow 0$  and identify the resulting basic hypergeometric beta integral.

**Exercise 14.19** (Elliptic integral operators and the star–triangle relation). For  $|t| < 1$  define

$$(\mathcal{M}_t f)(w) = \frac{\kappa_{p,q}}{\Gamma(t^2; p, q)} \int_{\mathbb{T}} \frac{\Gamma(tw^{\pm 1}z^{\pm 1}; p, q)}{\Gamma(z^{\pm 2}; p, q)} f(z) \frac{dz}{2\pi i z},$$

and

$$\mathcal{D}_s(y, w) = \Gamma\left((pq)^{1/2} s^{-1} y^{\pm 1} w^{\pm 1}; p, q\right).$$

Apply the elliptic beta integral to the internal integration variable and prove the operator identity

$$\mathcal{M}_s \mathcal{D}_{st}(y, \cdot) \mathcal{M}_t = \mathcal{D}_t(y, \cdot) \mathcal{M}_{st} \mathcal{D}_s(y, \cdot)$$

whenever the contours separate reciprocal pole sequences and the parameters satisfy the corresponding modulus bounds. Interpret the kernels as edge weights and  $\mathcal{D}$  as a site weight of a continuous spin model. Prove that the two local triangular partition functions are equal and that the row transfer operators obtained by the two factorizations commute. For a finite periodic strip, deduce equality of the transfer spectra and partition functions under a star-triangle move; state the positivity regime and contour assumptions under which these are statistical-mechanical observables.

## 15 Double Affine Hecke Algebras

**Exercise 15.1** (Symmetric polynomials). For  $N \geq 1$ , let

$$\Lambda_N = \mathbb{C}[x_1, \dots, x_N]^{S_N}.$$

Define

$$e_r = \sum_{1 \leq i_1 < \dots < i_r \leq N} x_{i_1} \cdots x_{i_r}, \quad h_r = \sum_{1 \leq i_1 \leq \dots \leq i_r \leq N} x_{i_1} \cdots x_{i_r},$$

$$p_r = \sum_{i=1}^N x_i^r, \quad e_0 = h_0 = 1.$$

Set  $e_r = 0$  for  $r > N$  and  $h_r = 0$  for  $r < 0$ .

- (a) Prove that every element of  $\Lambda_N$  is uniquely a polynomial in  $e_1, \dots, e_N$ .
- (b) Prove

$$\sum_{r=0}^N e_r t^r = \prod_{i=1}^N (1 + x_i t), \quad \sum_{r \geq 0} h_r t^r = \prod_{i=1}^N \frac{1}{1 - x_i t},$$

and deduce

$$\left( \sum_{r \geq 0} h_r t^r \right) \left( \sum_{r=0}^N (-1)^r e_r t^r \right) = 1, \quad r h_r = \sum_{j=1}^r p_j h_{r-j}.$$

**Exercise 15.2** (The Sutherland model and Jack polynomials). For  $N$  identical particles of mass  $m$  on a ring of radius  $R$  and  $g \geq 1$ , let

$$\text{Conf}_N(S^1) = \{(\theta_1, \dots, \theta_N) : \theta_i \neq \theta_j \text{ for } i \neq j\}.$$

On  $C_c^\infty(\text{Conf}_N(S^1))^{S_N}$  define the nonnegative quadratic form

$$q_g[\psi] = \frac{\hbar^2}{2mR^2} \int_{(S^1)^N} \left[ \sum_{j=1}^N |\partial_{\theta_j} \psi|^2 + \frac{g(g-1)}{2} \sum_{i < j} \frac{|\psi|^2}{\sin^2((\theta_i - \theta_j)/2)} \right] d\theta.$$

Use as theorem input [36] that  $q_g$  is closable for  $g \geq 1$ . Define  $\hat{H}_g$  to be the unique self-adjoint operator associated with its closure in  $L^2((S^1)^N)^{S_N}$ ; this is the Friedrichs realization of the displayed differential expression. Define

$$\Psi_0(\theta) = \prod_{i < j} |e^{i\theta_i} - e^{i\theta_j}|^g.$$

For partitions of equal size, write  $\mu < \lambda$  in dominance order when

$$\sum_{i=1}^k \mu_i \leq \sum_{i=1}^k \lambda_i$$

for every  $k \geq 1$ , with missing parts interpreted as zero, and with at least one inequality strict. For a partition  $\lambda$  of length at most  $N$ , let  $m_\lambda$  be the sum of the distinct permutations of  $z_1^{\lambda_1} \cdots z_N^{\lambda_N}$ .

The Jack polynomial  $P_\lambda^{(1/g)}$  is the symmetric polynomial normalized by

$$P_\lambda^{(1/g)} = m_\lambda + \sum_{\mu < \lambda} c_{\lambda\mu}(g) m_\mu$$

and by satisfying

$$\left[ \sum_i (z_i \partial_{z_i})^2 + g \sum_{i < j} \frac{z_i + z_j}{z_i - z_j} (z_i \partial_{z_i} - z_j \partial_{z_j}) \right] P_\lambda^{(1/g)} = \left[ \sum_i (\lambda_i^2 + g(N + 1 - 2i) \lambda_i) \right] P_\lambda^{(1/g)};$$

here  $<$  is dominance order.

Conjugate the quantum Hamiltonian by  $\Psi_0$ , identify its commuting quantum integrals, and prove that

$$\Psi_\lambda = \Psi_0 P_\lambda^{(1/g)}(e^{i\theta_1}, \dots, e^{i\theta_N})$$

is a joint eigenfunction with

$$E_\lambda = \frac{\hbar^2}{2mR^2} \left[ \frac{g^2 N(N^2 - 1)}{12} + \sum_{j=1}^N (\lambda_j^2 + g(N + 1 - 2j) \lambda_j) \right].$$

**Exercise 15.3** (Jack transition spectrum). Let  $\rho_r = \sum_{j=1}^N e^{ir\theta_j}$  be a density mode. Derive the Jack Pieri identity in orthonormal normalization for  $\rho_1 P_\lambda^{(1/g)}$  and show that only nonincreasing tuples obtained from  $\lambda$  by increasing one entry by one occur. For a weak periodic drive proportional to  $\rho_1 e^{-i\Omega t} + \rho_{-1} e^{i\Omega t}$ , deduce the allowed resonance frequencies

$$\Omega_{\mu\lambda} = \frac{|E_\mu - E_\lambda|}{\hbar}$$

and express their relative intensities as squared normalized Jack Pieri coefficients. State how spectroscopy of particles confined to a ring determines  $g$  from the line shifts and tests the Jack selection rule from the missing lines.

**Exercise 15.4** (Double-affine braid groups). For  $r \geq 2$ , the *double-affine braid group*  $\tilde{B}_r$  of type  $GL_r$  is generated by  $T_i$  for  $1 \leq i < r$ , commuting invertible elements  $X_j$ , commuting invertible elements  $Y_j$ , and a central invertible element  $\mathbf{q}$ , subject to the braid relations and

$$\begin{aligned} T_i X_i T_i &= X_{i+1}, & T_i X_j &= X_j T_i \quad (j \neq i, i+1), \\ T_i^{-1} Y_i T_i^{-1} &= Y_{i+1}, & T_i Y_j &= Y_j T_i \quad (j \neq i, i+1), \\ Y_1 X_1 \cdots X_r &= \mathbf{q} X_1 \cdots X_r Y_1, & X_1^{-1} Y_2 &= Y_2 X_1^{-1} T_1^{-2}. \end{aligned}$$

Let  $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ . Interpret  $T_i$  as an adjacent braid and  $X_j, Y_j$  as motions of the  $j$ th point around the two cycles of  $T^2$ . Derive the displayed relations from these motions and identify  $\tilde{B}_r$  with the central extension of

$$\pi_1(\text{Conf}_r(T^2)/S_r)$$

whose total cycle commutator is  $\mathbf{q}$ . Prove that deleting either the  $X$ -family or the  $Y$ -family leaves an affine braid group.

**Exercise 15.5** (Rank-one DAHA and the Askey–Wilson operator). For nonzero parameters  $q, t$ , the *double affine Hecke algebra*  $H_{q,t}(GL_r)$  is the quotient of  $\mathbb{C}(q, t)[\tilde{B}_r]$  by

$$\mathfrak{q} = q, \quad (T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

For nonzero  $k_0, k_1, u_0, u_1$  and a choice of  $q^{1/2}$ , the rank-one DAHA  $H_q(C_1^\vee, C_1)$  is generated by  $T_0^{\pm 1}, T_1^{\pm 1}, (T_0^\vee)^{\pm 1}, (T_1^\vee)^{\pm 1}$  subject to

$$(T_i - k_i)(T_i + k_i^{-1}) = 0, \quad (T_i^\vee - u_i)(T_i^\vee + u_i^{-1}) = 0 \quad (i = 0, 1),$$

$$T_1^\vee T_1 T_0 T_0^\vee = q^{-1/2}.$$

For the finite Hecke generator  $T_1$ , define

$$e = \frac{T_1 + k_1^{-1}}{k_1 + k_1^{-1}}.$$

Fix nonzero  $k_0, k_1, u_0, u_1$  and choose  $q^{1/2}$ . On  $\mathcal{L} = \mathbb{C}[z^{\pm 1}]$  put

$$(s_1 f)(z) = f(z^{-1}), \quad (s_0 f)(z) = f(qz^{-1}),$$

$$(T_1 f)(z) = k_1 f(z^{-1}) + \frac{(k_1 - k_1^{-1}) + (u_1 - u_1^{-1})z}{1 - z^2} (f(z) - f(z^{-1})),$$

$$(T_0 f)(z) = k_0 f(qz^{-1}) + \frac{(k_0 - k_0^{-1}) + (u_0 - u_0^{-1})q^{1/2}z^{-1}}{1 - qz^{-2}} (f(z) - f(qz^{-1})).$$

Let  $X$  act by multiplication by  $z$ . The Askey–Wilson parameters in this convention are

$$a = k_1 u_1, \quad b = -k_1 u_1^{-1}, \quad c = q^{1/2} k_0 u_0, \quad d = -q^{1/2} k_0 u_0^{-1}.$$

Put

$$T_1^\vee = X^{-1} T_1^{-1}, \quad T_0^\vee = q^{-1/2} T_0^{-1} X.$$

For  $0 < q < 1$  and nonzero  $a, b, c, d$ , define

$$A(z) = \frac{(1 - az)(1 - bz)(1 - cz)(1 - dz)}{(1 - z^2)(1 - qz^2)}$$

and, on  $f(z) = f(z^{-1})$ ,

$$(\mathcal{D}_{AW} f)(z) = A(z)(f(qz) - f(z)) + A(z^{-1})(f(q^{-1}z) - f(z)).$$

- (a) Verify that the displayed reflection–difference operators preserve  $\mathcal{L}$ , that  $T_0, T_1, T_0^\vee, T_1^\vee$  satisfy the four quadratic Hecke relations and the product relation, and that they give the rank-one DAHA polynomial representation with the stated parameter map.
- (b) Prove

$$e\mathcal{L} = \mathcal{L}^{z \mapsto z^{-1}} = \mathbb{C}[z + z^{-1}],$$

and prove that the spherical action contains multiplication by  $z + z^{-1}$  and  $\mathcal{D}_{AW}$ .



**Exercise 15.6** (Askey–Wilson spectral theorem). For  $x = (z + z^{-1})/2$ , define the Askey–Wilson polynomial by

$$p_n(x; a, b, c, d \mid q) = a^{-n} (ab, ac, ad; q)_n \times {}_4\phi_3 \left( \begin{matrix} q^{-n}, abcdq^{n-1}, az, az^{-1} \\ ab, ac, ad \end{matrix}; q, q \right).$$

Its weight on the unit circle is

$$w_{AW}(z) = \frac{(z^2, z^{-2}; q)_\infty}{(az, a/z, bz, b/z, cz, c/z, dz, d/z; q)_\infty}.$$

Prove

$$\mathcal{D}_{AW} p_n = (q^{-n} - 1)(1 - abcdq^{n-1})p_n.$$

For  $a, b, c, d \in (-1, 1)$ , prove that  $\mathcal{D}_{AW}$  is symmetric for

$$\langle f, g \rangle_{AW} = \int_{|z|=1} f(z) \overline{g(z)} w_{AW}(z) \frac{dz}{2\pi i z},$$

and deduce the orthogonality and three-term recurrence of the Askey–Wilson polynomials.

**Exercise 15.7** (Wilson polynomials and degeneration). For positive  $a, b, c, d$ , define

$$W_n(x^2; a, b, c, d) = (a+b)_n (a+c)_n (a+d)_n \times {}_4F_3 \left( \begin{matrix} -n, n+a+b+c+d-1, a+ix, a-ix \\ a+b, a+c, a+d \end{matrix}; 1 \right).$$

Set  $q = e^{-\varepsilon}$  and scale the Askey–Wilson parameters and spectral variable by

$$(a, b, c, d, z) = (q^A, q^B, q^C, q^D, q^{ix}).$$

After the scalar normalization by  $(1-q)^{-3n}$ , prove that the Askey–Wilson polynomial tends to  $W_n(x^2; A, B, C, D)$  as  $\varepsilon \downarrow 0$ . Derive the Wilson difference equation, weight, orthogonality, and three-term recurrence as limits of the corresponding Askey–Wilson identities.

**Exercise 15.8** (Type- $A$  polynomial representations). Let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_r^{\pm 1}],$$

let  $X_j$  act by multiplication by  $x_j$ , and let  $s_i$  exchange  $x_i$  and  $x_{i+1}$ . Define

$$T_i = t^{1/2} s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i/x_{i+1} - 1}, \quad 1 \leq i < r,$$

and

$$(\pi f)(x_1, \dots, x_r) = f(x_2, \dots, x_r, qx_1).$$

For generic independent parameters  $q, t$ , call the operator algebra generated by  $X_j^{\pm 1}$ ,  $T_i$ , and  $\pi^{\pm 1}$  the type- $A$  polynomial operator model.

Prove that the displayed operators preserve  $\mathcal{P}_r$  and satisfy the braid and Hecke relations. Prove

$$\pi X_i \pi^{-1} = X_{i+1} \quad (1 \leq i < r), \quad \pi X_r \pi^{-1} = q X_1,$$

and identify the two affine-Hecke subalgebras generated respectively by  $T_i, X_j$  and by  $T_i, Y_j$ . Prove that this representation is obtained from the double-affine braid-group algebra by imposing

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$$

and specializing its central parameter to  $q$ . Conclude that the polynomial operator model is a realization of  $\mathbf{H}_{q,t}(GL_r)$ .

**Exercise 15.9** (Macdonald operators and polynomials). Let

$$(T_{q,x_i} f)(x_1, \dots, x_r) = f(x_1, \dots, qx_i, \dots, x_r).$$

The first type- $A$  Macdonald operator is

$$D_{q,t} = \sum_{i=1}^r \left( \prod_{j \neq i} \frac{tx_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

Define

$$Y_i = T_i \cdots T_{r-1} \pi T_1^{-1} \cdots T_{i-1}^{-1}.$$

Use the displayed polynomial representation and the affine braid relations to prove that the  $Y_i$  commute. Prove that their symmetric Laurent polynomials preserve  $\mathcal{P}_r^{S_r}$  and identify  $t^{-(r-1)/2}(Y_1 + \cdots + Y_r)$  with  $D_{q,t}$ .

**Exercise 15.10** (Macdonald spectral theory). For generic  $q, t$ , the Macdonald polynomial  $P_\lambda(x; q, t)$  is the symmetric polynomial triangular in the monomial symmetric functions and satisfying

$$D_{q,t} P_\lambda = \left( \sum_{i=1}^r q^{\lambda_i} t^{r-i} \right) P_\lambda, \quad \lambda_1 \geq \cdots \geq \lambda_r \geq 0.$$

For  $0 < q, t < 1$ , define

$$\Delta_{q,t}(x) = \prod_{i \neq j} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty},$$

and

$$\langle f, g \rangle_{q,t} = \frac{1}{r!} \int_{(S^1)^r} f(x) \overline{g(x)} \Delta_{q,t}(x) \prod_{j=1}^r \frac{dx_j}{2\pi i x_j}.$$

Order monomial symmetric functions by dominance and prove that  $D_{q,t}$  is triangular with the displayed diagonal entries. For generic  $q, t$ , deduce existence and uniqueness of  $P_\lambda$ . Prove directly from the torus integral that  $D_{q,t}$  is symmetric when  $0 < q, t < 1$ , and deduce orthogonality and joint diagonalization for symmetric Laurent polynomials in the commuting  $Y_i$ .

**Exercise 15.11** (DAHA Fourier duality). Define the DAHA Fourier assignment on the torus, winding, and Hecke generators by

$$\mathcal{F}(X_j) = Y_j^{-1}, \quad \mathcal{F}(Y_j) = X_j, \quad \mathcal{F}(T_i) = T_i.$$

For  $\rho = (r-1, r-2, \dots, 0)$  and a partition  $\lambda$ , put

$$\xi_\lambda = (q^{\lambda_1} t^{r-1}, q^{\lambda_2} t^{r-2}, \dots, q^{\lambda_r}).$$

Verify the DAHA relations after applying  $\mathcal{F}$  and prove that it extends to an automorphism. Use as theorem input the polynomial intertwiner theorem [10]: for generic  $q, t$ , this automorphism is implemented between the polynomial representation and its spectral realization by a nondegenerate transform whose kernel is normalized to equal 1 at  $(t^\rho, t^\rho)$ . Apply the intertwining relations to the commuting  $X$ - and  $Y$ -operators and prove the evaluation symmetry

$$\frac{P_\lambda(\xi_\mu; q, t)}{P_\lambda(t^\rho; q, t)} = \frac{P_\mu(\xi_\lambda; q, t)}{P_\mu(t^\rho; q, t)}.$$

Derive the duality between multiplication recurrences in  $\lambda$  and the commuting Macdonald difference equations in the spectral variable.

**Exercise 15.12** (Macdonald pairings and the Cauchy identity). Let  $\Lambda_{\text{sym}} = \mathbb{C}[p_1, p_2, \dots]$  be graded by  $\deg p_r = r$ . For a partition  $\mu$ , let  $m_r(\mu)$  be the number of its parts equal to  $r$ , and define

$$z_\mu = \prod_{r \geq 1} r^{m_r(\mu)} m_r(\mu)!, \quad p_\mu = \prod_j p_{\mu_j}, \quad p_r(x) = \sum_i x_i^r.$$

Define the symmetric-function Macdonald pairing by

$$\langle p_\mu, p_\nu \rangle_{q,t}^{\text{sym}} = \delta_{\mu\nu} z_\mu \prod_j \frac{1 - q^{\mu_j}}{1 - t^{\mu_j}}.$$

For a cell  $s = (i, j)$  in the diagram of  $\lambda$ , define its arm and leg by

$$a_\lambda(s) = \lambda_i - j, \quad \ell_\lambda(s) = \lambda'_j - i,$$

and put

$$b_\lambda(q, t) = \prod_{s \in \lambda} \frac{1 - q^{a_\lambda(s)} t^{\ell_\lambda(s)+1}}{1 - q^{a_\lambda(s)+1} t^{\ell_\lambda(s)}}.$$

Prove that the finite-variable Macdonald polynomials are stable under adjoining a zero variable and hence determine elements  $P_\lambda$  of  $\Lambda_{\text{sym}}$ . Put  $Q_\lambda = b_\lambda(q, t) P_\lambda$ . Use the adjointness of the DAHA difference operators and triangularity to prove

$$\langle P_\lambda, P_\mu \rangle_{q,t}^{\text{sym}} = \delta_{\lambda\mu} b_\lambda(q, t)^{-1}.$$

Deduce that  $P_\lambda$  and  $Q_\lambda$  are dual. Expand the reproducing kernel in the power sums and use the  $q$ -binomial identity to prove

$$\sum_\lambda P_\lambda(a; q, t) Q_\lambda(b; q, t) = \prod_{i,j} \frac{(ta_i b_j; q)_\infty}{(a_i b_j; q)_\infty}$$

for finite alphabets with  $|a_i b_j| < 1$ .

**Exercise 15.13** (Finite Macdonald measures). Let  $0 \leq q, t < 1$ , let  $a = (a_1, \dots, a_N)$  and  $b = (b_1, \dots, b_M)$  be nonnegative with  $a_i b_j < 1$ , and let  $P_\lambda, Q_\lambda$  be the Macdonald functions. Define

$$\Pi(a; b) = \sum_\lambda P_\lambda(a; q, t) Q_\lambda(b; q, t), \quad \mathbb{P}_{q,t}(\lambda) = \frac{P_\lambda(a; q, t) Q_\lambda(b; q, t)}{\Pi(a; b)}.$$

Apply the first Macdonald difference operator in the  $a$  variables to the Cauchy identity. For distinct  $a_i$ , prove

$$\mathbb{E}_{q,t} \left[ \sum_{i=1}^N q^{\lambda_i} t^{N-i} \right] = \sum_{i=1}^N \prod_{j \neq i} \frac{ta_i - a_j}{a_i - a_j} \prod_{k=1}^M \frac{1 - a_i b_k}{1 - ta_i b_k},$$

and extend the identity to coincident parameters by continuity. Verify the specialization  $q = t$  directly and the  $q$ -Whittaker specialization  $t = 0$ . For samples from the finite measure, identify the displayed expectation as a directly estimable statistic and state the positivity, truncation, and sampling assumptions required for comparison with the formula.

**Exercise 15.14** (The Calogero–Sutherland limit). Write  $x_i = e^{z_i}$ , set  $q = e^{-\epsilon}$  and  $t = e^{-g\epsilon}$ , and expand  $D_{q,t}$  through order  $\epsilon^2$ . After subtracting its scalar value on 1 and removing the center-of-mass drift, obtain

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i < j} \coth\left(\frac{z_i - z_j}{2}\right) (\partial_{z_i} - \partial_{z_j}).$$

On the circle, put  $z_i = i\theta_i$  and conjugate by

$$\Psi_0(\theta) = \prod_{i < j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g.$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = - \sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i < j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Compute the eigenfunctions of degrees zero and one and, for  $r = 2$ , degree two. Identify their Sutherland eigenvalues.

**Exercise 15.15** (Hypergeometric degeneration). If no denominator parameter  $b_j$  is a nonpositive integer, define

$${}_rF_s \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; z \right) = \sum_{m=0}^{\infty} \frac{(a_1)_m \cdots (a_r)_m}{(b_1)_m \cdots (b_s)_m} \frac{z^m}{m!}.$$

The series is finite when a numerator parameter is a nonpositive integer.

Set  $q = e^{-\epsilon}$ . Prove

$$\lim_{\epsilon \rightarrow 0^+} \frac{(q^a; q)_m}{(1 - q)^m} = (a)_m, \quad \lim_{q \rightarrow 1^-} (1 - q)^{1-a} \frac{(q; q)_{\infty}}{(q^a; q)_{\infty}} = \Gamma(a).$$

Deduce

$$\lim_{q \rightarrow 1^-} {}_r\phi_{r-1} \left( \begin{matrix} q^{a_1}, \dots, q^{a_r} \\ q^{b_1}, \dots, q^{b_{r-1}} \end{matrix}; q, z \right) = {}_rF_{r-1} \left( \begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_{r-1} \end{matrix}; z \right).$$

Derive Gauss's summation from the  $q$ -Gauss summation. Prove the corresponding degeneration of the basic hypergeometric  $q$ -difference equation to the Gauss equation.

**Exercise 15.16** ( $BC_r$  weights and the Koornwinder operator). Write  $f(z^{\pm m}) = f(z^m)f(z^{-m})$  and

$$f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon, \delta = \pm 1} f(z_i^{\epsilon} z_j^{\delta}).$$

For  $0 < q, t < 1$  and  $a_1, a_2, a_3, a_4 \in (-1, 1)$ , define

$$\Delta_{BC_r}(z) = \prod_{i=1}^r \frac{(z_i^{\pm 2}; q)_\infty}{\prod_{s=1}^4 (a_s z_i^{\pm 1}; q)_\infty} \times \prod_{1 \leq i < j \leq r} \frac{(z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}.$$

Use the normalized torus measure

$$\frac{1}{2^r r!} \prod_{i=1}^r \frac{dz_i}{2\pi i z_i}.$$

For  $z = (z_1, \dots, z_r)$  put

$$A_i(z) = \frac{\prod_{s=1}^4 (1 - a_s z_i)}{(1 - z_i^2)(1 - q z_i^2)} \prod_{j \neq i} \frac{(1 - t z_i z_j)(1 - t z_i / z_j)}{(1 - z_i z_j)(1 - z_i / z_j)}.$$

On Laurent polynomials invariant under permutations and inversions define

$$\mathcal{D}_K = \sum_{i=1}^r \left[ A_i(z)(T_{q, z_i} - 1) + A_i(z^{-1})(T_{q, z_i}^{-1} - 1) \right].$$

Prove that  $\Delta_{BC_r}$  is invariant under permutations and independent inversions of the  $z_i$ . Identify its factors with the roots  $\pm e_i$ ,  $\pm 2e_i$ , and  $\pm e_i \pm e_j$ . For  $r = 1$ , recover the Askey–Wilson weight and its order-two Weyl normalization, and identify  $\mathcal{D}_K$  with  $\mathcal{D}_{AW}$ .

**Exercise 15.17** (Koornwinder spectral theory). The Koornwinder polynomial  $K_\lambda$  is the Weyl-invariant Laurent polynomial triangular in orbit sums and satisfying

$$\mathcal{D}_K K_\lambda = \left( \sum_{i=1}^r \left[ a_1 a_2 a_3 a_4 q^{-1} t^{2r-i-1} (q^{\lambda_i} - 1) + t^{i-1} (q^{-\lambda_i} - 1) \right] \right) K_\lambda,$$

with the scalar eigenvalue fixed by the displayed convention.

Use as theorem input Koornwinder's symmetry theorem [28]:  $\mathcal{D}_K$  is symmetric for the  $\Delta_{BC_r}$  inner product. Deduce that its triangular eigenfunctions are mutually orthogonal when the displayed eigenvalues are distinct.

**Exercise 15.18** (Koornwinder spectral values and  $BC_r$  simulator frequencies). Assume  $r \geq 2$ ,  $M \geq 2$ ,  $0 < q, t < 1$ ,  $a_1, a_2, a_3, a_4 \in (-1, 1)$ , and the displayed eigenvalues are distinct on

$$\Lambda_M = \{\lambda : \ell(\lambda) \leq r, |\lambda| \leq M\}.$$

Let  $\mathcal{H}_M$  be the span of the normalized Koornwinder polynomials  $K_\lambda$ ,  $\lambda \in \Lambda_M$ , and define the finite programmable  $q$ -difference Hamiltonian

$$\hat{H}_M = \varepsilon \mathcal{D}_K|_{\mathcal{H}_M}, \quad \varepsilon > 0.$$

Prove that  $\mathcal{H}_M$  is invariant and that  $\hat{H}_M$  is self-adjoint with spectral values

$$E_\lambda = \varepsilon \sum_{i=1}^r \left[ a_1 a_2 a_3 a_4 q^{-1} t^{2r-i-1} (q^{\lambda_i} - 1) + t^{i-1} (q^{-\lambda_i} - 1) \right].$$

For prepared normalized eigenstates  $K_\lambda$  and  $K_\mu$ , derive the spectral spacing

$$\Omega_{\mu\lambda} = \frac{|E_\mu - E_\lambda|}{\hbar}$$

and compute all pairwise gaps among the four levels indexed by  $0, (1), (2), (1, 1)$ . State the positivity, finite-truncation, state-preparation, resolved-partition, energy-calibration, and nondegeneracy assumptions under which the observed lines test the Koornwinder eigenvalue formula.

**Exercise 15.19** (Askey–Wilson spectrum). Fix  $M \geq 4$  and parameters  $a, b, c, d \in (-1, 1)$  for which the Askey–Wilson weight is positive and the first  $M + 1$  eigenvalues are distinct. Let  $\mathcal{H}_M$  be the span of the normalized Askey–Wilson polynomials  $p_0, \dots, p_M$  with their positive Askey–Wilson inner product. Use the self-adjoint Hamiltonian  $\hat{H}_M = \varepsilon \mathcal{D}_{AW}|_{\mathcal{H}_M}$ , where  $\varepsilon > 0$  is a calibrated energy scale. Prove that its spectral values are

$$E_n = \varepsilon(q^{-n} - 1)(1 - abcdq^{n-1}).$$

Write the three-term recurrence in the form

$$(z + z^{-1})p_n = A_n p_{n+1} + B_n p_n + C_n p_{n-1}$$

and compute the corresponding normalized matrix elements. Deduce that a periodic perturbation proportional to the compression of  $z + z^{-1}$  to  $\mathcal{H}_M$  produces only neighboring transitions, with resonance frequencies  $|E_{n+1} - E_n|/\hbar$  and intensities determined by  $A_n$  and  $C_{n+1}$ . Assuming  $\varepsilon$  is known and at least three nondegenerate lines are resolved, derive the equations satisfied by  $q$  and the product  $abcd$ . Compute the Jacobian of two independent gap ratios and state the nonvanishing and parameter-range hypotheses under which these two quantities are locally identifiable.

Form the data map from  $(a, b, c, d)$  to four selected squared normalized neighboring matrix elements. Prove that it is invariant under permutation of the parameters. At a chosen parameter point, compute its differential in the four elementary symmetric functions and state the full-rank hypothesis under which the measured strengths locally determine the unordered set  $\{a, b, c, d\}$ . When this rank condition fails, identify the remaining fiber instead of asserting identifiability; state also that squared strengths cannot distinguish any additional discrete ambiguity in that fiber. State that these are predictions for the engineered simulator.

**Exercise 15.20** (Macdonald–Ruijsenaars spectrum). Let  $0 < q, t < 1$ , fix  $M \geq 0$ , and define

$$\mathcal{H}_{q,t}^{(M)} = \text{span}\{P_\lambda(x; q, t) : |\lambda| \leq M\}$$

with the positive Macdonald inner product. Put

$$d_0 = \sum_{i=1}^r t^{r-i}, \quad \hat{H}_{q,t}^{(M)} = \varepsilon(d_0 - D_{q,t})|_{\mathcal{H}_{q,t}^{(M)}}, \quad \varepsilon > 0.$$

Identify  $D_{q,t}$  with the first symmetric function of the commuting DAHA winding operators and derive its compatibility with the Hecke exchange relations. Prove that  $\mathcal{H}_{q,t}^{(M)}$  is invariant, that  $\hat{H}_{q,t}^{(M)}$  is self-adjoint, and that the normalized Macdonald states have energies

$$E_\lambda = \varepsilon \sum_{i=1}^r t^{r-i} (1 - q^{\lambda_i}).$$

**Exercise 15.21** (Macdonald transition spectrum). Model a finite programmable  $q$ -difference simulator by this operator. Use the Macdonald Pieri identity for  $\rho_1 = x_1 + \cdots + x_r$  and its adjoint to determine every nonzero matrix element of the compression of  $\rho_1 + \rho_1^*$  to  $\mathcal{H}_{q,t}^{(M)}$ . For a periodic drive proportional to this compressed observable, derive the resonance frequencies

$$\Omega_{\mu\lambda} = |E_\mu - E_\lambda|/\hbar$$

and the relative intensities as squared normalized Macdonald Pieri coefficients. Explain that recovery of  $q, t$  requires known  $\varepsilon$  and  $r$ , resolved partition labels, sufficiently many nondegenerate transitions, and a fixed convention for any discrete parameter ambiguity, and that the prediction concerns the engineered simulator.

**Exercise 15.22** (Macdonald–Jack degeneration). Set  $q = e^{-\delta}$  and  $t = e^{-g\delta}$ ; after the scalar and center-of-mass normalizations in the Calogero–Sutherland limit exercise, prove that

$$\lim_{\delta \downarrow 0} P_\lambda(x; e^{-\delta}, e^{-g\delta}) = P_\lambda^{(1/g)}(x)$$

in the monic normalization. Prove that the rescaled spectral gaps and normalized Macdonald Pieri coefficients converge to the Sutherland energies and Jack Pieri coefficients. State how measured line locations and strengths test this degeneration. Explain that recovery of the deformation parameter requires resolved partition labels and sufficiently many nondegenerate transitions.

**Exercise 15.23** (Elliptic Ruijsenaars spectral tori). Let  $0 < p < 1$  and choose  $\tau \in i\mathbb{R}_{>0}$  with  $p = e^{2\pi i\tau}$ .

- (a) Prove that multiplication by  $p$  acts freely and properly discontinuously on  $\mathbb{C}^\times$  and that the exponential map induces a biholomorphism

$$E_p := \mathbb{C}^\times / p^\mathbb{Z} \cong \mathbb{C} / (\mathbb{Z} + \tau\mathbb{Z}).$$

Call this compact complex torus the *multiplicative elliptic torus*. For fixed  $t \in \mathbb{C}^\times$ , compute

$$\frac{\theta(tpz; p)}{\theta(pz; p)} = t^{-1} \frac{\theta(tz; p)}{\theta(z; p)}$$

and interpret this automorphy as a meromorphic section on  $E_p$ .

- (b) Define the Jacobi theta constants by

$$\vartheta_2(\tau) = \sum_{n \in \mathbb{Z}} e^{\pi i \tau (n+1/2)^2}, \quad \vartheta_3(\tau) = \sum_{n \in \mathbb{Z}} e^{\pi i \tau n^2},$$

and put

$$m(\tau) = \frac{\vartheta_2(\tau)^4}{\vartheta_3(\tau)^4}.$$

Use as theorem input the analytic Legendre uniformization theorem [19, 16]: the torus  $E_p$  is biholomorphic to the smooth compactification of

$$C_m : \quad y^2 = x(x-1)(x-m(\tau)),$$

and its invariant differential corresponds to a nonzero scalar multiple of  $dx/y$ . Prove from the theta series that

$$m(\tau) = 16p^{1/2} + O(p) \quad (p \downarrow 0),$$

so the elliptic torus degenerates toward the double-root cubic  $y^2 = x^2(x - 1)$ . Write  $m = m(\tau)$ , put  $k = \sqrt{m}$  and  $k' = \sqrt{1 - k^2}$ , and define

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}.$$

Use the same uniformization theorem to prove that

$$m = k^2, \quad \tau = i \frac{K(k')}{K(k)}, \quad p = e^{-2\pi K(k')/K(k)}.$$

With the positive branch of  $y$  over  $0 < x < m$ , use  $x = m \sin^2 \phi$  to derive the real period identity

$$2 \int_0^m \frac{dx}{\sqrt{x(x-1)(x-m)}} = 4K(k).$$

(c) For type  $A_{r-1}$  and  $0 < q, t < 1$ , define

$$D_{p,q,t}^{\text{ell}} = \sum_{i=1}^r \left( \prod_{j \neq i} \frac{\theta(tx_i/x_j; p)}{\theta(x_i/x_j; p)} \right) T_{q,x_i}.$$

Determine the automorphy of each coefficient under multiplication of a ratio  $x_i/x_j$  by  $p$ . Prove directly from  $\theta(z; 0) = 1 - z$  that

$$D_{p,q,t}^{\text{ell}} \longrightarrow D_{q,t} \quad (p \rightarrow 0)$$

locally uniformly away from the polar divisors. After taking this trigonometric degeneration, set  $q = e^{-\varepsilon}$  and  $t = e^{-g\varepsilon}$ , derive the Heckman–Opdam differential operator from the second-order term as  $\varepsilon \rightarrow 0$ , and take its rational scaling limit to obtain the rational Calogero–Moser operator.

(d) Fix a finite multiplicative grid

$$\mathcal{G}_M = \{(q^{n_1} x_1^0, \dots, q^{n_r} x_r^0) : -M \leq n_i \leq M\}$$

whose ratios avoid every displayed polar divisor for  $0 \leq p \leq p_0$ . Let  $C_p^{(M)}$  be the compression of  $D_{p,q,t}^{\text{ell}}$  to functions on  $\mathcal{G}_M$ , discarding a shifted value when it leaves the grid, and define the self-adjoint programmable Hamiltonian

$$\hat{H}_p^{(M)} = \frac{\epsilon}{2} \left( C_p^{(M)} + (C_p^{(M)})^* \right), \quad \epsilon > 0.$$

Prove entrywise and operator-norm convergence

$$\hat{H}_p^{(M)} \longrightarrow \hat{H}_0^{(M)} \quad (p \downarrow 0),$$

and hence convergence of every ordered spectral value and transition frequency. At a parameter where a selected nondegenerate gap has nonzero derivative with respect to  $p$ , prove that its measured value locally determines  $p$  and therefore the Legendre modulus  $m(\tau)$ . State the finite-grid, boundary-compression, coupling-calibration, pole-avoidance, state-preparation, and spectral-resolution assumptions under which the measured hopping amplitudes and resonance gaps test the theta-function deformation and its nodal degeneration.



## 16 Algebraic Curves

**Exercise 16.1** (Pendulum periods on the Legendre curve). On  $T^*S^1$  consider

$$H(\theta, p) = \frac{p^2}{2M\ell^2} + Mg\ell(1 - \cos \theta).$$

For an oscillation of amplitude  $0 < \theta_0 < \pi$ , set  $k = \sin(\theta_0/2)$  and  $m = k^2$ . Consider the complex plane curve

$$C_m : y^2 = x(x-1)(x-m).$$

Use energy conservation and the real period of  $dx/y$  to prove

$$T(\theta_0) = 4\sqrt{\frac{\ell}{g}} K(k).$$

Apply the generalized binomial theorem with  $a = \frac{1}{2}$ . For

$$I_n = \int_0^{\pi/2} \sin^{2n} \phi \, d\phi,$$

use integration by parts to prove

$$I_0 = \frac{\pi}{2}, \quad I_n = \frac{2n-1}{2n} I_{n-1} \quad (n \geq 1),$$

and obtain

$$K(k) = \frac{\pi}{2} \sum_{n=0}^{\infty} \left( \frac{(1/2)_n}{n!} \right)^2 k^{2n}, \quad |k| < 1.$$

Deduce

$$T(\theta_0) = 2\pi\sqrt{\frac{\ell}{g}} \left( 1 + \frac{\theta_0^2}{16} + O(\theta_0^4) \right).$$

Deduce the experimentally testable increase of period with amplitude and the logarithmic divergence as  $\theta_0 \uparrow \pi$ .

**Exercise 16.2** (Affine schemes and the Legendre family). Replace  $m$  by an indeterminate and form the finitely generated  $\mathbb{C}[m]$ -algebra

$$A = \mathbb{C}[m, x, y]/(y^2 - x(x-1)(x-m)).$$

For a commutative ring  $R$ , an ideal  $\mathfrak{p} \subsetneq R$  is *prime* when  $ab \in \mathfrak{p}$  implies  $a \in \mathfrak{p}$  or  $b \in \mathfrak{p}$ . Define

$$\text{Spec } R = \{\mathfrak{p} : \mathfrak{p} \text{ is prime}\}, \quad V(I) = \{\mathfrak{p} : I \subseteq \mathfrak{p}\}, \quad D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}.$$

Prove that the sets  $V(I)$  are the closed sets of a topology and that the principal open sets  $D(f)$  are closed under finite intersections and form a basis for the topology.

For  $f \in R$ , define the localization  $R_f$  by the universal property that  $f$  becomes invertible and every homomorphism from  $R$  in which  $f$  is invertible factors uniquely through  $R_f$ . A *presheaf* of rings on a space  $X$  assigns a ring  $\mathcal{F}(U)$  to each open set and compatible restriction maps. It is a *sheaf* when compatible local sections on an open cover glue uniquely. Set

$$\mathcal{O}_{\text{Spec } R}(D(f)) = R_f$$

with the localization restriction maps. Prove the sheaf gluing condition on principal-open covers, extend this prescription uniquely to a sheaf on  $\operatorname{Spec} R$ , and prove

$$R \cong \Gamma(\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R}).$$

For a sheaf of rings  $\mathcal{F}$  and  $p \in X$ , define its stalk by

$$\mathcal{F}_p = \varinjlim_{p \in U} \mathcal{F}(U).$$

A commutative ring is *local* when it has a unique maximal ideal. A *locally ringed space* is a space  $X$  with a sheaf of commutative rings  $\mathcal{O}_X$  whose stalks are local rings. Prove

$$\mathcal{O}_{\operatorname{Spec} R, \mathfrak{p}} \cong R_{\mathfrak{p}}, \quad \mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}R_{\mathfrak{p}}.$$

For a continuous map  $f : X \rightarrow Y$ , set  $(f_*\mathcal{O}_X)(U) = \mathcal{O}_X(f^{-1}(U))$ . A morphism of locally ringed spaces

$$(f, f^\#) : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

consists of  $f$  and a morphism of sheaves  $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  whose induced homomorphisms on stalks are local. Here a morphism of sheaves of rings is a family of ring homomorphisms on open sets commuting with restriction maps, and a homomorphism of local rings is *local* when the inverse image of the maximal ideal is the maximal ideal.

An *affine scheme* is a locally ringed space isomorphic to  $(\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$ . A *scheme* is a locally ringed space admitting an open cover by affine schemes, and a *scheme morphism* is a morphism of locally ringed spaces. A *complex scheme* is a scheme equipped with a morphism to  $\operatorname{Spec} \mathbb{C}$ , and its morphisms are required to commute with these structure morphisms. It is *of finite type* when it admits a finite affine open cover by spectra of finitely generated commutative  $\mathbb{C}$ -algebras.

For a homomorphism  $\varphi : R \rightarrow S$ , prove that inverse image of prime ideals and localization induce a scheme morphism

$$\varphi^* : \operatorname{Spec} S \longrightarrow \operatorname{Spec} R$$

and establish the affine anti-equivalence

$$\operatorname{Hom}_{\operatorname{Sch}}(\operatorname{Spec} S, \operatorname{Spec} R) \cong \operatorname{Hom}_{\operatorname{Ring}}(R, S).$$

The inclusion  $\mathbb{C}[m] \rightarrow A$  therefore defines the Legendre-family morphism

$$\pi : \mathcal{E}_{\text{aff}} = \operatorname{Spec} A \longrightarrow \mathbb{A}_m^1 = \operatorname{Spec} \mathbb{C}[m].$$

Prove that both schemes are complex and of finite type and that  $\pi$  commutes with their structure morphisms to  $\operatorname{Spec} \mathbb{C}$ . For  $m_0 \in \mathbb{C}$ , define its algebraic fiber by

$$\mathcal{E}_{m_0}^{\text{aff}} = \operatorname{Spec} \left( A \otimes_{\mathbb{C}[m]} \mathbb{C}[m]/(m - m_0) \right)$$

Use as theorem input the weak Nullstellensatz [24]: every maximal ideal of  $\mathbb{C}[z_1, \dots, z_n]$  is the kernel of evaluation at a unique point of  $\mathbb{C}^n$ . Identify the closed points of the fiber with

$$y^2 = x(x - 1)(x - m_0).$$

Show that replacing an ideal by its radical preserves the closed points but can change the structure sheaf. Compare  $\operatorname{Spec} \mathbb{C}[t]/(t)$  with  $\operatorname{Spec} \mathbb{C}[t]/(t^2)$ , identify the nonzero nilpotent class in the latter ring, and compute its square.

**Exercise 16.3** (Local rings, tangent spaces, and singular fibers). For  $p \in X = \operatorname{Spec} R$  corresponding to  $\mathfrak{p}$ , define the residue field, cotangent space, and tangent space by

$$\kappa(p) = R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}}, \quad T_p^*X = \mathfrak{p}R_{\mathfrak{p}}/(\mathfrak{p}R_{\mathfrak{p}})^2, \quad T_pX = \operatorname{Hom}_{\kappa(p)}(T_p^*X, \kappa(p)).$$

Define the Krull dimension of a ring as the supremum of the lengths of strict chains of prime ideals and define the local dimension of  $X$  at  $p$  as  $\dim R_{\mathfrak{p}}$ .

For a  $\mathbb{C}$ -algebra  $R$ , define the module of Kähler differentials  $\Omega_{R/\mathbb{C}}^1$  by a derivation  $d : R \rightarrow \Omega_{R/\mathbb{C}}^1$  universal among  $\mathbb{C}$ -linear derivations from  $R$  to  $R$ -modules. If

$$R = \mathbb{C}[x_1, \dots, x_n]/(f_1, \dots, f_r),$$

derive the conormal sequence and prove at a closed point  $p$  that

$$T_pX = \{v \in \mathbb{C}^n : df_j(p)(v) = 0 \text{ for } 1 \leq j \leq r\}.$$

Use as theorem input the Jacobian criterion over  $\mathbb{C}$  [24]:  $p$  is smooth exactly when  $\dim_{\mathbb{C}} T_pX = \dim \mathcal{O}_{X,p}$ .

Apply the criterion to the fibers  $\mathcal{E}_m^{\text{aff}}$  constructed above. Prove that they are smooth for  $m \notin \{0, 1\}$  and that the fibers at  $m = 0$  and  $m = 1$  have singular points  $(0, 0)$  and  $(1, 0)$ , respectively. Compute the local dimension at every closed point of these fibers and compare it with the tangent-space dimension at the two singular points. Define the tangent cone of a plane curve at a point by the lowest nonzero homogeneous part of a translated local equation and prove that both singular fibers have two distinct tangent directions over  $\mathbb{C}$ .

Determine the smooth and singular points of

$$y^2 - x^3, \quad y^2 - x^2(x + 1), \quad xy,$$

and distinguish the cusp, node, and crossing by their local rings and tangent cones. For the morphism

$$\operatorname{Spec} \mathbb{C}[u, t]/(t^2 - u) \longrightarrow \operatorname{Spec} \mathbb{C}[u],$$

show that every fiber over  $u \neq 0$  consists of two reduced points, whereas the fiber over  $u = 0$  is  $\operatorname{Spec} \mathbb{C}[t]/(t^2)$ , and compute its tangent space and length-two local algebra.

**Exercise 16.4** (Projective curves and the Legendre compactification). Let  $R = \bigoplus_{n \geq 0} R_n$  be a finitely generated graded  $\mathbb{C}$ -algebra generated by  $R_1$ , and let  $R_+ = \bigoplus_{n > 0} R_n$ . Define

$$\operatorname{Proj} R = \{\mathfrak{p} \subset R : \mathfrak{p} \text{ is homogeneous prime and } R_+ \not\subset \mathfrak{p}\}.$$

For homogeneous  $f$  of positive degree, define

$$D_+(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}, \quad \mathcal{O}_{\operatorname{Proj} R}(D_+(f)) = (R_f)_0.$$

Prove that these affine charts glue to a scheme. For a finite-dimensional complex vector space  $V$ , prove

$$\mathbb{P}(V) = \operatorname{Proj}(\operatorname{Sym} V^\vee)$$

and recover its interpretation as the space of complex lines in  $V$ .

For a homogeneous ideal  $I \subset R$ , define the projective subscheme  $\operatorname{Proj}(R/I)$ . Show that homogenization of an affine ideal gives its projective closure and identify the points added at infinity for

a parabola and a hyperbola. Call a complex scheme *projective* when it is isomorphic to a closed subscheme of some  $\mathbb{P}(V)$ . Define a *projective algebraic curve* to be a projective complex scheme of finite type whose irreducible components all have Krull dimension one. For a projective plane curve  $C = V(F)$  defined by a homogeneous polynomial, define  $\deg C = \deg F$ .

For  $m \in \mathbb{C} \setminus \{0, 1\}$ , construct the projective closure

$$E_m : Y^2 Z = X(X - Z)(X - mZ) \subset \mathbb{P}^2.$$

Prove that  $E_m$  is a projective algebraic curve of degree three. Use the Jacobian criterion on its affine charts to prove that  $E_m$  is nonsingular and that it has the unique point  $O = [0 : 1 : 0]$  over the line at infinity. Prove that the rational function  $x = X/Z$  extends to a morphism

$$\pi_m : E_m \longrightarrow \mathbb{P}^1, \quad \pi_m([X : Y : Z]) = [X : Z] \quad ([X : Y : Z] \neq O), \quad \pi_m(O) = [1 : 0].$$

**Exercise 16.5** (Intersection multiplicity, resultants, and discriminants). Let  $C = V(F)$  and  $D = V(G)$  be projective plane curves with no common component. At  $p \in C \cap D$ , let  $F_p, G_p \in \mathcal{O}_{\mathbb{P}^2, p}$  be local equations and define their local intersection multiplicity by

$$I_p(C, D) = \dim_{\mathbb{C}} \mathcal{O}_{\mathbb{P}^2, p} / (F_p, G_p).$$

Prove that this length is finite, is invariant under multiplication of  $F$  and  $G$  by local units, and equals one exactly for a transverse intersection of smooth branches.

For a two-dimensional vector space  $W$  and binary forms  $F \in \text{Sym}^m W^\vee$  and  $G \in \text{Sym}^n W^\vee$ , define their resultant through the determinant on the highest exterior powers of

$$\text{Sym}^{n-1} W^\vee \oplus \text{Sym}^{m-1} W^\vee \longrightarrow \text{Sym}^{m+n-1} W^\vee, \quad (A, B) \longmapsto AF + BG.$$

Prove that it vanishes exactly when  $F$  and  $G$  have a common projective zero. Define the discriminant of a one-variable polynomial by the resultant with its derivative and prove that

$$\text{Disc}(t^3 + at + b) = -4a^3 - 27b^2.$$

Compute also

$$\text{Disc}(x(x-1)(x-m)) = m^2(1-m)^2$$

and recover algebraically the two singular Legendre fibers found by the Jacobian criterion.

Use as theorem input Bézout's theorem for plane curves [24]:

$$\sum_{p \in C \cap D} I_p(C, D) = \deg C \deg D.$$

Derive the possible intersection patterns of a line with a smooth cubic and prove that tangency is detected by a repeated root and hence by the corresponding discriminant locus.

**Exercise 16.6** (Caustics and Airy–Pearcey diffraction). For a monochromatic scalar wave with wave number  $k = 2\pi/\lambda$ , define the fold and cusp canonical fields as oscillatory integrals

$$\mathcal{A}_{\text{fold}}(a; k) = \int_{\mathbb{R}}^{\text{osc}} e^{ik(t^3/3 - at)} dt,$$

$$\mathcal{A}_{\text{cusp}}(a, b; k) = \int_{\mathbb{R}}^{\text{osc}} e^{ik(t^4/4 + at^2/2 + bt)} dt.$$

For a phase function, call the image of its degenerate stationary points in parameter space its *caustic*. Prove that the fold caustic is  $a = 0$  and that the cusp caustic is the discriminant curve

$$4a^3 + 27b^2 = 0$$

of the stationary cubic  $t^3 + at + b$ .

Using the Airy function, prove

$$\mathcal{A}_{\text{fold}}(a; k) = 2\pi k^{-1/3} \text{Ai}(-k^{2/3}a).$$

Define the Pearcey function by

$$\text{Pe}(A, B) = \int_{\mathbb{R}}^{\text{osc}} e^{i(t^4/4 + At^2/2 + Bt)} dt$$

and derive

$$\mathcal{A}_{\text{cusp}}(a, b; k) = k^{-1/4} \text{Pe}(k^{1/2}a, k^{3/4}b).$$

Deduce that fold dark fringes occur at  $a = a_n k^{-2/3}$ , where  $\text{Ai}(-a_n) = 0$ , and that the two cusp control scales are  $k^{-1/2}$  and  $k^{-3/4}$ . Predict the normalized intensities  $|\text{Ai}|^2$  and  $|\text{Pe}|^2$  near fold and cusp caustics. State the scalar-wave, short-wavelength, canonical-form, finite-aperture, and detector-resolution assumptions under which measured fringe locations test the discriminant geometry and its wavelength scaling.

**Exercise 16.7** (Branched covers and the Riemann–Hurwitz formula). For a compact connected oriented surface  $S$ , define its genus  $g(S)$  by

$$\chi(S) = 2 - 2g(S).$$

Let  $f : C \rightarrow D$  be a nonconstant holomorphic map between compact connected Riemann surfaces. At  $p \in C$ , choose local parameters  $z$  at  $p$  and  $w$  at  $f(p)$ . Define the *ramification index*  $e_p \geq 1$  by

$$w \circ f = z^{e_p} u(z), \quad u(0) \neq 0,$$

and define the ramification divisor

$$R_f = \sum_{p \in C} (e_p - 1)[p].$$

Show that  $e_p$  is independent of the local parameters and that the number of points in a nearby regular fiber, counted with multiplicity, is a constant  $d$ , the degree of  $f$ .

Triangulate  $D$  so that every branch value is a vertex, lift the triangulation to  $C$ , and compare vertices, edges, and faces to prove the Riemann–Hurwitz formula

$$2g(C) - 2 = d(2g(D) - 2) + \deg R_f.$$

For a square-free polynomial  $q \in \mathbb{C}[x]$  of degree  $2g + 1$  or  $2g + 2$ , let  $C_q$  be the nonsingular projective curve obtained from  $y^2 = q(x)$ . Show that the projection  $x : C_q \rightarrow \mathbb{P}^1$  has degree two, determine its branch points including the point or points over infinity, and deduce that  $C_q$  has genus  $g$ .

**Exercise 16.8** (Legendre ramification and holomorphic differentials). For  $m \in \mathbb{C} \setminus \{0, 1\}$ , let  $\pi_m : E_m \rightarrow \mathbb{P}^1$  be the projective Legendre curve and morphism constructed above. Prove that it has degree two and is ramified exactly over  $0, 1, m, \infty$ . Use Riemann–Hurwitz to obtain

$$\chi(E_m) = 2\chi(\mathbb{P}^1) - 4 = 0$$

and hence that  $E_m$  has genus one.

On the affine chart  $Z = 1$ , put

$$\eta_m = \frac{dx}{y}, \quad y^2 = x(x-1)(x-m).$$

Check at each ramification point and at infinity that  $\eta_m$  extends to a nowhere-vanishing holomorphic one-form on  $E_m$ . For  $0 < m < 1$ , let  $\gamma_m$  be the closed real oval over  $[0, m]$ . With  $m = k^2$ , prove that its orientation may be chosen so that

$$\oint_{\gamma_m} \eta_m = 4K(k), \quad T(\theta_0) = \sqrt{\frac{\ell}{g}} \oint_{\gamma_m} \eta_m.$$

**Exercise 16.9** (Riemann bilinear identity and period lattices). For a compact genus-one Riemann surface  $C$  and a nowhere-vanishing holomorphic one-form  $\eta$ , define the period homomorphism and period subgroup by

$$\text{per}_\eta : H_1(C; \mathbb{Z}) \longrightarrow \mathbb{C}, \quad [\gamma] \longmapsto \int_\gamma \eta, \quad \Lambda_\eta = \text{im}(\text{per}_\eta).$$

Let  $\alpha, \beta$  be oriented simple closed curves meeting once positively and generating the first homology of  $C$ , and put

$$A = \int_\alpha \eta, \quad B = \int_\beta \eta.$$

Cut  $C$  along  $\alpha \cup \beta$ , choose a primitive of  $\eta$  on the resulting polygon, and apply Stokes' theorem to its boundary to prove the genus-one Riemann bilinear identity

$$\frac{i}{2} \int_C \eta \wedge \bar{\eta} = \text{Im}(\overline{A}B) > 0.$$

Deduce that  $A$  and  $B$  are linearly independent over  $\mathbb{R}$  and that

$$\Lambda_\eta = A\mathbb{Z} + B\mathbb{Z}$$

is a discrete subgroup of rank two. Apply this to  $(E_m, \eta_m)$  and identify the real generator  $A = 4K(\sqrt{m})$  for  $0 < m < 1$ .

**Exercise 16.10** (Abel maps and elliptic uniformization). Let  $\Lambda_{\eta_m}$  be the period lattice obtained above. For a fixed  $p_0 \in E_m$ , define

$$\mathcal{A}_{p_0}(p) = \left[ \int_{p_0}^p \eta_m \right] \in \mathbb{C} / \Lambda_{\eta_m}.$$

Show that this is independent of the path modulo periods, is holomorphic, and has nowhere-vanishing differential. Show that it is therefore a finite covering. Identify its induced map on first homology with  $\text{per}_{\eta_m} : H_1(E_m; \mathbb{Z}) \rightarrow \Lambda_{\eta_m}$  and use the period-pairing calculation to prove that the covering has one sheet. Conclude that it is a biholomorphism

$$E_m \cong \mathbb{C} / \Lambda_{\eta_m}.$$

Under this identification, show that translation on the complex torus defines an intrinsic group law on  $E_m$  with identity  $p_0$ .

**Exercise 16.11** (Picard–Fuchs equations, monodromy, and pendulum periods). Let  $U = \mathbb{C} \setminus \{0, 1\}$ . Along a path in  $U$ , transport a closed cycle continuously through the family  $E_m$  and call the resulting identification of the groups  $H_1(E_m; \mathbb{Z})$  *Gauss–Manin transport*. Transport around a loop defines its *monodromy*.

For a transported cycle  $\gamma_m$ , put

$$\Pi(m) = \oint_{\gamma_m} \eta_m.$$

Differentiate under the integral sign and reduce differentials modulo exact differentials on  $E_m$  to prove the Picard–Fuchs equation

$$m(1-m)\Pi''(m) + (1-2m)\Pi'(m) - \frac{1}{4}\Pi(m) = 0.$$

Identify this with the Gauss hypergeometric equation. Prove that the projective cubic has discriminant

$$\Delta(m) = 16m^2(1-m)^2$$

in the displayed normalization and that the family has singular fibers at  $m = 0, 1, \infty$ .

Show that local monodromy at  $m = 1$  fixes one analytic solution and sends a logarithmic companion to itself plus a multiple of the analytic solution. For the real oval, use  $\Pi(m) = 4K(\sqrt{m})$  and the hypergeometric connection formula to obtain

$$K(\sqrt{m}) = \log \frac{4}{\sqrt{1-m}} + O((1-m)|\log(1-m)|) \quad (m \uparrow 1).$$

For the pendulum of length  $\ell$ , where  $m = \sin^2(\theta_0/2)$ , deduce

$$T(\theta_0) = 4\sqrt{\frac{\ell}{g}} \log \frac{4}{\cos(\theta_0/2)} + O\left(\sqrt{\frac{\ell}{g}} \cos^2 \frac{\theta_0}{2} \left| \log \cos^2 \frac{\theta_0}{2} \right| \right).$$

State how a period-versus-amplitude measurement tests the location and logarithmic type of the singular fiber.

## 17 Riemann–Roch

**Exercise 17.1** (Divisors, line bundles, and first Chern classes). For a scheme  $X$ , an *algebraic line bundle* is an  $\mathcal{O}_X$ -module locally isomorphic to  $\mathcal{O}_X$ . For  $X = \text{Proj } R$  and  $n \in \mathbb{Z}$ , define the graded shift by  $R(n)_d = R_{n+d}$  and define the twisting sheaf  $\mathcal{O}_X(n)$  on the standard affine charts by

$$\mathcal{O}_X(n)(D_+(f)) = (R(n)_f)_0.$$

For projective space, prove intrinsically for  $n \geq 0$  that

$$\Gamma(\mathbb{P}(V), \mathcal{O}(n)) \cong \text{Sym}^n V^\vee,$$

and that  $\mathcal{O}(-1)$  is the tautological line bundle whose fiber over a line  $\ell \subset V$  is  $\ell$ . For a smooth projective plane curve  $C$ , define  $\mathcal{O}_C(n)$  by restriction. Use as theorem input [24, 19] that an algebraic line bundle on a smooth complex projective curve determines a holomorphic line bundle and that this correspondence preserves tensor products and global sections.

On a compact Riemann surface  $C$ , a *divisor* is a finite formal sum

$$D = \sum_{p \in C} n_p [p], \quad n_p \in \mathbb{Z}, \quad \deg D = \sum_p n_p.$$

For a nonzero meromorphic function  $f$ , define  $\operatorname{div}(f) = \sum_p \operatorname{ord}_p(f) [p]$ . Use the argument principle to prove

$$\deg \operatorname{div}(f) = 0.$$

A *holomorphic line bundle* on  $C$  is a complex line bundle with local product maps whose transition functions are nowhere-vanishing holomorphic functions  $g_{ij}$  satisfying  $g_{ij}g_{jk} = g_{ik}$ . For a divisor  $D = \sum_p n_p [p]$ , define  $\mathcal{O}(D)$  to be the sheaf of meromorphic functions  $h$  satisfying  $\operatorname{div}(h) + D \geq 0$  on each open set. Show that this sheaf is the sheaf of holomorphic sections of a holomorphic line bundle. Two divisors are *linearly equivalent* when their difference is principal, namely the divisor of a nonzero meromorphic function. Prove

$$\mathcal{O}(D) \cong \mathcal{O}(D') \iff D \text{ and } D' \text{ are linearly equivalent.}$$

For a nonzero meromorphic section  $s$ , define  $\operatorname{div}(s)$  from its local representatives. Use as theorem input the meromorphic-section theorem [19]: every holomorphic line bundle on a compact Riemann surface has a nonzero meromorphic section. Prove that

$$\deg L := \deg \operatorname{div}(s)$$

is independent of  $s$ , that tensor products add degrees, and that  $L \cong \mathcal{O}(\operatorname{div}(s))$ . Use Bézout's theorem to prove

$$\deg \mathcal{O}_C(1) = \deg C$$

for every smooth projective plane curve  $C$ .

For a Hermitian connection  $\nabla$  on  $L$ , define

$$c_1(L) = \left[ \frac{i}{2\pi} F_\nabla \right] \in H_{\text{dR}}^2(C; \mathbb{R}).$$

Show that this class is independent of the Hermitian connection and is natural under pullback. Compare connection forms on overlaps and use Stokes' theorem around the zeros and poles of a meromorphic section to prove

$$\int_C c_1(L) = \deg L.$$

**Exercise 17.2** (Čech cohomology and exact sequences). Let  $\mathcal{F}$  be a sheaf of complex vector spaces on a compact Riemann surface  $C$ , and let  $\mathfrak{U} = (U_i)_{i \in I}$  be a finite ordered open cover. Define the alternating Čech cochains and coboundary by

$$\begin{aligned} \check{C}^q(\mathfrak{U}, \mathcal{F}) &= \prod_{i_0 < \dots < i_q} \mathcal{F}(U_{i_0} \cap \dots \cap U_{i_q}), \\ (\delta s)_{i_0 \dots i_{q+1}} &= \sum_{r=0}^{q+1} (-1)^r s_{i_0 \dots \widehat{i_r} \dots i_{q+1}}|_{U_{i_0} \cap \dots \cap U_{i_{q+1}}}. \end{aligned}$$

Prove  $\delta^2 = 0$  and define  $\check{H}^q(\mathfrak{U}, \mathcal{F}) = \ker \delta / \operatorname{im} \delta$ . Show directly from the sheaf gluing axiom that

$$\check{H}^0(\mathfrak{U}, \mathcal{F}) = \mathcal{F}(C).$$



Use as theorem input [19] that a finite cover whose nonempty intersections are Stein computes the cohomology  $H^q(C, \mathcal{F})$  of a coherent analytic sheaf, that coherent sheaves on  $C$  have finite-dimensional cohomology and vanish in degrees  $q > 1$ , and that a short exact sequence of coherent sheaves induces a long exact cohomology sequence. For a holomorphic line bundle  $L$ , a point  $p \in C$ , and its inclusion  $i_p : \{p\} \rightarrow C$ , prove that evaluation at  $p$  gives an exact sequence

$$0 \longrightarrow L(-p) \longrightarrow L \longrightarrow (i_p)_* L_p \longrightarrow 0.$$

Writing  $h^q(L) = \dim H^q(C, L)$  and  $\chi(L) = h^0(L) - h^1(L)$ , use its long exact sequence to prove

$$\chi(L) = \chi(L(-p)) + 1.$$

Iterate this identity through the positive and negative parts of a divisor and use the meromorphic-section theorem to deduce

$$\chi(L) = \deg L + \chi(\mathcal{O}_C)$$

for every holomorphic line bundle on  $C$ .

**Exercise 17.3** (Serre duality and the Riemann–Roch theorem). Define the canonical line bundle  $K_C$  to be the holomorphic cotangent bundle of a compact connected Riemann surface  $C$ . Let  $g$  be its topological genus and use as theorem input [19] the analytic genus identity  $\dim H^0(C, K_C) = g$ . Use as theorem input Serre duality [19] in the form of a natural perfect pairing

$$H^1(C, L) \otimes H^0(C, K_C \otimes L^{-1}) \longrightarrow \mathbb{C}, \quad H^1(C, L) \cong H^0(C, K_C \otimes L^{-1})^\vee.$$

Show that every holomorphic function on  $C$  is constant and deduce

$$\chi(\mathcal{O}_C) = 1 - g.$$

Combine this with the Čech calculation above to prove the Riemann–Roch theorem

$$h^0(L) - h^0(K_C \otimes L^{-1}) = \deg L + 1 - g.$$

Apply it to  $L = K_C$  to prove  $\deg K_C = 2g - 2$ . Deduce that

$$h^0(L) = \deg L + 1 - g \quad \text{when } \deg L > 2g - 2,$$

and that a negative-degree line bundle has no nonzero holomorphic sections. Specialize to  $g = 1$  and prove that a line bundle of degree  $N > 0$  has  $h^0(L) = N$ .

**Exercise 17.4** (Polarized quantum spaces). Let  $(M, \omega)$  be prequantizable. A *complex polarization* is an involutive Lagrangian subbundle  $P \subset T_{\mathbb{C}}M$ . Define its annihilator and canonical line by

$$P^\circ = \{\alpha \in T_{\mathbb{C}}^*M : \alpha|_P = 0\}, \quad K_P = \Lambda^{\text{top}} P^\circ.$$

A *half-form bundle* is a line bundle  $\delta_P$  with  $\delta_P^{\otimes 2} \cong K_P$ . Use as theorem input [47] the induced partial connection on  $\delta_P$  along  $P$  and the half-form pairing. Define the corrected polarized quantum space to be the square-integrable sections of  $L \otimes \delta_P$  annihilated by the tensor-product partial connection along  $P$ .

For the vertical polarization on  $T^*V$ , defined as the tangent distribution to the fibers of  $T^*V \rightarrow V$ , recover  $L^2(V)$  and the operators

$$\psi(x) \longmapsto \ell(x)\psi(x), \quad \psi(x) \longmapsto -i\hbar d\psi_x(v)$$

for  $\ell \in V^*$  and  $v \in V$ .

A *Kähler manifold* is a complex manifold  $(M, J)$  with a symplectic form  $\omega$  satisfying

$$\omega(Ju, Jv) = \omega(u, v), \quad g(u, v) = \omega(u, Jv)$$

for a Riemannian metric  $g$ . For a compact Kähler manifold with  $P = T^{0,1}M$ , prove

$$P^\circ = T^{*1,0}M, \quad K_P = K_M := \Lambda^{\text{top}} T^{*1,0}M.$$

Call a Hermitian holomorphic line bundle *positive* when its first Chern form satisfies  $c_1(L, \nabla)(u, Ju) > 0$  for every nonzero tangent vector  $u$ . When the prequantum line bundle is positive and a square root  $K_M^{1/2}$  exists, identify the corrected quantum space with

$$H^0(M, L \otimes K_M^{1/2}),$$

where  $H^0$  denotes holomorphic sections. On an elliptic curve, use the trivialization by  $dz$  to show that the half-form correction does not change the theta-section count.

**Exercise 17.5** (Abel's theorem and theta divisors). Let  $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$  and put

$$\vartheta_1(z \mid \tau) = \vartheta \left[ \begin{smallmatrix} \frac{1}{2} \\ \frac{1}{2} \end{smallmatrix} \right] (z \mid \tau).$$

Use the theta product and automorphy laws to prove that  $\vartheta_1$  has one simple zero modulo the period lattice, at the origin.

For a degree-zero divisor  $D = \sum_p n_p [p]$ , define its Abel sum by

$$\mathfrak{a}(D) = \sum_p n_p p \in E_\tau.$$

Show that this is independent of the chosen lifts of the points to  $\mathbb{C}$ . If

$$D = [p_1] + \cdots + [p_N] - [q_1] - \cdots - [q_N]$$

and  $\mathfrak{a}(D) = 0$ , choose lifts with  $\sum p_j - \sum q_j = m + n\tau$  and prove that

$$f_D(z) = e^{-2\pi i n z} \frac{\prod_{j=1}^N \vartheta_1(z - p_j \mid \tau)}{\prod_{j=1}^N \vartheta_1(z - q_j \mid \tau)}$$

is elliptic and satisfies  $\text{div}(f_D) = D$ . Conversely, apply the argument principle to  $f'/f$  and  $zf'/f$  around a period parallelogram to prove that a principal divisor has degree zero and Abel sum zero. Conclude the genus-one Abel theorem

$$D \text{ is principal} \iff \deg D = 0 \text{ and } \mathfrak{a}(D) = 0.$$

Let  $\text{Pic}^d(E_\tau)$  denote the isomorphism classes of holomorphic line bundles of degree  $d$ , with tensor product as the operation on  $\text{Pic}^0(E_\tau)$ . Fix  $0 \in E_\tau$  and prove that

$$E_\tau \longrightarrow \text{Pic}^0(E_\tau), \quad p \longmapsto \mathcal{O}([p] - [0]),$$

is an isomorphism of complex groups. Show that tensoring with  $\mathcal{O}(N[0])$  identifies  $\text{Pic}^0(E_\tau)$  with  $\text{Pic}^N(E_\tau)$ . Identify effective divisors of degree  $N$  with unordered  $N$ -tuples of theta zeros and show that their Abel sum determines the corresponding line bundle.

**Exercise 17.6** (Theta sections and genus-one Riemann–Roch). For  $a, b \in \mathbb{R}$ ,  $z \in \mathbb{C}$ , and  $\tau$  in the upper half-plane, define

$$\vartheta \begin{bmatrix} a \\ b \end{bmatrix} (z \mid \tau) = \sum_{n \in \mathbb{Z}} \exp\left(\pi i(n+a)^2\tau + 2\pi i(n+a)(z+b)\right).$$

On  $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ , define the standard degree- $N$  theta line bundle  $L_N$  by representing its holomorphic sections as functions on  $\mathbb{C}$  satisfying

$$f(z+1) = f(z), \quad f(z+\tau) = e^{-\pi i N \tau - 2\pi i N z} f(z).$$

Use these automorphy factors and the argument principle to prove that every nonzero holomorphic section has a zero divisor of degree  $N$ . For  $r \in \mathbb{Z}/N\mathbb{Z}$ , define

$$f_r(z) = \vartheta \begin{bmatrix} r/N \\ 0 \end{bmatrix} (Nz \mid N\tau) = \theta_{N,r}(z \mid \tau)$$

and derive its two automorphy laws directly from the series.

For any line bundle  $L$  of degree  $N > 0$ , use Abel's theorem to write

$$L \cong \mathcal{O}((N-1)[0] + [p])$$

for a unique  $p \in E_\tau$ . Prove from  $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$  that multiplication by  $N$  is surjective, choose  $a \in E_\tau$  with  $-Na = p$ , and identify  $L$  with the pullback of the standard degree- $N$  theta bundle by translation by  $a$ . Translate the distinguished level- $N$  theta sections and use their distinct magnetic-translation characters to construct  $N$  linearly independent sections of  $L$ . Conversely, expand a section on the universal cover in Fourier modes and use its two automorphy laws to show that all coefficients are determined by  $N$  initial values. Deduce

$$\dim H^0(E_\tau, L) = N.$$

Prove that a nonzero holomorphic section of a degree-zero line bundle has no zeros and therefore trivializes the bundle. Deduce

$$\dim H^0(E_\tau, L) = \begin{cases} 1, & L \cong \mathcal{O}, \\ 0, & L \not\cong \mathcal{O} \end{cases} \quad (\deg L = 0).$$

Show that a negative-degree line bundle has no nonzero holomorphic sections. Since the nowhere-vanishing form  $dz$  trivializes the canonical line bundle, combine these cases to prove the genus-one Riemann–Roch identity

$$\dim H^0(E_\tau, L) - \dim H^0(E_\tau, L^{-1}) = \deg L.$$

Relate translation of theta-function zeros to  $\text{Pic}^N(E_\tau)$ .

**Exercise 17.7** (Theta-function Landau spectra). Let

$$Q = \mathbb{R}^2/(L_x\mathbb{Z} \oplus L_y\mathbb{Z}) \cong E_\tau, \quad \tau = i \frac{L_y}{L_x}, \quad A = L_x L_y,$$

and let a particle of charge  $e$  experience a magnetic field through the curvature convention

$$F_\nabla = -\frac{ie}{\hbar} B \, dx \wedge dy.$$

(a) Suppose  $B > 0$  is constant and put

$$N = \frac{eBA}{2\pi\hbar}.$$

Use prequantization to prove that a magnetic line bundle exists exactly when  $N \in \mathbb{Z}$ . For  $N > 0$ , use the automorphy laws above to identify

$$H^0(Q, L) = \bigoplus_{r \in \mathbb{Z}/N\mathbb{Z}} \mathbb{C} \theta_{N,r}(z \mid \tau), \quad z = \frac{x}{L_x} + \tau \frac{y}{L_y}.$$

Compute the magnetic-translation eigenvalues of the theta sections and relate their holomorphic-gauge representatives to physical wavefunctions by the common unitary-gauge factor. For translations by  $L_x/N$  and  $L_y/N$ , use holonomy around a fundamental magnetic cell to prove

$$UV = e^{2\pi i/N} VU.$$

Define the covariant Landau Hamiltonian by

$$\hat{H} = \frac{\hbar^2}{2m} \nabla^* \nabla.$$

Construct raising and lowering operators and prove

$$E_n = \hbar \frac{eB}{m} \left( n + \frac{1}{2} \right), \quad \dim \ker(\hat{H} - E_n) = N.$$

(b) Let  $B$  now be nonuniform, let  $L \rightarrow E_\tau$  be fixed with a unitary connection  $\nabla$  satisfying the displayed curvature convention, and assume

$$N = \frac{e}{2\pi\hbar} \int_{E_\tau} B \, dx \, dy > 0.$$

Use the complex structure to define the positive Pauli Hamiltonian

$$H_+ = \frac{\hbar^2}{2m} \left( \nabla^* \nabla - \frac{e}{\hbar} B \right) = \frac{\hbar^2}{m} \bar{\partial}_\nabla^* \bar{\partial}_\nabla.$$

Prove that its zero modes are the holomorphic sections of  $L$  and use the first-Chern calculation above and genus-one Riemann–Roch to prove

$$\dim \ker H_+ = N.$$

Deduce that this count is unchanged under smooth deformations of the unitary connection on the fixed line bundle.

Deduce the cyclotron-resonance spacing, degeneracy density  $eB/(2\pi\hbar)$  in the constant-field model, magnetic-translation interference phase, and robust Pauli zero-mode count. State separately the flat-torus, flux-integrality, spin-coupling, positivity, noninteracting, and spectral-resolution assumptions under which these measurements test the theta-function and Riemann–Roch predictions.

## 18 Finite-Gap Spectral Curves

**Exercise 18.1** (Floquet operators and Bloch spectral curves). Let

$$L = -\frac{d^2}{du^2} + V(u)$$

with real smooth potential of period  $P$ . For  $E \in \mathbb{C}$ , let  $\mathcal{S}_E$  be the two-dimensional solution space of  $L\psi = E\psi$ , and define the monodromy automorphism

$$M(E) : \mathcal{S}_E \longrightarrow \mathcal{S}_E, \quad \psi(u) \longmapsto \psi(u + P).$$

Use constancy of the Wronskian to prove  $\det M(E) = 1$ . Define

$$\Delta(E) = \operatorname{tr} M(E)$$

and call  $\rho$  a Bloch multiplier when  $M(E)\psi = \rho\psi$  for a nonzero solution. Prove

$$\rho + \rho^{-1} = \Delta(E)$$

Let  $H^2([0, P])$  be the space of  $L^2$  functions whose first two distributional derivatives belong to  $L^2$ , and use the one-dimensional Sobolev trace theorem [26] to evaluate a function and its first derivative at the endpoints. For  $\rho \in \mathbb{C}^\times$ , define the twisted Sobolev domain

$$H_\rho^2 = \{\psi \in H^2([0, P]) : \psi(P) = \rho\psi(0), \psi'(P) = \rho\psi'(0)\}.$$

Regard

$$\mathcal{L}_{\rho, E} : H_\rho^2 \longrightarrow L^2([0, P]), \quad \mathcal{L}_{\rho, E}\psi = (L - E)\psi,$$

as a bounded operator. For a bounded operator  $T : H_0 \rightarrow H_1$ , define its cokernel by

$$\operatorname{coker} T = H_1 / \operatorname{im} T.$$

Use variation of parameters to prove that the image of  $\mathcal{L}_{\rho, E}$  is closed and that its kernel and cokernel are finite-dimensional. Prove

$$\ker \mathcal{L}_{\rho, E} \cong \ker(M(E) - \rho I)$$

and that  $\mathcal{L}_{\rho, E}$  is invertible exactly when  $M(E) - \rho I$  is invertible. Deduce that the noninvertibility locus of the family  $\mathcal{L}_{\rho, E}$  is the Bloch spectral curve

$$\mathcal{B}_L : \quad \rho^2 - \Delta(E)\rho + 1 = 0.$$

For  $E \in \mathbb{R}$  and  $|\rho| = 1$ , use Green's identity to prove that  $\mathcal{L}_{\rho, E}$  is self-adjoint as an operator on  $L^2([0, P])$  with domain  $H_\rho^2$ , and deduce

$$\dim \ker \mathcal{L}_{\rho, E} - \dim \operatorname{coker} \mathcal{L}_{\rho, E} = 0.$$

Show that its branch points occur at the periodic and antiperiodic energies  $\Delta(E) = \pm 2$ , and that a real energy supports a propagating Bloch wave exactly when  $|\Delta(E)| \leq 2$ .

**Exercise 18.2** (Hyperelliptic curves and holomorphic differentials). Let

$$C : y^2 = \prod_{r=1}^{2g+1} (x - e_r), \quad e_r \neq e_s \quad (r \neq s),$$

denote its nonsingular projective completion. Use Riemann–Hurwitz to prove that  $C$  has genus  $g$ . Use the projective construction and Jacobian criterion to verify the asserted nonsingularity. Analyze the finite branch points and the point over infinity to show that the distinguished differentials

$$\eta_j = \frac{x^{j-1} dx}{y}, \quad 1 \leq j \leq g,$$

are holomorphic. Compare their vanishing orders at infinity and use  $\dim H^0(C, K_C) = g$  to show that they exhaust the holomorphic differentials.

Choose a symplectic marking  $(a_1, \dots, a_g, b_1, \dots, b_g)$  of  $H_1(C; \mathbb{Z})$ , define the period maps

$$A(\eta) = \left( \int_{a_1} \eta, \dots, \int_{a_g} \eta \right), \quad B(\eta) = \left( \int_{b_1} \eta, \dots, \int_{b_g} \eta \right),$$

and prove from the Riemann bilinear identity that  $A$  is an isomorphism. Let  $\omega_j$  be the uniquely normalized differentials satisfying  $\int_{a_i} \omega_j = \delta_{ij}$  and define

$$\Omega_{ij} = \int_{b_i} \omega_j.$$

Use the bilinear identity to prove

$$\Omega^T = \Omega, \quad \text{Im } \Omega > 0.$$

**Exercise 18.3** (Jacobians, Abel–Jacobi maps, and Riemann theta functions). For a compact Riemann surface  $C$ , define its Jacobian intrinsically by

$$\text{Jac}(C) = H^0(C, K_C)^\vee / \left\{ \eta \mapsto \int_\gamma \eta : \gamma \in H_1(C; \mathbb{Z}) \right\}.$$

Using the symplectic marking and normalized periods above, identify it with

$$\mathbb{C}^g / (\mathbb{Z}^g + \Omega \mathbb{Z}^g).$$

Define the Riemann theta function by the normally convergent series

$$\vartheta(z \mid \Omega) = \sum_{n \in \mathbb{Z}^g} \exp(\pi i n^T \Omega n + 2\pi i n^T z).$$

Prove its holomorphy and automorphy laws

$$\begin{aligned} \vartheta(z + m \mid \Omega) &= \vartheta(z \mid \Omega), \\ \vartheta(z + \Omega n \mid \Omega) &= e^{-\pi i n^T \Omega n - 2\pi i n^T z} \vartheta(z \mid \Omega). \end{aligned}$$

Fix  $p_0 \in C$  and define the Abel–Jacobi map by

$$\mathcal{A}_{p_0}(p) = \left[ \eta \mapsto \int_{p_0}^p \eta \right] \in \text{Jac}(C).$$

Prove that it is well defined and extend it additively to divisors. Use as theorem input the Abel–Jacobi theorem [19]: a degree-zero divisor is principal exactly when its Abel–Jacobi image vanishes, and every point of the Jacobian is represented by a degree-zero divisor. Deduce

$$\mathrm{Pic}^0(C) \cong \mathrm{Jac}(C).$$

Use as theorem input Riemann’s vanishing theorem [16] to identify a translate of the zero locus of  $\vartheta$  with the Abel–Jacobi images of effective divisors of degree  $g - 1$ . Recover the genus-one Abel and theta-divisor statements above when  $g = 1$ .

**Exercise 18.4** (Jacobi elliptic functions and elliptic uniformization). For  $0 < k < 1$ , define  $\mathrm{sn}(u; k)$  near  $u = 0$  by

$$u = \int_0^{\mathrm{sn}(u; k)} \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}$$

and extend it by analytic continuation. Define  $\mathrm{cn}(u; k)$  and  $\mathrm{dn}(u; k)$  by

$$\mathrm{cn}^2 = 1 - \mathrm{sn}^2, \quad \mathrm{dn}^2 = 1 - k^2 \mathrm{sn}^2, \quad \mathrm{cn}(0; k) = \mathrm{dn}(0; k) = 1.$$

Derive

$$\mathrm{sn}' = \mathrm{cn} \mathrm{dn}, \quad \mathrm{cn}' = -\mathrm{sn} \mathrm{dn}, \quad \mathrm{dn}' = -k^2 \mathrm{sn} \mathrm{cn}.$$

Put  $m = k^2$  and define

$$x(u) = m \mathrm{sn}^2(u; k), \quad y(u) = m \mathrm{sn}(u; k) \mathrm{cn}(u; k) \mathrm{dn}(u; k).$$

Prove

$$y(u)^2 = x(u)(x(u) - 1)(x(u) - m), \quad \frac{dx}{y} = 2 du.$$

Identify the period lattice of  $(x(u), y(u))$  and recover the uniformization of the Legendre curve constructed earlier.

**Exercise 18.5** (Lamé theta functions and the spectral curve). Let  $K = K(k)$ ,  $K' = K(\sqrt{1-k^2})$ , and  $\tau = iK'/K$ . Using the theta functions with characteristics, put

$$H(u) = \vartheta \left[ \begin{smallmatrix} \frac{1}{2} \\ \frac{1}{2} \end{smallmatrix} \right] \left( \frac{u}{2K} \mid \tau \right), \quad \Theta(u) = \vartheta \left[ \begin{smallmatrix} 0 \\ \frac{1}{2} \end{smallmatrix} \right] \left( \frac{u}{2K} \mid \tau \right), \quad Z(u) = \frac{d}{du} \log \Theta(u).$$

For the Lamé operator

$$L_k = -\frac{d^2}{du^2} + 2k^2 \mathrm{sn}^2(u; k),$$

use the theta addition identities to verify the Hermite solutions

$$\psi_{\pm}(u, a) = \frac{H(u \pm a)}{\Theta(u)} \exp(\mp u Z(a)),$$

$$L_k \psi_{\pm}(\cdot, a) = E(a) \psi_{\pm}(\cdot, a), \quad E(a) = 1 + k^2 - k^2 \mathrm{sn}^2(a; k).$$

From the theta automorphy laws, derive their Bloch multipliers over the potential period  $2K$ :

$$\rho_{\pm}(a) = -\exp(\mp 2K Z(a)).$$

Define

$$\mu(a) = \frac{i}{2} \frac{dE(a)}{da}.$$

Use the Jacobi differential identities to prove

$$\mu(a)^2 = (E(a) - k^2)(E(a) - 1)(E(a) - 1 - k^2).$$

Thus the parameter  $a$  uniformizes the Lamé spectral curve

$$\Sigma_k : \quad \mu^2 = (E - k^2)(E - 1)(E - 1 - k^2).$$

**Exercise 18.6** (Baker–Akhiezer functions and finite-gap reconstruction). Let

$$\Sigma : \quad \mu^2 = \prod_{j=0}^{2g} (E - E_j), \quad E_0 < E_1 < \cdots < E_{2g},$$

be a nonsingular real hyperelliptic curve with marked point  $\infty$ , and let  $D$  be an effective divisor of degree  $g$ . Let  $d\Omega_1$  be the unique meromorphic differential with zero  $a$ -periods, no pole away from  $\infty$ , and principal part  $d\sqrt{E}$  at  $\infty$ . Put

$$\Omega_1(p) = \int^p d\Omega_1, \quad U_j = \frac{1}{2\pi i} \int_{b_j} d\Omega_1,$$

take the Abel map  $\mathcal{A}$  to be based at  $\infty$ , and let  $Z \in \text{Jac}(\Sigma)$  be the theta translate determined by  $D$  through Riemann’s vanishing theorem.

For  $x \in \mathbb{R}$ , let  $F_x$  be the flat holomorphic line bundle whose holonomy around a closed curve  $\gamma$  is

$$\text{hol}_{F_x}(\gamma) = \exp \left( -x \int_{\gamma} d\Omega_1 \right).$$

Call  $(D, Z)$  *admissible* when, for every  $x \in \mathbb{R}$ ,

$$H^0(\Sigma, K_{\Sigma}(-D) \otimes F_x^{-1}) = 0, \quad \vartheta(Ux + Z \mid \Omega) \neq 0,$$

and  $\frac{d^2}{dx^2} \log \vartheta(Ux + Z \mid \Omega)$  is real. Since  $\deg F_x = 0$ , use Riemann–Roch to prove

$$\dim H^0(\Sigma, \mathcal{O}(D) \otimes F_x) = 1.$$

For admissible  $(D, Z)$ , define

$$\psi(x, p) = \frac{\vartheta(\mathcal{A}(p) + Ux + Z \mid \Omega) \vartheta(Z \mid \Omega)}{\vartheta(\mathcal{A}(p) + Z \mid \Omega) \vartheta(Ux + Z \mid \Omega)} e^{x\Omega_1(p)}.$$

Use the theta automorphy laws to prove that the apparent monodromy of the quotient cancels that of the exponential, so  $\psi$  is a function on  $\Sigma \setminus \{\infty\}$  with poles prescribed by  $D$ . Prove that  $\psi(x, p)e^{-x\Omega_1(p)}$  is the unique section of  $\mathcal{O}(D) \otimes F_x$  normalized to have value 1 at  $\infty$  in the local trivialization used in the theta formula. Deduce from the Riemann–Roch dimension count that  $\psi$  is the unique normalized Baker–Akhiezer function with these divisor, monodromy, and asymptotic data.

Use as theorem input the finite-gap reconstruction theorem [5]: there is a constant  $c$  determined by the expansion at infinity such that

$$u(x) = c - 2 \frac{d^2}{dx^2} \log \vartheta(Ux + Z \mid \Omega)$$



is real and smooth for admissible  $(D, Z)$ , and

$$\left(-\frac{d^2}{dx^2} + u(x)\right)\psi(x, p) = E(p)\psi(x, p).$$

Show from the two points over a regular  $E$  and their Bloch multipliers that the spectrum is

$$[E_0, E_1] \cup [E_2, E_3] \cup \cdots \cup [E_{2g-2}, E_{2g-1}] \cup [E_{2g}, \infty).$$

Specialize to  $g = 1$  and recover, up to translation and an additive energy constant, the Lamé potential and Hermite theta solutions above.

**Exercise 18.7** (Lamé band edges and attenuation prediction). Use the half-period values of the Jacobi functions to recover the three band-edge solutions

$$L_k \operatorname{dn} = k^2 \operatorname{dn}, \quad L_k \operatorname{cn} = \operatorname{cn}, \quad L_k \operatorname{sn} = (1 + k^2) \operatorname{sn}.$$

Determine which are periodic and antiperiodic over  $2K$ . Use the theta-function Bloch multipliers to prove

$$\operatorname{spec}(L_k) = [k^2, 1] \cup [1 + k^2, \infty)$$

and identify  $(1, 1 + k^2)$  as the unique open gap.

For an energy in the gap, choose  $a$  with  $E = E(a)$  and prove that a Bloch wave changes in magnitude by  $e^{-2K\kappa(E)}$  per cell, where

$$\kappa(E) = \frac{1}{2K} |\log |\rho_{\pm}(a)|| = |\operatorname{Re} Z(a)|.$$

Predict the three band-edge energies and show that, up to interface factors, the transmitted amplitude through  $M$  cells is suppressed by  $e^{-2KM\kappa(E)}$  and the transmitted intensity by  $e^{-4KM\kappa(E)}$ . Show that the gap closes as  $k \rightarrow 0$ , and state the periodic-medium, coherence, finite-sample, and energy-resolution assumptions under which a transmission experiment tests the spectral curve.

**Exercise 18.8** (Cnoidal KdV waves and finite-gap transport). For the normalized Korteweg–de Vries equation

$$u_t + 6uu_x + u_{xxx} = 0,$$

put  $\xi = x - vt$  and

$$u(x, t) = a + 2k^2 \kappa^2 \operatorname{cn}^2(\kappa\xi; k), \quad 0 < k < 1.$$

Derive

$$\frac{d^2}{dz^2} \operatorname{cn}^2(z; k) = 2(1 - k^2) + 4(2k^2 - 1) \operatorname{cn}^2(z; k) - 6k^2 \operatorname{cn}^4(z; k)$$

and prove that the displayed wave solves KdV exactly when

$$v = 6a + 4\kappa^2(2k^2 - 1).$$

Deduce its amplitude, wavelength, and speed:

$$A = 2k^2 \kappa^2, \quad \lambda = \frac{2K(k)}{\kappa}, \quad v = 6a + 4\kappa^2(2k^2 - 1).$$

Define the complete elliptic integral of the second kind and the elliptic nome by

$$E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta, \quad q = e^{-\pi K'(k)/K(k)}.$$

Prove the Fourier expansion

$$\operatorname{cn}^2(z; k) = \frac{E(k) - (1 - k^2)K(k)}{k^2 K(k)} + \frac{2\pi^2}{k^2 K(k)^2} \sum_{n \geq 1} \frac{nq^n}{1 - q^{2n}} \cos \frac{n\pi z}{K(k)}.$$

Use as theorem input the finite-gap KdV evolution theorem [5], which replaces  $Ux + Z$  in the reconstructed potential by  $Ux + Vt + Z$  for the normalized second-kind KdV period vector  $V$ . Show that the genus-one case gives the cnoidal wave above and that  $k \uparrow 1$  gives the solitary wave

$$u(x, t) = a + 2\kappa^2 \operatorname{sech}^2(\kappa(x - (6a + 4\kappa^2)t)).$$

Predict the wavelength, phase speed, and harmonic-amplitude ratios of a weakly dispersive one-dimensional wave train from  $K$ ,  $E$ , and the nome. State the KdV scaling, weak-nonlinearity, long-wavelength, negligible-loss, one-dimensional, and finite-window assumptions under which a wave-tank or transmission-line experiment tests these predictions.

## 19 Index Theory

**Exercise 19.1** (Fredholm operators and the Calkin algebra). Let  $H_0$  and  $H_1$  be complex Hilbert spaces. Write  $\mathcal{B}(H_0, H_1)$  for the bounded operators and define  $\mathcal{K}(H_0, H_1)$  to be the operator-norm closure of the finite-rank operators. Call its elements *compact operators*. Prove that composition on either side by a bounded operator preserves compactness and that  $\mathcal{K}(H) = \mathcal{K}(H, H)$  is a closed two-sided ideal in  $\mathcal{B}(H) = \mathcal{B}(H, H)$ . Define the *Calkin algebra* by

$$\mathcal{Q}(H) = \mathcal{B}(H)/\mathcal{K}(H).$$

A bounded operator  $T : H_0 \rightarrow H_1$  is *Fredholm* when its kernel and cokernel are finite-dimensional and its image is closed. Define

$$\operatorname{ind}_{\text{an}}(T) = \dim \ker T - \dim \operatorname{coker} T.$$

Use as theorem input Atkinson's theorem [11]:  $T$  is Fredholm exactly when there is a bounded  $S : H_1 \rightarrow H_0$  for which

$$ST - I_{H_0} \in \mathcal{K}(H_0), \quad TS - I_{H_1} \in \mathcal{K}(H_1).$$

For  $H_0 = H_1$ , identify this condition with invertibility of the image of  $T$  in the Calkin algebra.

Prove that compact perturbations preserve the Fredholm property and the index, that

$$\operatorname{ind}_{\text{an}}(T^*) = -\operatorname{ind}_{\text{an}}(T), \quad \operatorname{ind}_{\text{an}}(T_2 T_1) = \operatorname{ind}_{\text{an}}(T_2) + \operatorname{ind}_{\text{an}}(T_1),$$

and that the index is constant along a norm-continuous path of Fredholm operators. On  $\ell^2(\mathbb{N})$ , let  $S$  be the unilateral shift  $Se_n = e_{n+1}$ . Construct a parametrix modulo finite-rank operators and compute

$$\operatorname{ind}_{\text{an}}(S) = -1, \quad \operatorname{ind}_{\text{an}}(S^*) = 1.$$

For the twisted periodic Schrödinger operators

$$\mathcal{L}_{\rho, E} : H_\rho^2 \longrightarrow L^2([0, P])$$

with  $|\rho| = 1$  and  $E \in \mathbb{R}$ , use the closed-range and kernel-cokernel calculations to prove

$$\operatorname{ind}_{\text{an}}(\mathcal{L}_{\rho, E}) = 0, \quad \ker \mathcal{L}_{\rho, E} \neq 0 \iff \rho^2 - \Delta(E)\rho + 1 = 0.$$

**Exercise 19.2** (Elliptic operators and analytic index). Let  $E, F \rightarrow M$  be smooth complex vector bundles over a closed smooth manifold. For  $f \in C^\infty(M)$  let  $M_f$  denote multiplication by  $f$ . A linear map

$$D : \Gamma^\infty(E) \longrightarrow \Gamma^\infty(F)$$

is a *differential operator of order at most  $m$*  when every  $(m+1)$ -fold iterated commutator with multiplication operators vanishes. For an operator of order  $m$ ,  $x \in M$ ,  $\xi \in T_x^*M$ , and  $v \in E_x$ , choose  $f$  with  $df_x = \xi$  and a section  $s$  with  $s(x) = v$ , and define

$$\sigma_m(D)(x, \xi)v = \frac{1}{m!} \text{ad}_{M_f}^m(D)s(x), \quad \text{ad}_{M_f}(D) = [D, M_f].$$

Prove that this is independent of  $f$  and  $s$ , depends homogeneously of degree  $m$  on  $\xi$ , and defines the *principal symbol*

$$\sigma_m(D) : \pi^*E \longrightarrow \pi^*F \quad \text{over } T^*M.$$

Call  $D$  *elliptic* when  $\sigma_m(D)(x, \xi)$  is invertible for every  $\xi \neq 0$ .

Choose Hermitian and Riemannian metrics and a compatible connection on  $E$ . Using the spectral calculus for the nonnegative connection Laplacian  $\Delta_E = \nabla^* \nabla$ , define  $H^s(M; E)$  as the completion of smooth sections for the norm

$$\|u\|_{H^s} = \|(I + \Delta_E)^{s/2} u\|_{L^2}.$$

Use as theorem input [26] that different choices give equivalent norms. Use as theorem input [26] the coordinate-invariant elliptic regularity, symbol quantization, and parametrix theorem: for every real  $s$ , an elliptic  $D$  extends to

$$D_s : H^s(M; E) \longrightarrow H^{s-m}(M; F)$$

and admits a bounded parametrix whose two remainders are compact. Deduce from Atkinson's theorem that  $D_s$  is Fredholm, that its index is independent of  $s$ , and use symbol quantization to prove that it depends only on the homotopy class of the elliptic principal symbol. Define this integer to be  $\text{ind}_{\text{an}}(D)$  and prove

$$\text{ind}_{\text{an}}(D^*) = -\text{ind}_{\text{an}}(D).$$

For a holomorphic line bundle  $L$  on a compact Riemann surface  $C$ , show that

$$\bar{\partial}_L : \Omega^{0,0}(C; L) \longrightarrow \Omega^{0,1}(C; L)$$

is elliptic. Use as theorem input [19, 26] the Dolbeault identification of its kernel and cokernel with  $H^0(C, L)$  and  $H^1(C, L)$ . Recover from Riemann–Roch the analytic-index formula

$$\text{ind}_{\text{an}}(\bar{\partial}_L) = \deg L + 1 - g(C).$$

**Exercise 19.3** (Homotopy, suspension, and clutching). A *based space* is a space  $X$  with a distinguished point  $x_0$ . Define based maps, based homotopies, and the set  $[X, Y]_*$  of based homotopy classes. Define the reduced suspension and based loop space by

$$\Sigma X = X \times [0, 1] / (X \times \{0, 1\} \cup \{x_0\} \times [0, 1]),$$

$$\Omega Y = \{\gamma : [0, 1] \rightarrow Y : \gamma(0) = \gamma(1) = y_0\}.$$

Construct the suspension–loop adjunction

$$[\Sigma X, Y]_* \cong [X, \Omega Y]_*.$$

Let  $V$  be a finite-dimensional complex vector space and let  $g : X \rightarrow \text{Aut}(V)$  be a based map. Glue the trivial bundles on the two reduced cones on  $X$  by  $g$  along their common copy of  $X$  and denote the resulting bundle on  $\Sigma X$  by  $E_g$ . Prove that based homotopic clutching maps give isomorphic bundles and that direct sum of clutching maps corresponds to direct sum of bundles. Call bundles  $E$  and  $F$  *stably isomorphic* when

$$E \oplus \mathbb{C}^r \cong F \oplus \mathbb{C}^s$$

for some  $r, s$ . Use as theorem input the clutching theorem [1]: for compact Hausdorff  $X$ , stable bundles on  $\Sigma X$  are classified by stable homotopy classes of clutching maps.

Take  $X = S^1$ ,  $V = \mathbb{C}$ , and let  $g(z)$  be multiplication by  $z$ . Identify  $\Sigma S^1$  with  $S^2$ , identify  $E_g$  with the Hopf line bundle after choosing the orientation compatible with the clutching convention, and use the circle-cocycle calculation to prove

$$\int_{S^2} c_1(E_g) = 1.$$

**Exercise 19.4** (Connections, curvature, and Chern–Weil forms). Let  $E \rightarrow M$  be a smooth complex vector bundle of rank  $r$ . Define its direct sum, tensor product, dual, and pullback by the corresponding fiberwise operations. Call a connection *complex* when  $\nabla_X(is) = i\nabla_X s$ . For a complex connection, extend  $\nabla$  to  $E$ -valued forms by the graded Leibniz rule and write

$$F_\nabla = \nabla^2 \in \Omega^2(M; \text{End } E)$$

for its curvature.

Prove the Bianchi identity  $\nabla F_\nabla = 0$ . Using trace and determinant, define

$$\text{ch}(E, \nabla) = \text{Tr} \exp\left(\frac{iF_\nabla}{2\pi}\right), \quad c(E, \nabla) = \det\left(I + \frac{iF_\nabla}{2\pi}\right) = 1 + c_1(E, \nabla) + \cdots + c_r(E, \nabla).$$

Use Bianchi and cyclicity of trace to prove that the homogeneous components of  $\text{ch}(E, \nabla)$  are closed. For a path  $\nabla_t$  of connections, derive

$$\frac{d}{dt} \text{Tr} \exp\left(\frac{iF_{\nabla_t}}{2\pi}\right) = d \text{Tr} \left( \frac{i\dot{\nabla}_t}{2\pi} \exp\left(\frac{iF_{\nabla_t}}{2\pi}\right) \right).$$

Use as theorem input [7] the corresponding Chern–Weil transgression formula for the determinant invariant polynomials. Deduce that the de Rham classes  $\text{ch}(E)$  and  $c_j(E)$  are independent of the connection, natural under pullback, and satisfy

$$c(E \oplus F) = c(E)c(F), \quad \text{ch}(E \oplus F) = \text{ch}(E) + \text{ch}(F), \quad \text{ch}(E \otimes F) = \text{ch}(E) \text{ch}(F).$$

Prove also that

$$\text{ch}_q(E^\vee) = (-1)^q \text{ch}_q(E).$$

Recover the first Chern form of a line bundle used above.

**Exercise 19.5** (Spectral projections and Berry bundles). Let  $W$  be a finite-dimensional Hermitian vector space, let  $B$  be a smooth manifold, and let  $H : B \rightarrow \text{End}(W)$  be a smooth family of self-adjoint operators. Suppose a fixed contour  $\Gamma$  encloses an isolated spectral cluster of constant multiplicity. Define

$$P_b = \frac{1}{2\pi i} \oint_{\Gamma} (z - H(b))^{-1} dz, \quad \mathcal{E} = \coprod_{b \in B} \text{im } P_b.$$

Prove that  $P_b$  is a smooth family of orthogonal projections and that  $\mathcal{E} \rightarrow B$  is a smooth Hermitian vector bundle. Construct its local product maps from the restrictions of nearby projections to a fixed fiber, without choosing a frame. Define the Berry connection by

$$\nabla^B s = P ds$$

for sections satisfying  $Ps = s$ , and prove

$$F_B = P(dP \wedge dP)P, \quad c_1(\mathcal{E}) = \left[ \frac{i}{2\pi} \text{Tr } P(dP \wedge dP) \right].$$

**Exercise 19.6** (Two-level Berry degree and gap protection). For the Pauli endomorphisms and the distinguished two-band Hamiltonian on the Brillouin torus  $B = T^2$ ,

$$H(k) = d_1(k)\sigma_1 + d_2(k)\sigma_2 + d_3(k)\sigma_3, \quad d(k) \neq 0,$$

let  $\hat{d} = d/|d|$  and let  $\mathcal{E}_-$  be the negative-energy line bundle. From  $P_- = (I - \hat{d} \cdot \sigma)/2$ , derive

$$\int_{T^2} c_1(\mathcal{E}_-) = \frac{1}{4\pi} \int_{T^2} \hat{d} \cdot (\partial_{k_1} \hat{d} \times \partial_{k_2} \hat{d}) dk_1 dk_2 = \deg(\hat{d}).$$

Show that this integer cannot change under a smooth deformation that keeps the spectral gap open.

**Exercise 19.7** (Topological  $K$ -theory and Bott periodicity). For a compact Hausdorff space  $X$ , let  $\text{Vect}_{\mathbb{C}}(X)$  be the commutative monoid of isomorphism classes of finite-rank complex vector bundles under direct sum. Define  $K^0(X)$  as its Grothendieck group: elements are formal differences  $[E] - [F]$ , with

$$[E] - [F] = [E'] - [F']$$

when there is a bundle  $G$  for which

$$E \oplus F' \oplus G \cong E' \oplus F \oplus G.$$

Use tensor product to make  $K^0(X)$  a ring. For based spaces define the smash product

$$X \wedge Y = X \times Y / (X \vee Y), \quad X \vee Y = X \times \{y_0\} \cup \{x_0\} \times Y,$$

and define the exterior product in  $K^0(X \times Y)$  by

$$a \boxtimes b = \text{pr}_X^* a \cdot \text{pr}_Y^* b.$$

Define

$$\widetilde{K}^0(X) = \ker(K^0(X) \rightarrow K^0(\{x_0\})).$$

For a closed based subspace  $A \subset Z$ , define

$$K^0(Z, A) = \widetilde{K}^0(Z/A).$$

For  $a = [E] - [E'] \in \widetilde{K}^0(X)$ , stabilize and choose an isomorphism  $E_{x_0} \cong E'_{x_0}$ , and do the same for a representative of  $b \in \widetilde{K}^0(Y)$ . Show that  $a \boxtimes b$ , together with the induced stable trivialization on  $X \vee Y$ , determines a relative class in

$$K^0(X \times Y, X \vee Y) \cong \widetilde{K}^0(X \wedge Y),$$

independent of the stabilizations and isomorphisms. Deduce the bilinear reduced exterior product

$$\widetilde{K}^0(X) \otimes \widetilde{K}^0(Y) \longrightarrow \widetilde{K}^0(X \wedge Y).$$

Define

$$\widetilde{K}^{-n}(X) = \widetilde{K}^0(\Sigma^n X) \quad (n \geq 0), \quad K^{-n}(X) = \widetilde{K}^{-n}(X_+), \quad K^1(X) = K^{-1}(X).$$

Here  $X_+ = X \sqcup \{*\}$  for an unbased compact space  $X$ .

Let  $BU(r)$  be the classifying space characterized by the natural correspondence between rank- $r$  complex bundles on a compact space and homotopy classes of maps to  $BU(r)$ . Let  $BU$  and  $U$  denote the stable classifying and unitary spaces, and write  $[X, Y]$  for unbased homotopy classes. Use as theorem input [1] the stable classification

$$K^0(X) \cong [X, \mathbb{Z} \times BU], \quad K^1(X) \cong [X, U],$$

and complex Bott periodicity

$$\Omega(\mathbb{Z} \times BU) \simeq U, \quad \Omega U \simeq \mathbb{Z} \times BU.$$

Let

$$\beta = [E_g] - [\mathbb{C}] \in \widetilde{K}^0(S^2)$$

be the Bott class from the Hopf bundle above. Prove that exterior product with  $\beta$  gives

$$\widetilde{K}^0(\Sigma^n X) \cong \widetilde{K}^0(\Sigma^n X \wedge S^2) \cong \widetilde{K}^0(\Sigma^{n+2} X) \quad (n \geq 0).$$

Use this isomorphism to extend reduced and unreduced (K)-theory two-periodically to all degrees. Deduce

$$\widetilde{K}^0(S^{2r}) \cong \mathbb{Z}, \quad \widetilde{K}^0(S^{2r+1}) = 0, \quad K^1(S^{2r}) = 0, \quad K^1(S^{2r+1}) \cong \mathbb{Z}.$$

Define a *CW complex* inductively by attaching disks along their boundary spheres, and call an inclusion  $A \hookrightarrow X$  a *cofibration* when it has the homotopy-extension property. Use as theorem input [1] the six-term exact sequence in reduced  $K$ -theory for a cofibration and the standard cell attachment  $S^1 \vee S^1 \rightarrow T^2 \rightarrow S^2$  to compute

$$K^0(T^2) \cong \mathbb{Z}^2, \quad K^1(T^2) \cong \mathbb{Z}^2.$$

When  $X$  is a compact smooth manifold, use as theorem input [7] that the Chern–Weil classes  $c_j(E)$  are the real images of integral cohomology classes. Use their rational polynomial relation with the Chern character and the Chern–Weil identities to prove that

$$\text{ch}([E] - [F]) = \text{ch}(E) - \text{ch}(F)$$

defines a natural ring homomorphism

$$\text{ch} : K^0(X) \longrightarrow H^{\text{even}}(X; \mathbb{Q}).$$

Use as theorem input [1] that these classes define the same natural transformation for finite CW complexes and that it becomes an isomorphism after tensoring with  $\mathbb{Q}$ .

For a continuous finite-rank spectral projection  $P(x)$  of a gapped Hamiltonian over a compact parameter space  $X$ , form its image bundle and class in  $K^0(X)$ . Prove that this class is unchanged by a gap-preserving homotopy and identify, for  $X = T^2$ , its rank and first-Chern components with the two integers in the displayed computation.

**Exercise 19.8** (Spectral flow and Toeplitz index). Let  $A_t$ ,  $0 \leq t \leq 1$ , be a norm-continuous path of bounded self-adjoint Fredholm operators with invertible endpoints. Choose a partition  $0 = t_0 < \dots < t_N = 1$  and numbers  $\varepsilon_j > 0$  so that the finite-rank spectral projections

$$\mathbf{1}_{[-\varepsilon_j, \varepsilon_j]}(A_t)$$

vary continuously for  $t \in [t_{j-1}, t_j]$ . Define the *spectral flow* by

$$\text{sf}(A_t) = \sum_{j=1}^N \left( \text{rank } \mathbf{1}_{[0, \varepsilon_j]}(A_{t_j}) - \text{rank } \mathbf{1}_{[0, \varepsilon_j]}(A_{t_{j-1}}) \right).$$

Prove independence of the choices, additivity under concatenation, and invariance under endpoint-fixed homotopies through such paths.

Let  $W$  be a finite-dimensional Hermitian vector space, let  $u : S^1 \rightarrow U(W)$  be smooth, and define

$$\text{wind}(\det u) = \frac{1}{2\pi i} \int_{S^1} (\det u)^{-1} d(\det u).$$

Let  $H^2(S^1; W) \subset L^2(S^1; W)$  be the closed span of the nonnegative Fourier modes, let  $P_+$  be its orthogonal projection, and define the Toeplitz operator

$$T_u = P_+ M_u|_{H^2(S^1; W)}.$$

Use as theorem input [1] that  $T_u$  is Fredholm and that homotopies of  $u$  produce norm-continuous Fredholm paths. For  $W = \mathbb{C} \oplus W'$  compute the index when  $u(z) = z^n \oplus I_{W'}$  from the unilateral shift. Use as theorem input [1] that  $SU(r)$  is simply connected for  $r \geq 2$  and use the determinant fibration

$$SU(r) \longrightarrow U(r) \xrightarrow{\det} U(1)$$

to prove that determinant induces  $\pi_1(U(r)) \cong \mathbb{Z}$ . Combine this with index homotopy invariance and stabilization to prove

$$\text{ind}_{\text{an}}(T_u) = -\text{wind}(\det u).$$

Define the odd  $K$ -theory pairing on the circle by

$$\langle [u], [S^1] \rangle = \text{wind}(\det u) = -\text{ind}_{\text{an}}(T_u).$$

For a loop  $A_t$  of self-adjoint Fredholm operators based at an invertible operator, define its suspended operator on periodic paths by

$$\mathcal{D}_A = \frac{d}{dt} + A_t.$$

Use as theorem input the suspension theorem [1]

$$\text{ind}_{\text{an}}(\mathcal{D}_A) = \text{sf}(A_t)$$

and identify both integers with the corresponding  $K^1$  pairing.

**Exercise 19.9** (Thom classes and the topological index). For a locally compact space  $Y$ , define compactly supported  $K$ -theory by

$$K_c^j(Y) = \widetilde{K}^j(Y^+), \quad j \in \mathbb{Z}/2,$$

where  $Y^+$  is the one-point compactification. For a complex vector bundle  $E \rightarrow X$ , define its Thom space by

$$\text{Th}(E) = D(E)/S(E),$$

where  $D(E)$  and  $S(E)$  are its unit disk and sphere bundles for any Hermitian metric. Define a *Thom class*  $u_E \in K_c^0(E)$  to be a class whose restriction to each fiber is the Bott generator in  $K_c^0(\mathbb{C}^{\text{rank } E})$ . Use Bott periodicity as theorem input [1] for the  $K$ -theoretic Thom isomorphism

$$K^j(X) \longrightarrow K_c^j(E), \quad a \longmapsto \pi^* a \cdot u_E.$$

For an elliptic operator  $D : \Gamma(E_0) \rightarrow \Gamma(E_1)$  on a closed manifold  $M$ , its principal symbol is an isomorphism away from the zero section. Define its *symbol class*

$$[\sigma(D)] \in K_c^0(T^*M)$$

as the stable-homotopy class of the triple  $(\pi^*E_0, \pi^*E_1, \sigma(D))$ , where in general a compactly supported class is represented by two bundles together with an isomorphism outside a compact set. Verify compatibility of this description with direct sum, homotopy, and stabilization.

Use as theorem input [7] the Whitney embedding and tubular-neighborhood theorems and choose an embedding  $M \hookrightarrow \mathbb{R}^N$ . Show that the Euclidean orthogonal splitting induces an embedding

$$T^*M \hookrightarrow T^*\mathbb{R}^N \cong \mathbb{R}^{2N}$$

whose normal bundle is the pullback of  $\nu \oplus \nu$ , where  $\nu$  is the normal bundle of  $M$ , and give  $\nu \oplus \nu$  the complex structure  $J(v, w) = (-w, v)$ . Use its Thom class, extension by zero, and Bott periodicity to define

$$\text{ind}_{\text{top}} : K_c^0(T^*M) \longrightarrow K_c^0(\mathbb{R}^{2N}) \cong \mathbb{Z}.$$

Use as theorem input [1] isotopy invariance and stabilization of embeddings to prove that this map is independent of all choices.

**Exercise 19.10** (Atiyah–Singer index theorem). Use as theorem input the splitting principle [7]: after a pullback injective on rational cohomology, a complex vector bundle  $E$  splits as a sum of line bundles. If their first Chern classes are  $x_j$ , call the  $x_j$  the *formal Chern roots* of  $E$ , so  $c(E) = \prod_j (1 + x_j)$ . Define

$$\text{Td}(E) = \prod_j \frac{x_j}{1 - e^{-x_j}},$$

and for an oriented real bundle  $V$  whose complexification has formal roots  $\{\pm x_j\}$  define

$$\hat{A}(V) = \prod_j \frac{x_j/2}{\sinh(x_j/2)}.$$

Use the splitting principle to interpret these symmetric power series as characteristic classes. Use the  $K$ -theoretic Chern character constructed above and prove the multiplicativity of the two displayed classes under direct sum.



A *spin structure* on an oriented Riemannian  $n$ -manifold is a lift of its oriented orthonormal frame bundle through  $\text{Spin}(n) \rightarrow \text{SO}(n)$ . Let  $S = S^+ \oplus S^-$  be the associated complex spinor bundle in even dimension, characterized as a module over the complex Clifford bundle generated by  $v$  with  $v^2 = -\|v\|^2$ . For a Hermitian bundle  $E$  with connection, define the twisted chiral Dirac operator

$$\mathcal{D}_E^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E)$$

by composing the tensor-product connection with Clifford multiplication. Compute its principal symbol and prove ellipticity.

Define

$$\text{Spin}^c(n) = (\text{Spin}(n) \times U(1)) / \{(1, 1), (-1, -1)\}.$$

A *spin<sup>c</sup> structure* is a lift of the oriented frame bundle through  $\text{Spin}^c(n) \rightarrow \text{SO}(n)$ . Use as theorem input [30] that a Hermitian complex manifold has a canonical *spin<sup>c</sup>* structure with spinor bundle  $\Lambda^{0,*}T^*M$ . Define its chiral Dolbeault–Dirac operator

$$\mathcal{D}_{\bar{\partial},E}^+ = \sqrt{2}(\bar{\partial}_E + \bar{\partial}_E^*) : \Omega^{0,\text{even}}(M; E) \longrightarrow \Omega^{0,\text{odd}}(M; E)$$

and compute its principal symbol.

Use as theorem input the Atiyah–Singer index theorem [2]:

$$\text{ind}_{\text{an}}(D) = \text{ind}_{\text{top}}[\sigma(D)]$$

for every elliptic differential operator on a closed manifold. Use as theorem input [2] its cohomological consequences

$$\text{ind}_{\text{an}}(\mathcal{D}_{\bar{\partial},E}^+) = \int_M \text{ch}(E) \text{Td}(T^{1,0}M)$$

for a compact complex manifold and

$$\text{ind}_{\text{an}}(\mathcal{D}_E^+) = \int_M \hat{A}(TM) \text{ch}(E)$$

for a closed spin manifold.

For a compact Riemann surface  $C$ , use  $T^{1,0}C \cong K_C^\vee$  and the Riemann–Roch computation of  $\deg K_C$ , and identify the chiral Dolbeault–Dirac index with that of  $\bar{\partial}_L : \Omega^{0,0}(C; L) \rightarrow \Omega^{0,1}(C; L)$  to derive

$$\text{ind}_{\text{an}}(\bar{\partial}_L) = \int_C \left( c_1(L) + \frac{1}{2}c_1(T^{1,0}C) \right) = \deg L + 1 - g.$$

For a closed spin surface and a line bundle  $L$  of degree  $N$ , derive

$$\text{ind}_{\text{an}}(\mathcal{D}_L^+) = N$$

and identify this integer as the difference of the two chiral zero-mode counts.

**Exercise 19.11** (Monopole harmonics and spherical Landau levels). Let  $S_R^2$  be the round sphere of radius  $R$ , let  $N \in \mathbb{Z}$ , and let  $L_N \rightarrow S_R^2$  be a Hermitian line bundle with unitary connection of constant curvature

$$F_\nabla = -\frac{iN}{2R^2} dA, \quad \frac{i}{2\pi} \int_{S_R^2} F_\nabla = N.$$

For a particle of charge  $e$ , impose the convention  $F_{\nabla} = -ieB dA/\hbar$  and deduce the magnetic-flux quantization

$$\Phi_B = \int_{S_R^2} B dA = \frac{Nh}{e}.$$

On the northern and southern polar charts use the connection forms

$$A_N = -\frac{iN}{2}(1 - \cos \theta) d\phi, \quad A_S = \frac{iN}{2}(1 + \cos \theta) d\phi,$$

and determine their transition function on the overlap.

For  $\alpha, \beta > -1$ , define the Jacobi polynomial  $P_n^{(\alpha, \beta)}(x)$  as the degree- $n$  polynomial solution of

$$(1 - x^2)y'' + [\beta - \alpha - (\alpha + \beta + 2)x]y' + n(n + \alpha + \beta + 1)y = 0$$

normalized by  $P_n^{(\alpha, \beta)}(1) = \binom{n + \alpha}{n}$ . Prove its orthogonality for distinct degrees with respect to  $(1 - x)^\alpha(1 + x)^\beta dx$  on  $[-1, 1]$ .

Put  $q = |N|/2$ . Separate the covariant Laplacian in the two monopole gauges, extract the endpoint powers imposed by regularity and the transition function, and reduce the polar equation to the Jacobi equation. Call the normalized global eigensections obtained in this way the *monopole harmonics*. Identify sections of  $L_N$  with functions  $F : SU(2) \rightarrow \mathbb{C}$  satisfying

$$F(gh) = \chi_N(h)^{-1}F(g)$$

under the identification  $S^2 \cong SU(2)/U(1)$ , where

$$h_\theta = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \quad \chi_N(h_\theta) = e^{iN\theta}.$$

Use Peter–Weyl theory to prove their orthogonality and completeness and to show that

$$\ell = q + n, \quad n = 0, 1, 2, \dots, \quad \dim \mathcal{H}_\ell = 2\ell + 1 = |N| + 2n + 1.$$

For the scalar magnetic Hamiltonian

$$H_N = \frac{\hbar^2}{2m} \nabla^* \nabla,$$

derive

$$E_n = \frac{\hbar^2}{2mR^2} \left( (q + n)(q + n + 1) - q^2 \right), \quad E_{n+1} - E_n = \frac{\hbar^2}{mR^2} (n + q + 1).$$

For  $N \geq 0$ , equip  $S^2 \cong \mathbb{P}^1$  with its complex structure, identify  $L_N$  with  $\mathcal{O}(N)$ , and define the spin<sup>c</sup> Dolbeault–Pauli Hamiltonian

$$H_+ = \frac{\hbar^2}{m} \bar{\partial}_{\nabla}^* \bar{\partial}_{\nabla}.$$

Use Riemann–Roch and the Dolbeault index theorem to prove

$$\dim \ker H_+ = N + 1,$$

identify these zero modes with the lowest monopole-harmonic multiplet, and predict the complete level degeneracies and spacings above. State the uniform-field, round-sphere, fixed-charge, spin<sup>c</sup> coupling, noninteracting, and spectral-resolution assumptions under which spherical Landau-level spectroscopy tests the Jacobi-polynomial and index predictions.

**Exercise 19.12** (Family index and quantized Hall transport). Let  $B$  be compact and let  $T_b : H_0 \rightarrow H_1$  be a norm-continuous family of Fredholm operators. Use as theorem input [3] that there is a finite-dimensional subspace  $V \subset H_1$  for which

$$H_0 \oplus V \longrightarrow H_1, \quad (u, v) \longmapsto T_b u + v$$

is surjective for every  $b \in B$ . Prove that its kernels form a vector bundle  $\mathcal{K}_T \rightarrow B$  and define the *family index* by

$$\text{Ind}(T) = [\mathcal{K}_T] - [B \times V] \in K^0(B).$$

Prove independence of  $V$ , homotopy invariance, and additivity under direct sum. Show that the construction is local on  $B$  and therefore extends to continuous Fredholm morphisms between Hilbert bundles.

Let  $E_\tau$  carry a degree- $N > 0$  magnetic line bundle  $L_N$ , and let

$$\widehat{E}_\tau = \text{Hom}(H_1(E_\tau; \mathbb{Z}), U(1))$$

be the torus of boundary-condition twists. Let  $\mathcal{P} \rightarrow E_\tau \times \widehat{E}_\tau$  be the normalized Poincaré line bundle, whose restriction to  $E_\tau \times \{\chi\}$  is the flat line bundle  $F_\chi$ . Orient  $\widehat{E}_\tau$  so that

$$\frac{1}{2} \int_{E_\tau} c_1(\mathcal{P})^2$$

is its positive integral generator. Define the lowest-Landau-level bundle and filled-level line by

$$\mathcal{H}_N|_\chi = H^0(E_\tau, L_N \otimes F_\chi), \quad \mathcal{D}_N = \bigwedge^N \mathcal{H}_N.$$

Use the theta automorphy laws to realize the fibers by level- $N$  theta functions and Riemann–Roch to prove that  $\mathcal{H}_N$  has rank  $N$ . Show that the Sobolev completions of  $\bar{\partial}_{L_N \otimes F_\chi}$  form a Fredholm family and that Riemann–Roch identifies its family index with  $[\mathcal{H}_N]$ .

Use as theorem input the families Atiyah–Singer formula [3]: for a holomorphic submersion  $p : X \rightarrow B$  with compact fibers and a holomorphic bundle  $\mathcal{L} \rightarrow X$ ,

$$\text{ch Ind}(\bar{\partial}_{\mathcal{L}}) = \int_{X/B} \text{ch}(\mathcal{L}) \text{Td}(T^{1,0}(X/B)),$$

where the right side is integration along the compact complex fibers. Apply it to  $X = E_\tau \times \widehat{E}_\tau$ ,  $B = \widehat{E}_\tau$ , and  $\mathcal{L} = \text{pr}_1^* L_N \otimes \mathcal{P}$ . Use the triviality of the elliptic-curve tangent bundle to prove

$$\text{ch}(\mathcal{H}_N) = \int_{E_\tau} \exp(c_1(L_N) + c_1(\mathcal{P})).$$

Extract its degree-zero and degree-two parts to prove

$$\text{rank } \mathcal{H}_N = N, \quad \int_{\widehat{E}_\tau} c_1(\mathcal{D}_N) = 1.$$

Use as theorem input [42] the gapped noninteracting Kubo–Chern formula and the adiabatic theorem. Deduce for one completely filled Landau level

$$\sigma_{xy} = \frac{e^2}{h}, \quad Q_{\text{pump}} = e$$

for a unit winding of a boundary twist. Express the Berry connection of  $\mathcal{D}_N$  through the normalized theta sections and identify the same integer from their automorphy. Identify the pumped charge with the spectral-flow and  $K^1$ -pairing integer. State separately the spectral-gap, adiabatic, noninteracting, filled-level, mobility, contact, temperature, and finite-size assumptions under which Hall-voltage and charge-pump measurements test these predictions.

## 20 Cobordism

**Exercise 20.1** (Oriented, spin, and complex bordism rings). For an oriented manifold  $M$ , write  $-M$  for the same manifold with the opposite orientation. An *oriented bordism* from a closed oriented  $n$ -manifold  $M_0$  to a closed oriented  $n$ -manifold  $M_1$  is a compact oriented  $(n+1)$ -manifold  $W$  with an orientation-preserving identification

$$\partial W \cong (-M_0) \sqcup M_1,$$

product neighborhoods of its boundary components, and the boundary orientation determined by the outward-normal-first convention. Call  $M_0$  and  $M_1$  *oriented bordant* when such a  $W$  exists.

Prove that oriented bordism is an equivalence relation. Let  $\Omega_n^{SO}$  be the set of its equivalence classes. Prove that disjoint union makes  $\Omega_n^{SO}$  an abelian group with zero the empty manifold and inverse  $[M] \mapsto [-M]$ . Use the product orientation to make

$$\Omega_*^{SO} = \bigoplus_{n \geq 0} \Omega_n^{SO}$$

a graded-commutative ring with unit the positively oriented point. Prove

$$\Omega_0^{SO} \cong \mathbb{Z}, \quad \Omega_1^{SO} = 0, \quad \Omega_2^{SO} = 0$$

by signed point count and disks, and by using as theorem input the classification of closed oriented surfaces and constructing oriented handlebodies.

For a spin manifold with boundary, use the outward unit normal to induce a spin structure on its boundary. Define *spin bordism* by requiring the boundary identification to preserve these structures, and construct the graded ring  $\Omega_*^{\text{Spin}}$ . A *stable complex structure* on  $M$  is a complex-vector-bundle structure on  $TM \oplus \mathbb{R}^r$  for some  $r$ , considered modulo addition of trivial complex line bundles and homotopy. Define stable complex bordism and its ring  $\Omega_*^U$ . Prove that products inherit the corresponding structures and construct the forgetful homomorphisms

$$\Omega_*^{\text{Spin}} \longrightarrow \Omega_*^{SO}, \quad \Omega_*^U \longrightarrow \Omega_*^{SO}.$$

**Exercise 20.2** (Pontryagin–Thom collapse and Thom spectra). For  $G = SO, \text{Spin}$ , let  $BG(k)$  be the classifying space for rank- $k$  real vector bundles with  $G$ -structure and let  $\gamma_k^G \rightarrow BG(k)$  be the universal bundle. Put

$$MG(k) = \text{Th}(\gamma_k^G), \quad \pi_n(MG) = \varinjlim_k \pi_{n+k}(MG(k)),$$

where stabilization of the universal bundles induces the structure maps. For complex rank- $k$  bundles, put

$$MU(k) = \text{Th}(\gamma_k^U), \quad \pi_n(MU) = \varinjlim_k \pi_{n+2k}(MU(k)).$$

Call the resulting sequences of Thom spaces the spectra  $MSO$ ,  $M\text{Spin}$ , and  $MU$ .

Choose an embedding  $M^n \hookrightarrow \mathbb{R}^{n+k}$  of a closed  $G$ -manifold, give its normal bundle the stable  $G$ -structure complementary to that on  $TM$ , and let its classifying map pull back  $\gamma_k^G$  to the normal bundle. Collapse the complement of a tubular neighborhood to construct

$$S^{n+k} \longrightarrow \text{Th}(\nu_M) \longrightarrow MG(k).$$

Prove that isotopy, stabilization, and a  $G$ -bordism do not change the resulting stable homotopy class. Use as theorem input the transversality and regular-value construction in the Pontryagin–Thom theorem [41] to prove that collapse induces ring isomorphisms

$$\Omega_n^{SO} \cong \pi_n(MSO), \quad \Omega_n^{\text{Spin}} \cong \pi_n(M \text{Spin}), \quad \Omega_n^U \cong \pi_n(MU).$$

A *stable framing* of  $M$  is a trivialization of its stable normal bundle. Let  $\Omega_n^{\text{fr}}$  be framed bordism and define the  $n$ th stable stem by

$$\pi_n^S = \varinjlim_k [S^{n+k}, S^k]_*.$$

Show that a framing turns the collapse map into a map  $S^{n+k} \rightarrow S^k$  and recover

$$\Omega_n^{\text{fr}} \cong \pi_n^S.$$

**Exercise 20.3** (Pontryagin numbers and the signature genus). For a real vector bundle  $V$ , define its Pontryagin classes by

$$p_j(V) = (-1)^j c_{2j}(V \otimes_{\mathbb{R}} \mathbb{C}) \in H_{\text{dR}}^{4j}(M).$$

For a closed oriented  $4k$ -manifold and a partition  $I = (i_1, \dots, i_r)$  of  $k$ , define the Pontryagin number

$$p_I[M] = \int_M p_{i_1}(TM) \wedge \dots \wedge p_{i_r}(TM).$$

If  $M = \partial W$  as an oriented manifold, use

$$TW|_M \cong TM \oplus \underline{\mathbb{R}}$$

and choose a connection that is a product near the boundary. Use Stokes' theorem for its Chern–Weil representatives to prove  $p_I[M] = 0$ . Deduce that every Pontryagin number defines a homomorphism  $\Omega_{4k}^{SO} \rightarrow \mathbb{Q}$ . Use as theorem input Thom's rational bordism theorem [41]: two oriented manifolds determine the same class in  $\Omega_*^{SO} \otimes \mathbb{Q}$  exactly when all their Pontryagin numbers agree.

For a closed oriented  $4k$ -manifold, use as theorem input Poincaré duality [7] to prove that

$$([\alpha], [\beta]) \mapsto \int_M \alpha \wedge \beta$$

is a nondegenerate symmetric form on  $H_{\text{dR}}^{2k}(M)$ , and define  $\text{sign}(M)$  to be its signature. If the formal Pontryagin roots satisfy  $p(TM) = \prod_j (1 + x_j^2)$ , define

$$L(TM) = \prod_j \frac{x_j}{\tanh x_j}.$$

Use as theorem input the Hirzebruch signature theorem [25]

$$\text{sign}(M) = \int_M L(TM)$$

and deduce that the signature is a ring homomorphism  $\Omega_*^{SO} \rightarrow \mathbb{Z}$ . In dimension four derive

$$\text{sign}(M) = \frac{1}{3} \int_M p_1(TM).$$

For  $h = c_1(\mathcal{O}(1))$  on  $\mathbb{CP}^2$ , use as theorem input the Euler sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}(1)^{\oplus 3} \longrightarrow T\mathbb{CP}^2 \longrightarrow 0$$

and prove

$$c(T\mathbb{CP}^2) = (1 + h)^3, \quad \int_{\mathbb{CP}^2} p_1 = 3, \quad \text{sign}(\mathbb{CP}^2) = 1,$$

and conclude that  $[\mathbb{CP}^2]$  has infinite order in  $\Omega_4^{SO}$ .

**Exercise 20.4** (Dirac and Dolbeault bordism genera). For a commutative ring  $R$ , a *genus* is a ring homomorphism from a bordism ring to  $R$ . Given a formal power series  $Q(x) \in R[[x]]$  with  $Q(0) = 1$ , use the splitting principle to construct the multiplicative characteristic class  $\prod_j Q(x_j)$  from the appropriate formal roots. Prove that its top-degree characteristic number is additive under disjoint union and multiplicative under products, and determine the condition under which stabilization leaves it unchanged.

For a closed spin  $4k$ -manifold, define

$$\hat{A}[M] = \int_M \hat{A}(TM).$$

Use the Atiyah–Singer theorem and the product formula for spinor bundles to prove

$$\hat{A}[M] = \text{ind}_{\text{an}}(\mathcal{D}_M^+) \in \mathbb{Z}$$

and that  $\hat{A}$  is a genus  $\Omega_*^{\text{Spin}} \rightarrow \mathbb{Z}$ . Prove its bordism invariance directly from the stable tangent bundle of a boundary and Stokes' theorem.

For a closed stable complex manifold  $X$ , define

$$\text{Td}[X] = \int_X \text{Td}(TX).$$

Use as theorem input the canonical  $\text{spin}^c$  Dirac operator associated to a stable complex structure [30], and apply the index theorem to prove that  $\text{Td}[X]$  is integral and defines a genus  $\Omega_*^U \rightarrow \mathbb{Z}$ . When  $X$  is a compact complex manifold, identify it with

$$\text{Td}[X] = \sum_q (-1)^q \dim H^q(X, \mathcal{O}_X).$$

From the degree-zero and degree-two terms of the Todd class, prove

$$\text{Td}[\text{pt}] = 1, \quad \text{Td}[C] = 1 - g$$

for a compact Riemann surface  $C$  of genus  $g$ .

**Exercise 20.5** (Atiyah–Patodi–Singer index and spectral eta functions). Let  $W$  be a compact even-dimensional spin Riemannian manifold whose metric is a product near its boundary  $M$ . Let  $E \rightarrow W$  be a Hermitian bundle with unitary connection of product type there. Write  $\mathcal{D}_{W,E}^+$  for the chiral Dirac operator and  $A_{M,E}$  for its self-adjoint tangential boundary operator. For  $\text{Re } s$  sufficiently large, define the eta function

$$\eta_{A_{M,E}}(s) = \sum_{\lambda \neq 0} \text{sign}(\lambda) |\lambda|^{-s},$$

where the spectral values are repeated with multiplicity. Use as theorem input [4] that this function has a meromorphic continuation regular at  $s = 0$ .

Let  $P_{\geq 0}(A_{M,E})$  be the spectral projection onto the nonnegative spectrum and impose the Atiyah–Patodi–Singer boundary condition

$$P_{\geq 0}(A_{M,E})(\psi|_M) = 0.$$

Use as theorem input the Atiyah–Patodi–Singer index theorem [4]

$$\text{ind}_{\text{APS}}(\mathcal{D}_{W,E}^+) = \int_W \hat{A}(TW) \text{ch}(E) - \frac{\eta_{A_{M,E}}(0) + h_{M,E}}{2}, \quad h_{M,E} = \dim \ker A_{M,E}.$$

Define the reduced eta invariant by

$$\bar{\eta}(A) = \frac{\eta_A(0) + \dim \ker A}{2} \quad \text{in } \mathbb{R}/\mathbb{Z}.$$

Deduce

$$\bar{\eta}(A_{M,E}) = \int_W \hat{A}(TW) \text{ch}(E) \pmod{\mathbb{Z}}.$$

Prove

$$\bar{\eta}(-A) = -\bar{\eta}(A) \pmod{\mathbb{Z}}$$

from spectral symmetry under orientation reversal. When two product-type bordisms are glued along oppositely oriented copies of the same boundary, use additivity of the characteristic integrals to show that their reduced eta corrections cancel modulo  $\mathbb{Z}$ . Recover the closed Atiyah–Singer formula when the boundary is empty.

**Exercise 20.6** (Hurwitz zeta functions and flux spectral asymmetry). For  $0 < a < 1$ , let

$$A_a = -i \frac{d}{d\theta} + a$$

act on periodic functions on  $\mathbb{R}/2\pi\mathbb{Z}$ . Prove that its spectral values are  $n + a$ ,  $n \in \mathbb{Z}$ . Define the Hurwitz zeta function for  $\text{Re } s > 1$  by

$$\zeta(s, a) = \sum_{n \geq 0} (n + a)^{-s}.$$

Separate the positive and negative spectral values to prove

$$\eta_{A_a}(s) = \zeta(s, a) - \zeta(s, 1 - a).$$

Use as theorem input the analytic continuation identity

$$\zeta(0, a) = \frac{1}{2} - a$$

to derive

$$\eta_{A_a}(0) = 1 - 2a, \quad \bar{\eta}(A_a) = \frac{1}{2} - a \pmod{\mathbb{Z}}.$$

For a single chiral massless charged Dirac channel on a coherent ring of circumference  $\ell$  threaded by flux  $\Phi$ , put

$$a = \frac{\Phi}{\Phi_0} \pmod{\mathbb{Z}}, \quad \Phi_0 = \frac{h}{e}, \quad E_n = \frac{2\pi\hbar v}{\ell}(n + a).$$

Define the eta-regularized vacuum charge by

$$Q_{\text{vac}} = -\frac{e}{2}\eta_{A_a}(0).$$

Predict

$$\frac{Q_{\text{vac}}}{e} = a - \frac{1}{2} \pmod{\mathbb{Z}}$$

and prove that insertion of one flux quantum produces unit spectral flow. State the single-channel, linear-dispersion, coherent-ring, charge regularization, flux-calibration, temperature, and level-resolution assumptions under which charge sensing and flux-resolved spectroscopy test the prediction.

**Exercise 20.7** (Jacobi theta functions and the K3 elliptic genus). For a closed complex manifold  $X$  of complex dimension  $d$  with formal Chern roots  $x_1, \dots, x_d$ , define its two-variable elliptic genus by

$$\text{Ell}_X(\tau, z) = \int_X \prod_{j=1}^d x_j \frac{\vartheta_1\left(\frac{x_j}{2\pi i} - z \mid \tau\right)}{\vartheta_1\left(\frac{x_j}{2\pi i} \mid \tau\right)}, \quad \text{Im } \tau > 0.$$

Use as theorem input [46] that this characteristic series defines a stable complex bordism genus with values in holomorphic functions of  $(\tau, z)$ .

Define

$$\vartheta_2(z \mid \tau) = \vartheta \begin{bmatrix} 1 \\ 2 \end{bmatrix} (z \mid \tau), \quad \vartheta_3(z \mid \tau) = \vartheta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (z \mid \tau), \quad \vartheta_4(z \mid \tau) = \vartheta \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} (z \mid \tau).$$

A *K3 surface* is a compact connected complex surface  $X$  with  $K_X \cong \mathcal{O}_X$  and  $H^1(X, \mathcal{O}_X) = 0$ . Use Serre duality and the Dolbeault index formula to prove

$$\text{Td}[X] = 2, \quad \int_X c_2(TX) = 24.$$

Use the Chern-root formula, the theta addition identities, and the K3 conditions  $c_1 = 0$ ,  $c_2[X] = 24$  to derive the K3 identity from [14]

$$\text{Ell}_{K3}(\tau, z) = 8 \left[ \left( \frac{\vartheta_2(z \mid \tau)}{\vartheta_2(0 \mid \tau)} \right)^2 + \left( \frac{\vartheta_3(z \mid \tau)}{\vartheta_3(0 \mid \tau)} \right)^2 + \left( \frac{\vartheta_4(z \mid \tau)}{\vartheta_4(0 \mid \tau)} \right)^2 \right].$$

Put  $q = e^{2\pi i \tau}$  and  $y = e^{2\pi i z}$ . Derive

$$\text{Ell}_{K3}(\tau, z) = 2y + 20 + 2y^{-1} + O(q), \quad \text{Ell}_{K3}(\tau, 0) = 24.$$

For an  $\mathcal{N} = (4, 4)$  supersymmetric sigma model with K3 target, use as theorem input [46, 14] the Ramond-sector identity

$$\text{Ell}_{K3}(\tau, z) = \text{Tr}_{\mathcal{H}_{RR}} \left( (-1)^{F_L + F_R} y^{J_0} q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24} \right)$$

and prove that right-moving supersymmetry cancels every non-ground-state pair, making the trace independent of  $\bar{q}$ . Predict 24 protected Ramond ground states with charge-resolved multiplicities 2, 20, 2 at charges 1, 0,  $-1$ . State the exact-supersymmetry, Ramond-boundary, charge-normalization, unbroken-supersymmetry, continuum-scaling, finite-volume, temperature, and symmetry-resolved spectroscopy assumptions under which a programmable quantum system tests these protected counts.



**Exercise 20.8** (Bordism characters and invertible topological phases). Let  $G$  be a finite internal symmetry group and let  $H = SO$  or  $\text{Spin}$ . An  $H$ -bordism over  $BG$  from  $(M_0, A_0)$  to  $(M_1, A_1)$  is an  $H$ -bordism  $W$  together with a map  $A : W \rightarrow BG$  whose restrictions are  $A_0$  and  $A_1$ . Let  $\Omega_d^H(BG)$  be the resulting bordism group of closed  $d$ -manifolds equipped with maps to  $BG$ . Prove that disjoint union defines its addition and that forgetting the map to  $BG$  gives a homomorphism

$$\Omega_d^H(BG) \longrightarrow \Omega_d^H.$$

Define a  $U(1)$ -valued bordism character to be a homomorphism

$$Z : \Omega_d^H(BG) \longrightarrow U(1).$$

Write its value on  $(M, A)$  as  $Z(M, A)$ . Prove

$$Z(\emptyset) = 1, \quad Z((-M), A) = \overline{Z(M, A)}, \quad Z(M \sqcup N, A \sqcup B) = Z(M, A)Z(N, B).$$

Define stacking by pointwise multiplication of characters and prove that complex conjugation supplies the stacking inverse. Call a gapped  $G$ -symmetric phase *invertible* when, after adjoining decoupled product degrees of freedom, it has a stacking inverse, and call two such phases equivalent when a uniformly gapped  $G$ -symmetric path joins their Hamiltonians. Use as theorem input [20] that, under the reflection-positivity and low-energy topological-field-theory hypotheses, deformation classes of invertible phases are computed from the associated Thom spectrum and their closed-manifold partition functions give bordism characters. Distinguish this full stable-homotopy classification from the information retained by partition functions on closed manifolds.

A *normalized*  $U(1)$ -valued 2-cocycle on  $G$  is a function  $\omega : G \times G \rightarrow U(1)$  satisfying

$$\omega(e, g) = \omega(g, e) = 1, \quad \omega(g, h)\omega(gh, k) = \omega(h, k)\omega(g, hk).$$

For  $\beta : G \rightarrow U(1)$  with  $\beta(e) = 1$ , define

$$(\delta\beta)(g, h) = \frac{\beta(g)\beta(h)}{\beta(gh)},$$

and define  $H^2(G, U(1))$  to be normalized cocycles modulo multiplication by such coboundaries. Prove that operators satisfying

$$U_g U_h = \omega(g, h) U_{gh}, \quad U_e = 1,$$

associate precisely because  $\omega$  is a cocycle and that rephasing  $U_g$  changes  $\omega$  by a coboundary. Use as theorem input [9, 35] that one-dimensional bosonic gapped phases with a unique bulk ground state, unbroken onsite symmetry  $G$ , and finite correlation length are classified, under symmetric gapped-path equivalence, by  $H^2(G, U(1))$ , with boundary degrees of freedom carrying the corresponding projective representation.

For commuting  $g, h \in G$ , define the twisted torus response

$$Z_\omega(T^2; g, h) = \frac{\omega(g, h)}{\omega(h, g)}.$$

Prove that it depends only on the cohomology class of  $\omega$ . For  $G = \mathbb{Z}_2 \times \mathbb{Z}_2$ , written additively as pairs  $(r, s)$ , prove that

$$\omega((r, s), (r', s')) = (-1)^{sr'}.$$

is a normalized cocycle and that the two generating holonomies give

$$Z_\omega(T^2; (1, 0), (0, 1)) = -1.$$

Interpret the phase  $\pi$  as the quantized interference response of the nontrivial bosonic symmetry-protected phase, stating separately the onsite-symmetry, gap, unique-bulk-ground-state, flat-background-field, coherence, and phase-resolution assumptions.

**Exercise 20.9** (Spin bordism and the Majorana chain). For a closed spin surface  $(\Sigma, \sigma)$ , use as theorem input that  $\sigma$  determines a quadratic refinement

$$q_\sigma : H_1(\Sigma; \mathbb{Z}_2) \longrightarrow \mathbb{Z}_2, \quad q_\sigma(x + y) = q_\sigma(x) + q_\sigma(y) + x \cdot y,$$

of the mod-2 intersection pairing. Define its Arf invariant intrinsically by the Gauss sum

$$(-1)^{\text{Arf}(\sigma)} = 2^{-g} \sum_{x \in H_1(\Sigma; \mathbb{Z}_2)} (-1)^{q_\sigma(x)},$$

where  $g$  is the genus. Use as theorem input

$$\Omega_2^{\text{Spin}} \cong \mathbb{Z}_2$$

with Arf as generator of the character group. Prove that

$$Z_{\text{Arf}}(\Sigma, \sigma) = (-1)^{\text{Arf}(\sigma)}$$

is multiplicative under disjoint union, is unchanged by spin bordism, and squares to the trivial response under stacking.

For fermionic annihilation operators  $c_1, \dots, c_N$  satisfying

$$\{c_j, c_k\} = 0, \quad \{c_j, c_k^*\} = \delta_{jk},$$

define the Majorana operators

$$\gamma_{j,A} = c_j + c_j^*, \quad \gamma_{j,B} = -i(c_j - c_j^*).$$

Derive

$$\gamma_{j,\alpha}^* = \gamma_{j,\alpha}, \quad \{\gamma_{j,\alpha}, \gamma_{k,\beta}\} = 2\delta_{jk}\delta_{\alpha\beta}.$$

On an open chain define the distinguished Kitaev Hamiltonian from [27]

$$H_{\text{top}} = -J \sum_{j=1}^{N-1} i\gamma_{j,B}\gamma_{j+1,A}, \quad J > 0.$$

Prove that its  $N - 1$  bond operators commute, square to the identity, and leave  $\gamma_{1,A}$  and  $\gamma_{N,B}$  unpaired. Deduce the spectral values

$$E_m = -J(N - 1) + 2Jm, \quad 0 \leq m \leq N - 1,$$

with multiplicity  $2\binom{N-1}{m}$ , a bulk gap  $2J$ , and two ground states of opposite total fermion parity. Compare with

$$H_{\text{triv}} = -J \sum_{j=1}^N i\gamma_{j,A}\gamma_{j,B},$$

which has a unique ground state. Stack two copies of  $H_{\text{top}}$  and construct parity-even bilinear couplings between the two Majoranas at each end that remove the edge degeneracy without closing the bulk gap. Relate this  $\mathbb{Z}_2$  stacking law to  $Z_{\text{Arf}}$ .

Use as theorem input the locality estimate that a uniformly gapped parity-preserving local deformation splits the two edge levels by  $O(e^{-N/\xi})$  for some correlation length  $\xi$ . Predict an exponentially split pair of parity-resolved subgap levels and, for a parity-conserving weak link with phase difference  $\phi$ , use the low-energy coupling

$$H_{\text{link}}(\phi) = it \cos(\phi/2) \gamma_L \gamma_R$$

to derive the two branches

$$E_{\pm}(\phi) = \pm t \cos(\phi/2).$$

Show that a  $2\pi$  advance carries a fixed-parity state to the opposite energy branch, so that each fixed-parity branch has  $4\pi$  spectral flow. State separately the quadratic-model, open-boundary, bulk-gap, fermion-parity, quasiparticle-poisoning, coherence-length, temperature, and spectral-resolution assumptions under which tunneling spectroscopy and Josephson measurements test these predictions.

**Exercise 20.10** (Projective characters and the cluster chain). Let  $N \geq 4$  be even and put

$$K_j = Z_{j-1} X_j Z_{j+1}, \quad H_{\text{cl}} = -J \sum_{j=2}^{N-1} K_j, \quad J > 0,$$

on an open chain of  $N$  qubits, where  $X_j$  and  $Z_j$  are the Pauli operators on the  $j$ th tensor factor. Define the onsite  $G = \mathbb{Z}_2 \times \mathbb{Z}_2$  symmetry operators

$$S_e = \prod_{j \text{ even}} X_j, \quad S_o = \prod_{j \text{ odd}} X_j.$$

Prove that the  $K_j$  are independent commuting involutions and that both symmetries commute with every  $K_j$ . Derive the complete spectrum

$$E_m = -J(N-2) + 2Jm, \quad 0 \leq m \leq N-2,$$

with multiplicity  $4 \binom{N-2}{m}$ , including a four-dimensional ground space and gap  $2J$ .

On the ground space prove

$$S_e = Z_1 (Z_{N-1} X_N), \quad S_o = (X_1 Z_2) Z_N.$$

Show that the two left factors anticommute, as do the two right factors, although  $S_e$  and  $S_o$  commute. With

$$U_{(r,s)} = Z^r X^s,$$

derive

$$U_{(r,s)} U_{(r',s')} = (-1)^{sr'} U_{(r+r', s+s')}$$

and prove that each edge realizes the nontrivial projective class whose twisted torus response is  $-1$ . Show that a symmetry-preserving operator supported at one edge acts as a scalar on its irreducible projective doublet. Deduce that the projective multipliers at the two ends cancel in the linear global symmetry action and identify this cancellation as anomaly inflow from the bulk response.

For the irreducible edge representation, define its projective character by

$$\psi_{\omega}(r, s) = \text{Tr } U_{(r,s)}.$$

Compute

$$\psi_\omega(0,0) = 2, \quad \psi_\omega(r,s) = 0 \quad \text{for } (r,s) \neq (0,0),$$

and verify the projective-character orthogonality normalization

$$\frac{1}{4} \sum_{r,s \in \mathbb{Z}_2} |\psi_\omega(r,s)|^2 = 1.$$

Define the four group characters

$$\chi_{p,q}(r,s) = (-1)^{pr+qs}, \quad (p,q), (r,s) \in \mathbb{Z}_2^2.$$

Prove their orthogonality and completeness identities

$$\frac{1}{4} \sum_{r,s \in \mathbb{Z}_2} \chi_{p,q}(r,s) \overline{\chi_{p',q'}(r,s)} = \delta_{p,p'} \delta_{q,q'},$$

$$\frac{1}{4} \sum_{p,q \in \mathbb{Z}_2} \chi_{p,q}(r,s) \overline{\chi_{p,q}(r',s')} = \delta_{r,r'} \delta_{s,s'}.$$

Use the character projectors

$$P_{p,q} = \frac{1}{4} \sum_{r,s \in \mathbb{Z}_2} \chi_{p,q}(r,s) S_e^r S_o^s$$

to prove that every energy  $E_m$  occurs with multiplicity  $\binom{N-2}{m}$  in each symmetry sector. If an operator  $O$  transforms by  $\chi_{u,v}$ , derive the selection rule

$$P_{p,q} O P_{p',q'} = 0 \quad \text{unless} \quad (p,q) = (p' + u, q' + v).$$

For  $2 \leq r \leq s \leq N-1$  with  $r \equiv s \pmod{2}$ , prove the exact string-correlation identity

$$\left\langle Z_{r-1} \left( \prod_{k=0}^{(s-r)/2} X_{r+2k} \right) Z_{s+1} \right\rangle = 1$$

in every ground state. Predict four symmetry-resolved ground levels, the gap  $2J$ , the character selection rule, unit string correlation, and the edge-pulse commutator phase  $\pi$ . State separately the commuting-stabilizer, open-boundary, onsite-symmetry, coupling-calibration, state-preparation, temperature, coherence, symmetry-resolved-readout, and finite-size assumptions under which qubit-array spectroscopy, string measurements, and edge interferometry test these predictions; for small symmetry-preserving perturbations, distinguish exact quantization of the thermodynamic projective class from finite-size edge splitting and nonuniversal deformation of the string correlator.

## 21 Fusion Categories

**Exercise 21.1** (Temperley–Lieb categories and Jones–Wenzl projectors). For  $d \in \mathbb{C}$ , the *Temperley–Lieb category*  $\text{TL}_d$  has objects  $n \in \mathbb{Z}_{\geq 0}$ . The morphism space  $\text{Hom}_{\text{TL}_d}(m, n)$  is the complex span of planar nonintersecting pairings of  $m$  lower and  $n$  upper boundary points. Composition is vertical stacking, with every resulting closed loop removed and replaced by the scalar  $d$ ; tensor product is horizontal juxtaposition and has tensor unit 0. For  $d \in \mathbb{R}$ , reflection across a horizontal line defines  $f^*$ . The cup and cap diagrams give coevaluation and evaluation morphisms.

The *Temperley–Lieb algebra* is  $TL_n(d) = \text{End}_{\mathbf{TL}_d}(n)$ . It is generated by  $U_1, \dots, U_{n-1}$  with

$$U_i^2 = dU_i, \quad U_i U_{i\pm 1} U_i = U_i, \quad U_i U_j = U_j U_i \quad (|i - j| > 1).$$

Its diagram trace is normalized by

$$\text{tr}_0(1) = 1, \quad \text{tr}_{n+1}(x) = d \text{tr}_n(x), \quad \text{tr}_{n+1}(xU_n) = \text{tr}_n(x),$$

where  $x \in TL_n(d)$  is embedded in  $TL_{n+1}(d)$ .

Put

$$\zeta = e^{\pi i/(k+2)}, \quad d = \zeta + \zeta^{-1}, \quad [n]_\zeta = \frac{\zeta^n - \zeta^{-n}}{\zeta - \zeta^{-1}}.$$

With  $f_0 = f_1 = 1$ , define recursively in  $TL_{n+1}(d)$

$$f_{n+1} = f_n - \frac{[n]_\zeta}{[n+1]_\zeta} f_n U_n f_n \quad (1 \leq n \leq k).$$

(a) Use the Temperley–Lieb relations to prove inductively

$$f_n^2 = f_n, \quad f_n^* = f_n, \quad U_i f_n = f_n U_i = 0 \quad (1 \leq i < n), \quad \text{tr}(f_n) = [n+1]_\zeta.$$

(b) Deduce  $\text{tr}(f_{k+1}) = [k+2]_\zeta = 0$ .

**Exercise 21.2** (Temperley–Lieb semisimplification and the positive quotient). At  $d = \zeta + \zeta^{-1}$ , a morphism  $f : m \rightarrow n$  in  $\mathbf{TL}_d$  is negligible when

$$\text{tr}_m(gf) = 0 \quad \text{for every } g : n \rightarrow m.$$

Let  $\mathcal{N}(m, n)$  be the space of negligible morphisms. A tensor ideal  $\mathcal{I}$  is a collection of morphism subspaces closed under composition and tensoring. The quotient has morphisms

$$\text{Hom}_{\mathcal{C}/\mathcal{I}}(X, Y) = \text{Hom}_{\mathcal{C}}(X, Y)/\mathcal{I}(X, Y).$$

The  $C^*$ -idempotent completion has objects  $(X, p)$  with  $p = p^2 = p^*$  and morphisms  $q \text{Hom}(X, Y)p$ . Let  $\mathcal{C}_k$  be the idempotent completion of  $\mathbf{TL}_d/\mathcal{N}$ , and let  $V_a$  be the image of  $f_a$ .

Use as theorem input the root-of-unity Markov-form theorem [43]: the forms

$$\langle f, g \rangle = \text{tr}_m(f^*g), \quad f, g : m \rightarrow n,$$

are positive semidefinite, their radicals form the  $*$ -closed tensor ideal  $\mathcal{N}$  generated by  $f_{k+1}$ , and every negligible idempotent lies in that ideal. Deduce that composition and tensor product descend to positive-definite quotient morphism spaces, that the quotient removes every trace-zero summand, and that its  $C^*$ -idempotent completion has faithful positive categorical trace.

**Exercise 21.3** (Truncated fusion). Derive

$$V_a \otimes V_b \cong \bigoplus_{\substack{|a-b| \leq c \leq \min(a+b, 2k-a-b) \\ a+b+c \equiv 0 \pmod{2}}} V_c$$

and

$$d_a = [a+1]_\zeta = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))}.$$

**Exercise 21.4** (Unitary fusion structure). A nonzero object is simple when its endomorphism algebra is the scalar multiples of its identity. A  $C^*$ -tensor category is rigid when every object has a conjugate.

A  $C^*$ -category is semisimple when every object is a finite orthogonal direct sum of simple objects. A unitary fusion category is a rigid semisimple  $C^*$ -tensor category with finite-dimensional morphism spaces and finitely many simple objects up to unitary isomorphism.

Use left composition on the positive quotient spaces and diagram reflection to equip  $\mathcal{C}_k$  with  $C^*$ -norms. Use the cup and cap diagrams to construct standard conjugates. Prove from truncated fusion that every simple object is unitarily isomorphic to exactly one of

$$V_0, V_1, \dots, V_k,$$

and deduce that  $\mathcal{C}_k$  is a unitary fusion category. Prove that its standard categorical trace agrees with the quotient diagram trace and that the recoupling maps between simple-summand decompositions are unitary and satisfy the pentagon identity. With  $\zeta^{1/2} = e^{\pi i/(2(k+2))}$ , prove that

$$c_i = \zeta^{1/2} \text{id} - \zeta^{-1/2} U_i$$

is unitary, satisfies the braid relations, preserves  $\mathcal{N}$ , and descends to a unitary braiding on  $\mathcal{C}_k$ .

**Exercise 21.5** (Morita equivalence, Drinfeld centers, and the cobordism hypothesis). Let  $\mathcal{C}$  and  $\mathcal{D}$  be unitary fusion categories. A  $\mathcal{C}$ – $\mathcal{D}$  *bimodule category* is a finite semisimple  $C^*$ -category  $\mathcal{M}$  with commuting unitary left  $\mathcal{C}$ - and right  $\mathcal{D}$ -actions and unitary associativity and unit constraints satisfying the pentagon and triangle identities. For a  $\mathcal{C}$ – $\mathcal{D}$  bimodule  $\mathcal{M}$  and a  $\mathcal{D}$ – $\mathcal{E}$  bimodule  $\mathcal{N}$ , define their *relative tensor product* by the universal property

$$\text{Fun}_{\mathcal{C}\text{--}\mathcal{E}}(\mathcal{M} \boxtimes_{\mathcal{D}} \mathcal{N}, \mathcal{A}) \cong \text{Fun}_{\mathcal{D}\text{--bal}}(\mathcal{M} \times \mathcal{N}, \mathcal{A})$$

for every  $\mathcal{C}$ – $\mathcal{E}$  module category  $\mathcal{A}$ . Call  $\mathcal{C}$  and  $\mathcal{D}$  *Morita equivalent* when there is a  $\mathcal{C}$ – $\mathcal{D}$  bimodule  $\mathcal{M}$  with a  $\mathcal{D}$ – $\mathcal{C}$  bimodule  $\mathcal{M}^\vee$  and equivalences

$$\mathcal{M} \boxtimes_{\mathcal{D}} \mathcal{M}^\vee \simeq \mathcal{C}, \quad \mathcal{M}^\vee \boxtimes_{\mathcal{C}} \mathcal{M} \simeq \mathcal{D}$$

of regular bimodule categories. Use the universal property and associativity of relative tensor products to prove that Morita equivalence is an equivalence relation.

The *Drinfeld center*  $Z(\mathcal{C})$  has objects  $(X, \gamma)$ , where  $X \in \mathcal{C}$  and  $\gamma_Y : X \otimes Y \rightarrow Y \otimes X$  is a unitary natural isomorphism satisfying

$$\gamma_1 = \text{id}_X, \quad \gamma_{Y \otimes Z} = (\text{id}_Y \otimes \gamma_Z)(\gamma_Y \otimes \text{id}_Z),$$

with associators understood; its morphisms intertwine the half-braidings. Construct the tensor product and braiding in  $Z(\mathcal{C})$ . Use as theorem input [15, 43] that  $Z(\mathcal{C})$  is a unitary modular tensor category and that

$$\mathcal{C} \simeq_{\text{Morita}} \mathcal{D} \iff Z(\mathcal{C}) \simeq Z(\mathcal{D})$$

by a braided unitary equivalence. Deduce that the center is an invariant of the bulk rather than of a chosen boundary presentation.

Define the symmetric monoidal Morita 3-category **Fus** to have unitary fusion categories as objects, finite semisimple bimodule categories as 1-morphisms, module functors as 2-morphisms, and natural transformations as 3-morphisms, with Deligne tensor product as its monoidal product. An object in

a symmetric monoidal 3-category is *fully dualizable* when it has a tensor dual and its evaluation, coevaluation, and successive adjunction data admit left and right adjoints through level three. Use as theorem input [13] that every unitary fusion category is fully dualizable in **Fus**.

Let  $\text{Bord}_3^{\text{fr}}$  be the symmetric monoidal 3-category of compact framed manifolds and bordisms with corners through dimension three. A *fully extended framed 3-dimensional topological field theory* valued in **Fus** is a symmetric monoidal 3-functor

$$\mathcal{Z} : \text{Bord}_3^{\text{fr}} \longrightarrow \mathbf{Fus}.$$

Use as theorem input the cobordism hypothesis [32]: evaluation on the positively framed point identifies the 3-groupoid of such theories with the 3-groupoid of fully dualizable objects of **Fus**. Use the canonical spherical structure of a unitary fusion category as theorem input to obtain its oriented Turaev–Viro theory, and use as theorem input that its bulk line operators form  $Z(\mathcal{C})$ .

Let  $\text{Pic}(\mathbf{Fus})$  be the sub-3-groupoid of tensor-invertible objects and invertible morphisms. Prove that the cobordism hypothesis restricts on  $\text{Pic}(\mathbf{Fus})$  to the classification of invertible theories, whereas all fully dualizable objects classify the non-invertible theories as well. Define the global dimension, with  $\text{Irr}(\mathcal{C})$  the set of unitary isomorphism classes of simple objects, by

$$\text{Dim}(\mathcal{C}) = \sum_{[X] \in \text{Irr}(\mathcal{C})} d_X^2.$$

Use as theorem input

$$\text{Dim } Z(\mathcal{C}) = \text{Dim}(\mathcal{C})^2, \quad \text{Dim}(\mathcal{C} \boxtimes \mathcal{D}) = \text{Dim}(\mathcal{C}) \text{Dim}(\mathcal{D}),$$

and prove that a unitary fusion category is tensor-invertible in **Fus** only when it is Morita equivalent to  $\text{Hilb}_{\text{fd}}$ , the unitary fusion category of finite-dimensional Hilbert spaces. For  $\mathcal{C}_k$ , derive

$$\text{Dim}(\mathcal{C}_k) = \sum_{a=0}^k d_a^2 = \frac{k+2}{2 \sin^2(\pi/(k+2))} > 1 \quad (k \geq 1),$$

and conclude that its associated state-sum phase is non-invertible. State explicitly that this classifies the fully extended topological low-energy theories and does not by itself prove a classification of all microscopic gapped Hamiltonians.

**Exercise 21.6** (Categorical Fourier transform, modular data, and Verlinde diagonalization). For  $0 \leq a, b \leq k$ , define

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \quad \theta_a = \exp\left(\frac{\pi i a(a+2)}{2(k+2)}\right).$$

Let  $N_{ab}{}^c$  be the multiplicity of  $V_c$  in  $V_a \otimes V_b$ .

Use sine orthogonality to prove that  $S$  is real, symmetric, and unitary. For each  $a$ , let  $(N_a)_{bc} = N_{ab}{}^c$ . Use the truncated Chebyshev recursion to prove

$$N_a = S \text{diag}\left(\frac{S_{ax}}{S_{0x}}\right)_{x=0}^k S^{-1}$$

and hence

$$N_{ab}{}^c = \sum_{x=0}^k \frac{S_{ax} S_{bx} \overline{S_{cx}}}{S_{0x}}.$$

For functions  $f, g$  on the simple-object labels, define fusion convolution and the categorical Fourier transform by

$$(f *_{\text{fus}} g)(c) = \sum_{a,b=0}^k f(a)g(b)N_{ab}^c, \quad \widehat{f}(x) = \sum_{a=0}^k f(a) \frac{S_{ax}}{S_{0x}}.$$

Prove the convolution, inversion, and Plancherel identities

$$\widehat{f *_{\text{fus}} g}(x) = \widehat{f}(x)\widehat{g}(x), \quad f(a) = \sum_{x=0}^k S_{ax}S_{0x}\widehat{f}(x),$$

$$\sum_{a=0}^k |f(a)|^2 = \sum_{x=0}^k |S_{0x}|^2 |\widehat{f}(x)|^2.$$

Deduce that the number of fusion channels from  $a_1, \dots, a_n$  to  $c$  is  $(N_{a_1} \cdots N_{a_n})_{0c}$  and express it as a finite sum of products of sine functions. Use as theorem input the quantum-trace evaluation of the Hopf link to identify the displayed matrix  $S$  with the categorical  $S$ -matrix and conclude from its unitarity that the braiding is nondegenerate. Let  $\mathcal{C}_k^{\text{rev}}$  have the inverse braiding. Use as theorem input the canonical braided equivalence

$$\mathcal{C}_k \boxtimes \mathcal{C}_k^{\text{rev}} \simeq Z(\mathcal{C}_k)$$

and deduce that the simple bulk charges are indexed by pairs  $(x, y)$  with  $0 \leq x, y \leq k$ . Conclude that the Turaev–Viro state space of the torus has dimension  $(k+1)^2$ .

**Exercise 21.7** (Levin–Wen string-net Hamiltonians). Let  $\Gamma$  be a trivalent cellulation of a closed oriented surface and let  $\mathcal{C}_k$  be the unitary fusion category with simple objects  $V_0, \dots, V_k$ , fusion multiplicities  $N_{ab}^c$ , standard categorical trace, and quantum dimensions  $d_a$ . Write  $a^\vee$  for the unique label satisfying  $\overline{V_a} \cong V_{a^\vee}$ . Put

$$\mathcal{H}_\Gamma = \ell^2(\{0, \dots, k\}^{E(\Gamma)}), \quad \mathcal{D}^2 = \sum_{a=0}^k d_a^2.$$

For each vertex  $v$ , let  $Q_v$  be the projection onto edge labelings admissible under the fusion rules. For a face  $p$ , let  $B_p^a$  insert a coevaluation–evaluation loop labelled  $V_a$  inside  $p$  and resolve it into the boundary edges by the associator and the quantum recoupling maps; set it equal to zero off the subspace on which the vertices incident to  $p$  are admissible. Define

$$B_p = \frac{1}{\mathcal{D}^2} \sum_{a=0}^k d_a B_p^a$$

and, for  $\Delta_v, \Delta_p > 0$ ,

$$\widehat{H}_\Gamma = \Delta_v \sum_v (I - Q_v) + \Delta_p \sum_p (I - B_p).$$

Use the snake identities, the standard categorical trace, and the quantum recoupling pentagon to prove

$$(B_p^a)^* = B_p^{a^\vee}, \quad B_p^a B_p^b = \sum_c N_{ab}^c B_p^c.$$

Use

$$d_a d_b = \sum_c N_{ab}^c d_c$$



to prove

$$Q_v^2 = Q_v = Q_v^*, \quad B_p^2 = B_p = B_p^*, \quad [Q_v, B_p] = [B_p, B_{p'}] = 0.$$

Deduce that  $\hat{H}_\Gamma$  is positive and frustration-free and that its ground space is the simultaneous  $+1$  space of all  $Q_v$  and  $B_p$ . Prove that recoupling across an edge gives a unitary identification of the ground spaces before and after the corresponding local change of cellulation.

For  $k = 1$ , identify the labels with  $\mathbb{Z}/2\mathbb{Z}$ , let  $Z_e$  act on a label  $a_e$  by  $(-1)^{a_e}$ , and let  $X_e$  reverse that label. Show that

$$Q_v = \frac{1}{2} \left( I + \prod_{e \ni v} Z_e \right), \quad B_p = \frac{1}{2} \left( I + \prod_{e \subset \partial p} X_e \right).$$

On a torus, count the two independent relations among the vertex and face constraints and prove that the ground-state degeneracy is 4. Identify the two pairs of noncontractible string operators and their mutual sign, and state the boundary conditions and gap-preserving perturbation regime under which the fourfold low-energy multiplicity is a measurable prediction of a qubit realization.

**Exercise 21.8** (String-net loop operators and spectroscopy). On a torus let  $W_a^{\alpha,+}$  and  $W_a^{\alpha,-}$  insert a noncontractible  $V_a$  loop along a cycle  $\alpha$ , using respectively the braiding and inverse braiding of  $\mathcal{C}_k$  when the loop is resolved through the string net. Define  $W_a^{\beta,+}$  and  $W_a^{\beta,-}$  for a cycle  $\beta$  with intersection number one with  $\alpha$ .

Use as theorem input [31] the Levin–Wen–Turaev–Viro identification: local recoupling identifies the string-net ground space with the Turaev–Viro state space of the torus. Use the central equivalence to identify its one-dimensional sectors with pairs  $(x, y)$  such that  $0 \leq x, y \leq k$ , and recover

$$\dim \ker \hat{H}_\Gamma = (k+1)^2.$$

Prove the loop fusion relations

$$W_a^{\alpha,\pm} W_b^{\alpha,\pm} = \sum_c N_{ab}^c W_c^{\alpha,\pm}$$

and use Verlinde diagonalization to show that their eigenvalues in the sector  $(x, y)$  are

$$W_a^{\alpha,+} : \frac{S_{ax}}{S_{0x}}, \quad W_a^{\alpha,-} : \frac{\overline{S_{ay}}}{S_{0y}}.$$

For a weak calibrated loop perturbation

$$\hat{V} = - \sum_a \left( \epsilon_a^+ W_a^{\alpha,+} + \epsilon_a^- W_a^{\alpha,-} + \text{adjoints} \right),$$

derive the first-order splitting

$$\delta E_{x,y} = -2 \operatorname{Re} \sum_a \left( \epsilon_a^+ \frac{S_{ax}}{S_{0x}} + \epsilon_a^- \frac{\overline{S_{ay}}}{S_{0y}} \right).$$

Insert the sine formula for  $S$  and list the distinct splittings for a chosen  $k$ . State the topological-gap, system-size, loop-support, calibration, and sector-resolution assumptions under which these spectral lines test the doubled  $SU(2)_k$  string-net model.

**Exercise 21.9** (Anyon interference and fusion counting). Prove that the phases  $\theta_a$  define a ribbon twist on  $\mathcal{C}_k$ . On every simple fusion channel  $V_c \subseteq V_a \otimes V_b$ , prove the balancing identity

$$c_{V_b, V_a} c_{V_a, V_b} |_{V_c} = \frac{\theta_c}{\theta_a \theta_b} \text{id}_{V_c}.$$

For simple charges  $a, b$ , prove that the normalized monodromy amplitude is

$$M_{ab} = \frac{S_{ab} S_{00}}{S_{0a} S_{0b}}.$$

Identify

$$\frac{S_{ab}}{S_{0b}} = U_a \left( \cos \frac{(b+1)\pi}{k+2} \right)$$

with the specialized  $SU(2)$  Weyl character, where  $U_a$  is the Chebyshev family. In a two-path anyon interferometer with path intensities  $I_1, I_2$ , derive

$$I_b(\phi) = I_1 + I_2 + 2\sqrt{I_1 I_2} \Re(e^{i\phi} M_{ab}).$$

Prove that a calibrated full framing twist of the probe path replaces  $M_{ab}$  by  $\theta_a M_{ab}$  and shifts the interference phase by  $\arg \theta_a$ . Insert the sine formula and predict the visibility and phase for every enclosed charge. For a collection of prepared charges, use the Verlinde formula to predict the exact number of fusion outcomes in each total-charge sector. Identify indistinguishable charges for one probe and prove that the full family of probes distinguishes them. State the coherence, topological-gap, charge-preparation, and readout assumptions separately from the predicted fringes and integer state counts.

## References

- [1] M. F. Atiyah, *K-Theory*, W. A. Benjamin, 1967.
- [2] M. F. Atiyah and I. M. Singer, *The index of elliptic operators I*, Annals of Mathematics 87 (1968), 484–530. doi:10.2307/1970715.
- [3] M. F. Atiyah and G. B. Segal, *The index of elliptic operators II*, Annals of Mathematics 87 (1968), 531–545. doi:10.2307/1970716.
- [4] M. F. Atiyah, V. K. Patodi, and I. M. Singer, *Spectral asymmetry and Riemannian geometry I*, Mathematical Proceedings of the Cambridge Philosophical Society 77 (1975), 43–69. doi:10.1017/S0305004100049410.
- [5] E. D. Belokolos, A. I. Bobenko, V. Z. Enolskii, A. R. Its, and V. B. Matveev, *Algebro-Geometric Approach to Nonlinear Integrable Equations*, Springer, 1994.
- [6] R. E. Borcherds, *Monstrous moonshine and monstrous Lie superalgebras*, Inventiones Mathematicae 109 (1992), 405–444. doi:10.1007/BF01232032.
- [7] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology*, Graduate Texts in Mathematics 82, Springer, 1982.
- [8] S. Chandrasekhar, *The Mathematical Theory of Black Holes*, Clarendon Press, 1983.

- [9] X. Chen, Z.-C. Gu, Z.-X. Liu, and X.-G. Wen, *Symmetry protected topological orders and the group cohomology of their symmetry group*, Physical Review B 87 (2013), 155114. doi:10.1103/PhysRevB.87.155114.
- [10] I. Cherednik, *Double Affine Hecke Algebras*, London Mathematical Society Lecture Note Series 319, Cambridge University Press, 2005.
- [11] J. B. Conway, *A Course in Functional Analysis*, 2nd ed., Graduate Texts in Mathematics 96, Springer, 1990.
- [12] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der mathematischen Wissenschaften 290, Springer, 1999.
- [13] C. L. Douglas, C. Schommer-Pries, and N. Snyder, *Dualizable tensor categories*, Memoirs of the American Mathematical Society 268 (2020), no. 1308. doi:10.1090/memo/1308.
- [14] T. Eguchi, H. Ooguri, A. Taormina, and S.-K. Yang, *Superconformal algebras and string compactification on manifolds with  $SU(n)$  holonomy*, Nuclear Physics B 315 (1989), 193–221. doi:10.1016/0550-3213(89)90454-9.
- [15] P. Etingof, D. Nikshych, and V. Ostrik, *On fusion categories*, Annals of Mathematics 162 (2005), 581–642. doi:10.4007/annals.2005.162.581.
- [16] J. D. Fay, *Theta Functions on Riemann Surfaces*, Lecture Notes in Mathematics 352, Springer, 1973.
- [17] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.
- [18] G. B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999.
- [19] O. Forster, *Lectures on Riemann Surfaces*, Graduate Texts in Mathematics 81, Springer, 1981.
- [20] D. S. Freed and M. J. Hopkins, *Reflection positivity and invertible topological phases*, Geometry & Topology 25 (2021), 1165–1330. doi:10.2140/gt.2021.25.1165.
- [21] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, 1991.
- [22] W. M. Goldman, *Invariant functions on Lie groups and Hamiltonian flows of surface group representations*, Inventiones Mathematicae 85 (1986), 263–302. doi:10.1007/BF01389091.
- [23] G. H. Hardy and S. Ramanujan, *Asymptotic formulae in combinatory analysis*, Proceedings of the London Mathematical Society (2) 17 (1918), 75–115. doi:10.1112/plms/s2-17.1.75.
- [24] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977.
- [25] F. Hirzebruch, *Topological Methods in Algebraic Geometry*, 3rd ed., Grundlehren der mathematischen Wissenschaften 131, Springer, 1966. doi:10.1007/978-3-642-62018-8.
- [26] L. Hörmander, *The Analysis of Linear Partial Differential Operators III*, Grundlehren der mathematischen Wissenschaften 274, Springer, 1985.
- [27] A. Y. Kitaev, *Unpaired Majorana fermions in quantum wires*, Physics-Uspekhi 44 (2001), 131–136. doi:10.1070/1063-7869/44/10S/S29.

- [28] T. H. Koornwinder, *Askey–Wilson polynomials for root systems of type BC*, Contemporary Mathematics 138 (1992), 189–204. doi:10.1090/conm/138/1199128.
- [29] V. E. Korepin, N. M. Bogoliubov, and A. G. Izergin, *Quantum Inverse Scattering Method and Correlation Functions*, Cambridge University Press, 1993.
- [30] H. B. Lawson, Jr. and M.-L. Michelsohn, *Spin Geometry*, Princeton University Press, 1989.
- [31] M. A. Levin and X.-G. Wen, *String-net condensation: A physical mechanism for topological phases*, Physical Review B 71 (2005), 045110. doi:10.1103/PhysRevB.71.045110.
- [32] J. Lurie, *On the classification of topological field theories*, Current Developments in Mathematics 2008 (2009), 129–280. doi:10.4310/CDM.2008.v2008.n1.a3.
- [33] G. W. Mackey, *Induced representations of locally compact groups I*, Annals of Mathematics 55 (1952), 101–139. doi:10.2307/1969423.
- [34] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Texts in Applied Mathematics 17, Springer, 1999.
- [35] F. Pollmann and A. M. Turner, *Detection of symmetry-protected topological phases in one dimension*, Physical Review B 86 (2012), 125441. doi:10.1103/PhysRevB.86.125441.
- [36] M. Reed and B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*, Academic Press, 1975.
- [37] T. Regge and J. A. Wheeler, *Stability of a Schwarzschild singularity*, Physical Review 108 (1957), 1063–1069. doi:10.1103/PhysRev.108.1063.
- [38] E. Seneta, *Non-negative Matrices and Markov Chains*, 2nd ed., Springer Series in Statistics, Springer, 1981.
- [39] V. P. Spiridonov, *On the elliptic beta function*, Russian Mathematical Surveys 56 (2001), 185–186. doi:10.1070/RM2001v056n01ABEH000374.
- [40] M. E. Taylor, *Partial Differential Equations I: Basic Theory*, 2nd ed., Applied Mathematical Sciences 115, Springer, 2011.
- [41] R. Thom, *Quelques propriétés globales des variétés différentiables*, Commentarii Mathematici Helvetici 28 (1954), 17–86. doi:10.1007/BF02566923.
- [42] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, *Quantized Hall conductance in a two-dimensional periodic potential*, Physical Review Letters 49 (1982), 405–408. doi:10.1103/PhysRevLett.49.405.
- [43] V. G. Turaev, *Quantum Invariants of Knots and 3-Manifolds*, de Gruyter Studies in Mathematics 18, de Gruyter, 1994.
- [44] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.
- [45] E. P. Wigner, *On unitary representations of the inhomogeneous Lorentz group*, Annals of Mathematics 40 (1939), 149–204. doi:10.2307/1968551.
- [46] E. Witten, *Elliptic genera and quantum field theory*, Communications in Mathematical Physics 109 (1987), 525–536. doi:10.1007/BF01208956.

- [47] N. M. J. Woodhouse, *Geometric Quantization*, 2nd ed., Clarendon Press, 1992.
- [48] S. L. Woronowicz, *Compact matrix pseudogroups*, Communications in Mathematical Physics 111 (1987), 613–665. doi:10.1007/BF01219077.